AFTER FRAME

RC11_2025/2026:
ODD-TWINS

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SPACE FOLDING

SPATIAL EDITING

MARKOV CHAINS

KULESHOV EFFECT

BRIEF

This project begins with the idea of Rotterdam as an edited city. Rather than reading the city as a continuous whole, it understands Rotterdam as a spatial montage shaped by destruction, reconstruction, infrastructure and fragmented urban conditions. Drawing on Tschumi, the Kuleshov Effect and creative geography, the project explores how meaning emerges through sequence, contrast and transition.

A film database is divided into individual clips, which are classified by Scene Category, Basic Component, Colour, Shot Size, Movement and Duration. These labels are used to train Markov Chains, which learn transition probabilities from the original films and generate new sequences. A further curatorial logic guides the selection, branching and organisation of these sequences.

The generated clip labels are then translated into spatial parameters. Using one 50-clip sequence as an example, each clip first becomes a basic spatial potential before consecutive clips with the same Scene Category are grouped into segments. Through the spatial logic of the Kuleshov Effect, relationships between clips and segments transform their geometry, connections, openness and internal organisation. The resulting skeleton is further architecturalised through walls, glazing, openings and architectural elements, while materials extracted from the original film clips are applied back to the model.

Finally, the system is tested across 12 sites in Rotterdam selected through the network of the International Film Festival Rotterdam (IFFR). Different curated sequences are deployed across these locations, forming a distributed urban field that connects film, computation, curation and architectural design.

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CHAPTER 01

Project Origin: Starting from Film

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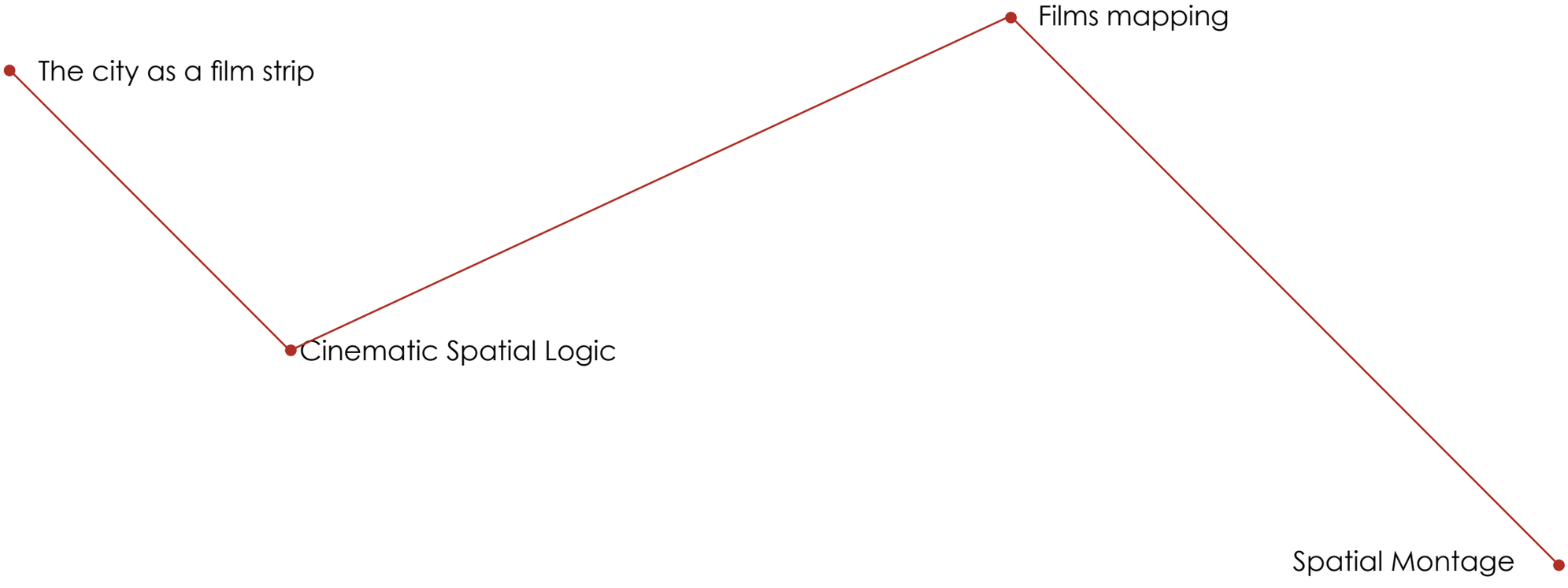


The city as a film strip

We chose cinematic editing as our entry point because our Term 1 shared databases already included film and documentary materials. More importantly, Rotterdam itself resembles an edited city: rather than growing organically, it was assembled, cut, and reconstructed through history.

We have chosen to use cinematic editing techniques as our primary point of entry. This decision is rooted in the fact that our shared databases from the term 1 already incorporated film and documentary materials. Furthermore, there is a profound structural similarity between cinematic editing and the historical development of Rotterdam.

Rotterdam is not a city that "grew" organically; rather, it is a city that was "edited" into existence.

This perspective is based on the following:

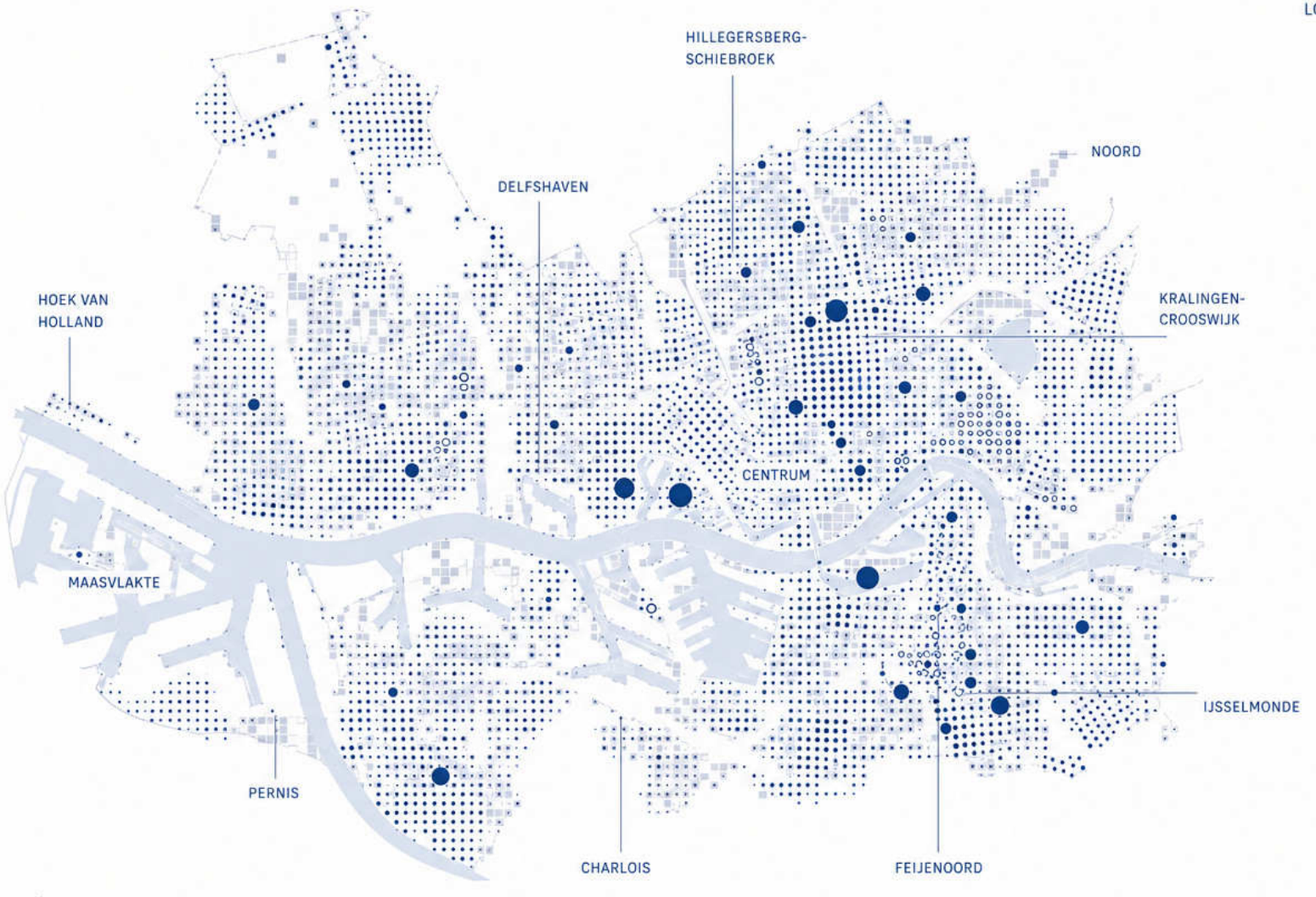
Historical Rupture: The war in 1940 caused a total fracture in the city's history. This destruction forced the city to abandon its original, natural evolution.

Spatial Montage: The post-war reconstruction process was, in essence, a form of spatial montage. Urban planners took unrelated architectural fragments, avant-garde infrastructure, and radical experimental housing, and "spliced" them together into the current city.

By adopting cinematic editing as our lens, we are no longer merely reacting to the existing characteristics of Rotterdam. Instead, we assume the role of the "Director," actively re-editing and re-composing the inherent meaning of these urban fragments.

FILMED IN ROTTERDAM

LOCATION OF FILMS SHOT IN ROTTERDAM



NUMBER OF FILMS SHOT IN THIS AREA

- 9746—4515
- 4515—672
- 671—277
- 119—42
- 41—1

AREA WITH DECREASE (COMPARED TO 2010–2015)

- 4514—1672
- 1671—812
- 811—214
- 276—120
- 213—1

NO CHANGE

This map shows the number of films shot in different areas of Rotterdam (2016–2023).

ROTTERDAM.
MAAKT
HET.
een film over liefde,
verlies en de stad
die altijd doorgaat.
51°55'N 4°29'E
ROTTERDAM
licht.
schaduw,
water,
wind.
Rotterdam.
DE AANVAL (1961)
EEN FILM VAN
FONS RADEMAKERS
DE AANVAL
SCENE
7
TAKE
2
DIRECTOR: FONS RADEMAKERS
CAMERA:
DATE: 1961
NEDERLAND
WEDEROPBOUW
VERNIEUWING
MENSEN
NL FILM
MAASHAVEN

CHAPTER 02

From Montage to Effect

→ → → →

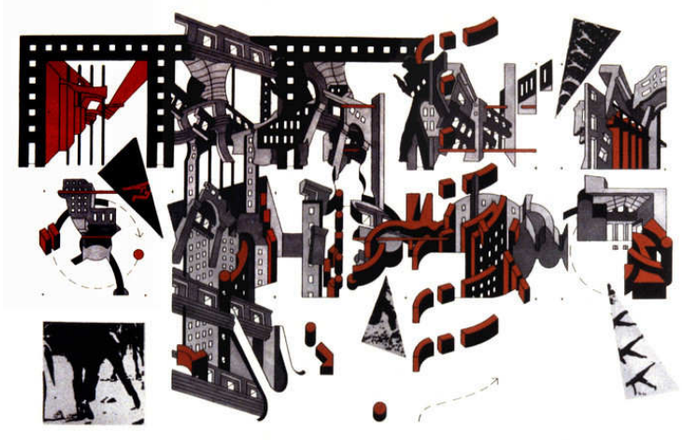
Cinematic Sequence

Bernard Tschumi
The Manhattan Transcripts
1976–1981

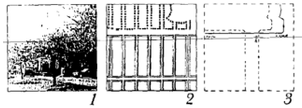
"Architecture is not simply about space and form, but also about event, action, and what happens in space."

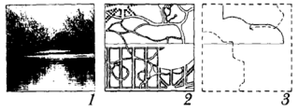
The Manhattan Transcripts differ from most architectural drawings insofar as they are neither real projects nor mere fantasies. Developed in the late 1970s, they proposed an architectural interpretation of reality. To achieve this, they employed a particular structure involving photographs that direct or "witness" events, while plans, sections, and diagrams outline spaces and indicate the movements of different protagonists within the architectural "stage set."

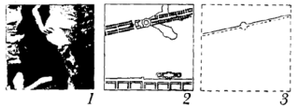
The explicit purpose of the Transcripts was to represent what is normally absent from conventional architectural drawing: the complex relationship between spaces and their use, between the set and the script, between "type" and "program," and between objects and events. Their implicit purpose, meanwhile, was closely related to the 20th-century city.



In this work, Tschumi abandoned traditional architectural plans in favor of cinematic storyboarding. He decomposed space into three parallel levels: Photographs (representing Events), Plan Sections (representing Physical Space), and Vector Lines (representing Human Movement).





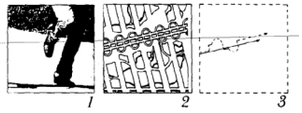


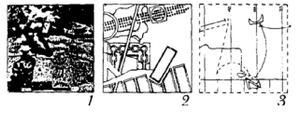
From Keyframes to Space:

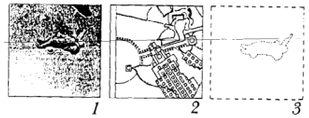
Tschumi transformed architectural design from static representation into temporal narrative. In The Manhattan Transcripts, each page works like a cinematic frame, recording space through events and movement. Architecture is no longer a fixed plan, but something unfolding through time. Similar to my project, spatial meaning is produced through the sequencing and reordering of visual fragments. Keyframes are therefore not just images, but narrative units that reconstruct space through montage and transition.

Action Editing:

He showed that spatial meaning is shaped through human movement and the sequencing of actions in space. Like a Point-of-View shot in film, space becomes not just a setting but a narrative device that directs perception and experience. Architecture is therefore understood through events, trajectories, and transitions, where movement actively "edits" space and produces meaning over time rather than through form alone.

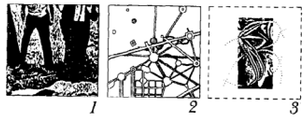


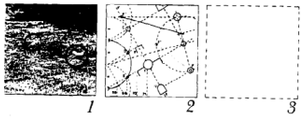




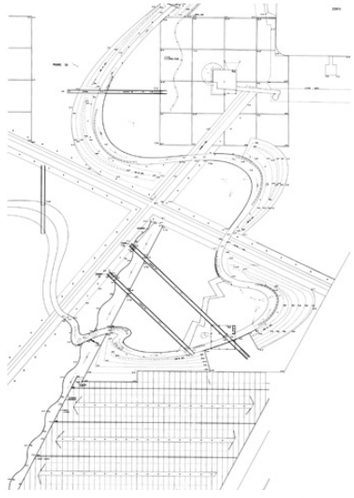
Parallel Narrative:

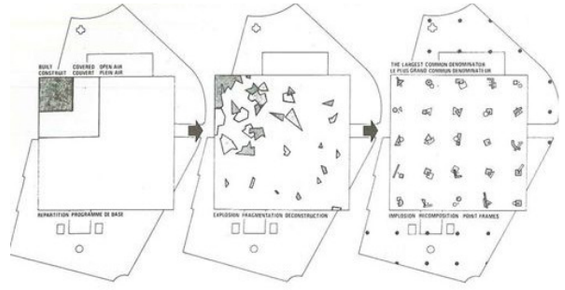
He showed that spatial meaning can emerge through parallel structures of events, spaces, and movements. By aligning these layers on the same timeline, he created a logic similar to cinematic cross-cutting, where separate elements begin to generate new relationships. Architecture is therefore read not as a single continuous form, but as a constructed narrative, in which parallel sequences produce association, tension, and meaning across time.

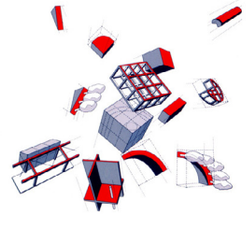


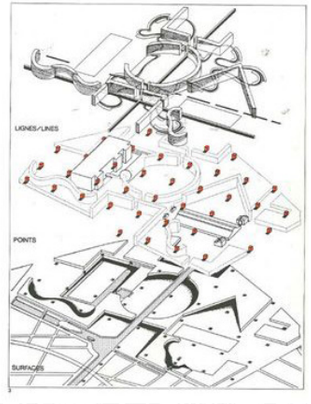


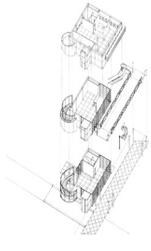
Parc de la Villette
Paris 1982-1998

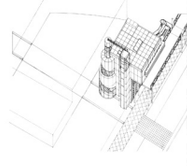


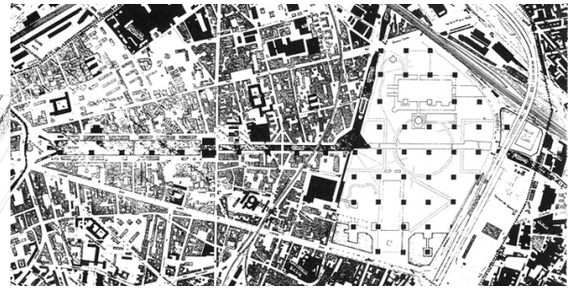












Superimposition: Tschumi forcibly overlaid three entirely independent systems: Points (the red Folies), Lines (the paths), and Surfaces (the lawns). While these systems do not interfere with one another, their intersections produce unexpected programmatic conflicts.

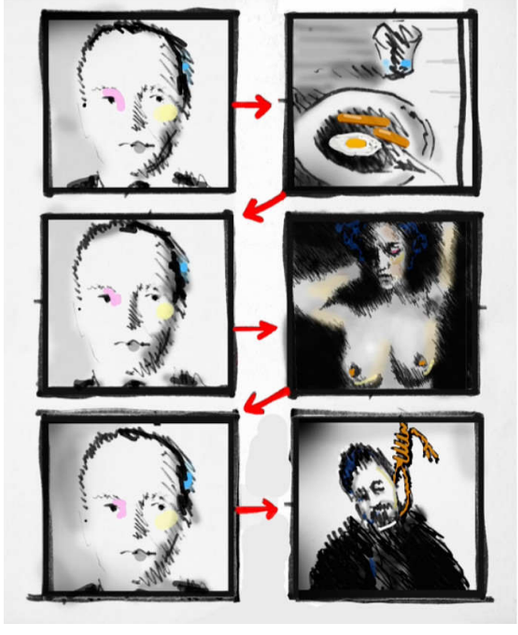
Neutral Nodes (Folies): The grid of red buildings, or Folies, are inherently "expressionless." Their meaning is entirely dependent on the context into which they are "edited" (e.g., a Folie by the water becomes a laundry/pier; one by the grass becomes a pavilion).
Spatial Montage: By rejecting the continuity of the landscape in favor of superimposition, Tschumi created a spatial version of double exposure or cinematic juxtaposition, using the forced overlap of different systems to manufacture conflict.

Architecturalization of the Kuleshov Effect: The red Folie plays the role of the "neutral face" from the Kuleshov experiment. It possesses no fixed meaning; its emotional and functional attributes depend entirely on the environment with which it is juxtaposed.

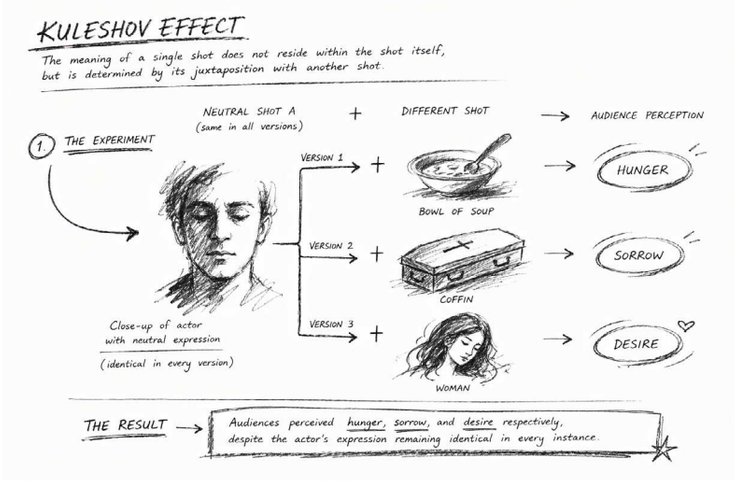
Timeline Editing: The walking paths throughout the park are designed as a Cinematic Promenade. This acts as a timeline that strings together a series of discontinuous "frames," turning the visitor's experience into a real-time "film playback".

The Kuleshov Effect

Director Lev Kuleshov edited a close-up of an actor with a completely neutral expression (Neutral Shot A).



He juxtaposed Shot A with three different images: a bowl of soup, a coffin, and a woman.



The Result: Audiences perceived hunger, sorrow, and desire respectively, despite the actor's expression remaining identical in every instance.

Core Methodology:

1 + 1 = X

In cinematic editing, the resulting meaning (Output C) exists neither in Shot A nor in Shot B. Instead, it is the third meaning generated by the viewer's brain as it connects these two fragments.

This effect explains how editors or researchers can manipulate and innovate new emotional narratives by simply altering the combination and sequence of shots.

The most central logic of this theory is 1 + 1 = X. It teaches us that when we place two unrelated images together, a third, entirely new narrative is produced.

This book is not a purely theoretical work, but rather a text popularizing and disciplining film structure.

That document presented several well-known viewpoints:

Kuleshov understands a single shot as a shot-sign, which can be imagined as words grammatically organized into a larger structure. Sequences and segments can be compared to sentences. That is, viewing film as a building block constructed from shot units.

The audience's understanding or perception of the meaning of shots comes more from the relationships between shots than from the details of the actors themselves.

Creative geography.

Creative Geography

Creative geography is a valuable perspective, roughly meaning that through editing, two physically disparate places are stitched together into a continuous space in the viewer's mind.

This led to a series of experiments. The space in film is not recorded by the camera, but created by the editor.

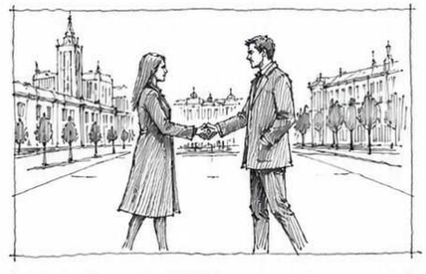
Scene 1: The female actor walks on Petro Street in Moscow. The male actor walks on the riverbank two miles away.
Scene 2: They see each other, smile, and then approach.
Scene 3: The two meet and shake hands. Filmed in another part of the city.
Scene 4: Then cuts to a shot of the White House from an American film.
Scene 5: Cuts back to Moscow, finally showing them climbing the steps of the Cathedral of Christ the Saviour.

Through editing, it is presented as climbing the steps of the White House. The footage of the two meeting is edited together, with different streets in Moscow and the White House in the background. The viewer's brain will automatically ignore the fact that these scenes are thousands of miles apart in reality, and firmly believe that they are happening in the same square.

Scene 1:
The female actor walks on Petro Street in Moscow.

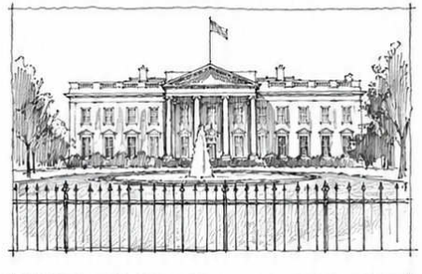
The male actor walks on the riverbank two miles away.

Scene 2:
They see each other, smile, and then approach.



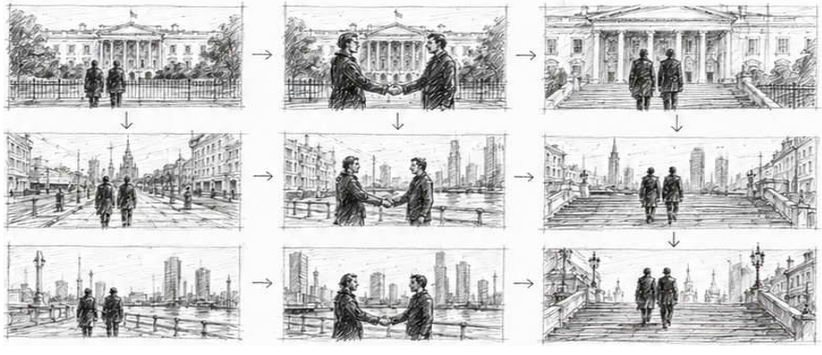
Scene 3:
The two meet and shake hands.

Filmed in another part of the city.



Scene 4:
Then cuts to a shot of the White House from an American film.

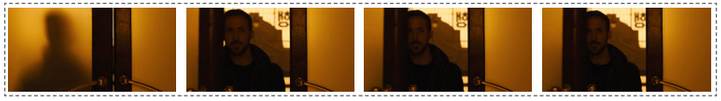
Scene 5:
Cuts back to Moscow, finally showing them climbing the steps of the Cathedral of Christ the Saviour.



Through editing, it is presented as climbing the steps of the White House. The footage of the two meeting is edited together, with different streets in Moscow and the White House in the background. The viewer's brain will automatically ignore the fact that these scenes are thousands of miles apart in reality, and firmly believe that they are happening in the same square.

Examples of the Kuryushov effect

***Scene A**: The man opens the door.*



Scene B:

Scene B:

1.The aftermath of the explosion evokes images of a recent battle or a kidnapping.

2.The presence of an unconscious person in the room creates a chilling and eerie atmosphere, instantly shifting the overall mood of the scene.

3.A man turns and smiles, making the audience feel as if they've encountered an old friend.

Examples of the Kuryushov effect

***Scene A**: The man*

Scene B:

Scene B:

1.Witnessing a subway accident prompts the audience to wonder if it's related to the man.
2.It could be a very romantic encounter.
3.The man witnessing robbers rushing into the bank evokes feelings of tension and unease.

Examples of the Kuryushov effect

***Scene A**: What is the old man looking at?*

Scene B:

Scene B:

1.A well-assembled armed force; the camera switch connects these two groups, making the viewer feel that the old man might be their target or their leader.
2.Encountering an acquaintance evokes a warm feeling.
3.The old man seeing everything around him exploding makes the viewer feel that this is a fantastical space, possibly a dream.

***Scene A**: The man looks up.*

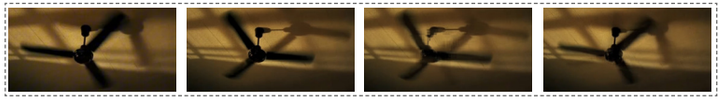
Scene B:

Scene B:

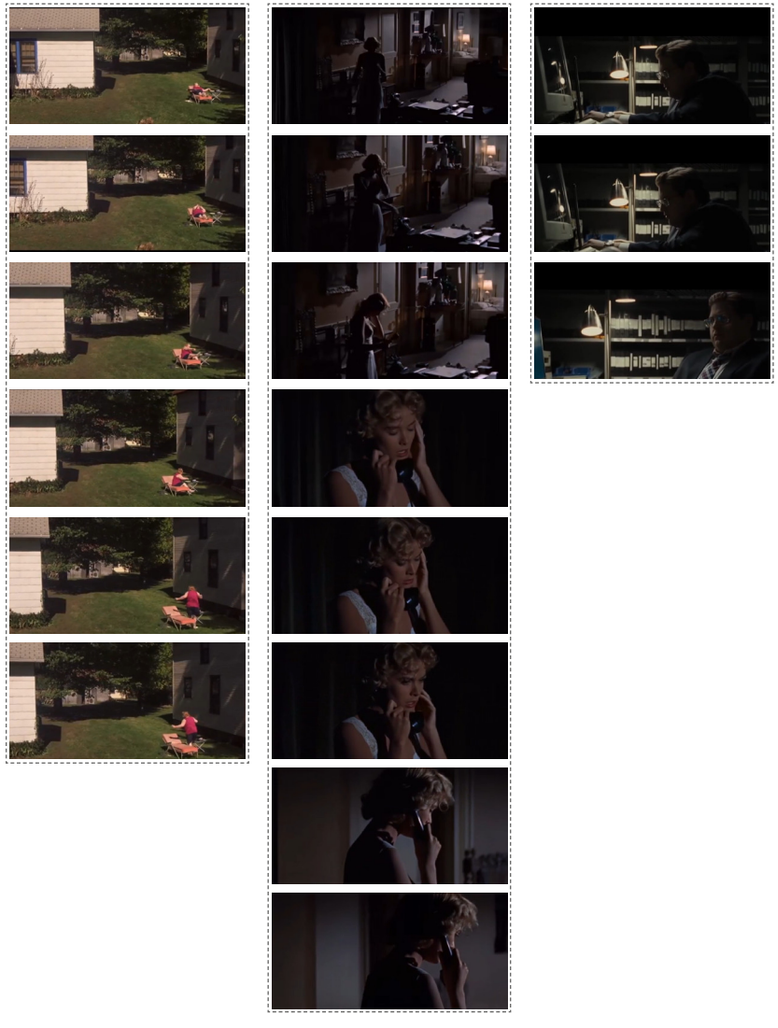
1.He sees the sea overflowing Mount Everest, foreshadowing a sense of apocalyptic terror and unease.
2.A girl is beckoning the man to come quickly, seemingly depicting a warm family outdoor activity.
3.A futuristic scene, perhaps showing a future where an aircraft comes to pick up the man.
4.Someone rides a horse towards him, perhaps the homeless man has finally been found.
5.The man looks up and sees paratroopers, evoking a sense of helplessness and sorrow among ordinary people in war.

However, we found that many neutral shots (Shot A) primarily focused on people and their facial expressions. To better study the Kuleshov effect, we considered changing the shot, shifting from the person's facial expression to another object, such as an electric fan. This led to the following experiment.

***Scene A**: The camera focuses on the ceiling. The electric fan occupies the absolute center of the composition, with a yellowish hotel ceiling in the background.*



Scene B:



Scene B:

1. A person lies on the lawn, sunbathing.

2. In a dimly lit room, a women tightly grips the receiver of an old-fashioned black telephone, repeatedly saying "Hello," but receives no response.

3. A man is working overtime in the office.

It can be felt that the focus of the film's narrative is not on what the camera captures, but on how it combines elements. Through shot B, the audience's perception of shot A can be completely altered, even creating a psychological dislocation. For example, the first group might feel comfortable, the second group might experience some psychological torment, and the third group might feel some social repression.

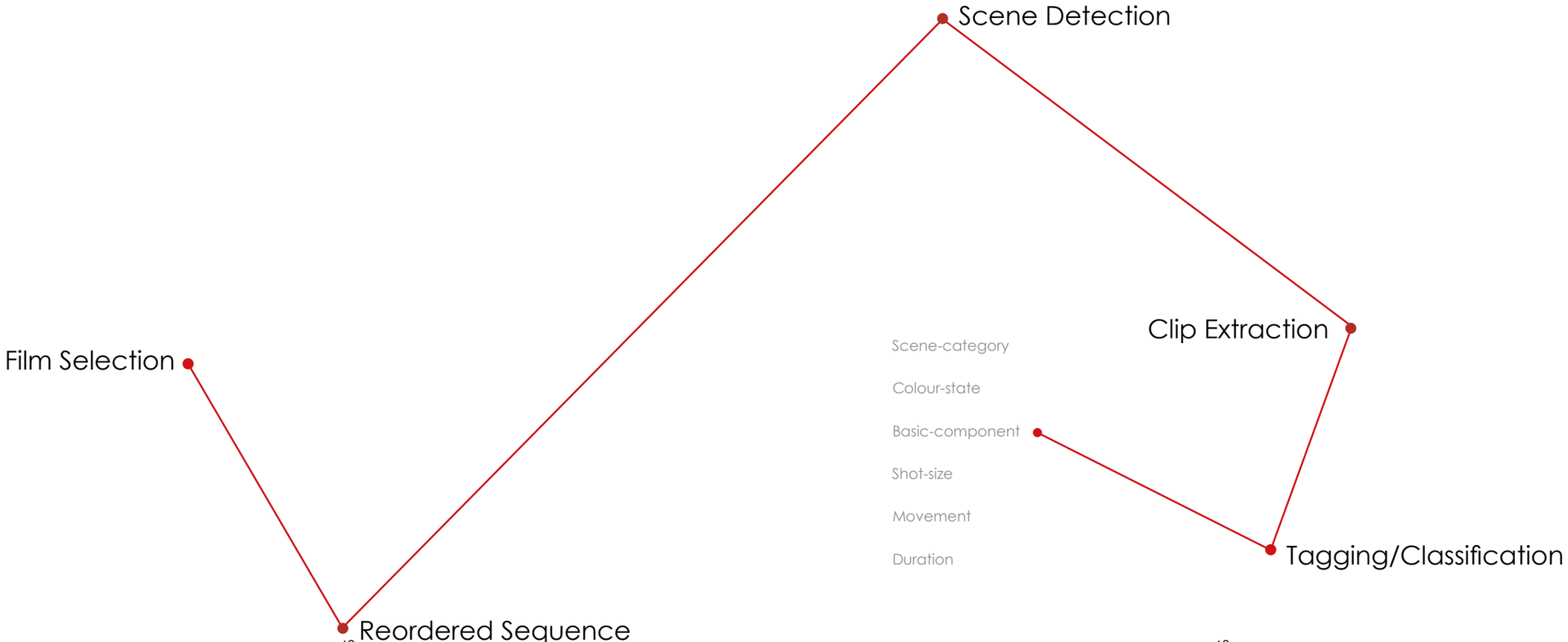
Conclusion

The Kuleshov Effect shows that the meaning of a film shot is not determined by a single image alone, but is constructed through juxtaposition, editing, and the order in which shots are viewed. The same neutral facial expression, when placed after images of food, a coffin, or a woman, can lead the audience to interpret it as hunger, sadness, or desire. In other words, viewers do not passively receive images; they actively complete meaning based on the relationship between shots. The Kuleshov Effect reveals the core logic of montage: cinematic space and emotion are not simply recorded by the camera, but reorganised, connected, and produced through editing. Therefore, it demonstrates that the power of film narrative exists not only within the image itself, but also in the relationships between shots.

CHAPTER 03

Building And Analysing The Film Corpus

→ → → →



Film Selection

The project begins by constructing a film corpus as a visual archive.

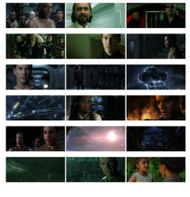
The selected films are not treated as narratives alone, but as collections of spatial, chromatic, and kinetic sequences that can be analysed and reorganised.

Tomorrowland

Metropolis

The_truman_show

The matrix1



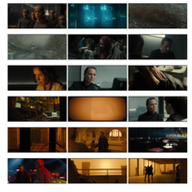
In The Mood For Love

The eight films were selected as a curated visual corpus rather than as a representative sample of cinema. Each film offers a distinct but comparable visual world in which space, atmosphere, bodies, objects, and movement are strongly organised through cinematic sequencing. This makes them suitable for studying how clips can be extracted, classified, and reconstructed as ordered visual states.

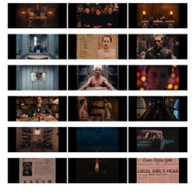
The selection brings together different spatial and cinematic conditions: the urban darkness and civic spectacle of the Batman films, the artificial and technological environments of Blade Runner 2049 and The Matrix, the constructed spatial logic of Inception, and the more intimate or interface-driven conditions of Swan Song and Who Am I. Together, they create a varied dataset across colour, scale, scene type, and visual composition.

Inception

Bladerunner 2049



The.grand.budapest.hotel



The.shining

Still_life

Swansong

Who am i

The aim is not to compare the films as narratives, nor to claim that the system understands their meaning. Instead, the films are used as a controlled archive of visual sequences. Their differences provide enough variation for classification, while their shared reliance on montage, atmosphere, and spatial progression makes them useful for testing how cinematic clips can become sequence data for later transition analysis.

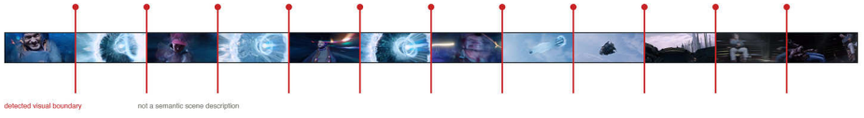
Visual Boundaries to Clip Units

The film is divided by detected visual discontinuities, not by narrative interpretation.

CONTINUOUS FILM

A selected sequence window is read as continuous image flow.

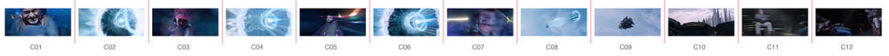
Red marks indicate detected boundaries between adjacent visual states.



CONTINUOUS FILM

Each interval between two boundaries becomes one discrete clip.

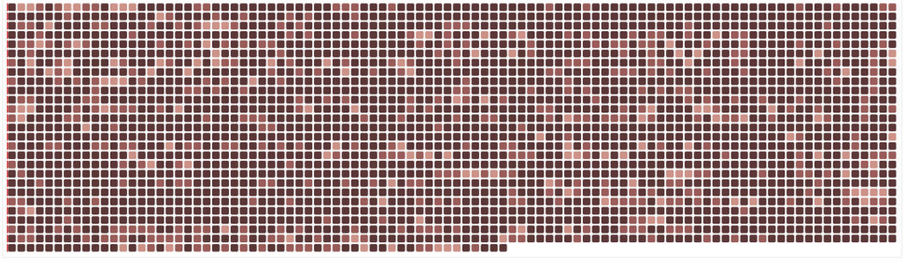
The clip remains a temporal unit with duration, filename, and sequence index.



CONTINUOUS FILM

Tomorrowland is reconstructed as 2,550 ordered clip units.

Every mark below is one extracted clip; the order reads left to right, then wraps to the next row.



2,550	2.0s	02:10:03
clips	median duration	total duration

Full-Film State Transition Matrix

A complete film sequence is converted from clip order into local state-to-state transitions.

Tomorrowland	2,550	2,549	1,427
film	clips	transitions	six-label states

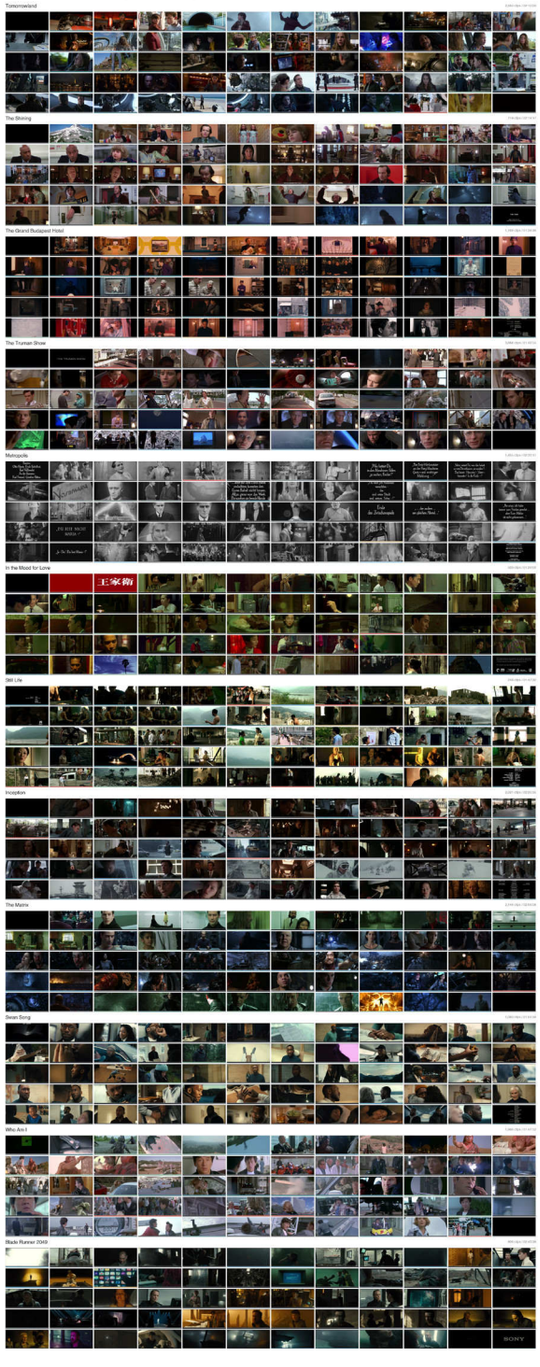
Complete six-label state sequence

All 2,550 current clips are retained in source order; each vertical slice is one clip.

Extracted clip library overview

60 sampled 21:9 clip frames per film

Extracted clip library overview

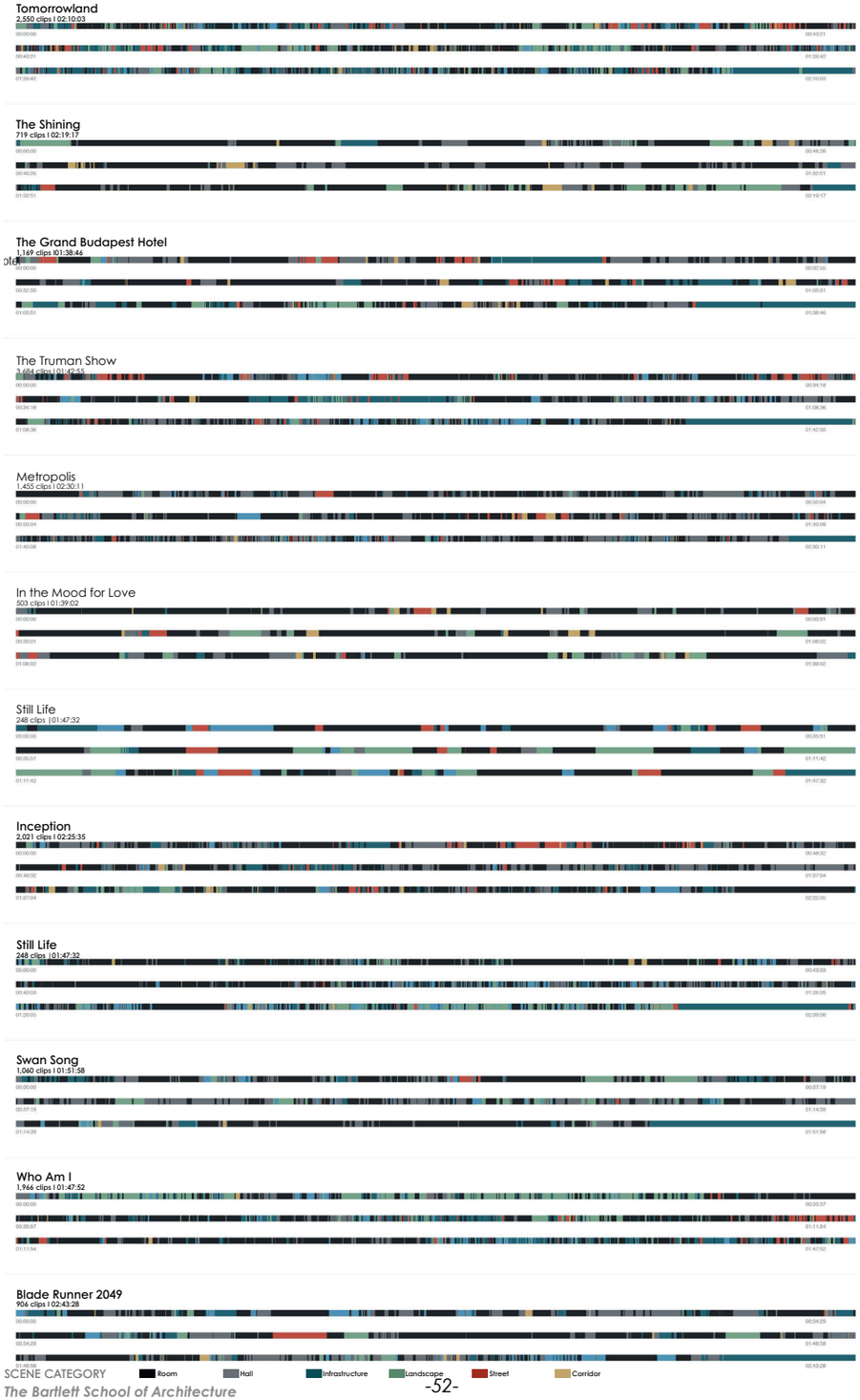


The project begins with the construction of a film corpus as a visual archive.

This portrait overview samples 60 middle frames from the extracted clips of each film, with every frame cropped to a consistent 21:9 format. It presents the corpus before classification, showing the visual material that later becomes labelled, sequenced, and translated into colour-state diagrams.

Twelve-film extradted clip timelines

Segment length follows its real duration.Every scene-coloured segment is one clip

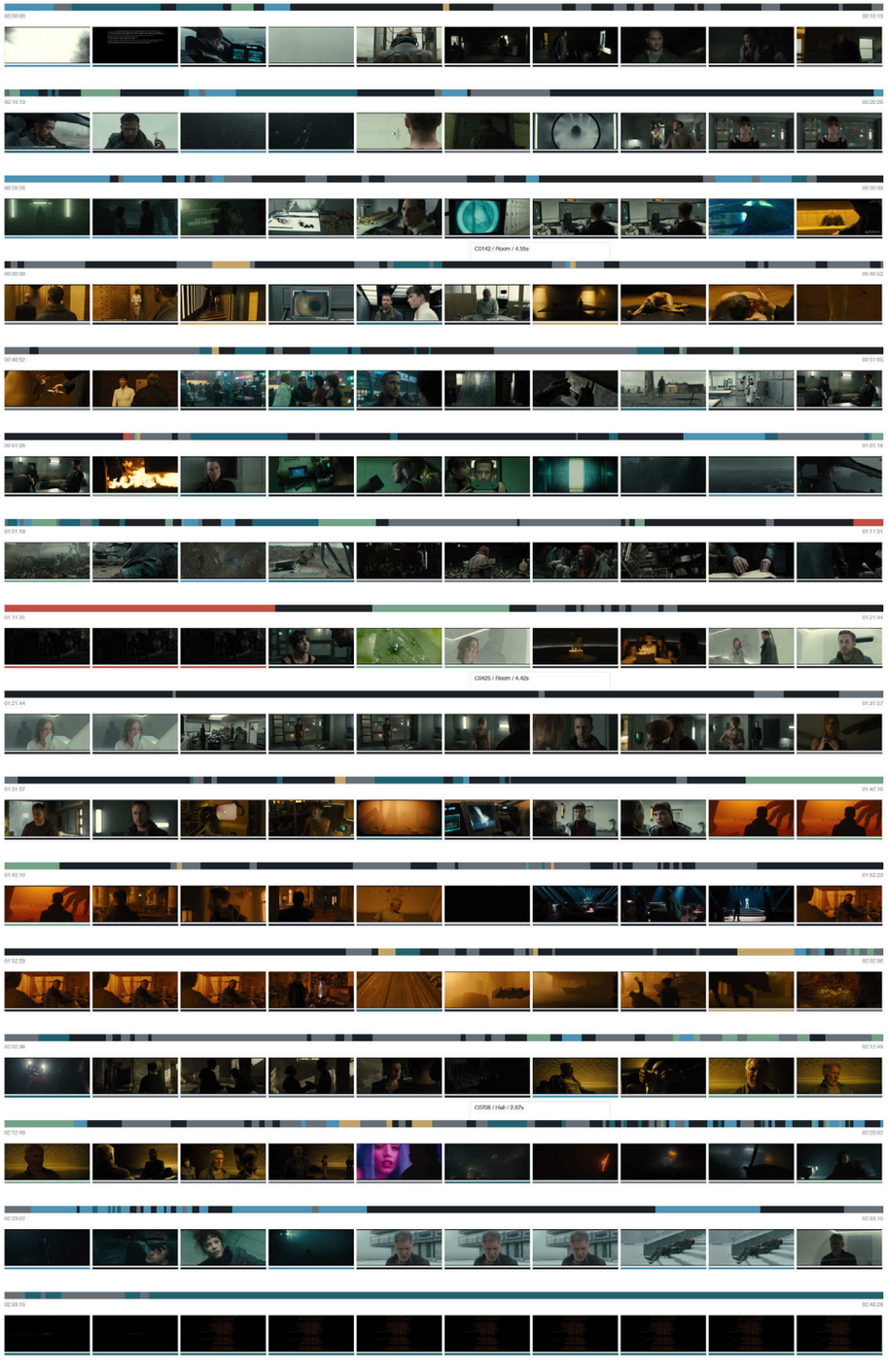


Scene-category

Scene Category establishes the primary spatial condition of each clip through seven recurrent settings: Room, Hall, Infrastructure, Landscape, Waterfront, Street, and Corridor. The classification does not attempt to reconstruct a complete location or narrative world; it identifies the dominant environmental envelope visible within the shot. Read sequentially, these labels reveal how a film moves between enclosure, circulation, exposure, and edge conditions. Within the spatial translation, Scene Category provides the base state from which a prototype can be selected. The other labels then modify that state, while adjacency between clips records how one spatial condition is reinterpreted through its position beside another—a relation that keeps the Kuleshov effect at the centre of the method.

Blade Runner 2049 clip timeline path

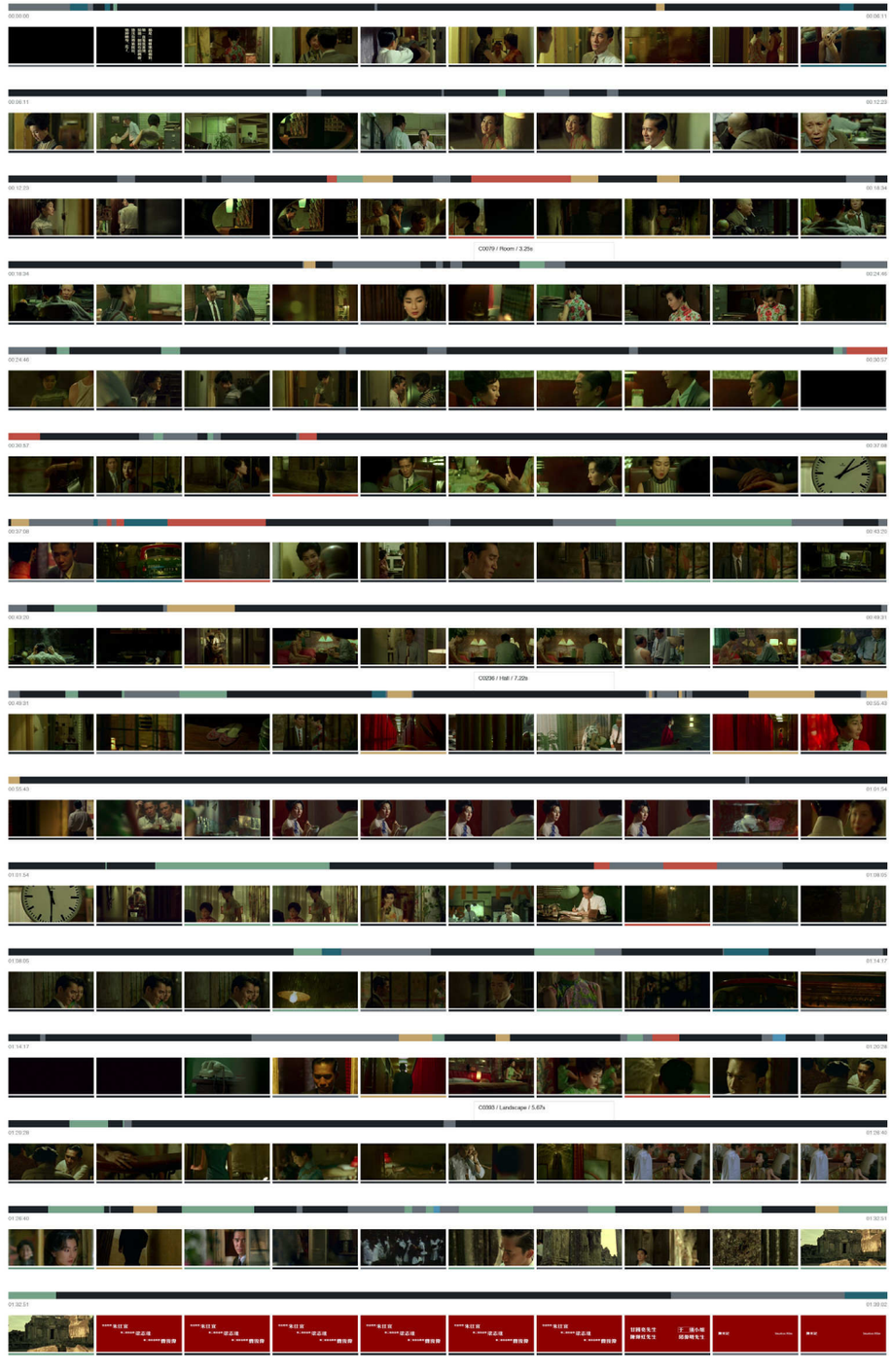
906 clips I total sequence time 02:43:28 I median clip 4.96s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

In the Mood for Love clip timeline path

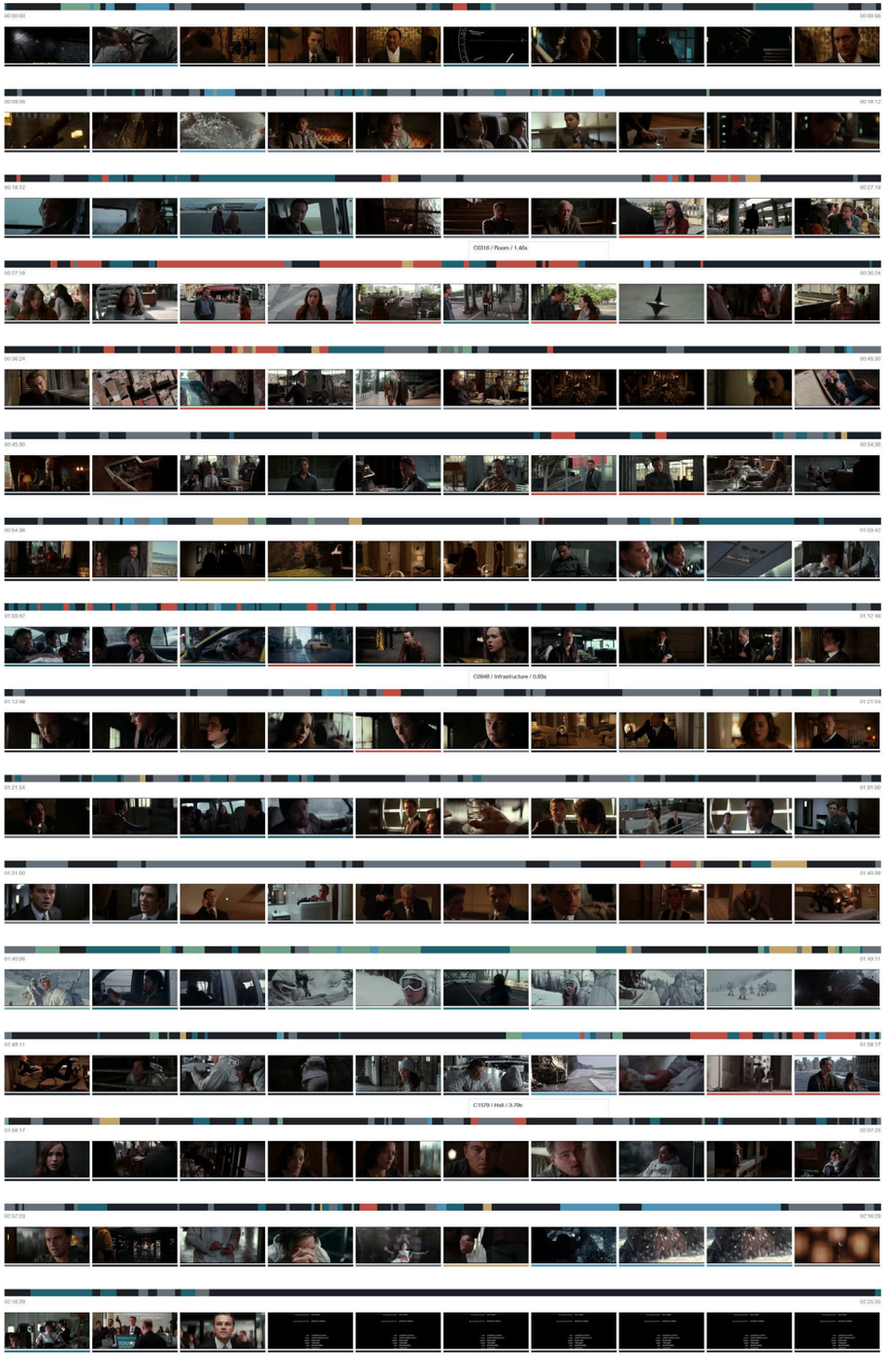
503 clips I total sequence time 01:39:02 I median clip 7.05s I scene-coloured duration pathduration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

Inception clip timeline path

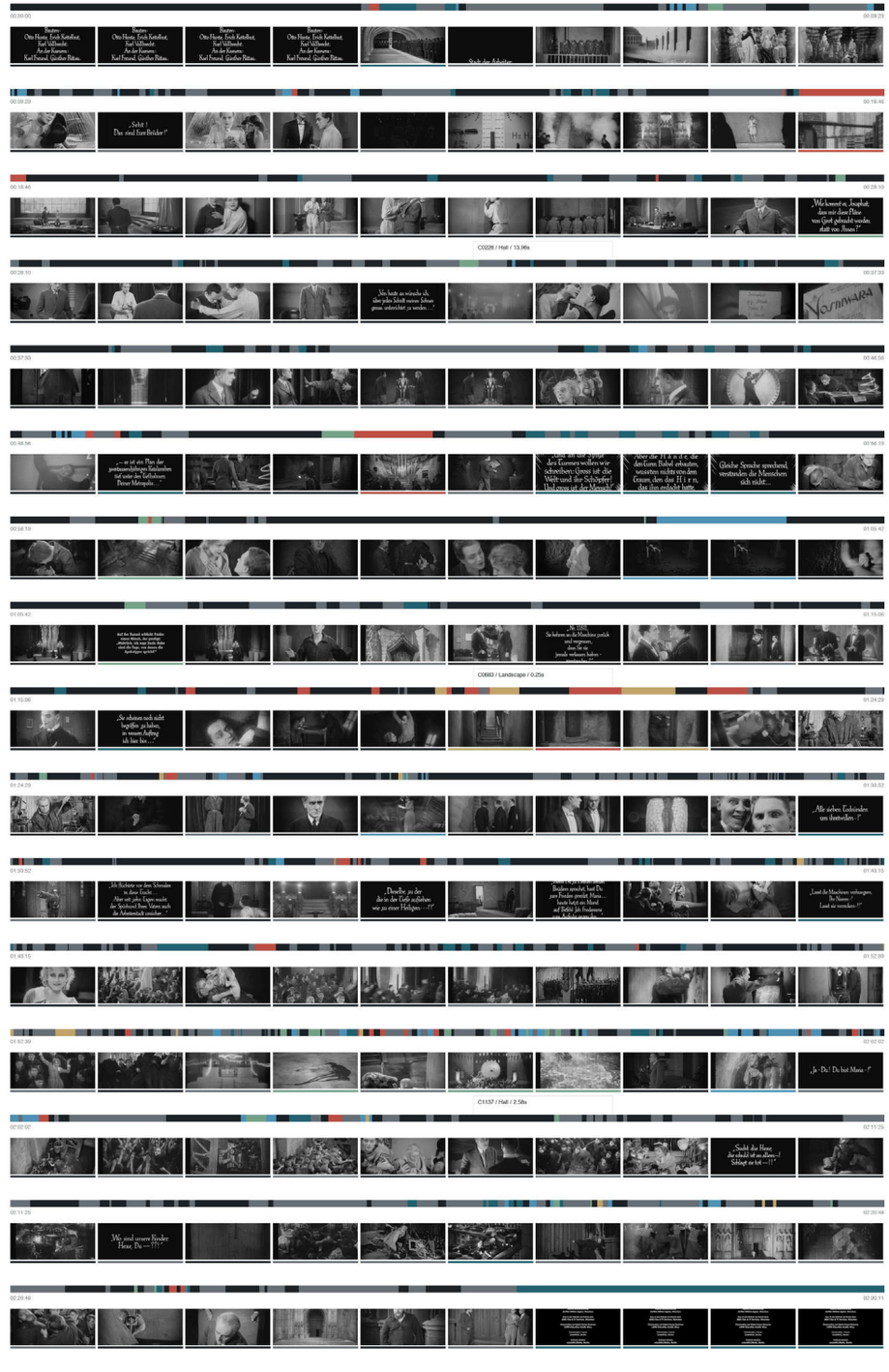
2,021 clips I total sequence time 02:25:35 I median clip 2.63s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

Metropolis clip timeline path

1,455 clips I total sequence time 02:30:11 I median clip 3.75s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

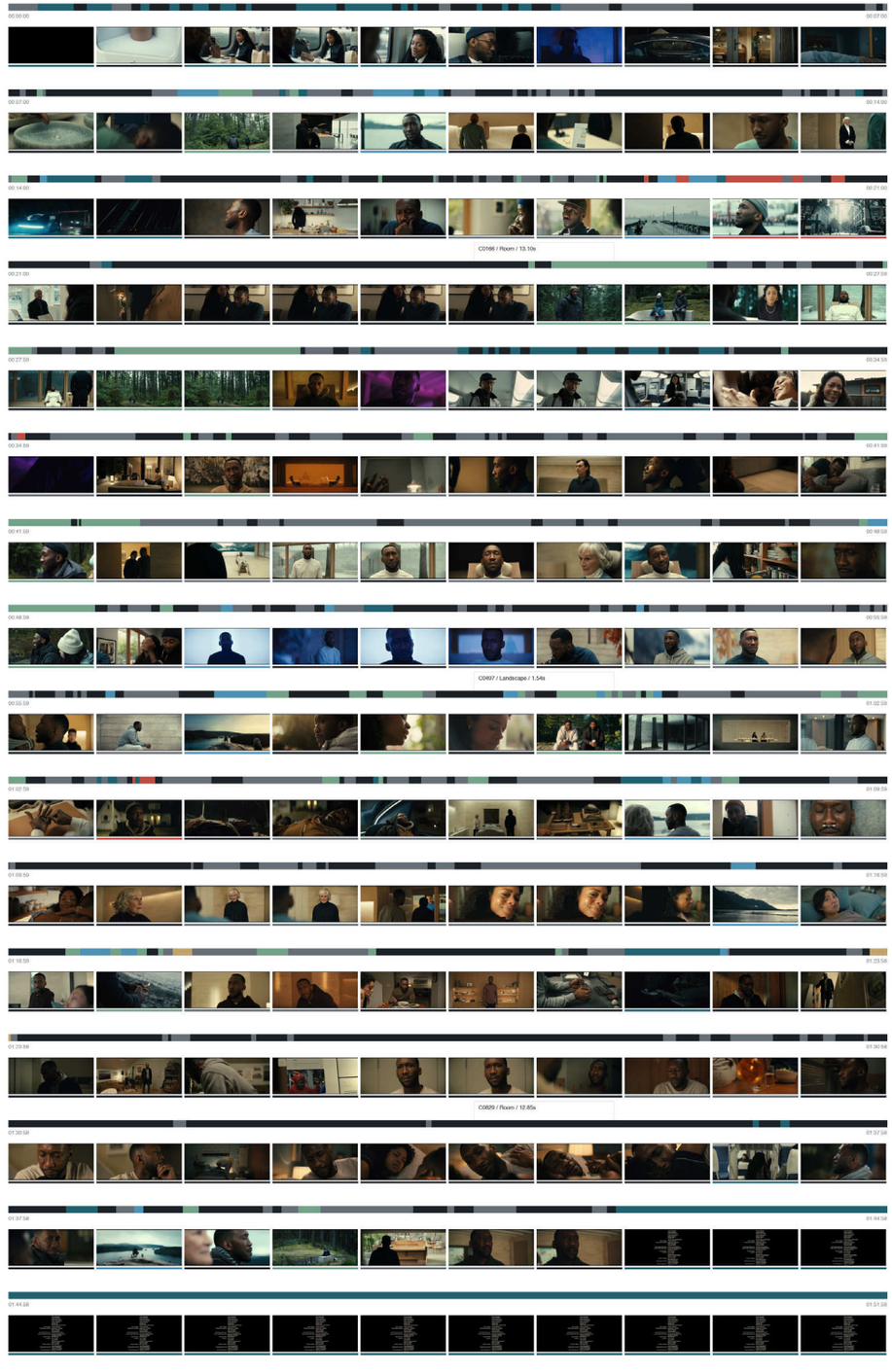
Still Life clip timeline path

248 clips I total sequence time 01:47:32 I median clip 12.72s I scene-coloured duration path

SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

Swan Songclip timeline path

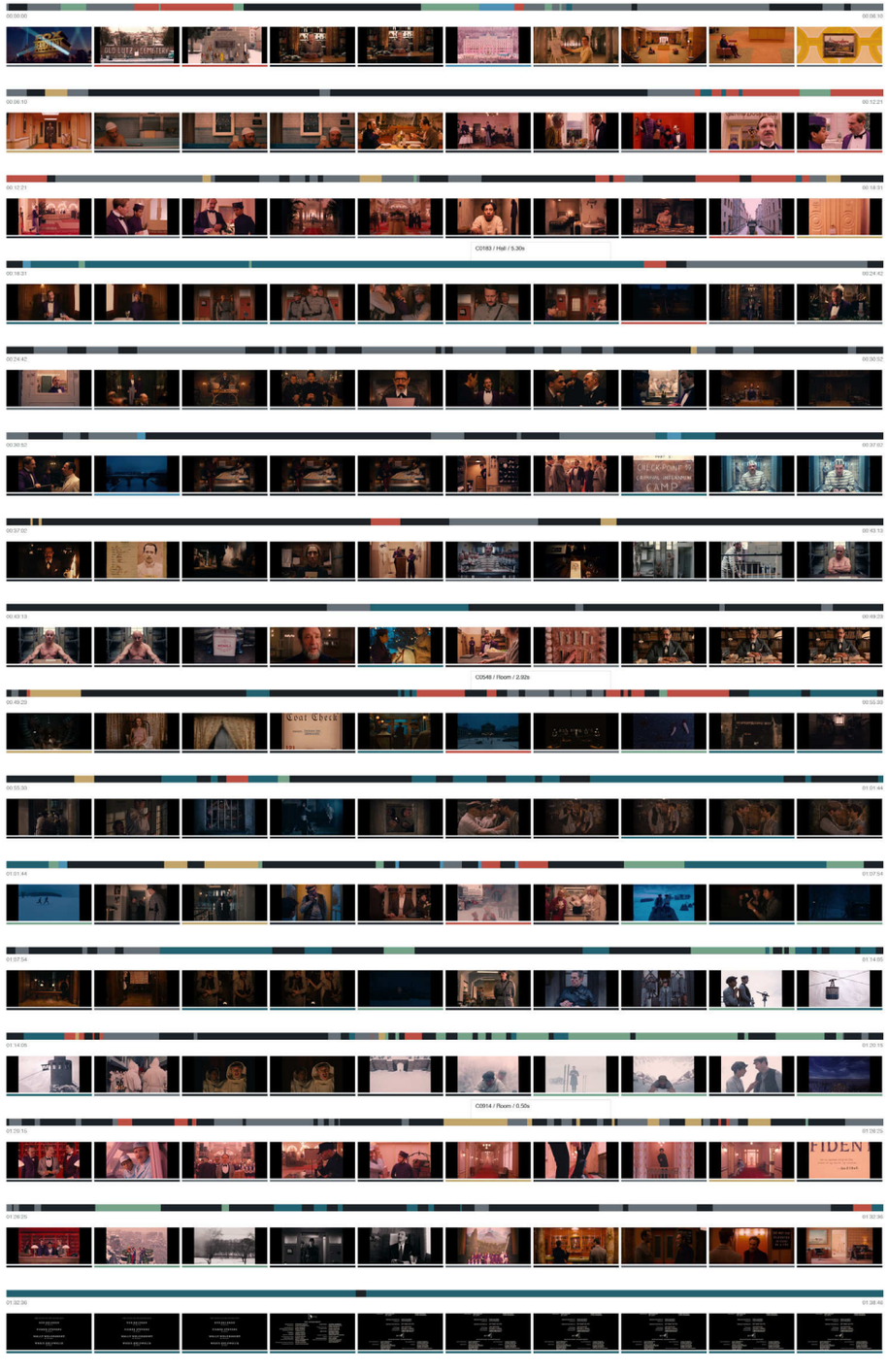
1,060 clips I total sequence time 01:51:58I median clip 3.75s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

The Grand Budapest Hotel clip timeline path

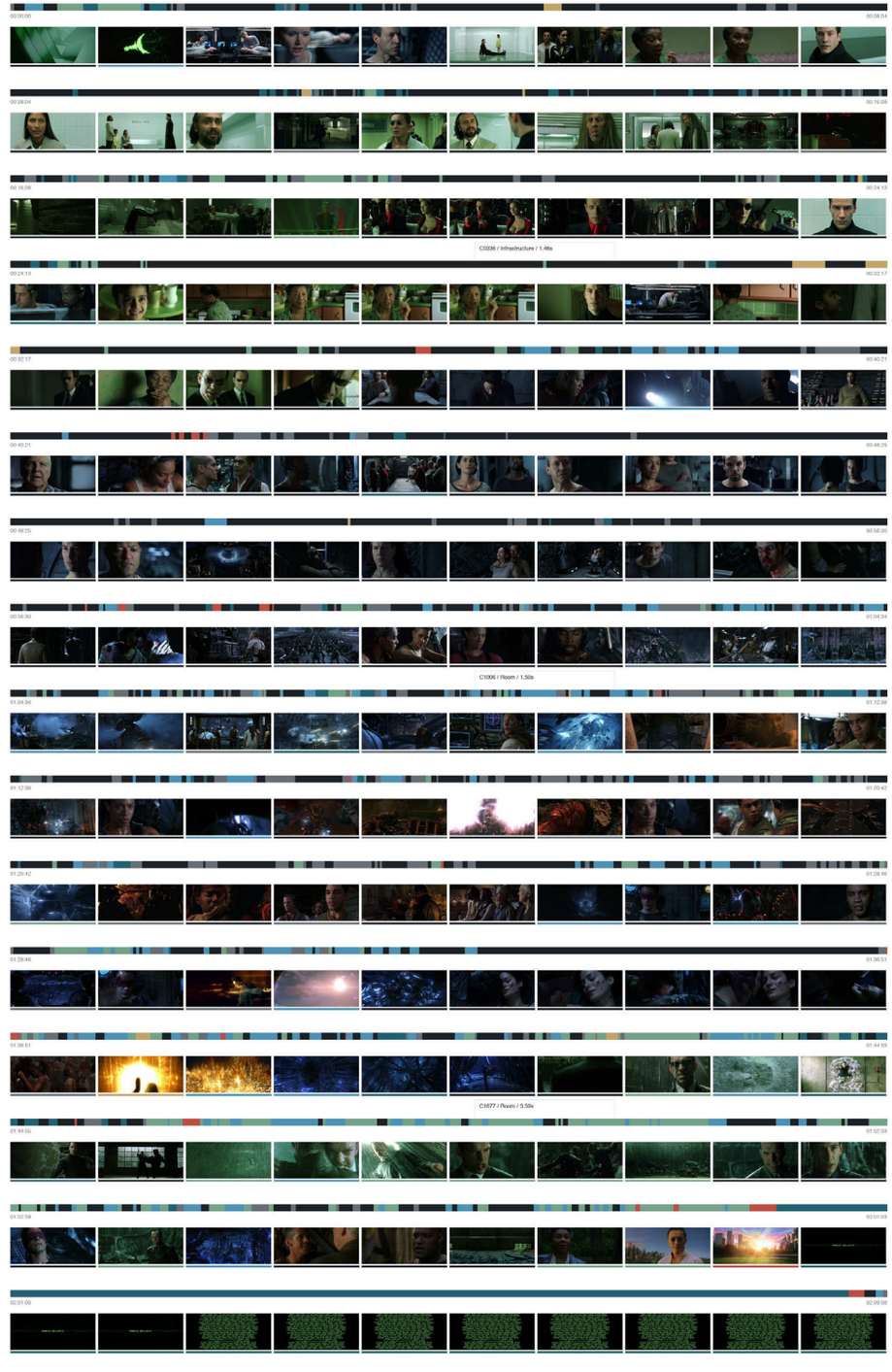
1,169 clips total sequence time 01:38:46 I median clip 2.92s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

The Matrix clip timeline path

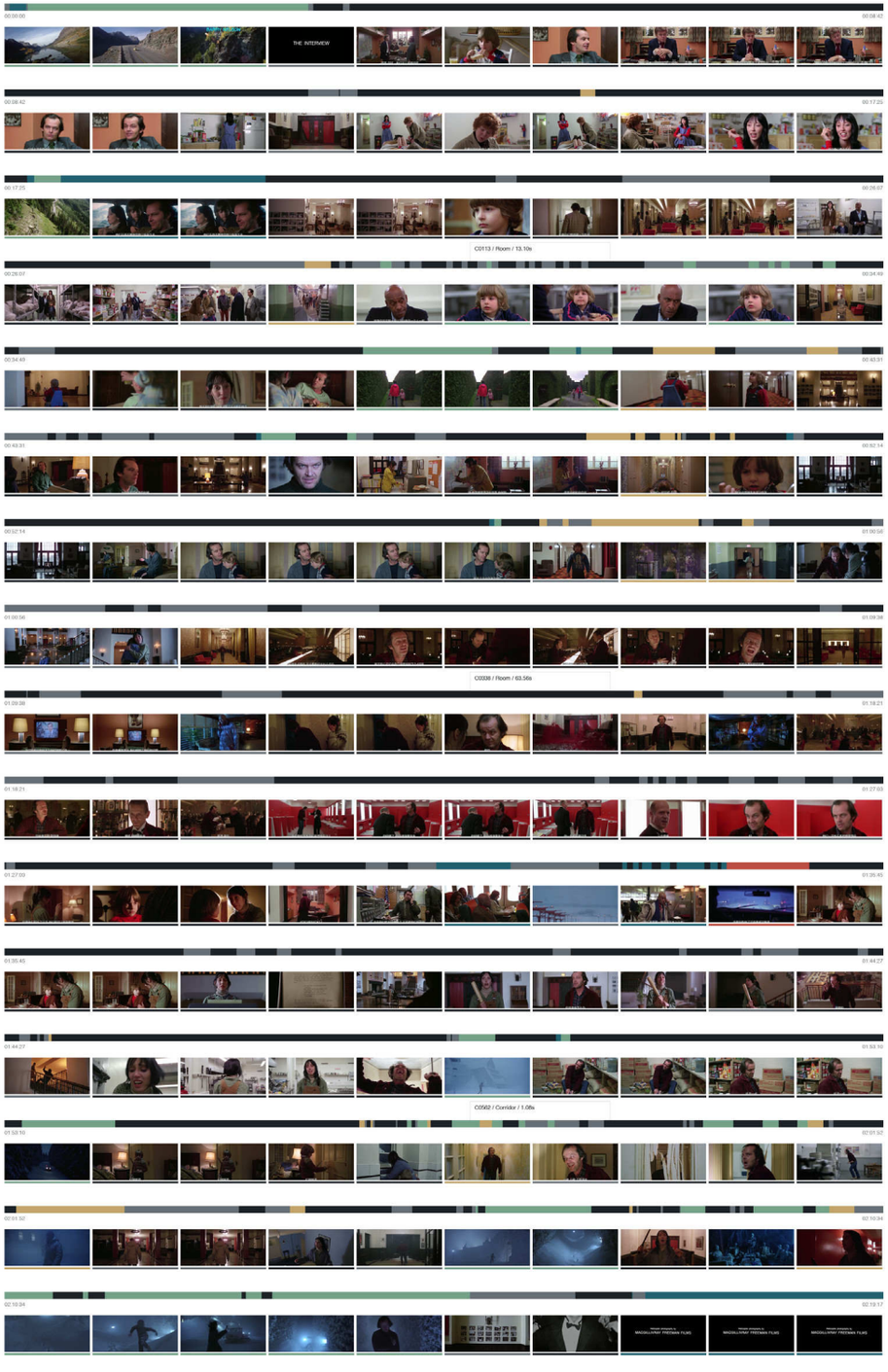
2,146 clips I total sequence time 02:09:08 I median clip 2.50s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

The Shining clip timeline path

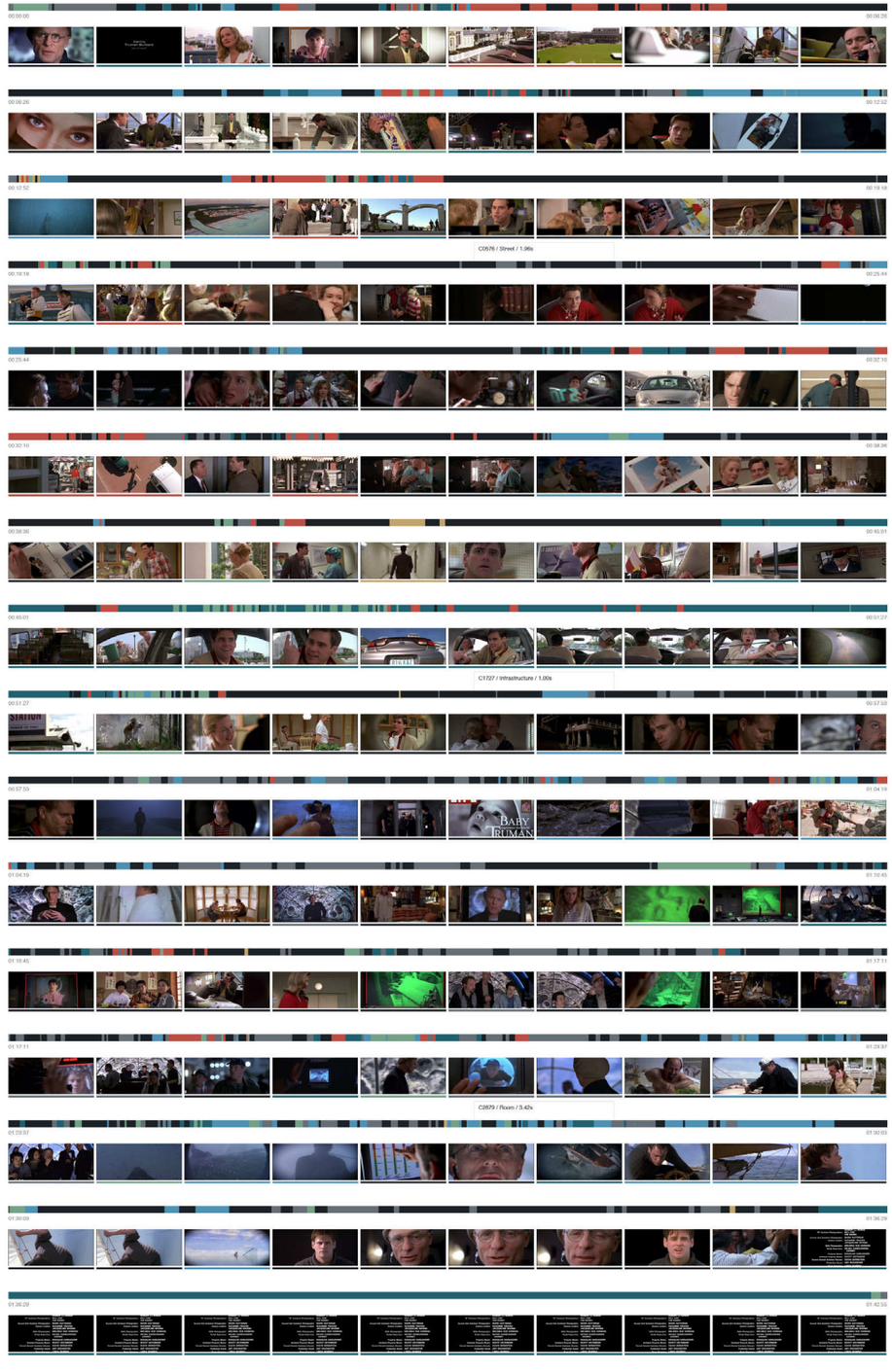
719 clips I total sequence time 02:19:17 I median clip 6.26s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

The Truman Show clip timeline path

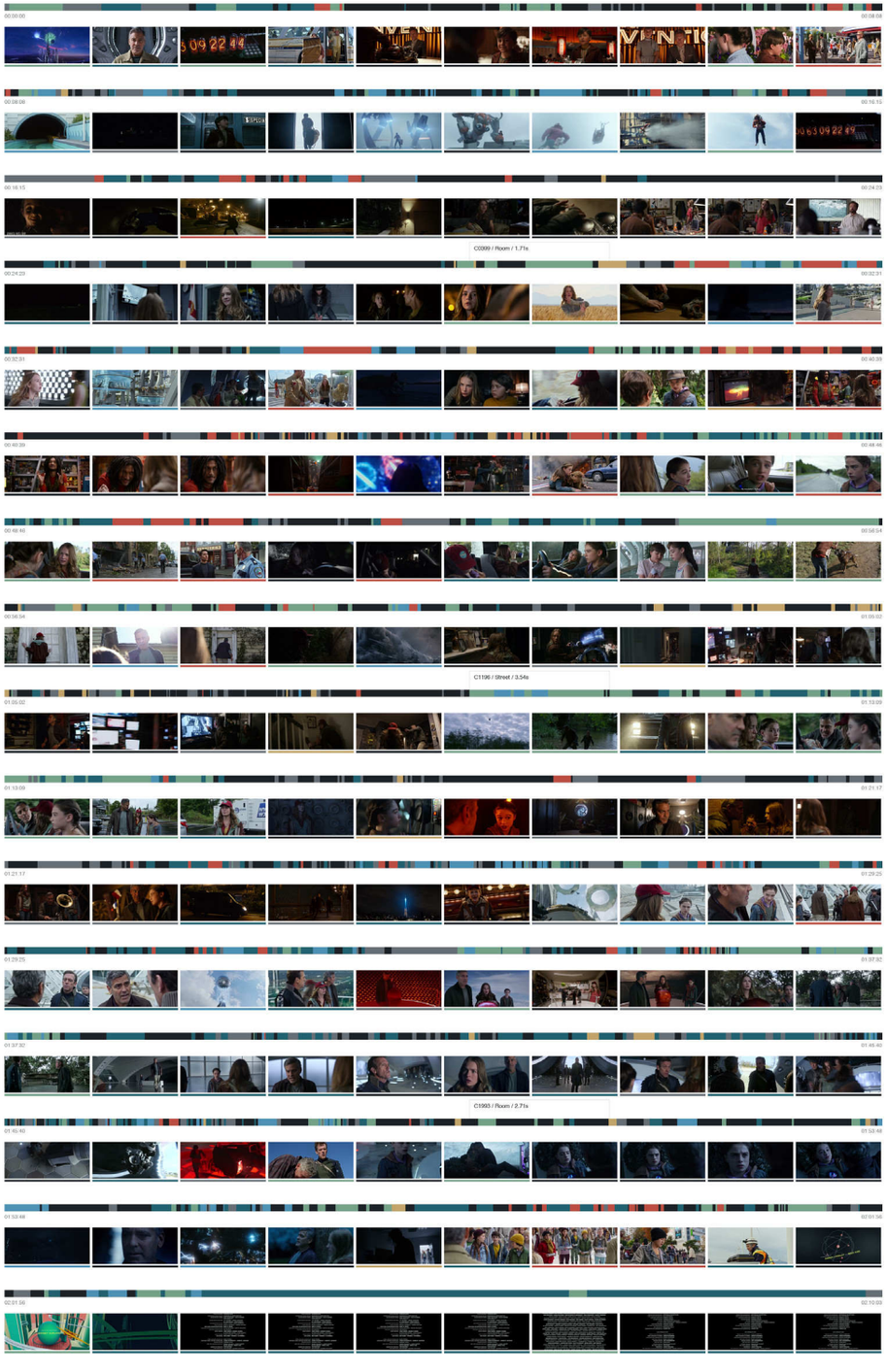
3,684 clips I total sequence time 01:42:55 I median clip 1.00s I scene-coloured duration path



SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

Tomorrowland clip timeline path

2,550 clips I total sequence time 02:10:03 I median clip 2.00s I scene-coloured duration path

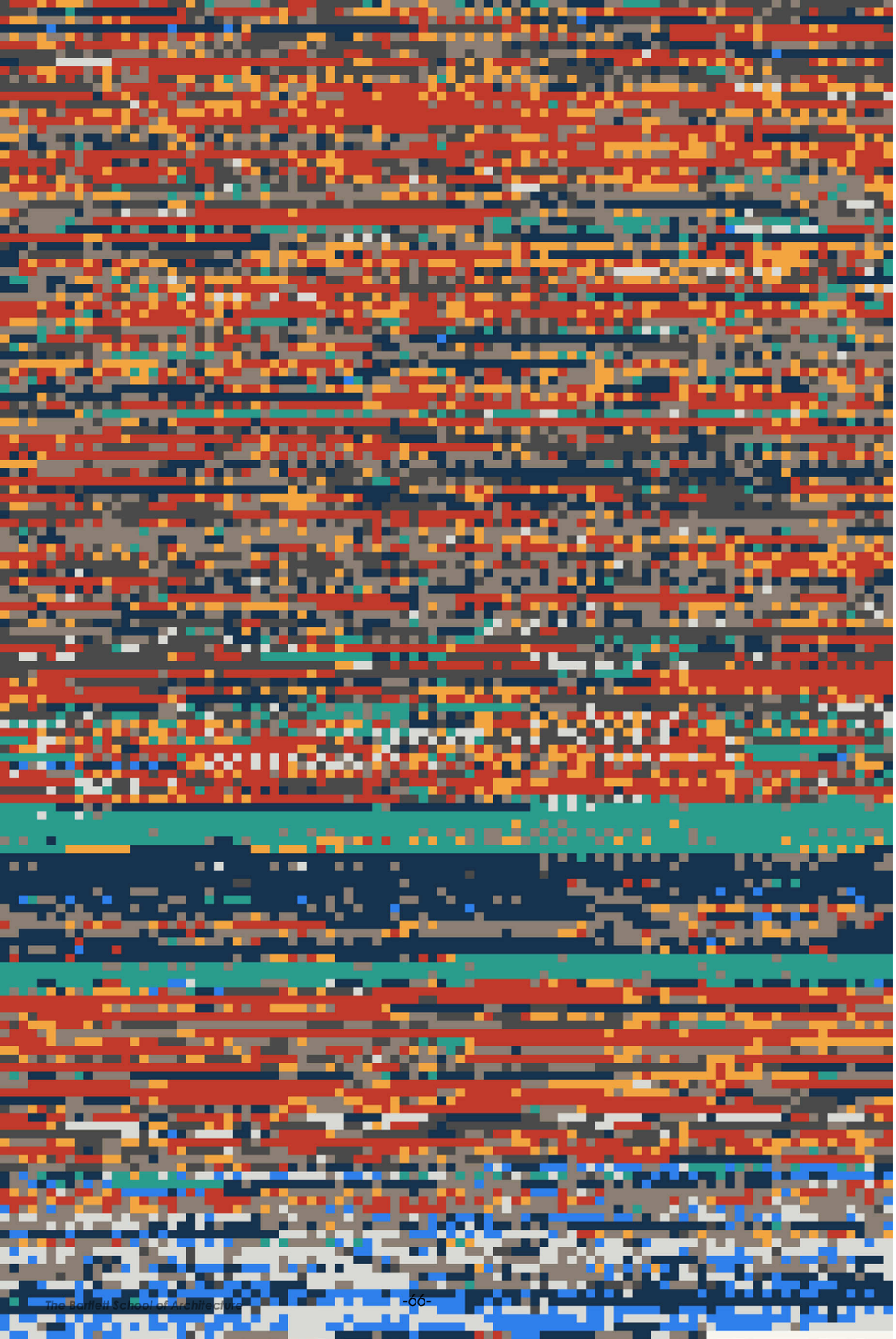


SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor

Who Am I clip timeline path

1,966 clips Itotal sequence time 01:47:52I median clip 2.38s I scene-coloured duration path

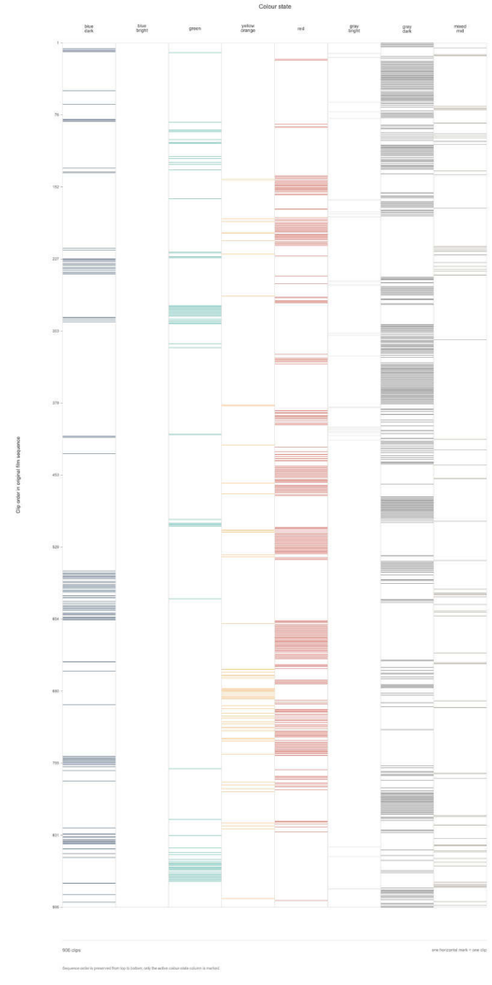
SCENE CATEGORY Room Hall Infrastructure Landscape Street Corridor



Colour-state

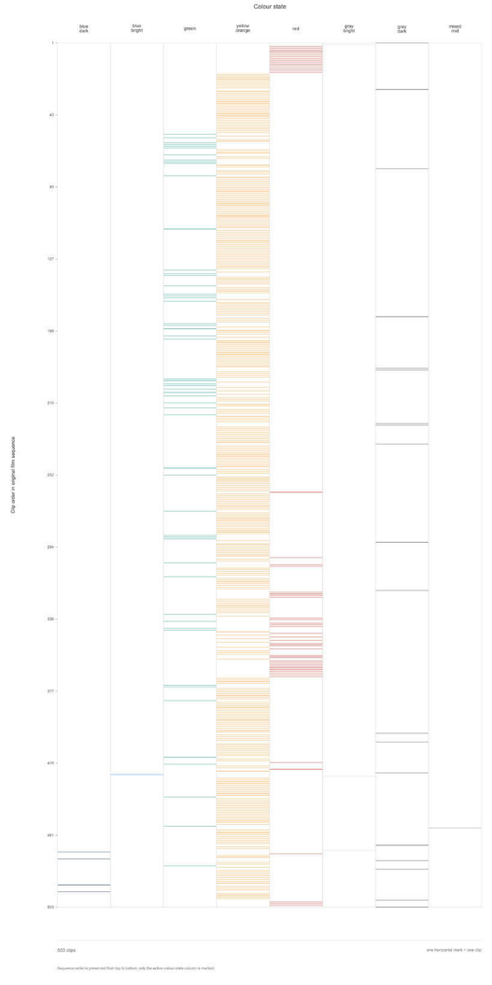
Colour translates each clip into one of eight chromatic states: dark blue, bright blue, green, yellow–orange, red, bright grey, dark grey, and mixed mid-tone. Two complementary readings are retained. The ordered sequence shows where colour states occur and how they cluster, repeat, or interrupt one another, while the count strip removes temporal order to compare their overall weight within a film. Colour is therefore treated as a relational atmosphere and material parameter rather than as a fixed emotional code. In spatial terms, it can adjust light, reflectance, material temperature, and visual density, but its meaning remains dependent on the states that precede and follow it.

ChromaticSequence



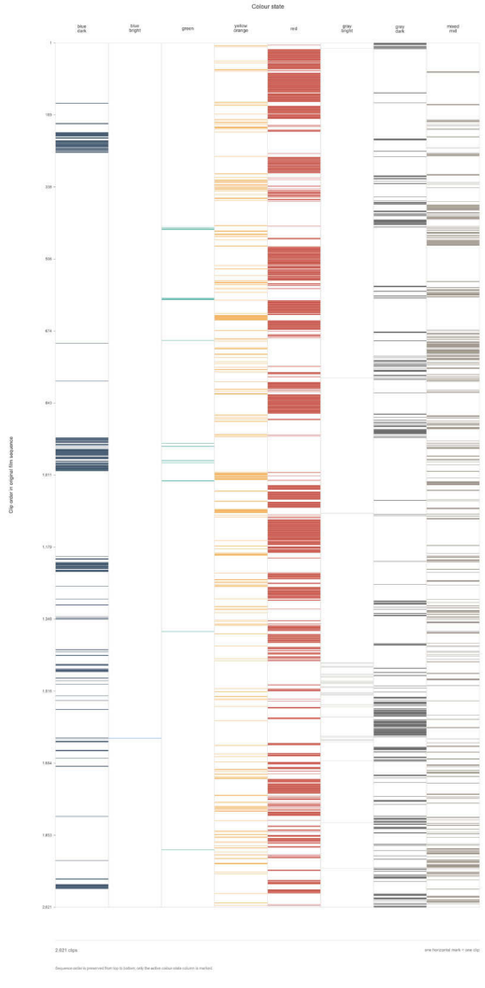
Across 906 ordered clips, dark grey is the most frequent colour label (342; 37.7%), followed by red (245; 27.0%) and dark blue (101; 11.1%). The label changes across 42.0% of adjacent clip transitions, producing a balanced alternation between recurring colour groups; the longest uninterrupted run contains 42 clips classified as dark grey. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

ChromaticSequence



Across 503 ordered clips, yellow–orange is the most frequent colour label (367; 73.0%), followed by green (55; 10.9%) and red (53; 10.5%). The label changes across 29.9% of adjacent clip transitions, producing clustered passages interrupted by periodic changes; the longest uninterrupted run contains 30 clips classified as yellow–orange. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

ChromaticSequence

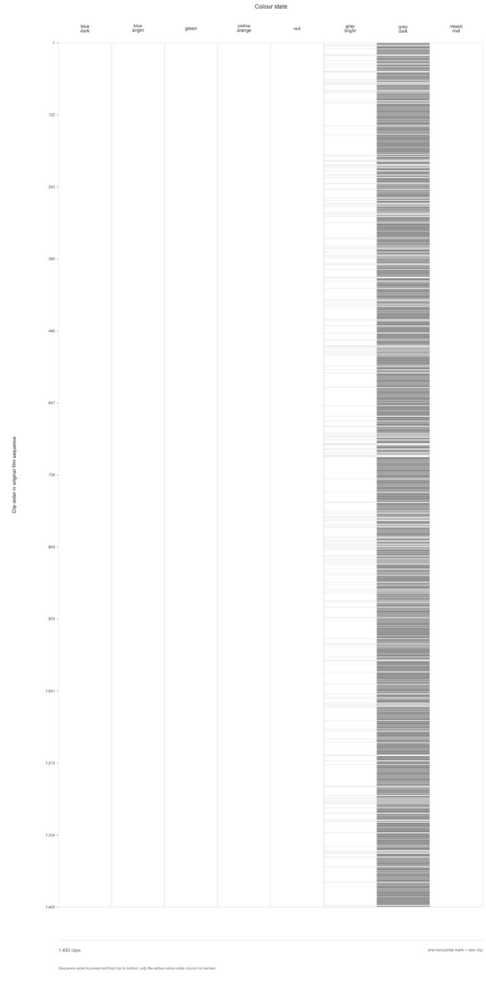


Inception

Across 2,021 ordered clips, red is the most frequent colour label (827; 40.9%), followed by mixed mid-tone (387; 19.1%) and dark grey (304; 15.0%). The label changes across 32.5% of adjacent clip transitions, producing clustered passages interrupted by periodic changes; the longest uninterrupted run contains 49 clips classified as red. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

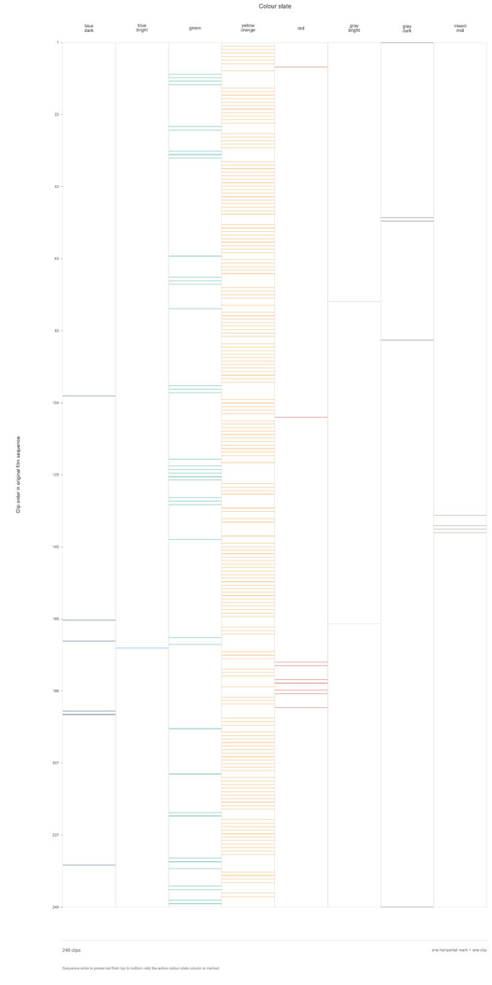
Metropolis

ChromaticSequence



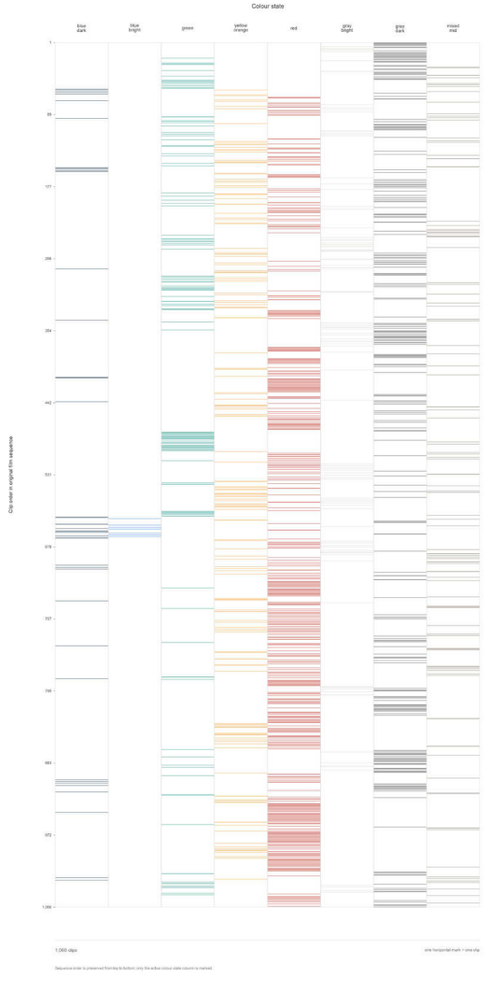
Across 1,455 ordered clips, dark grey is the most frequent colour label (1,275; 87.6%), followed by bright grey (180; 12.4%). The label changes across 23.6% of adjacent clip transitions, producing long, comparatively stable clusters; the longest uninterrupted run contains 38 clips classified as dark grey. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

ChromaticSequence



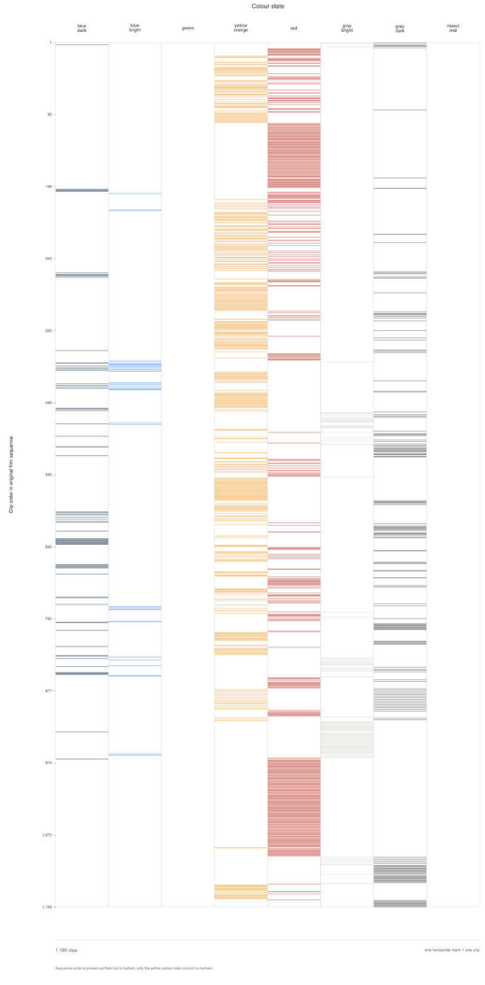
Across 248 ordered clips, yellow–orange is the most frequent colour label (181; 73.0%), followed by green (40; 16.1%) and red (9; 3.6%). The label changes across 27.9% of adjacent clip transitions, producing clustered passages interrupted by periodic changes; the longest uninterrupted run contains 22 clips classified as yellow–orange. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

ChromaticSequence



Across 1,060 ordered clips, red is the most frequent colour label (360; 34.0%), followed by dark grey (213; 20.1%) and yellow–orange (143; 13.5%). The label changes across 66.7% of adjacent clip transitions, producing frequent alternation between short colour-state runs; the longest uninterrupted run contains 17 clips classified as red. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

ChromaticSequence

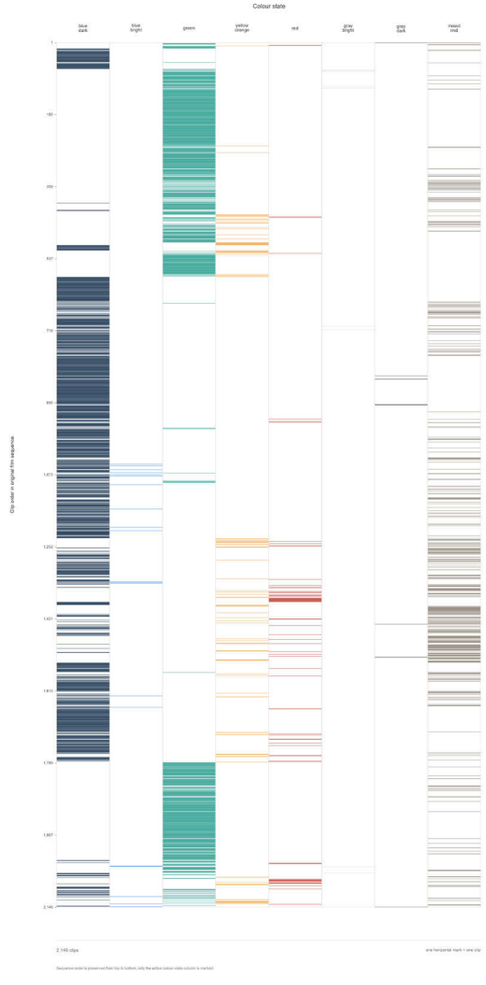


The Grand Budapest Hotel

Across 1,169 ordered clips, red is the most frequent colour label (411; 35.2%), followed by yellow–orange (395; 33.8%) and dark grey (165; 14.1%). The label changes across 35.8% of adjacent clip transitions, producing clustered passages interrupted by periodic changes; the longest uninterrupted run contains 119 clips classified as red. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

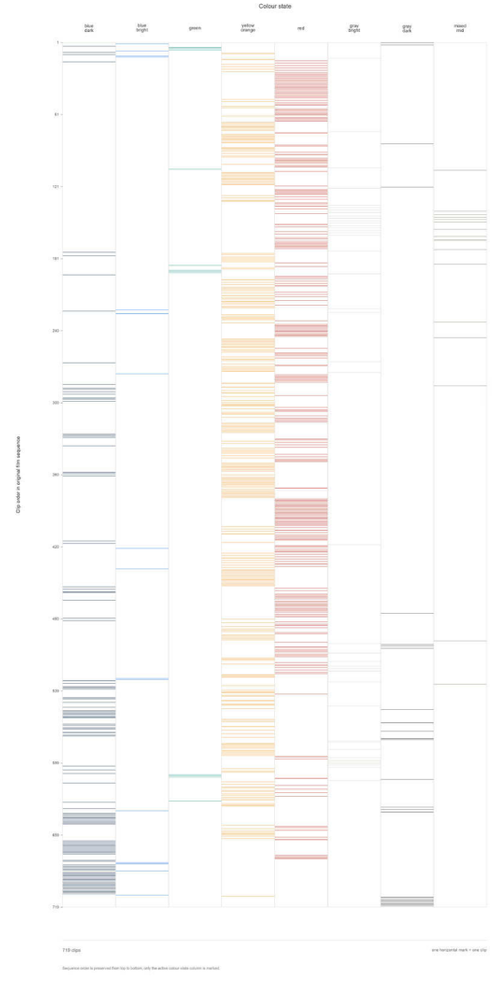
The Matrix

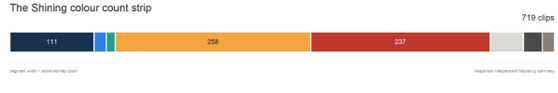
ChromaticSequence



Across 2,146 ordered clips, dark blue is the most frequent colour label (914; 42.6%), followed by green (699; 32.6%) and mixed mid-tone (340; 15.8%). The label changes across 29.4% of adjacent clip transitions, producing clustered passages interrupted by periodic changes; the longest uninterrupted run contains 140 clips classified as green. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

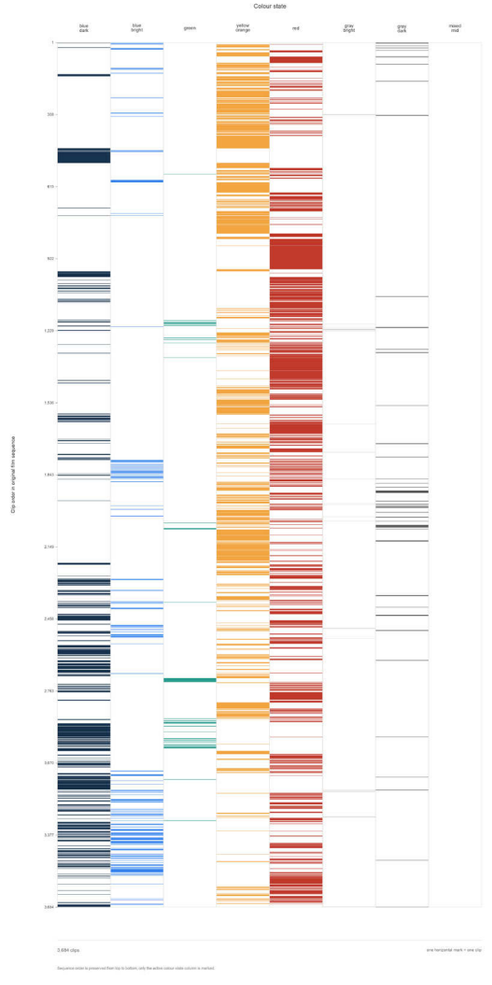
ChromaticSequence





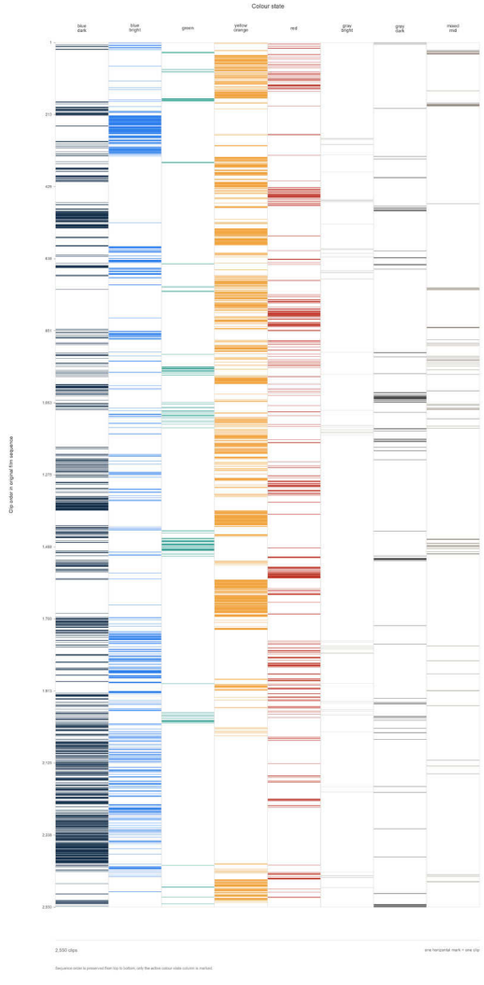
Across 719 ordered clips, yellow–orange is the most frequent colour label (258; 35.9%), followed by red (237; 33.0%) and dark blue (111; 15.4%). The label changes across 49.0% of adjacent clip transitions, producing a balanced alternation between recurring colour groups; the longest uninterrupted run contains 23 clips classified as red. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

ChromaticSequence



Across 3,684 ordered clips, red is the most frequent colour label (1,357; 36.8%), followed by yellow–orange (1,210; 32.8%) and dark blue (694; 18.8%). The label changes across 25.3% of adjacent clip transitions, producing clustered passages interrupted by periodic changes; the longest uninterrupted run contains 100 clips classified as red. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

ChromaticSequence

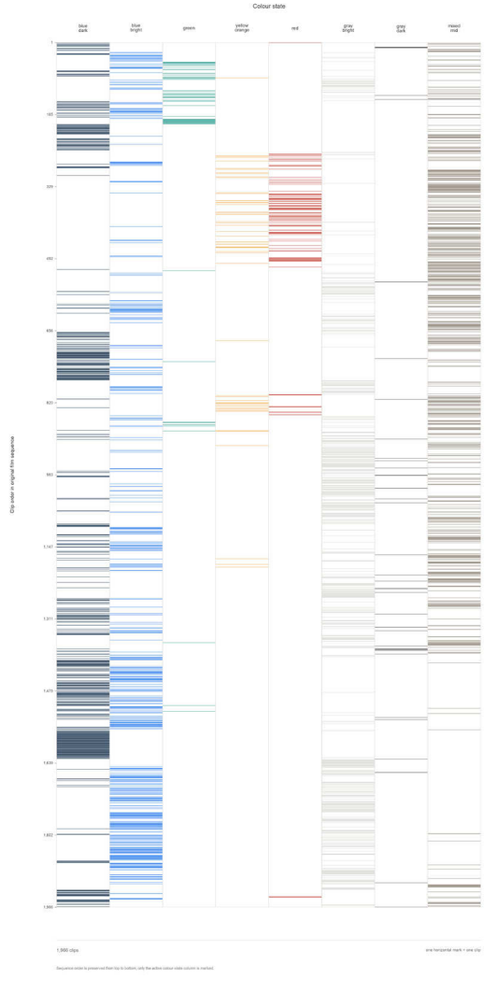


Across 2,550 ordered clips, yellow–orange is the most frequent colour label (730; 28.6%), followed by dark blue (710; 27.8%) and bright blue (440; 17.3%). The label changes across 46.4% of adjacent clip transitions, producing a balanced alternation between recurring colour groups; the longest uninterrupted run contains 32 clips classified as yellow–orange. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

Tomorrowland

Who am i

ChromaticSequence



Across 1,966 ordered clips, mixed mid-tone is the most frequent colour label (483; 24.6%), followed by bright grey (452; 23.0%) and dark blue (422; 21.5%). The label changes across 53.6% of adjacent clip transitions, producing a balanced alternation between recurring colour groups; the longest uninterrupted run contains 60 clips classified as dark blue. The vertical field retains where each state occurs, while the strip removes temporal order to show its overall weight.

Corpus clip-cell colour grid

12 films / 18,427 clips / each cell is one ordered clip

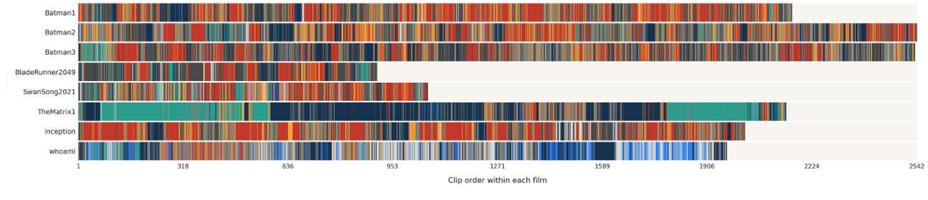
Each cell = one clip. Sequence reads left to right, then continues on the next row. Wrap width: 96 clips.

Each cell represents one ordered clip from the current 18,427-clip corpus and is filled with one of eight colour labels. The twelve films are separated into individual blocks. Within each block, the sequence reads from left to right and continues on the next row after every 96 clips. This wrapping method compresses films of different lengths into a common vertical format while preserving clip order. Sustained colour runs, abrupt changes, and recurring chromatic passages can therefore be read without treating the grid as a narrative interpretation.

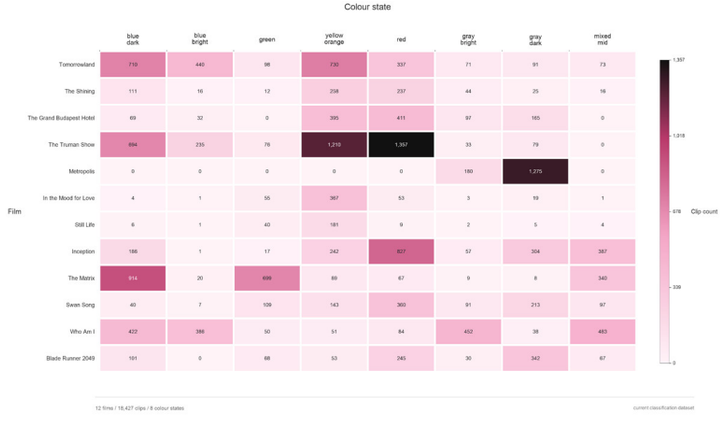
The companion linear heatmap places the same twelve films on a shared horizontal clip-order axis. Each row begins at clip 1, and every coloured cell represents one clip in its original order. A row stops when its film ends, so the remaining pale field records differences in film length rather than missing data. This view supports direct comparison of chromatic rhythm and run length across the corpus, while the wrapped grid above provides a denser reading of each individual sequence.

Corpus Clip-cell color Heatmap

12 films/ 18427 Clips within each film

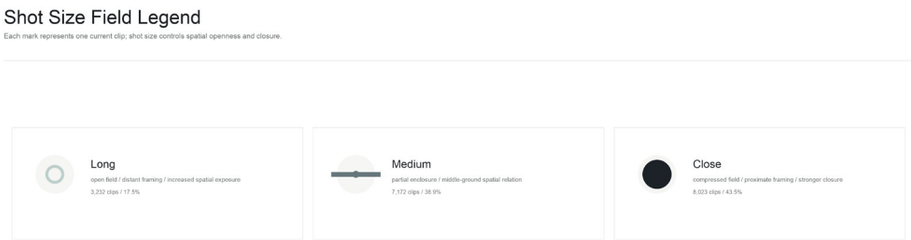


Clip count distribution by film and colour state



This matrix aggregates the same 18,427 clips into absolute counts for twelve films and eight colour states. Rows represent films and columns represent colour categories. Each printed number is the observed clip count, and darker tone indicates a larger value on a shared scale. Temporal order is intentionally removed. The strongest concentrations include dark grey in Metropolis, red and yellow-orange in The Truman Show, dark blue and green in The Matrix, and yellow-orange in In the Mood for Love and Still Life. Because the values are counts rather than within-film percentages, the matrix should be read as a corpus distribution and an indication of film-scale weight, not as a length-normalised comparison.

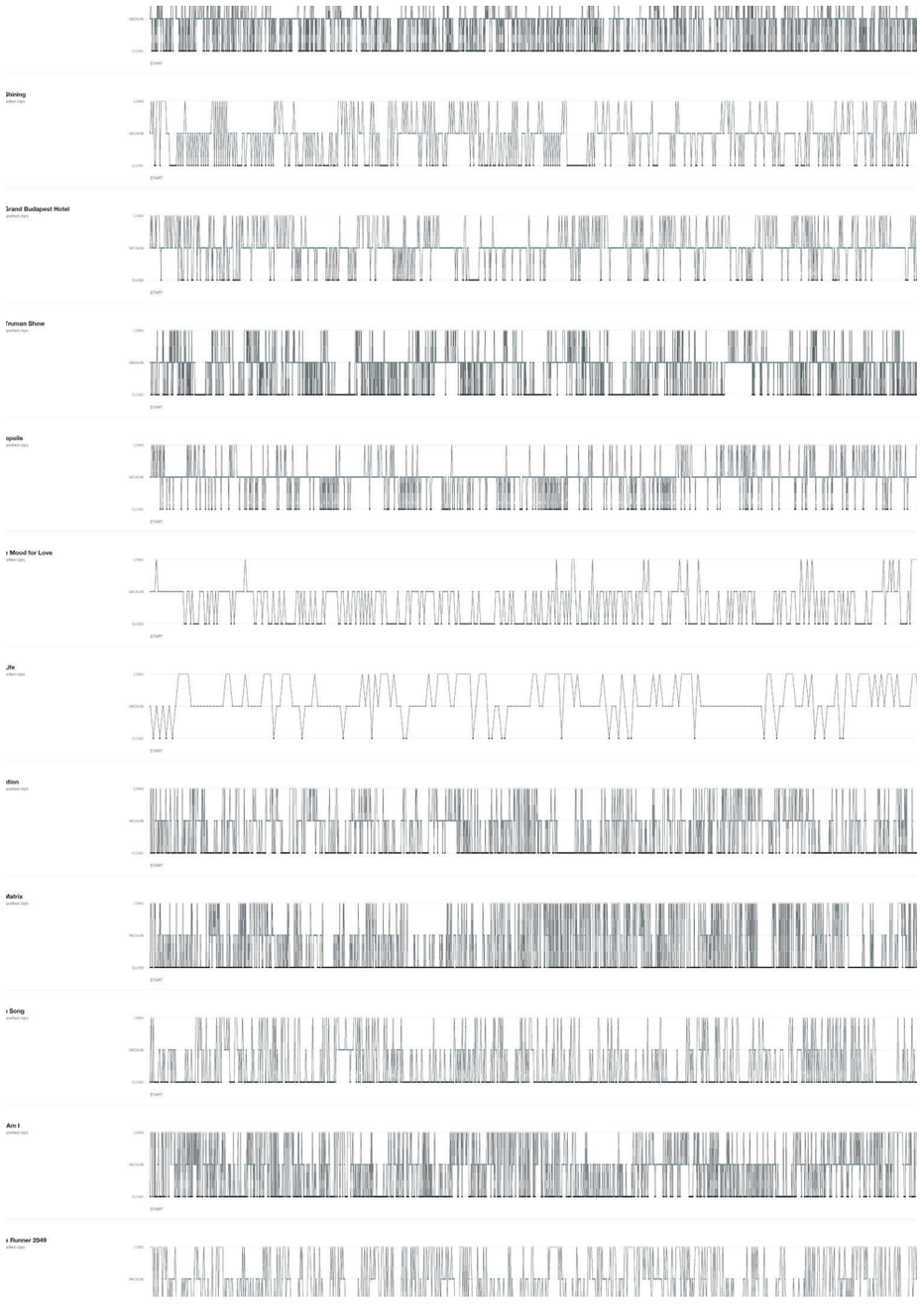
Shot-size



Shot Size classifies each clip as Long, Medium, or Close and links the abstract label back to the visible framing from which it was derived. It describes the relative distance between viewpoint, subject, and surrounding field: Long shots preserve more environmental context, Medium shots balance figure and setting, and Close shots compress attention around a limited visual zone. The frame-sequence diagram makes these changes legible as a continuous rhythm rather than as isolated examples. In spatial translation, Shot Size can regulate enclosure, porosity, viewing distance, and the amount of context available around an element; it should not be read as a literal one-to-one measure of physical room size.

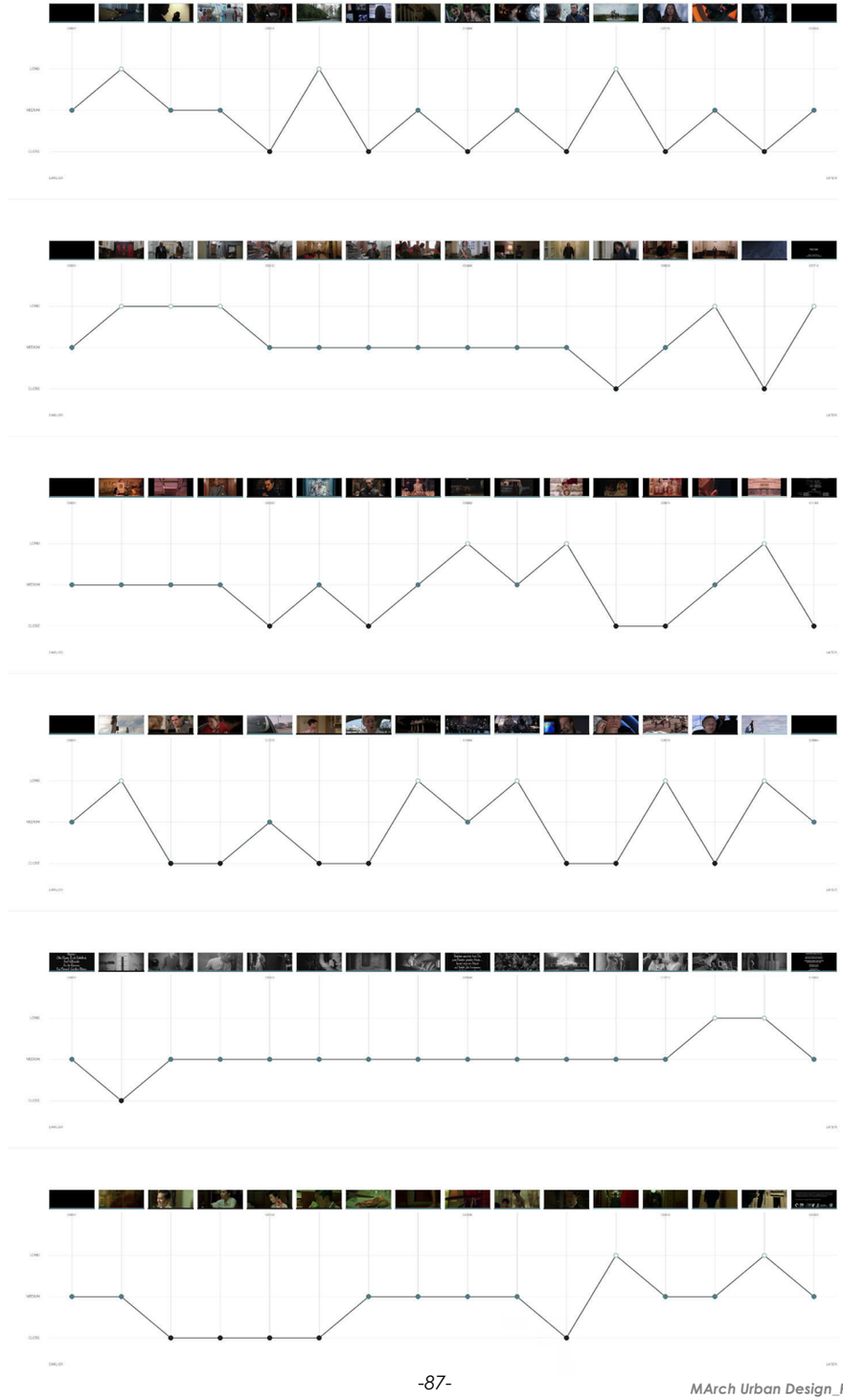
Camera Distance Grammar

Shot size is treated as a changing viewing distance across each film sequence.

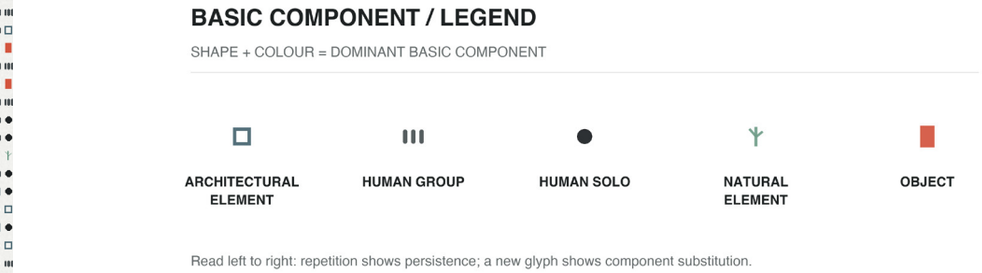


Camera Distance+ Clip Frames

Extracted clip frames paired with their shot-distance sequence.



Basic-component



Basic Components identifies the principal visible element within each clip through five categories: human solo, human group, object, architectural element, and natural element. This layer distinguishes what occupies or activates the scene without treating the detected component as a complete account of the shot. Across a sequence, changes in component type describe shifts between individual presence, collective occupation, material focus, constructed structure, and non-human field conditions. In the spatial workflow, these labels operate as an element vocabulary inserted into or withheld from the Scene Category base state. They differentiate clip-specific state instances, but they do not by themselves determine a finished architectural form.

BASIC COMPONENT EVIDENCE ATLAS

Three real clip instances converge on one component state and one provisional spatial operation.

01

ARCHITECTURAL ELEMENT

5,573 clips / 30.2%

The Shining
C0280

Grand Budapest
C1035

In the Mood
C0491

FRAME / WALL / SLAB INSERTION

Built edges and structural surfaces are read as an inserted component frame.

02

HUMAN GROUP

5,722 clips / 31.1%

The Shining
C0712

In the Mood
C0472

Still Life
C0176

CLUSTER / PARALLEL REPETITION

Several bodies produce a repeated or clustered component field.

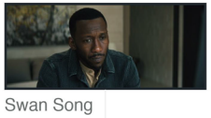
03

HUMAN SOLO

3,655 clips / 19.8%

Grand Budapest
C0469

In the Mood
C0090


Swan Song
C0296

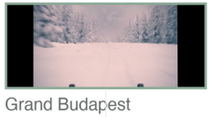
SINGULAR CORE / FOCAL ELEMENT

One body becomes a locally dominant component within the visual field.

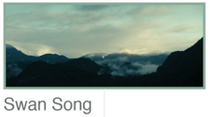
04

NATURAL ELEMENT

1,921 clips / 10.4%


Grand Budapest
C0923

Still Life
C0105


Swan Song
C0575

DISTRIBUTED / BRANCHING FIELD

Natural matter is translated as dispersed, branching, or continuous components.

05

OBJECT

1,556 clips / 8.4%

The Shining
C0211

Who Am I
C1002

Blade Runner 2049
C0293

LOCAL INSERTION / ATTACHED BLOCK

A compact object becomes a local attachment rather than the primary scene mass.

BASIC COMPONENT SEQUENCE TRANSCRIPTION / A

Two high-frequency passages expose reciprocal change between spatial frame, collective field, and isolated body.

01 THE SHINING

C0133–C0174 / 42 clips / 38 state changes

CONTAINER ↔ COLLECTIVE OCCUPATION

C0133 / GROUP C0142 / SOLO C0150 / GROUP C0158 / GROUP C0166 / ARCH C0174 / ARCH

ARCH 16 / GROUP 19 / SOLO 7 / NATURAL 0 / OBJECT 0

02 THE MATRIX

C0444–C0485 / 42 clips / 37 state changes

COLLECTIVE FIELD ↔ ISOLATED BODY

C0444 / SOLO C0453 / SOLO C0460 / GROUP C0468 / GROUP C0477 / GROUP C0485 / SOLO

ARCH 2 / GROUP 17 / SOLO 23 / NATURAL 0 / OBJECT 0

BASIC COMPONENT SEQUENCE TRANSCRIPTION / B

One mixed passage is followed by a comparative index of the three selected sequences.

03 TOMORROWLAND

C0946–C0987 / 42 clips / 36 state changes

MIXED COMPONENT SUBSTITUTION

ARCH 5 / GROUP 11 / SOLO 6 / NATURAL 9 / OBJECT 11

B / COMPARATIVE SEQUENCE INDEX

The same five-state vocabulary produces different persistence and substitution rhythms.

01 THE SHINING

38 state changes

ARCH 16 / GROUP 19 / SOLO 7 / NATURAL 0 / OBJECT 0

02 THE MATRIX

37 state changes

ARCH 2 / GROUP 17 / SOLO 23 / NATURAL 0 / OBJECT 0

03 TOMORROWLAND

36 state changes

ARCH 5 / GROUP 11 / SOLO 6 / NATURAL 9 / OBJECT 11

BASIC COMPONENT TRANSITION MATRIX

The full corpus records persistence and substitution between five component states; film boundaries are excluded.

A / ROW-NORMALISED TRANSITION PROBABILITY

18,415 WITHIN-FILM ADJACENCIES

	ARCH	GROUP	SOLO	NATURAL	OBJECT
ARCH	43.9% n=2,442	25.0% n=1,389	17.2% n=957	7.0% n=391	6.9% n=385
GROUP	24.0% n=1,371	44.1% n=2,522	20.8% n=1,193	5.1% n=293	6.0% n=343
SOLO	27.1% n=992	31.5% n=1,152	28.1% n=1,026	7.3% n=265	6.0% n=220
NATURAL	19.4% n=373	17.2% n=331	13.9% n=267	44.1% n=848	5.3% n=102
OBJECT	24.7% n=384	21.1% n=328	13.7% n=212	7.9% n=123	32.6% n=506

B / CORPUS COMPOSITION

CLIP COUNT AND SHARE

ARCHITECTURAL ELEMENT	5,573 / 30.2%
HUMAN GROUP	5,722 / 31.1%
HUMAN SOLO	3,655 / 19.8%
NATURAL ELEMENT	1,921 / 10.4%
OBJECT	1,556 / 8.4%

C / STRONGEST CROSS-LABEL ADJACENCIES

ARCH → GROUP	n=1,389 / 25.0%
GROUP → ARCH	n=1,371 / 24.0%
GROUP → SOLO	n=1,193 / 20.8%
SOLO → GROUP	n=1,152 / 31.5%

Movement

MOVEMENT / LEGEND

ORIENTATION + GEOMETRY + COLOUR = MOVEMENT STATE

⊙ STATIC	* RANDOM VARIABLE	► RIGHTWARD	◄ LEFTWARD
▲ UPWARD	▼ DOWNWARD	⊙ FORWARD	◇ BACKWARD

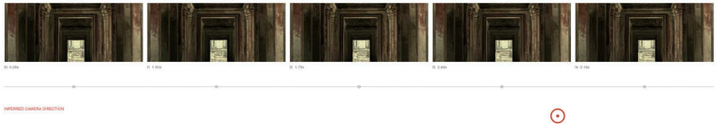
Read left to right: repetition shows directional persistence; a new glyph shows a movement-state change.

Movement records the dominant directional or stability condition of the image through Static, Random Variable, Leftward, Rightward, Upward, Downward, Forward, and Backward states. The analysis is grounded in visible frame-to-frame displacement and is therefore presented together with representative film frames, directional arrows, trajectories, and restrained motion symbols. This prevents the label from becoming detached from the actual cinematic evidence. As a spatial parameter, Movement introduces orientation, approach, retreat, crossing, ascent, descent, or pause into a state instance and into the transition between instances. It describes how a visual field changes across time; it does not claim to identify the narrative intention of that motion.

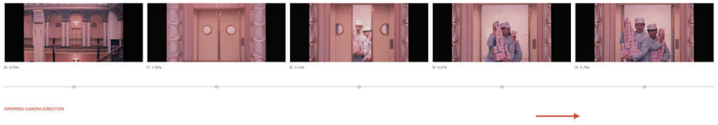
MOVEMENT/ FRAME EVIDENCE

Five sampled frames expose the movement carried by each representative clip

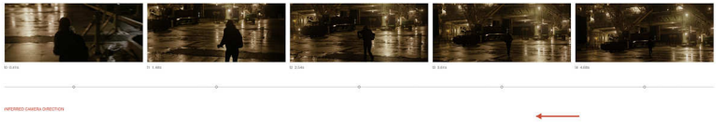
STATIC
In the Mood for Love
C0492 3.46s



RIGHTWARD
The Grand Budapest Hotel
C1043 6.30s



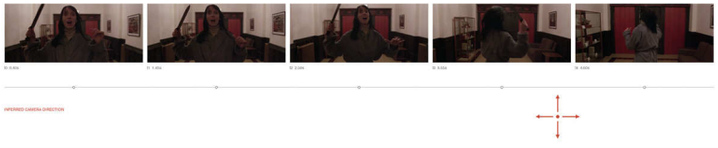
LEFTWARD
Tomorrowland
C0367 5.09s



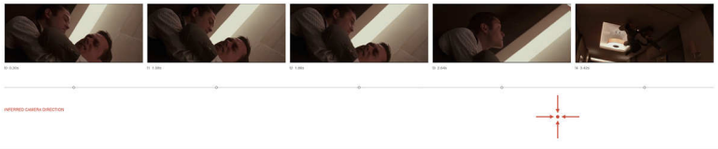
UPWARD
The Matrix
C0013 2.79s

DOWNWARD
Blade Runner 2049
C0263 4.50s

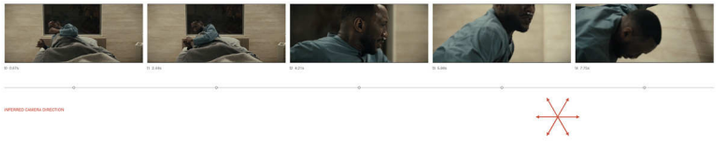
FORWARD
The Shining
C0662 5.00s



BACKWARD
Inception
C1516 3.71s



RANDOM VARIABLE
Swan Song
C0830 8.43s



MOVEMENT / VECTOR TRACES

Tracked image features reveal the measured displacement beneath each directional label.

Rightward

The Grand Budaoest Hotel/ C1043

OBSERVED IMAGE DISPLACEMENT
dx +34.4px dy +18.5px zoom +36.2px

INFERRED CAMERA DIRECTION
11 stable tracked features / confidence 1.00

Leftward

Tomorrowland/C0367

OBSERVED IMAGE DISPLACEMENT
dx -25.9px dy -4.7px zoom +0.5px
INFERRED CAMERA DIRECTION
7 stable tracked features / confidence 1.00

Upward

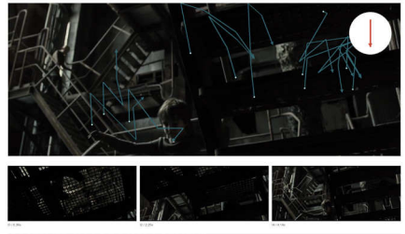
The Matrix/C0013

OBSERVED IMAGE DISPLACEMENT
dx -11.6px dy -40.8px zoom -10.7px

INFERRED CAMERA DIRECTION
8 stable tracked features / confidence 1.00

Downward

Blade Runner 2049/C0263



OBSERVED IMAGE DISPLACEMENT
dx +0.8px dy +12.8px zoom +1.0px

INFERRED CAMERA DIRECTION
11 stable tracked features / confidence 1.00

Forward

The shinning/ C0662

OBSERVED IMAGE DISPLACEMENT
dx +5.0px dy +3.1px zoom +8.2px

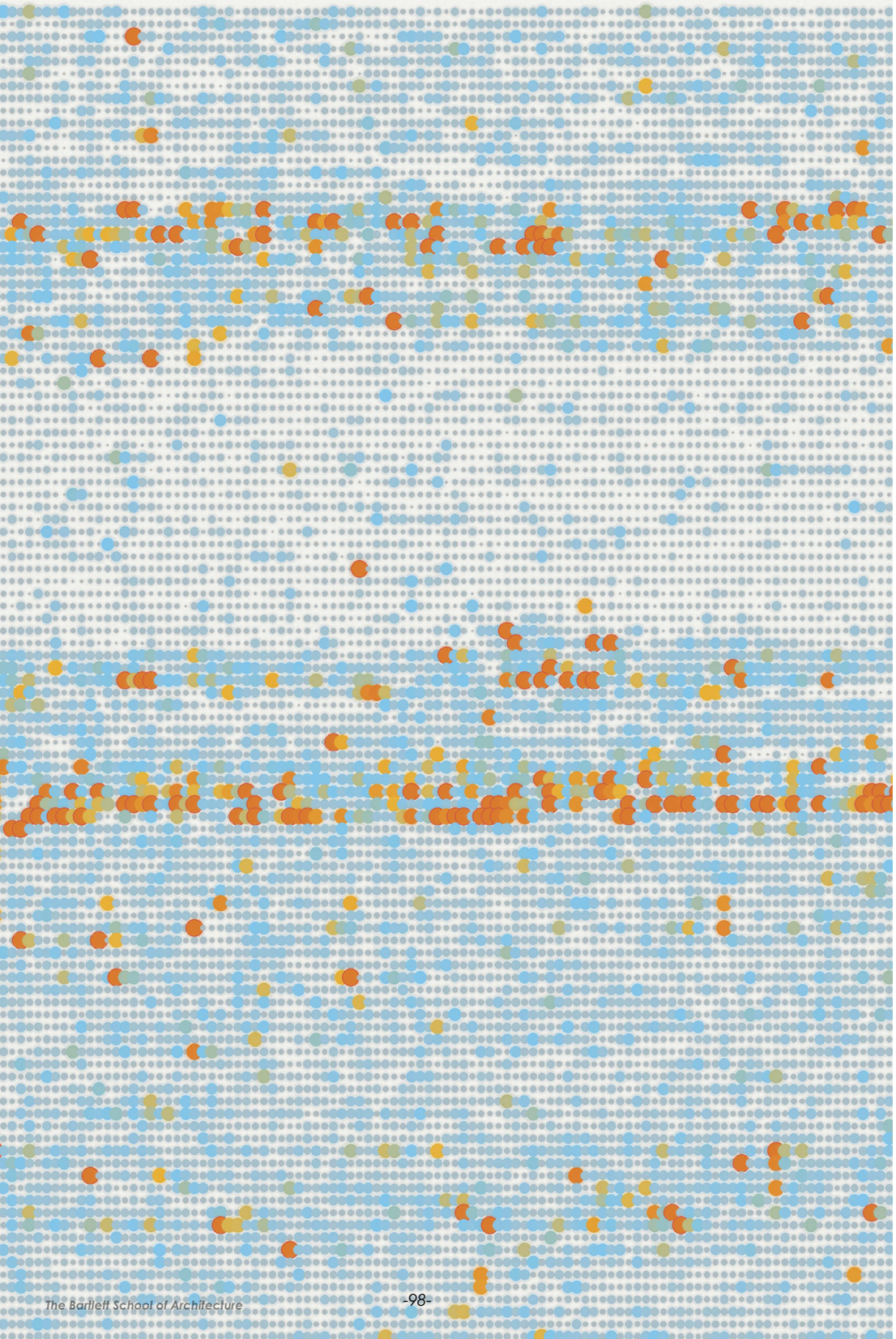
INFERRED CAMERA DIRECTION
10 stable tracked features / confidence 1.00

Backward

Inception/C1516

OBSERVED IMAGE DISPLACEMENT
dx -2.1px dy +1.0px zoom -4.1px

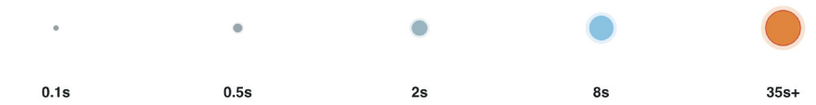
INFERRED CAMERA DIRECTION
3 stable tracked features / confidence 0.72



Duration

DURATION / LEGEND

REAL DURATION > LOG1P NORMALISATION > CIRCLE AREA + COLOUR

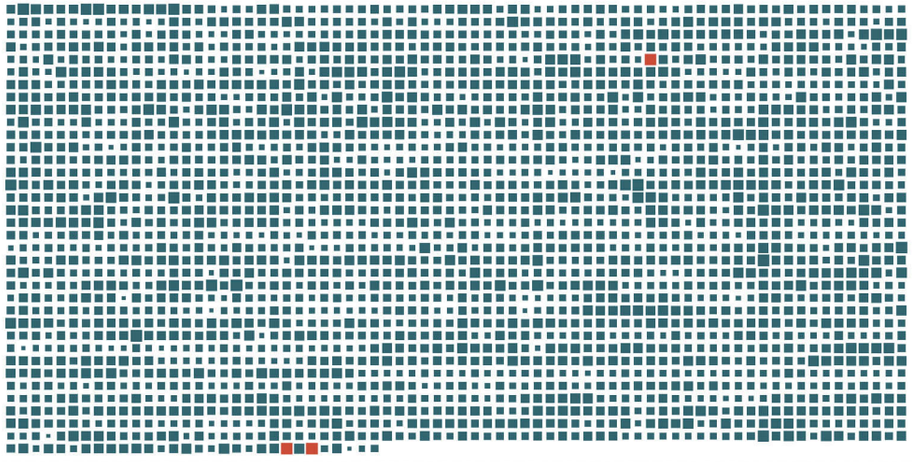


Small/cool = shorter; large/warm = longer; red outline = 35s+; halo is visual emphasis only.

Duration is a continuous value measured in seconds rather than a discrete visual category. It is read first as sequence rhythm: short clips generate dense temporal pulses, while longer clips create pauses, plateaus, or exceptional peaks. Because the corpus contains a wide range of durations, the visual mapping uses logarithmic normalisation so that common clips remain legible without erasing unusually long examples; annotations retain the measured seconds. In the model grammar, Duration acts only after Scene Category, Colour, Basic Components, Shot Size, and Movement have been composed. It uniformly scales the complete state instance while preserving its XYZ proportions and internal relationships—global scale, not local deformation.

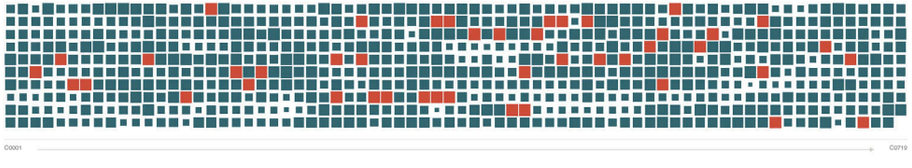
Tomprrowland

2550 clips/ median 2.00s



The Shining

719 clips/ media 6.26s



The Grand Budapest Hotel

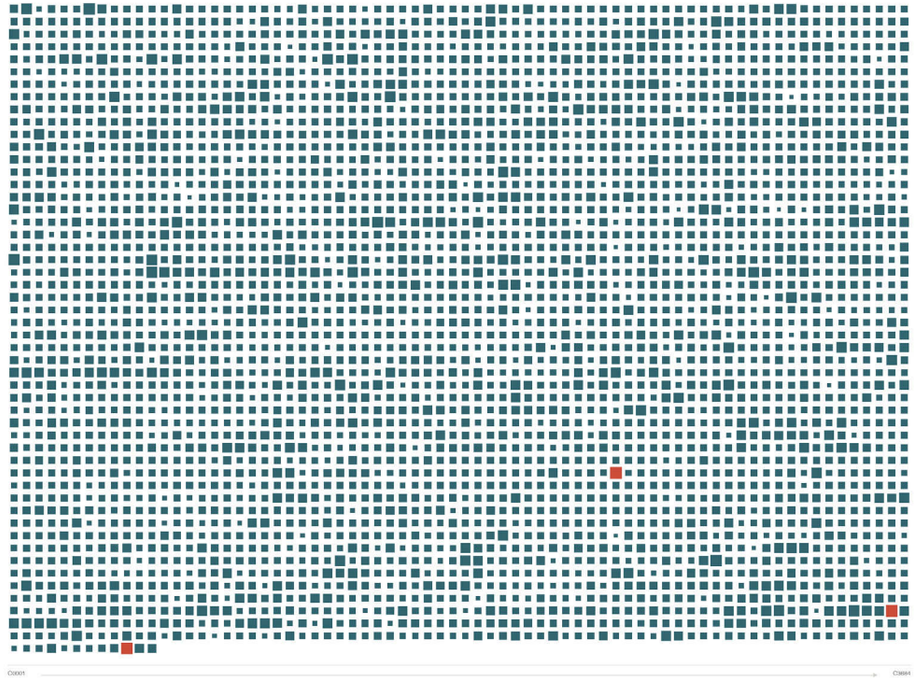
1169 clips/ media 2.92s

Every rectangle is one clip; area follows log-normalised duration and order runs left to right, top to bottom

DURATION /CLIP CELL FIELD

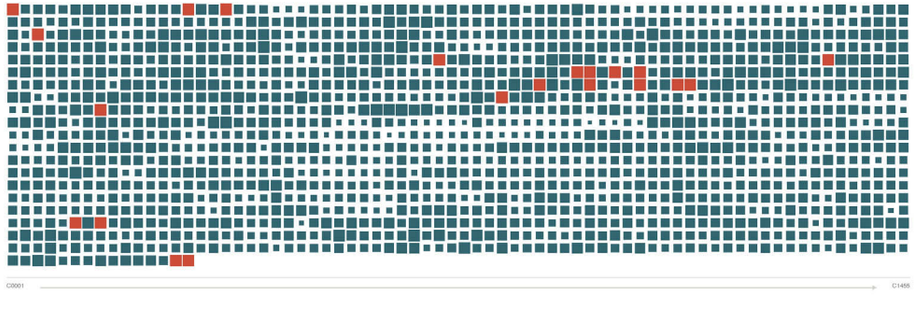
The Truman Show

3684 clips/ median 1.00s



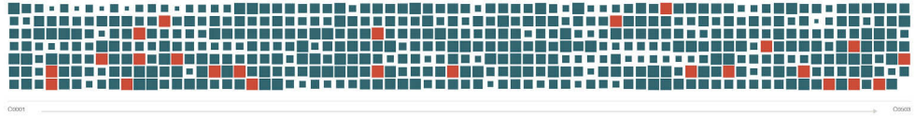
Metropolis

1455 clips/ median 3.75s



In the Mood for Love

503 clips/ median 7.05s



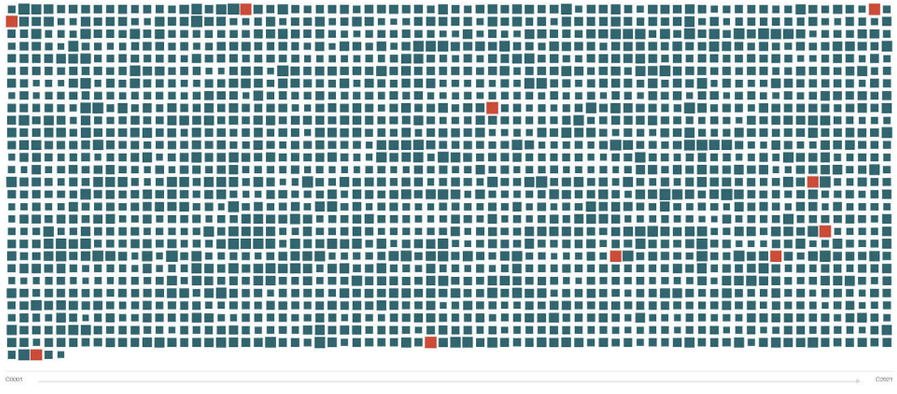
DURATION /CLIP CELL FIELD

Still Life

248 clips/ media 12.72s

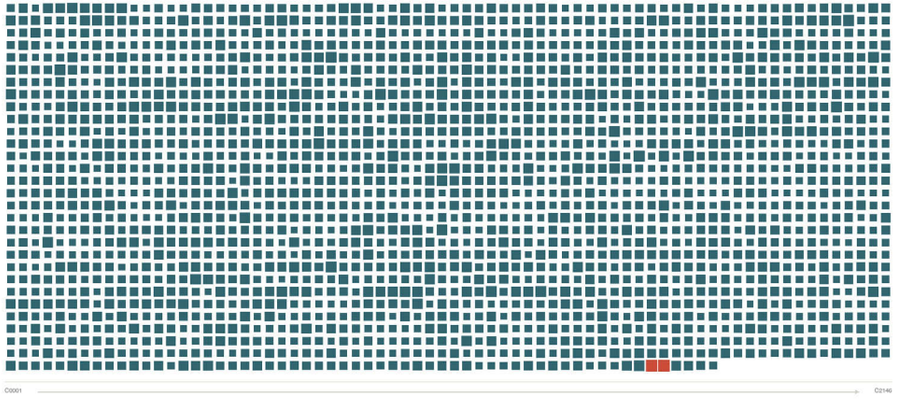
Inception

2021 clips/ media 2.63s



The Matrix

2146 clips/ median 2.50s



CELLAREA/DURATION ■ 0.3s ■ 2.4s ■ 10s ■ 35s

TEAL/ continuous log scale ■ RED/ 35s visual-scale cap reached

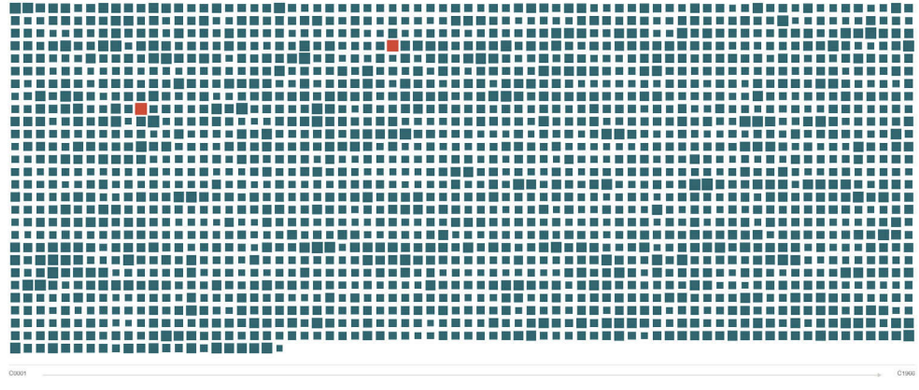
DURATION /CLIP CELL FIELD

Swan Song

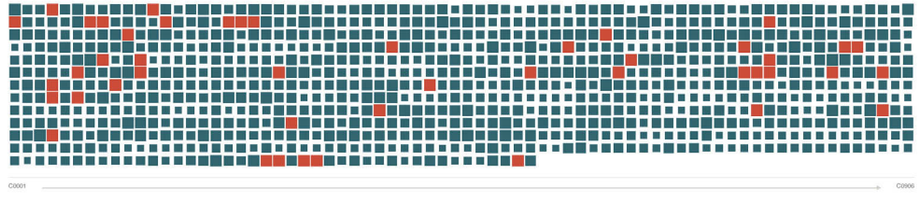
1060 clips/ media 3.75s

Who Am I

1966 clips/ media 2.38s



12 Blade Runner 2049 906 clips / median 4.96s

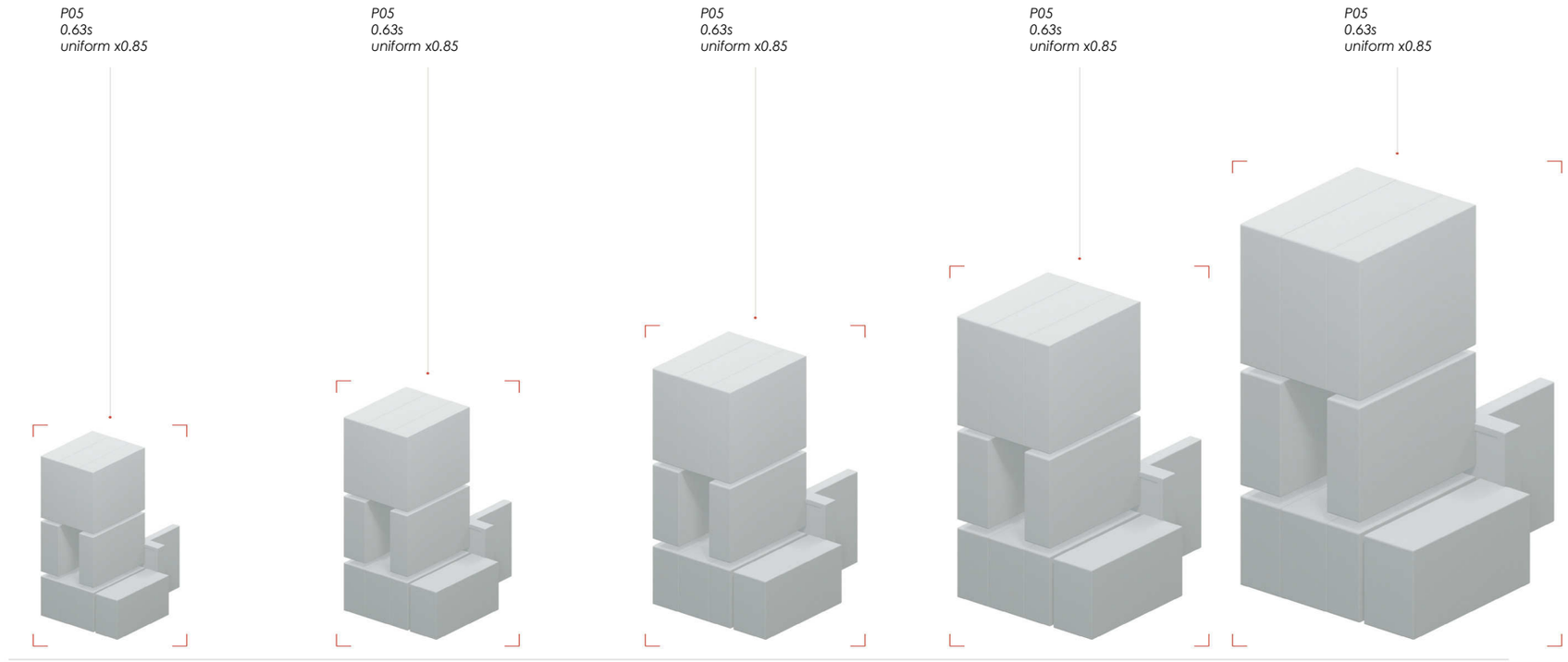


CELLAREA/DURATION ■ 0.3s ■ 2.4s ■ 10s ■ 35s

TEAL/ continuous log scale ■ RED/ 35s visual-scale cap reached

DURATION /GLOBAL MODEL SCALE

One composed clip model is repeated at five observed duration quantiles; only uniform XYZ scale changes.



S_X=S_Y=S_Z internal proportions and label relations remain unchanged

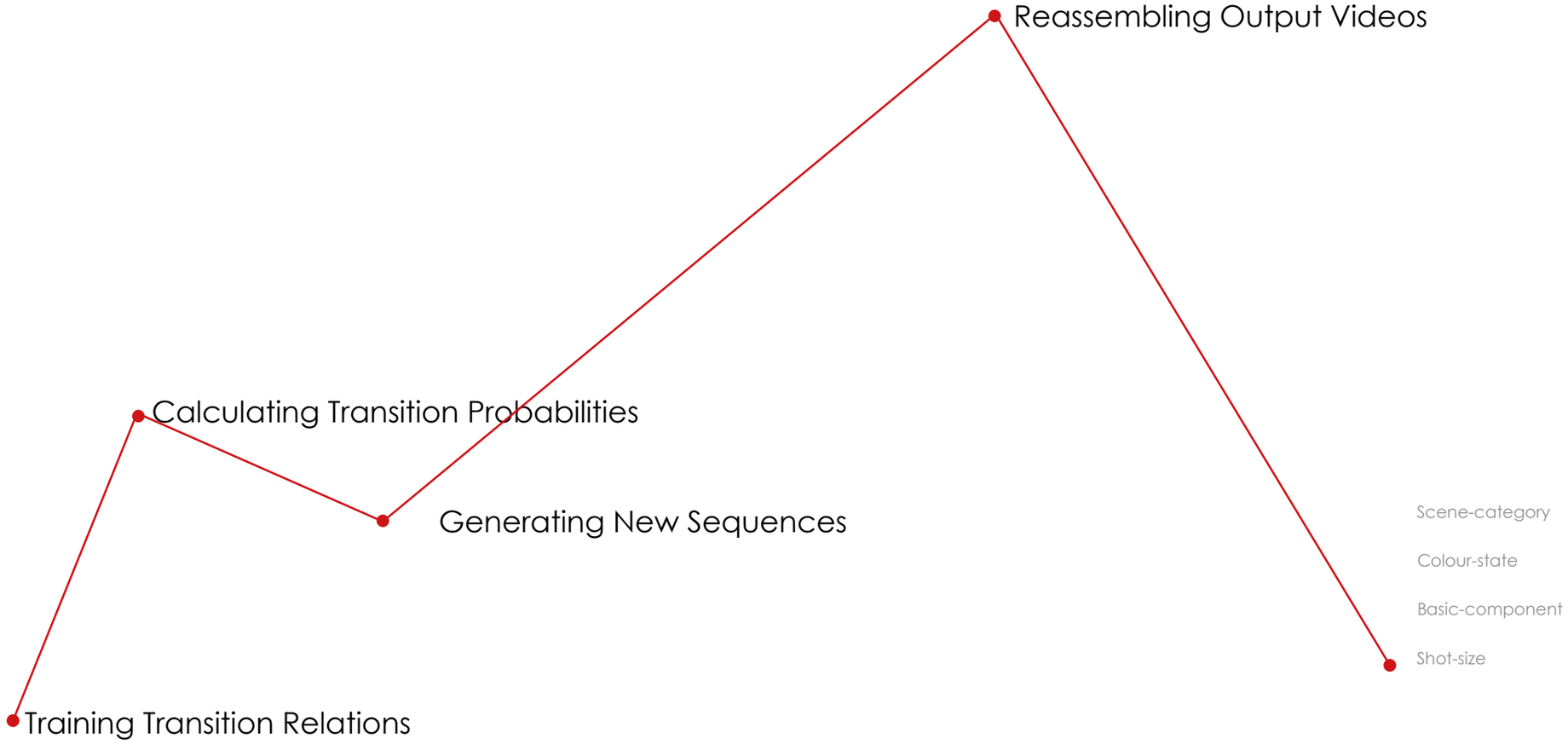
Implemented Mapping

Composed clip model is repeated at five observed duration quantles; only uniform XYZ scale changes

CHAPTER 04

Markov Chain and Duration Logic

→ → → →



Markov Chain

Based on the Wikipedia definition, a Markov chain is a discrete-time stochastic process that describes how a system moves from one state to another within a state space. Its key feature is the Markov property, meaning that the probability of the next state depends only on the current state, not on the full sequence of previous states. In each step, the system may either move to a new state or remain in the same state, and the probabilities of these changes are called transition probabilities. Therefore, a Markov chain can be understood as a probabilistic model for analysing patterns of change and predicting possible future states based on the present condition.

A probabilistic model for generating shot sequences

1. WHAT IS A MARKOV CHAIN?

A Markov chain is a discrete-time stochastic process that describes how a system moves from one state to another.

Key idea (Markov Property):
The next state depends only on the current state, not on the full history before it.

In other words:
Knowing the present is enough to predict the future.
The past can be forgotten.

Mathematically:

$$P(X_{(t+1)} \mid X_t, X_{(t-1)}, \ldots, X_0) = P(X_{(t+1} \mid X_t)$$

(The probability of the next state depends only on the current state.)

2. SIMPLE LOGIC

Current State -> (Transition Probability) -> Next State

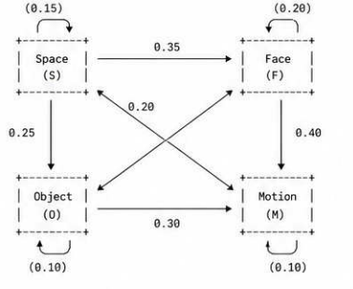
Current State (at time t)	--->	Transition Probability to possible states	-->	Next State (at time t+1)

Example (Film Shots):
If the current shot is SPACE (S):

Space (S)	--->	40% -> Face (F) 35% -> Object (O) 25% -> Motion (M)	---->	Next Shot (?)

The next shot is chosen according to these probabilities.

3. STATE DIAGRAM (EXAMPLE)



S = Space F = Face O = Object M = Motion
Each arrow shows the probability of moving from one shot type to another.

4. HOW IT WORKS IN THIS PROJECT

1. Classify film clips into shot types
   Space (S) / Face (F) / Object (O) / Motion (M)
2. Read the original film sequence
   Convert the film into a sequence of states
   e.g., S -> F -> F -> O -> M -> S
3. Count how often each transition happens
   Count transitions such as S->F, F->F, F->O ...
4. Calculate transition probabilities
   Convert counts into probabilities to form the transition matrix
5. Generate a new sequence
   Start from a state and generate the next state step by step based on the probabilities

5. EXAMPLE SEQUENCE

Original Film Sequence:

`S -> F -> O -> M -> S -> F -> O -> M`

Learned Transition Pattern (intuition):

- Space -> Face
- Face -> Object
- Object -> Motion
- Motion -> Face or Space

Generated New Sequence (Example):

`S -> F -> O -> M -> F -> S -> O -> M`

6. TRANSITION MATRIX (EXAMPLE)

Rows: from current state | Cols: to next state

From \ To	S (Space)	F (Face)	O (Object)	M (Motion)
S (Space)	0.15	0.35	0.25	0.25
F (Face)	0.20	0.20	0.40	0.20
O (Object)	0.20	0.20	0.10	0.30
M (Motion)	0.20	0.20	0.30	0.10

Each row sums to 1.0 (100% probability).
Example: From SPACE, 35% goes to FACE.

7. SUMMARY

Markov chains help us learn the hidden editing logic between different shot types.
By learning transition probabilities from existing films, we can generate new film sequences that follow similar patterns.

Markov State Diagram

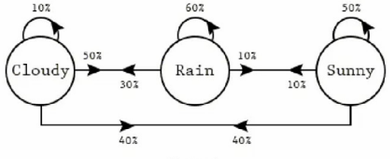


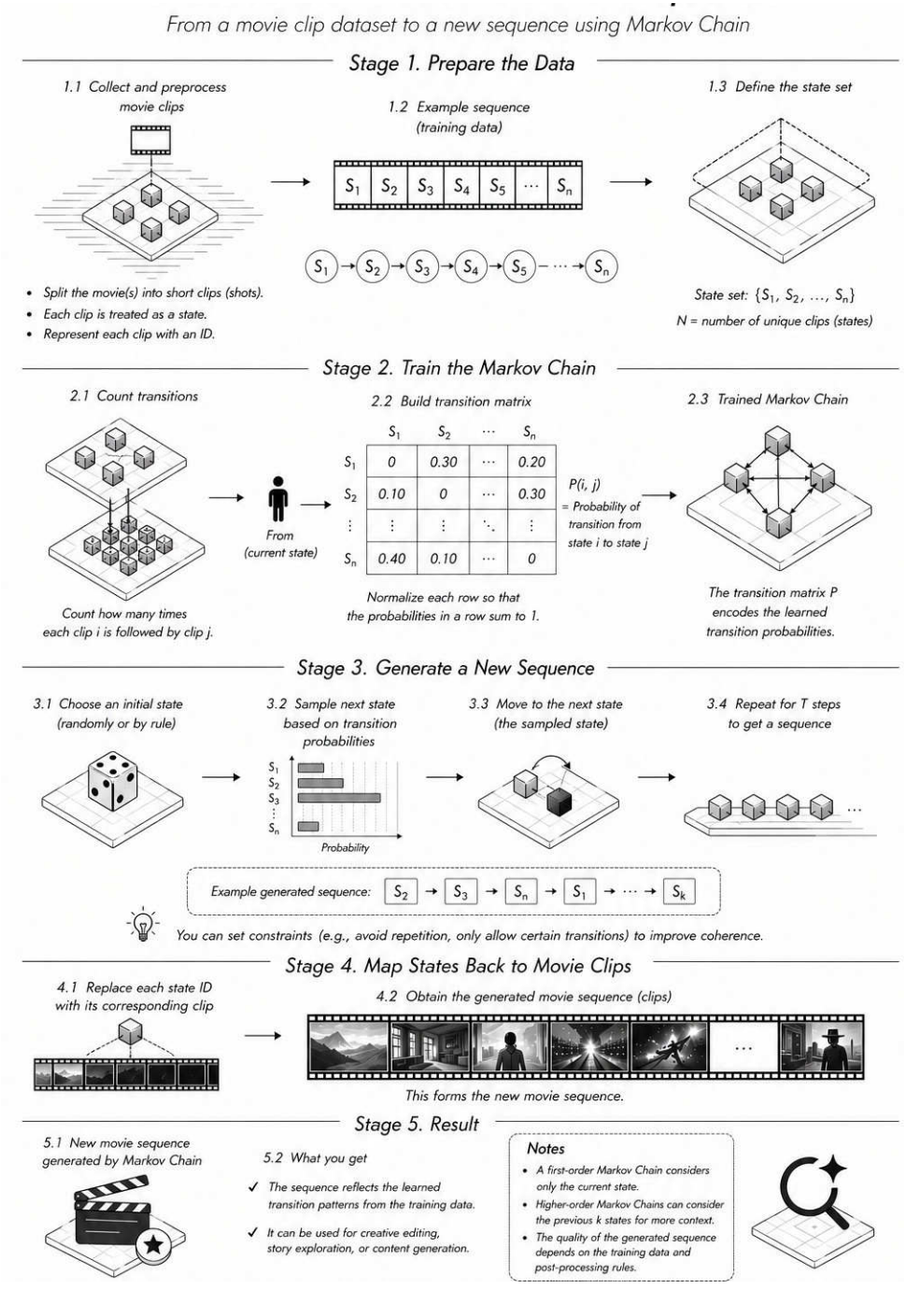
Figure 2

Transition Matrix

	C	R	S
C	0.1	0.5	0.4
R	0.3	0.6	0.1
S	0.4	0.1	0.5

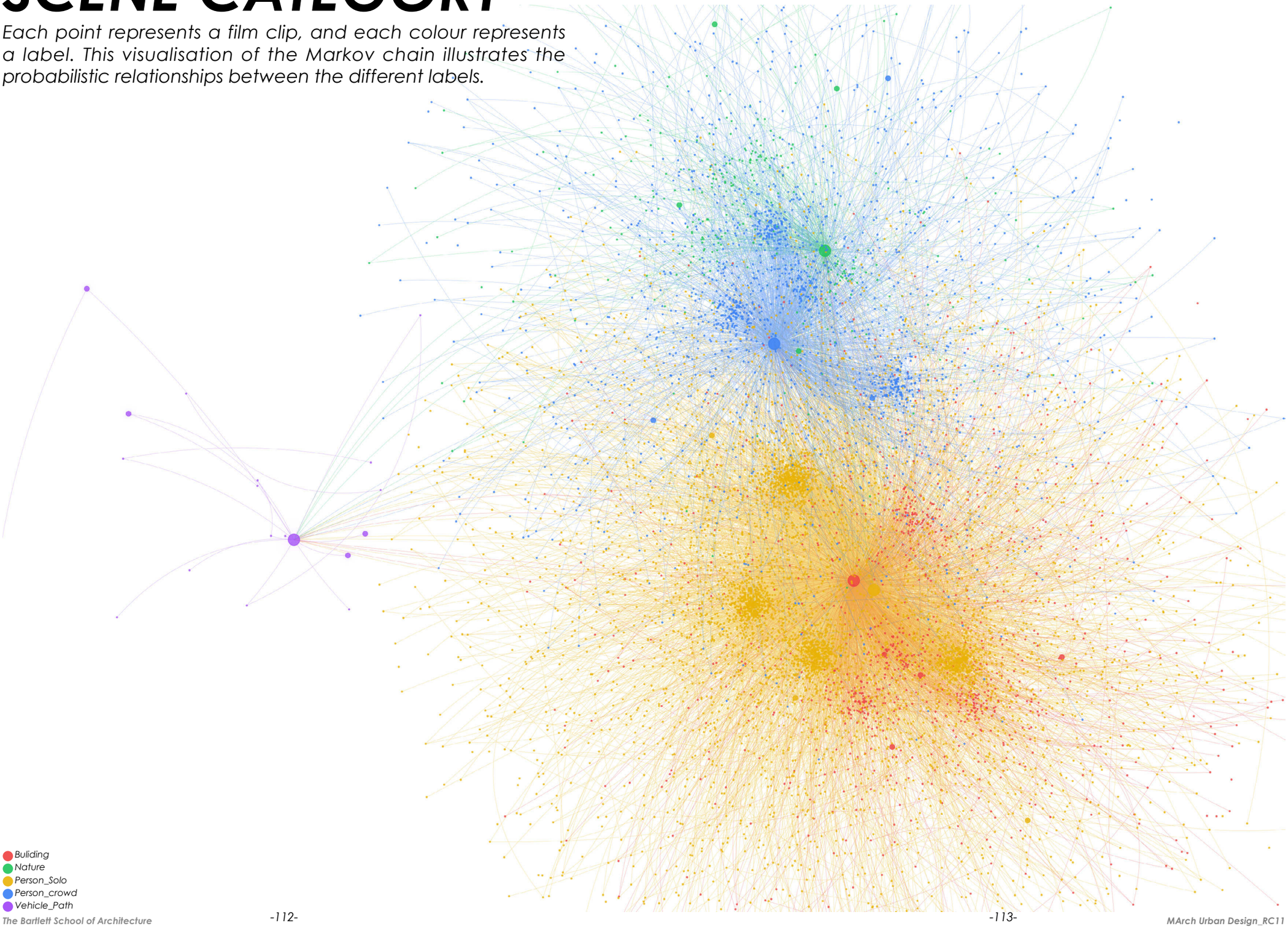
Figure 3

How to Use a Markov chain to Generate a New Moive Sequence



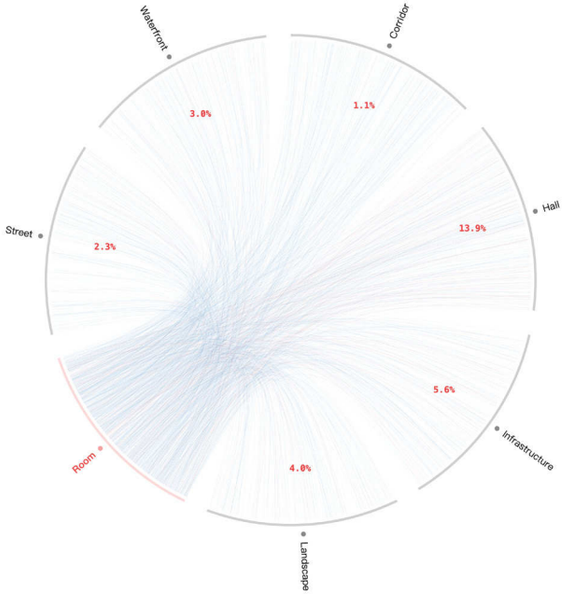
SCENE CATEGORY

Each point represents a film clip, and each colour represents a label. This visualisation of the Markov chain illustrates the probabilistic relationships between the different labels.



SCENE CATEGORY

Visualization of Transition Probabilities for Each Label



SOURCE STATE

Room

OUTBOUND TRANSITIONS

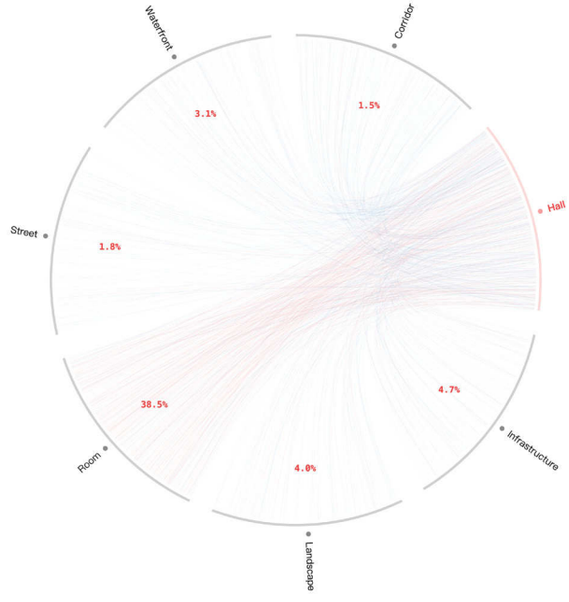
→ Room	70.1%	350 OCCURRENCES IN TRAINING DATA
→ Hall	13.9%	70 OCCURRENCES IN TRAINING DATA
→ Infrastructure	5.6%	28 OCCURRENCES IN TRAINING DATA
→ Landscape	4.0%	20 OCCURRENCES IN TRAINING DATA
→ Waterfront	3.0%	15 OCCURRENCES IN TRAINING DATA
→ Street	2.3%	12 OCCURRENCES IN TRAINING DATA
→ Corridor	1.1%	5 OCCURRENCES IN TRAINING DATA

▷ *Sequence Prediction*

CLEAR SEQUENCE

1 Infrastructure 2 Waterfront 3 Landscape 4 Landscape 5 Landscape 6 Landscape 7 Landscape 8 Landscape 9 Room 10 Landscape 11 Hall 12 Hall 13 Hall 14 Hall 15 Hall

Generate Next Clip Label →



SOURCE STATE

Hall

OUTBOUND TRANSITIONS

→ Hall	46.4%	232 OCCURRENCES IN TRAINING DATA
→ Room	38.5%	192 OCCURRENCES IN TRAINING DATA
→ Infrastructure	4.7%	24 OCCURRENCES IN TRAINING DATA
→ Landscape	4.0%	20 OCCURRENCES IN TRAINING DATA
→ Waterfront	3.1%	15 OCCURRENCES IN TRAINING DATA
→ Street	1.8%	9 OCCURRENCES IN TRAINING DATA
→ Corridor	1.5%	7 OCCURRENCES IN TRAINING DATA

▷ *Sequence Prediction*

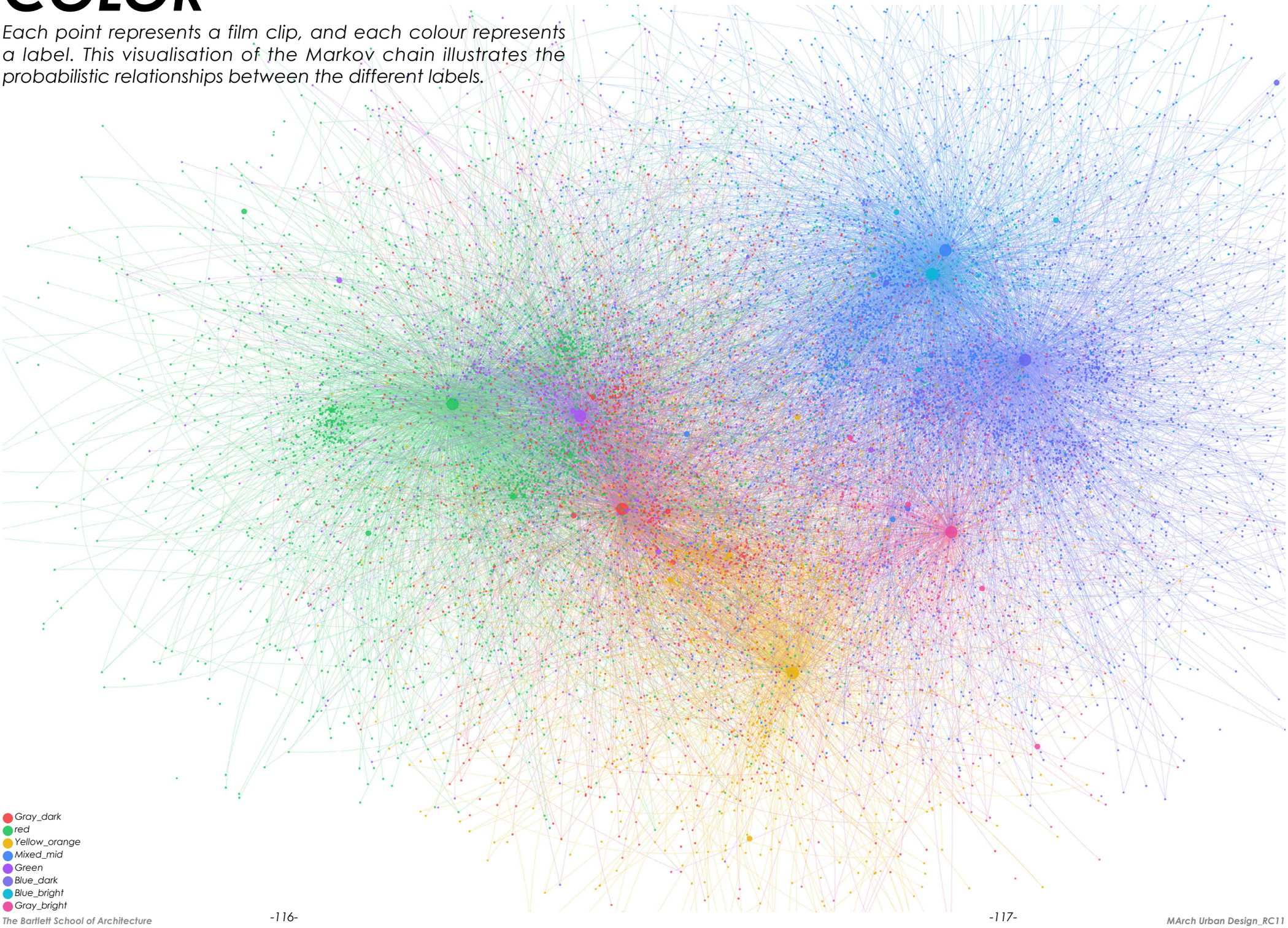
CLEAR SEQUENCE

1 Room 2 Landscape 3 Landscape 4 Landscape 5 Room 6 Room 7 Room 8 Street 9 Infrastructure 10 Infrastructure 11 Hall 12 Room

Generate Next Clip Label →

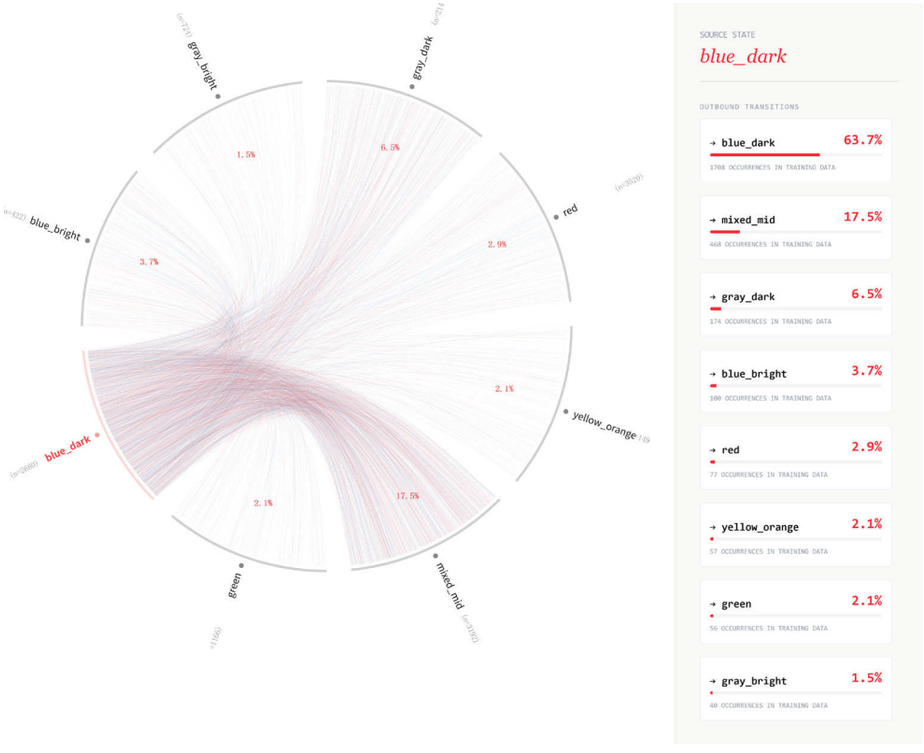
COLOR

Each point represents a film clip, and each colour represents a label. This visualisation of the Markov chain illustrates the probabilistic relationships between the different labels.



COLOR

Visualization of Transition Probabilities for Each Label



SOURCE STATE

blue_dark

OUTBOUND TRANSITIONS

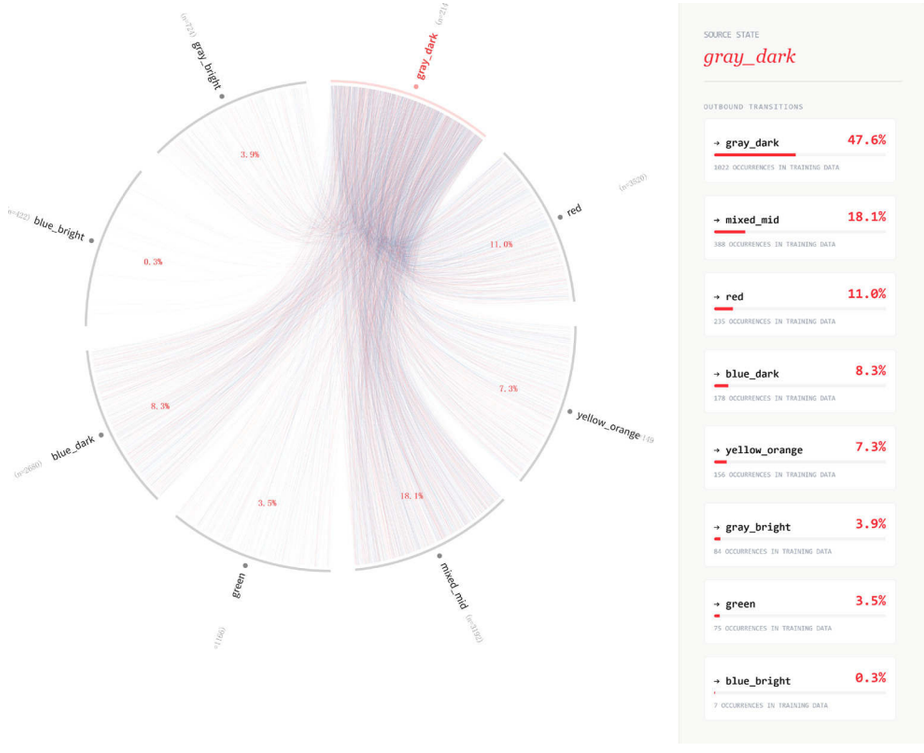
→ blue_dark	63.7%	1708 OCCURRENCES IN TRAINING DATA
→ mixed_mid	17.5%	468 OCCURRENCES IN TRAINING DATA
→ gray_dark	6.5%	174 OCCURRENCES IN TRAINING DATA
→ blue_bright	3.7%	100 OCCURRENCES IN TRAINING DATA
→ red	2.9%	77 OCCURRENCES IN TRAINING DATA
→ yellow_orange	2.1%	57 OCCURRENCES IN TRAINING DATA
→ green	2.1%	56 OCCURRENCES IN TRAINING DATA
→ gray_bright	1.5%	40 OCCURRENCES IN TRAINING DATA

▷ *Sequence Prediction*

CLEAR SEQUENCE

1 blue_dark 2 blue_dark 3 blue_dark 4 blue_dark 5 blue_dark 6 mixed_mid 7 green 8 red
9 red 10 mixed_mid 11 mixed_mid 12 gray_dark 13 mixed_mid 14 gray_dark 15 gray_bright
16 gray_dark 17 yellow_orange 18 yellow_orange 19 red 20 red 21 gray_dark 22 gray_dark

Generate Next Clip Label →



SOURCE STATE

gray_dark

OUTBOUND TRANSITIONS

→ gray_dark	47.6%	1022 OCCURRENCES IN TRAINING DATA
→ mixed_mid	18.1%	388 OCCURRENCES IN TRAINING DATA
→ red	11.0%	235 OCCURRENCES IN TRAINING DATA
→ blue_dark	8.3%	178 OCCURRENCES IN TRAINING DATA
→ yellow_orange	7.3%	156 OCCURRENCES IN TRAINING DATA
→ gray_bright	3.9%	84 OCCURRENCES IN TRAINING DATA
→ green	3.5%	75 OCCURRENCES IN TRAINING DATA
→ blue_bright	0.3%	7 OCCURRENCES IN TRAINING DATA

▷ *Sequence Prediction*

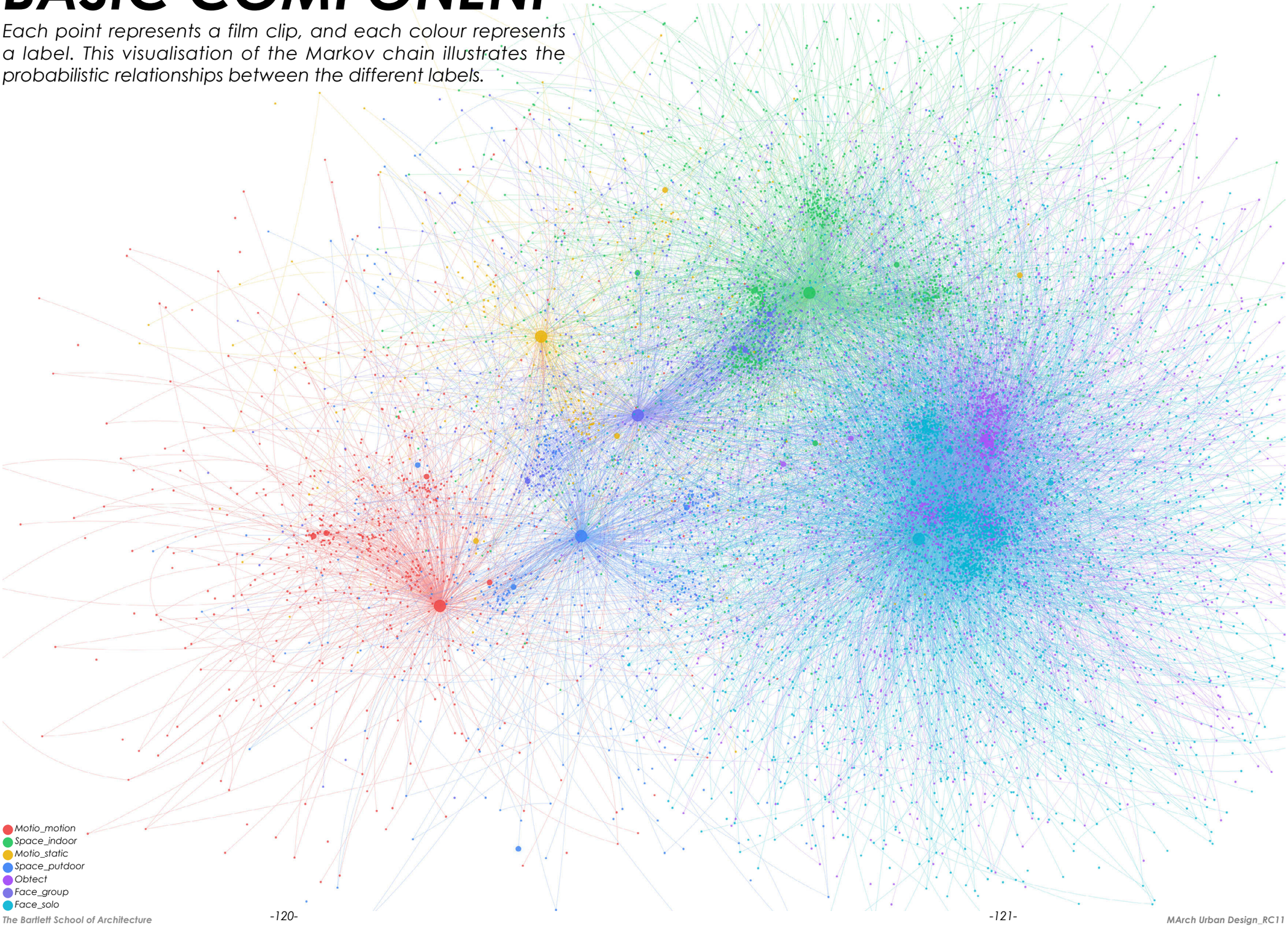
CLEAR SEQUENCE

1 gray_dark 2 blue_dark 3 blue_dark 4 blue_bright 5 blue_bright 6 blue_bright
7 blue_bright 8 gray_bright 9 blue_bright 10 blue_bright 11 gray_dark 12 red 13 red
14 yellow_orange 15 red 16 red 17 red

Generate Next Clip Label →

BASIC COMPONENt

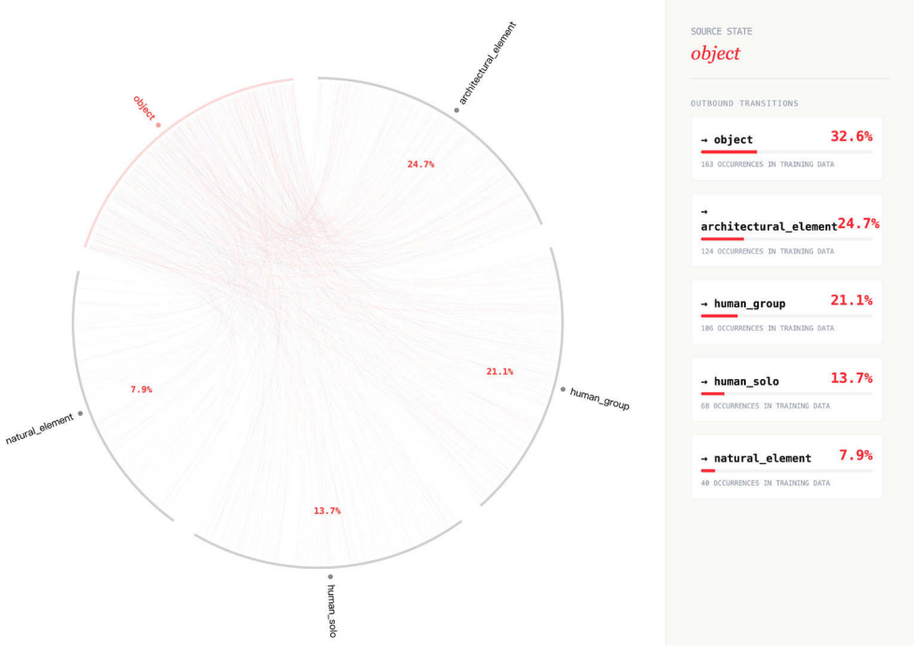
Each point represents a film clip, and each colour represents a label. This visualisation of the Markov chain illustrates the probabilistic relationships between the different labels.

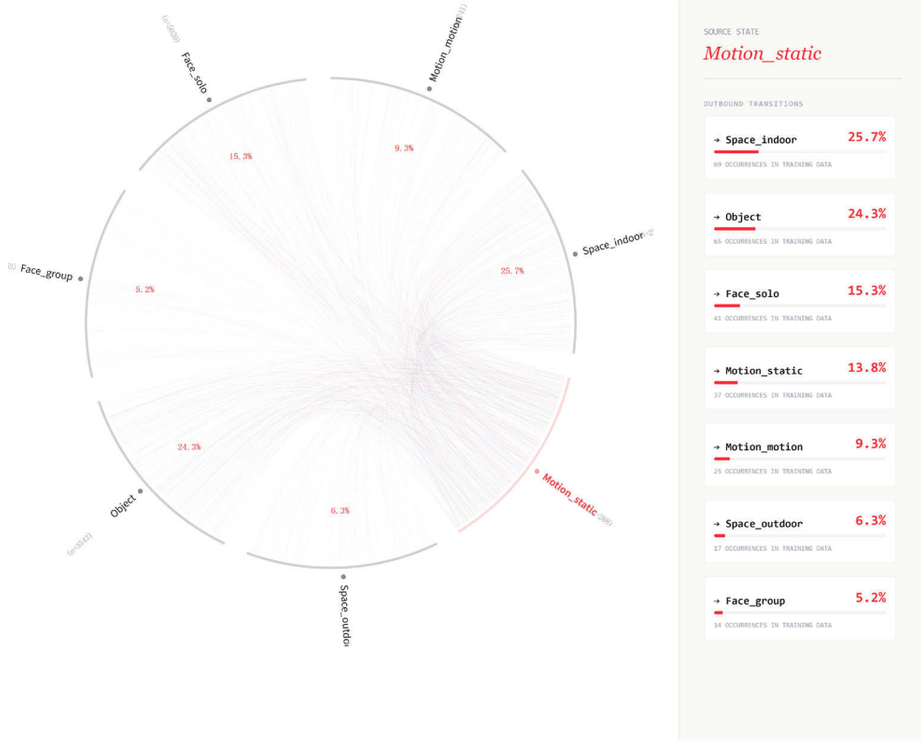


BASIC COMPONENT

Visualization of Transition Probabilities for Each Label

https://app-markov-chain-a97mq1.flyploy.com/





▷ *Sequence Prediction*

CLEAR SEQUENCE

1 architectural_element 2 human_solo 3 human_solo 4 architectural_element
5 architectural_element 6 human_solo 7 architectural_element 8 human_group 9 natural_element
10 human_solo 11 architectural_element 12 human_group

Generate Next Clip Label →

▷ *Sequence Prediction*

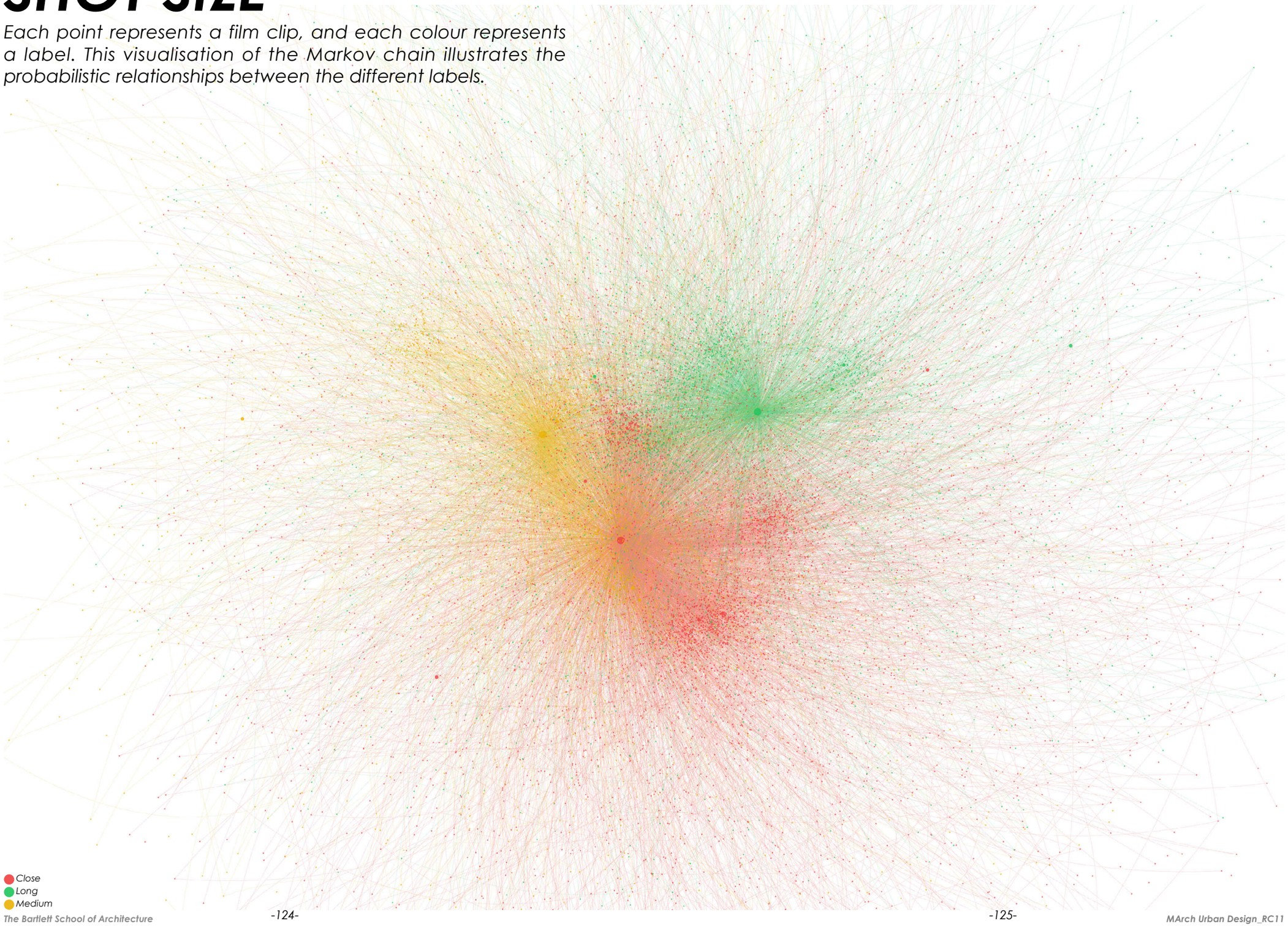
CLEAR SEQUENCE

1 architectural_element 2 object 3 object 4 object 5 object 6 architectural_element
7 natural_element 8 object 9 human_solo 10 architectural_element 11 object 12 object
13 object

Generate Next Clip Label →

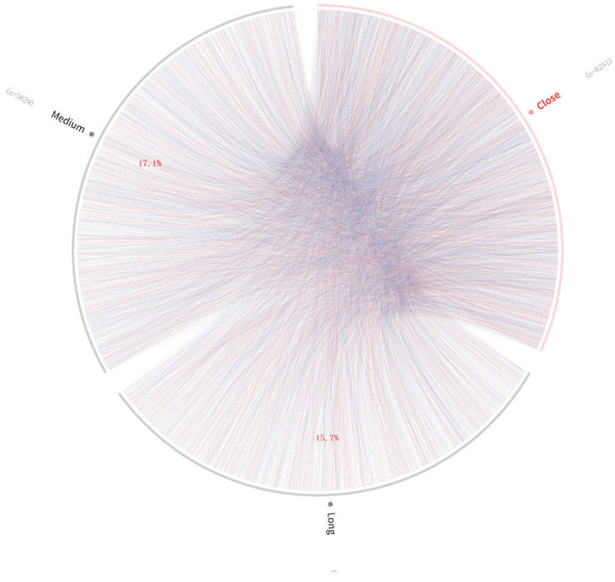
SHOT SIZE

Each point represents a film clip, and each colour represents a label. This visualisation of the Markov chain illustrates the probabilistic relationships between the different labels.



SHOT SIZE

Visualization of Transition Probabilities for Each Label



SOURCE STATE

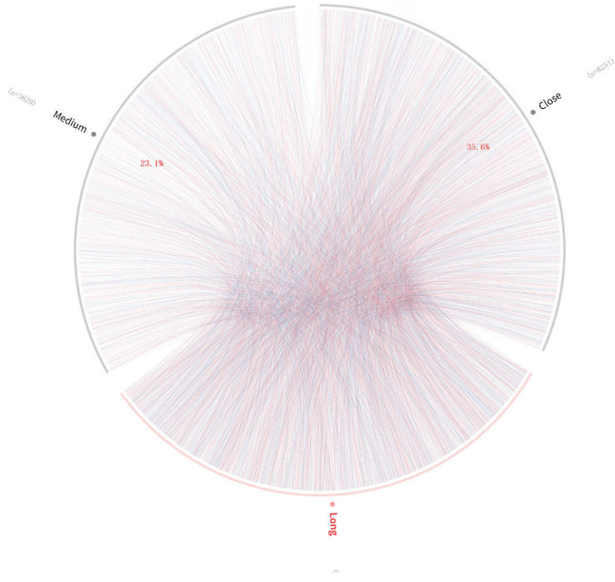
Close

OUTBOUND TRANSITIONS

→ Close 67.1%
5540 OCCURRENCES IN TRAINING DATA

→ Medium 17.1%
1415 OCCURRENCES IN TRAINING DATA

→ Long 15.7%
1296 OCCURRENCES IN TRAINING DATA



SOURCE STATE

Long

OUTBOUND TRANSITIONS

→ Long 41.3%
1429 OCCURRENCES IN TRAINING DATA

→ Close 35.6%
1232 OCCURRENCES IN TRAINING DATA

→ Medium 23.1%
799 OCCURRENCES IN TRAINING DATA

▷ *Sequence Prediction*

CLEAR SEQUENCE

1 Close 2 Close 3 Long 4 Long 5 Close 6 Close 7 Close 8 Close 9 Close
10 Close 11 Medium 12 Long 13 Close 14 Close 15 Close 16 Long 17 Medium 18 Long
19 Medium 20 Close

Generate Next Clip Label →

▷ *Sequence Prediction*

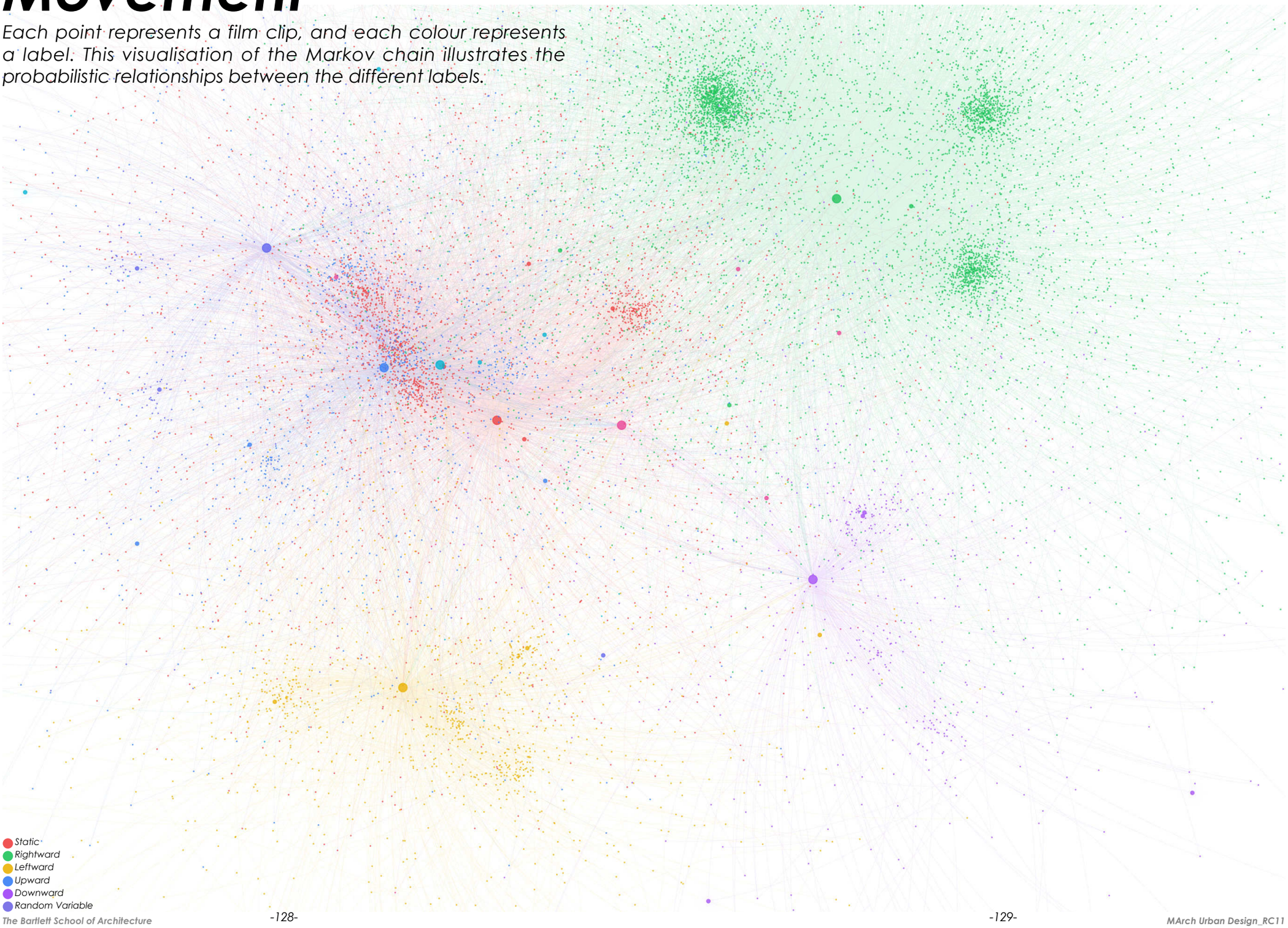
CLEAR SEQUENCE

1 Close 2 Long 3 Medium 4 Close 5 Long 6 Close 7 Close 8 Close 9 Close
10 Close 11 Close 12 Close 13 Close 14 Close 15 Long 16 Long 17 Medium 18 Close
19 Close 20 Long 21 Close 22 Close 23 Long 24 Medium 25 Medium 26 Close

Generate Next Clip Label →

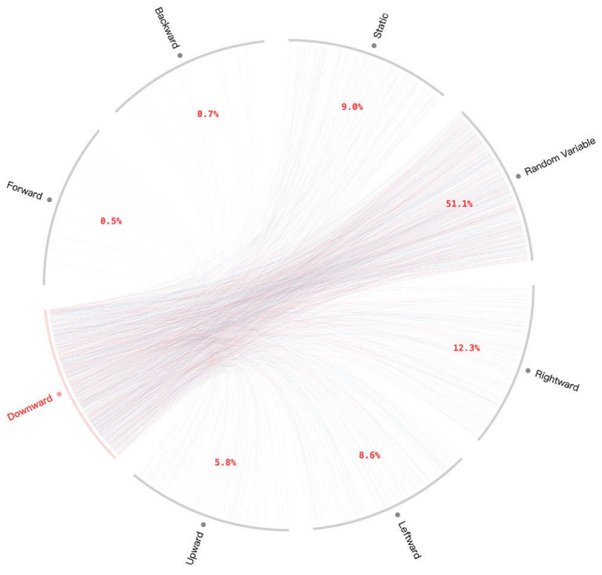
Movement

Each point represents a film clip, and each colour represents a label. This visualisation of the Markov chain illustrates the probabilistic relationships between the different labels.



MOVEMENT

Visualization of Transition Probabilities for Each Label



SOURCE STATE

Downward

OUTBOUND TRANSITIONS

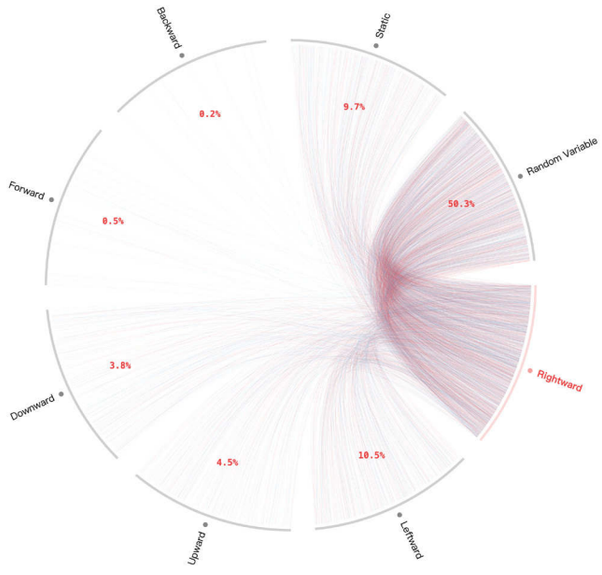
→ Random Variable	51.1%	308 OCCURRENCES IN TRAINING DATA
→ Rightward	12.3%	74 OCCURRENCES IN TRAINING DATA
→ Downward	12.1%	73 OCCURRENCES IN TRAINING DATA
→ Static	9.0%	54 OCCURRENCES IN TRAINING DATA
→ Leftward	8.6%	52 OCCURRENCES IN TRAINING DATA
→ Upward	5.8%	35 OCCURRENCES IN TRAINING DATA
→ Backward	0.7%	4 OCCURRENCES IN TRAINING DATA
→ Forward	0.5%	3 OCCURRENCES IN TRAINING DATA

▷ *Sequence Prediction*

CLEAR SEQUENCE

1 Rightward | 2 Random Variable | 3 Leftward | 4 Random Variable | 5 Static | 6 Static
7 Static | 8 Static | 9 Random Variable | 10 Random Variable | 11 Random Variable
12 Random Variable

Generate Next Clip Label →



SOURCE STATE

Rightward

OUTBOUND TRANSITIONS

→ Random Variable	50.3%	829 OCCURRENCES IN TRAINING DATA
→ Rightward	20.6%	339 OCCURRENCES IN TRAINING DATA
→ Leftward	10.5%	173 OCCURRENCES IN TRAINING DATA
→ Static	9.7%	160 OCCURRENCES IN TRAINING DATA
→ Upward	4.5%	74 OCCURRENCES IN TRAINING DATA
→ Downward	3.8%	62 OCCURRENCES IN TRAINING DATA
→ Forward	0.5%	9 OCCURRENCES IN TRAINING DATA
→ Backward	0.2%	3 OCCURRENCES IN TRAINING DATA

▷ *Sequence Prediction*

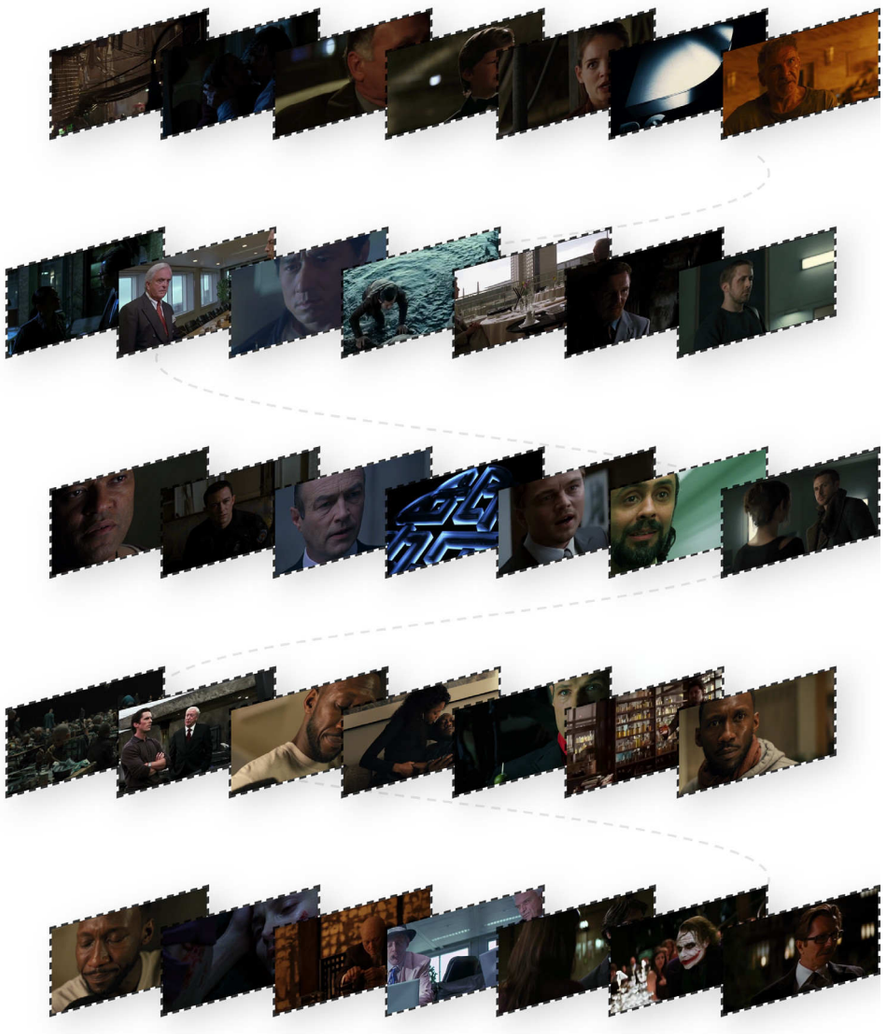
CLEAR SEQUENCE

1 Rightward | 2 Random Variable | 3 Leftward | 4 Random Variable | 5 Static | 6 Static
7 Static | 8 Static | 9 Random Variable | 10 Random Variable | 11 Random Variable
12 Random Variable

Generate Next Clip Label →

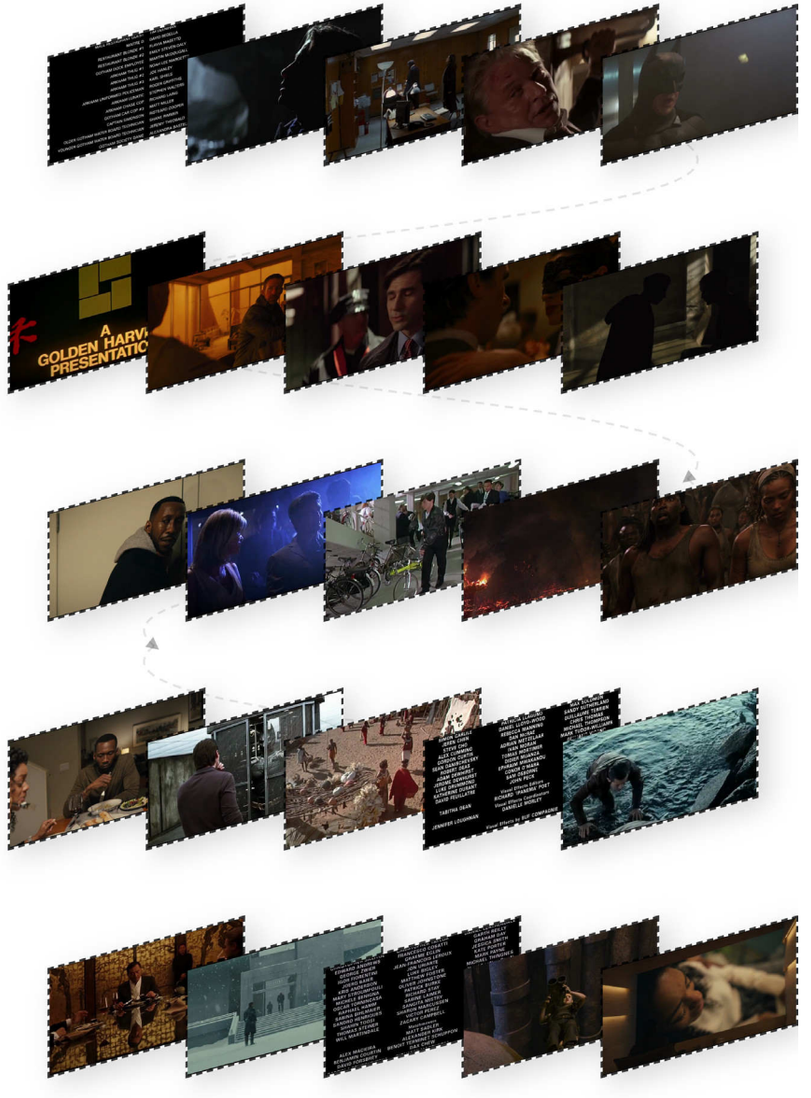
Scene Category

Building → Person_Solo → Building → Building → Person_Crowd → Person_Solo → Person_Solo → Person_Solo → Person_Solo → Person_Solo → Person_Crowd → Person_Solo



COLOR

gray_dark → gray_dark → yellow_orange → yellow_orange → red → red → red → red → gray_dark → gray_dark → gray_dark → mixed_mid → red → red → red → mixed_mid → mixed_mid → blue_dark → red → red → red → red → mixed_mid → gray_dark → gray_dark



Basic Components

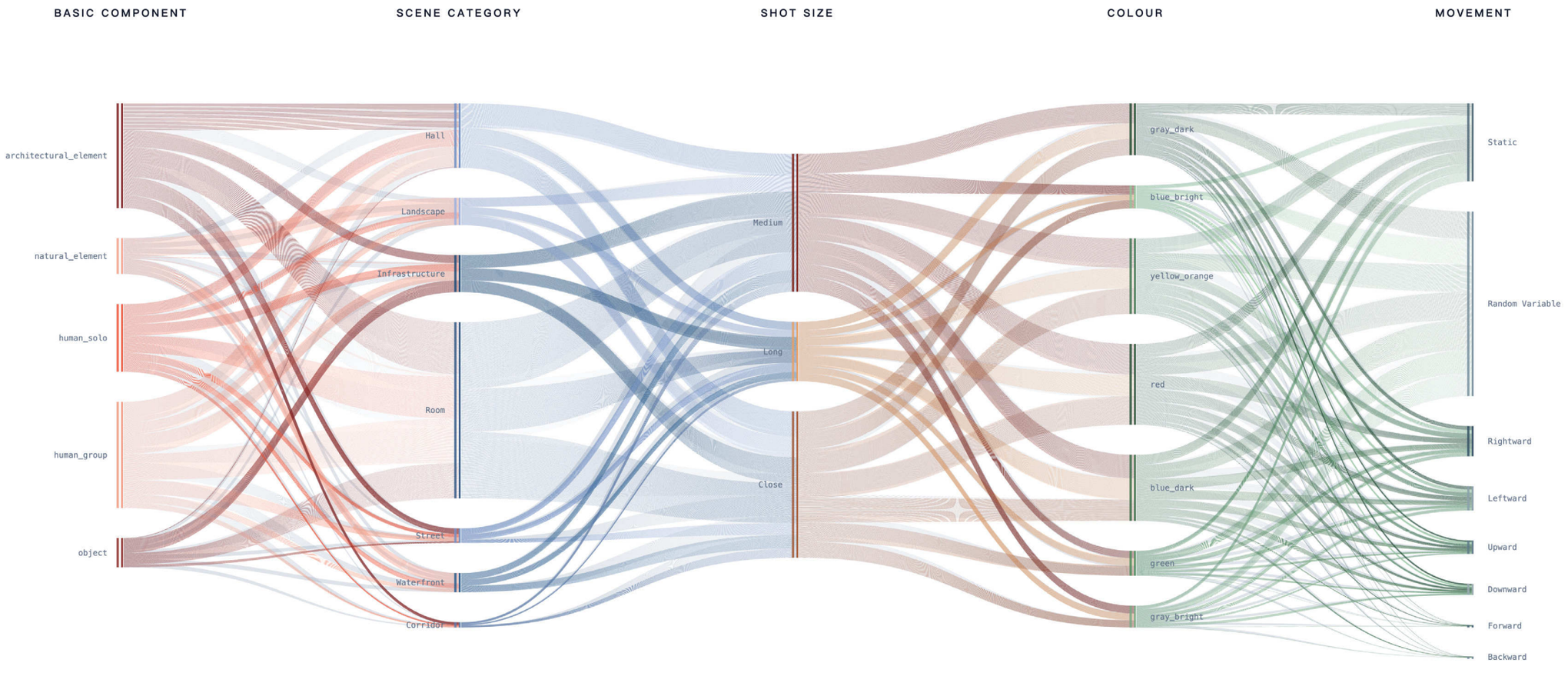
Space_outdoor → Space_outdoor → Motion_static → Object → Object → Face_solo → Space_indoor → Face_group → Face_group → Face_solo → Face_solo → Face_solo → Object → Face_solo → Face_solo → Object → Space_outdoor → Space_outdoor → Motion_static → Motion_motion

SHOT SIZE

Close → Medium → Close → Close → Close → Close → Medium → Long → Close → Close → Close → Close → Close → Close → Long → Close

Aggregate Markov Chain

The grey lines represent all possible combinations; the highlighted ones are those that actually exist in the film library.

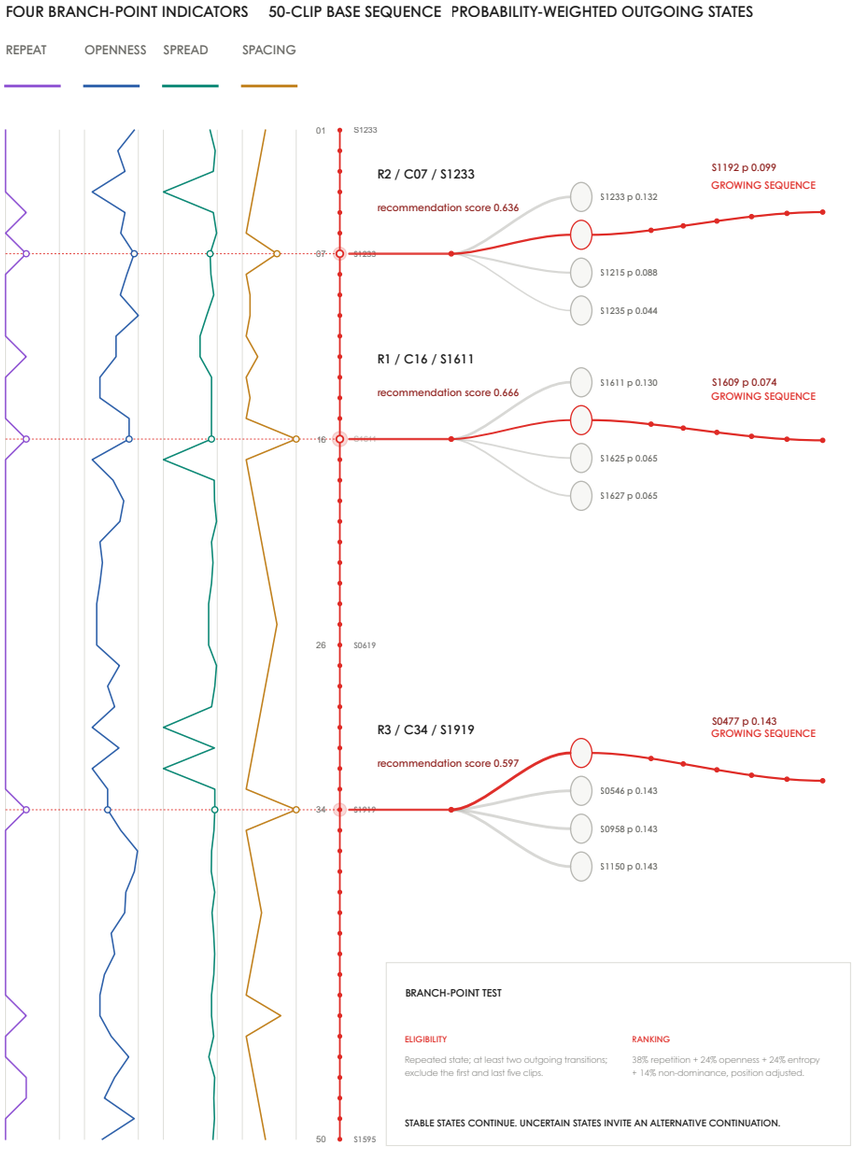


Total clips: 16959
Theoretical Combinations: 5880
Valid five-tag combinations: 1480

Curation Logic

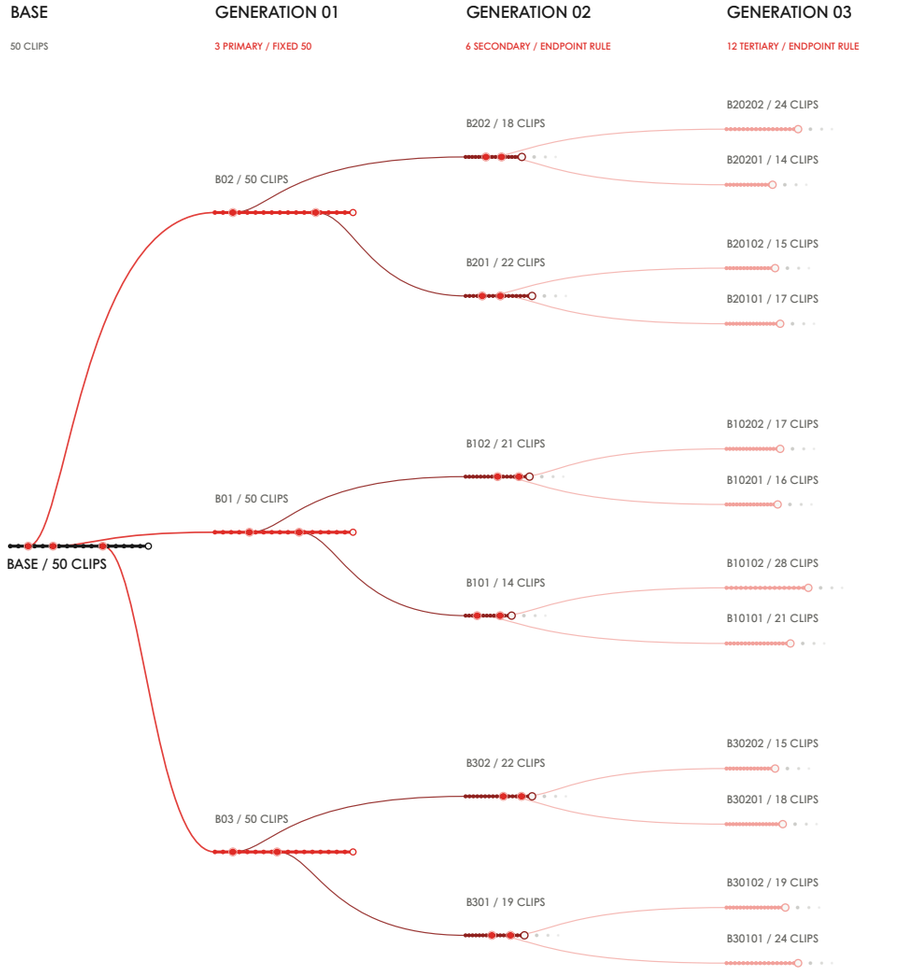
Sequence / Branch Formation Field

LOCAL UNCERTAINTY, CANDIDATE TRANSITIONS, AND THE START OF AN ALTERNATIVE CONTINUATION



RECURSIVE BRANCH GROWTH FIELD

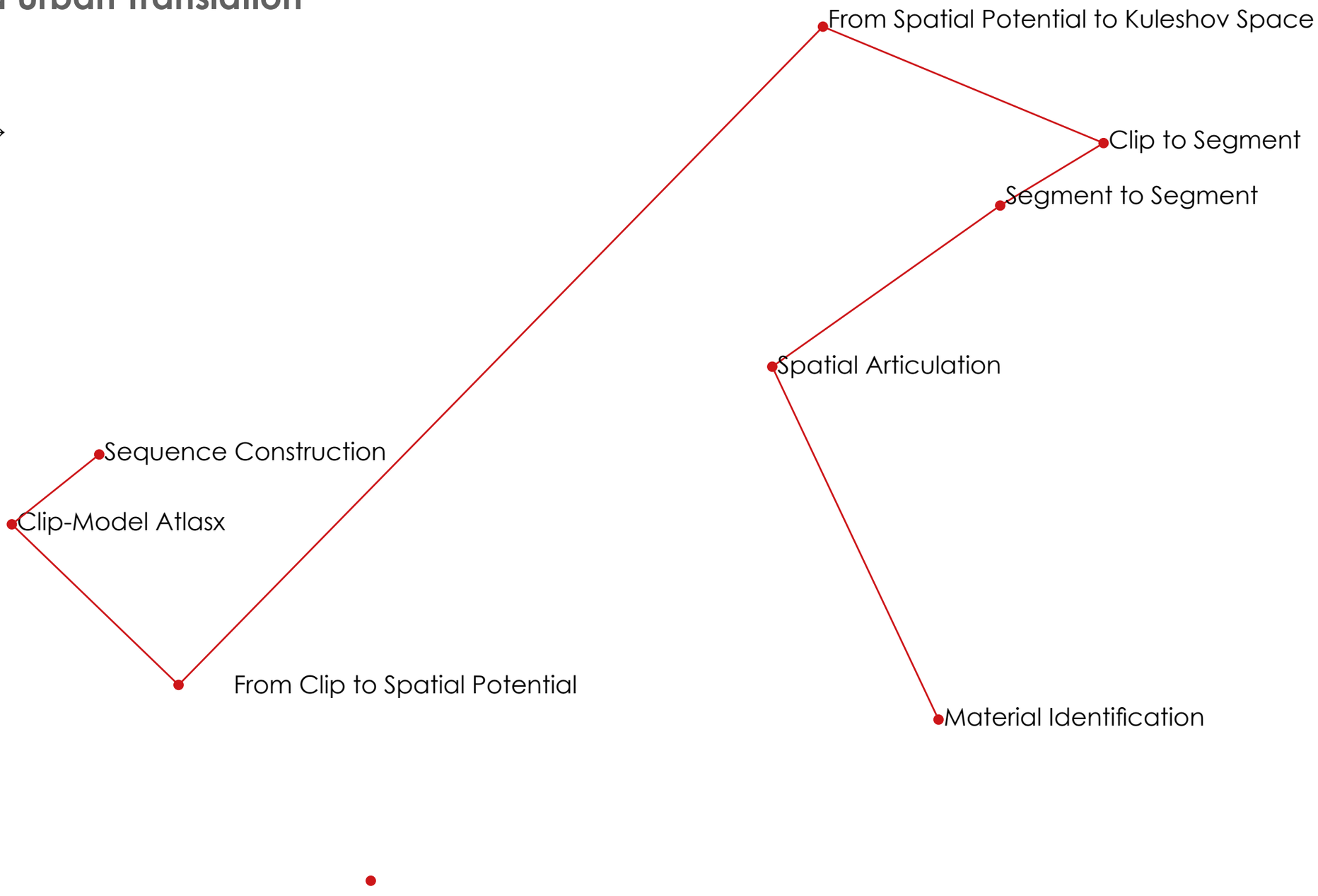
MULTI-GENERATION CONTINUATION, SOURCE-CLIP ATTACHMENT, AND NATURAL TERMINATION



CHAPTER 05

Toward Urban Translation

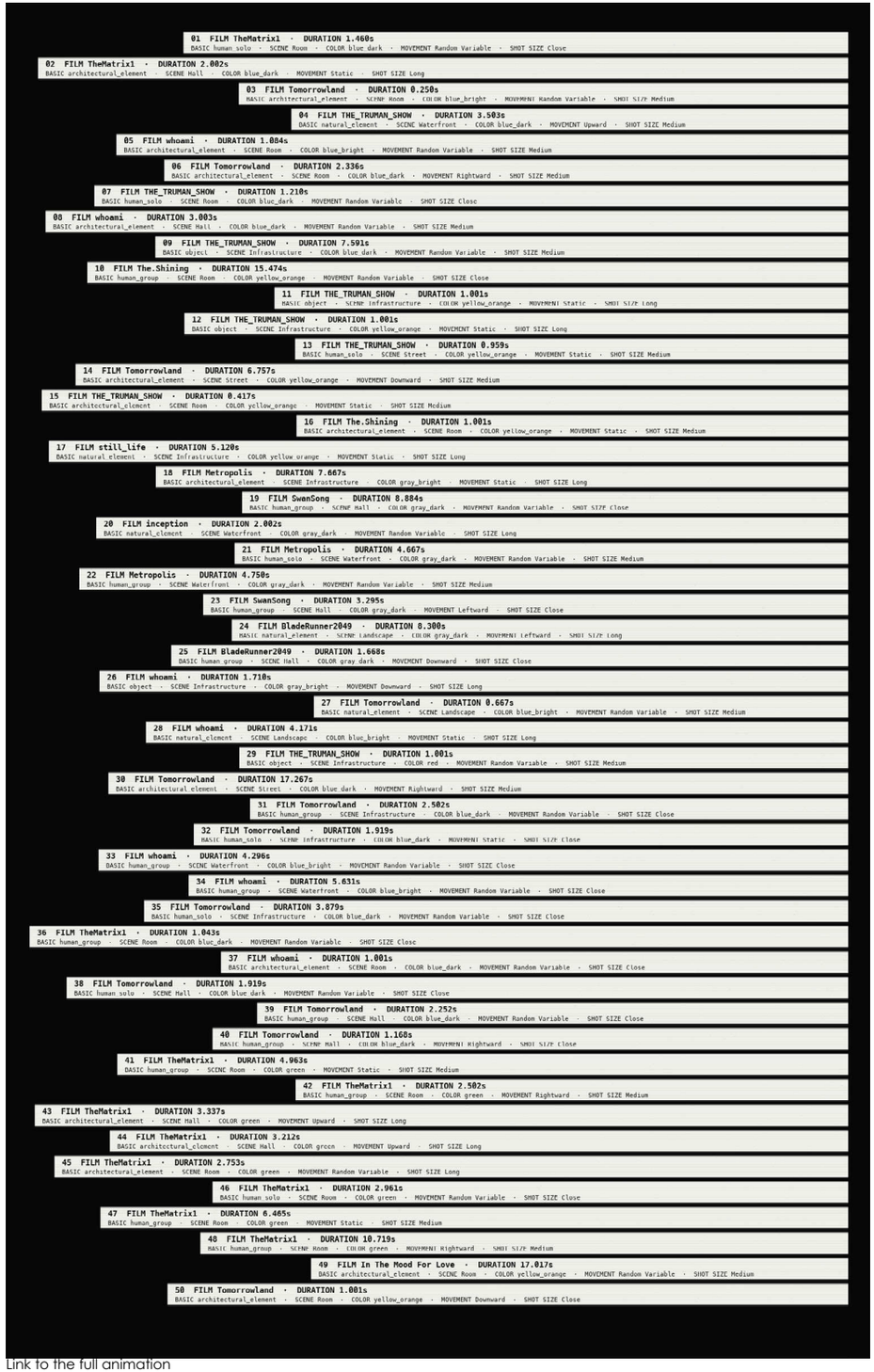
→ → → →



Sequence Construction

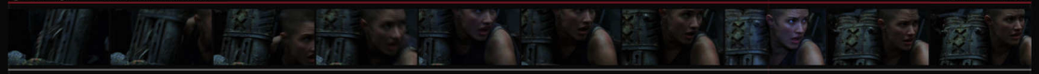
Following the Markov-chain and curatorial framework developed in the previous chapter, the system can now generate multiple alternative clip sequences while remaining consistent with the transition probabilities learned from the original films. Rather than producing a single fixed outcome, different sequences can be generated and curated according to specific relationships and turning points. This chapter uses one 50-clip sequence as an example to demonstrate the complete process of translating a generated sequence from film into spatial and architectural form.

A sequence contains 50 clips

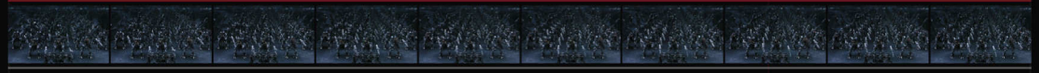


Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQAxHqguvjcoQLr2EJ9PNhLhAU2wqnxlTViB4lj2iqpV-vDA?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Z-m9ybSI6IldlYilsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=jn86N8

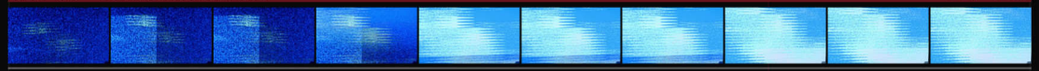
CLIP 01 · THE MATRIX — CLIP 1100



CLIP 02 · THE MATRIX — CLIP 0982



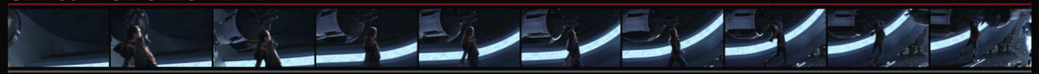
CLIP 03 · TOMORROWLAND — CLIP 1269



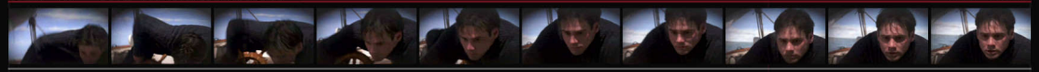
CLIP 04 · THE TRUMAN SHOW — CLIP 3404

CLIP 05 · WHO AM I — CLIP 1547

CLIP 06 · TOMORROWLAND — CLIP 2169



CLIP 07 · THE TRUMAN SHOW — CLIP 3519



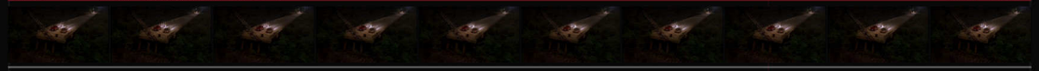
CLIP 08 · WHO AM I — CLIP 1438

CLIP 09 · THE TRUMAN SHOW — CLIP 1208

CLIP 10 · THE SHINING — CLIP 0442

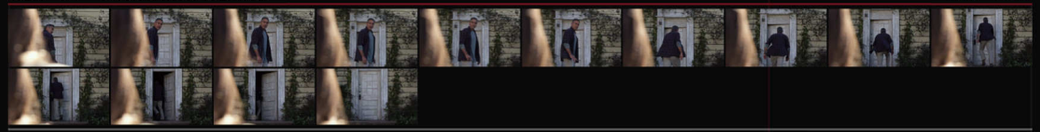
CLIP 11 · THE TRUMAN SHOW — CLIP 0367

CLIP 12 · THE TRUMAN SHOW — CLIP 0450

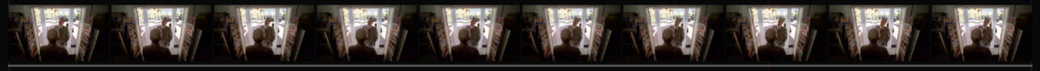


CLIP 13 · THE TRUMAN SHOW — CLIP 0103

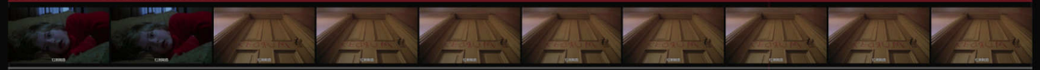
CLIP 14 · TOMORROWLAND — CLIP 1159



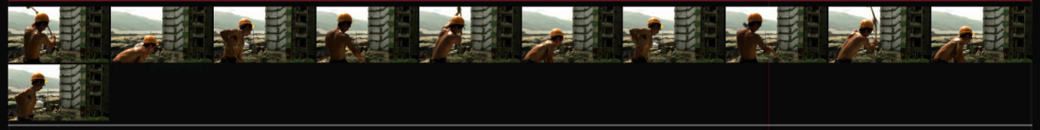
CLIP 15 · THE TRUMAN SHOW — CLIP 0146



CLIP 16 · THE SHINING — CLIP 0344



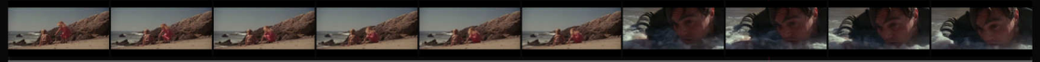
CLIP 17 · STILL LIFE — CLIP 0074



CLIP 18 · METROPOLIS — CLIP 0844

CLIP 19 · SWAN SONG — CLIP 0733

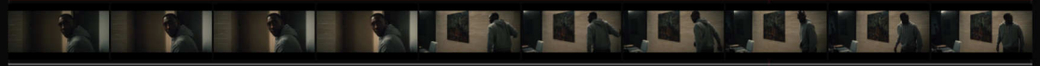
CLIP 20 · INCEPTION — CLIP 0009



CLIP 21 · METROPOLIS — CLIP 0028

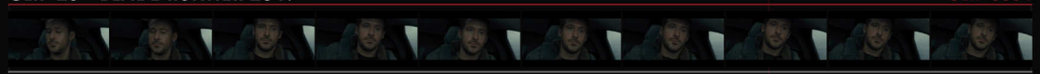
CLIP 22 · METROPOLIS — CLIP 1355

CLIP 23 · SWAN SONG — CLIP 0912



CLIP 24 · BLADE RUNNER 2049 — CLIP 0303

CLIP 25 · BLADE RUNNER 2049 — CLIP 0304

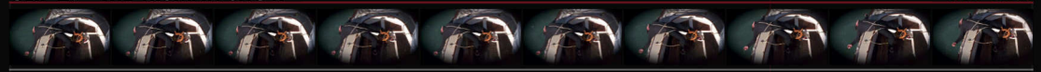


CLIP 26 · WHO AM I — CLIP 1803

CLIP 27 · TOMORROWLAND — CLIP 2119

CLIP 28 · WHO AM I — CLIP 0451

CLIP 29 · THE TRUMAN SHOW — CLIP 3480



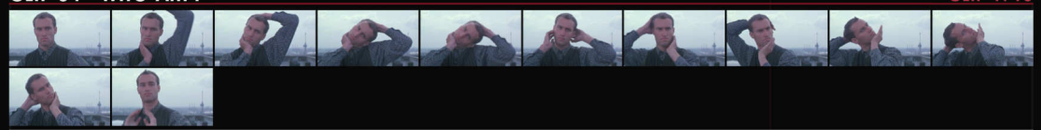
CLIP 30 · TOMORROWLAND — CLIP 1009

CLIP 31 · TOMORROWLAND — CLIP 2096

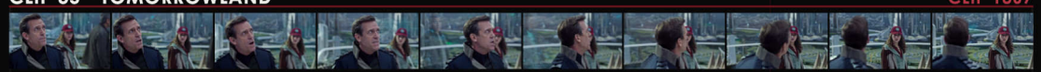
CLIP 32 · TOMORROWLAND — CLIP 1807

CLIP 33 · WHO AM I — CLIP 1753

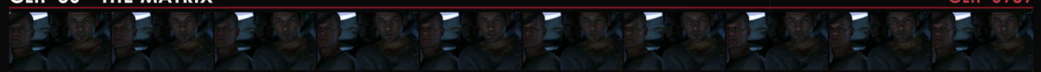
CLIP 34 · WHO AM I — CLIP 1710



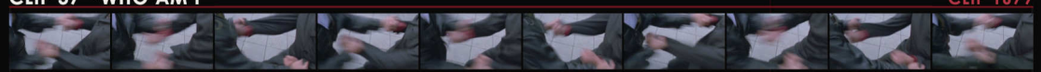
CLIP 35 · TOMORROWLAND — CLIP 1867



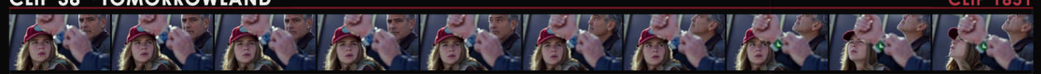
CLIP 36 · THE MATRIX — CLIP 0989



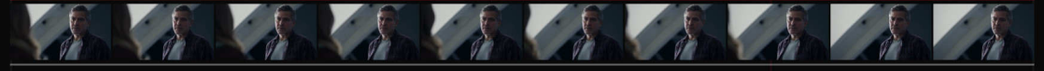
CLIP 37 · WHO AM I — CLIP 1679



CLIP 38 · TOMORROWLAND — CLIP 1851



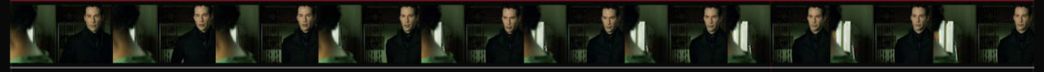
CLIP 39 · TOMORROWLAND — CLIP 2027



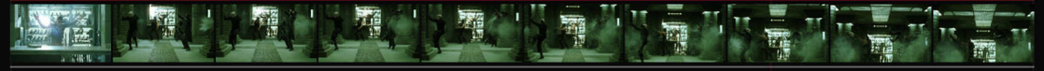
CLIP 40 · TOMORROWLAND — CLIP 2118

CLIP 41 · THE MATRIX — CLIP 0366

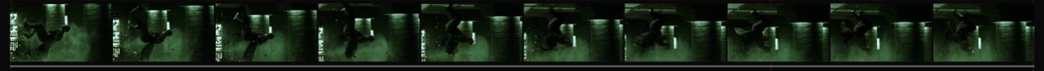
CLIP 42 · THE MATRIX — CLIP 0442



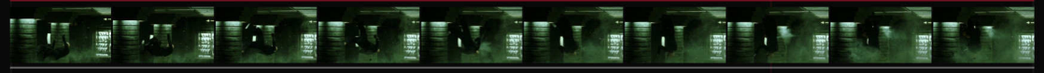
CLIP 43 · THE MATRIX — CLIP 0282



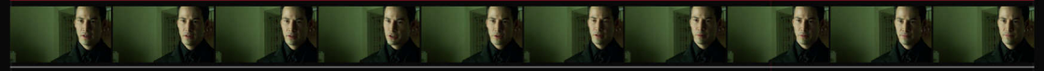
CLIP 44 · THE MATRIX — CLIP 0283



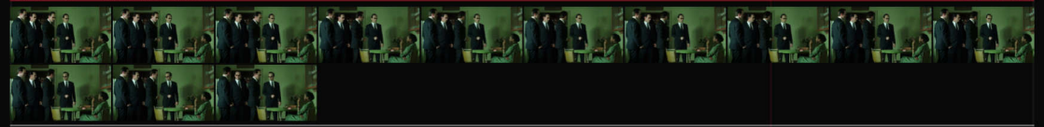
CLIP 45 · THE MATRIX — CLIP 0284



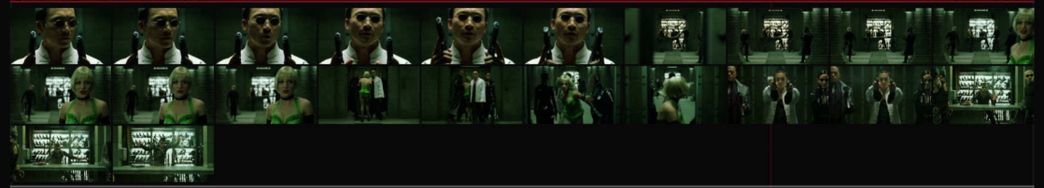
CLIP 46 · THE MATRIX — CLIP 0467



CLIP 47 · THE MATRIX — CLIP 0564

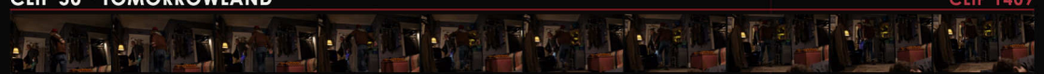


CLIP 48 · THE MATRIX — CLIP 0281



CLIP 49 · IN THE MOOD FOR LOVE — CLIP 0043

CLIP 50 · TOMORROWLAND — CLIP 1409



Clip-Model Atlas

This chapter presents a rule-based system for translating cinematic shot data into three-dimensional form. Each clip is encoded through a combination of scene category, shot size, basic component, movement, duration, and colour. These labels control the model's spatial envelope, degree of openness, embedded elements, orientation, and overall scale. Representative models illustrate how different label combinations produce distinct geometries, while controlled comparisons isolate the influence of individual variables. Rather than reconstructing literal objects from film footage, the resulting model library provides an abstract spatial interpretation of cinematic information, making patterns across thousands of clips visible, comparable, and physically legible.

Scene Category → *Spatial envelope and base form*

Shot Size → *Scale, openness, and void pattern*

Basic Component → *Embedded subject or architectural element*

Movement → *Position, direction, and orientation*

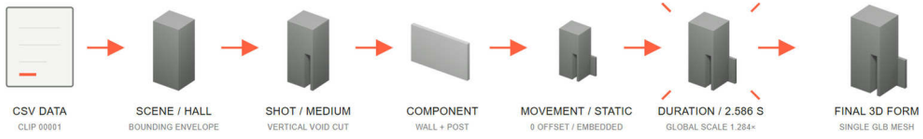
Duration → *Logarithmic global scale*

Colour → *Material information / metadata*

From Film Clip Labels To Three-Dimensional Form

01 GENERATIVE WORKFLOW

CLIP 00001 / ACTUAL MODEL-DERIVED STAGES



02 LABEL–GEOMETRY MAPPING

REPRESENTATIVE MODELS FROM THE LIBRARY

A SCENE CATEGORY

B BASIC COMPONENT

C SHOT SIZE

ROOM + HUMAN_GROUP + RANDOM VARIABLE / ≈ 2 S

03 CONTROLLED VARIABLE TESTS

ACTUAL MODELS / MATCHED LABEL SETS

A SHOT SIZE

ROOM + HUMAN_GROUP + RANDOM VARIABLE

ONLY SHOT SIZE CHANGES / DURATION ≈ 2 S

B MOVEMENT

HALL + ARCHITECTURAL + MEDIUM / ≈ 1 S

ONLY MOVEMENT CHANGES / OFFSET + ORIENTATION

C DURATION

ROOM + HUMAN_SOLO + MEDIUM + STATIC

ONLY DURATION CHANGES / LOGARITHMIC SCALE

CONTROL PRINCIPLE

Each comparison holds scene, component and the remaining labels constant; the highlighted variable alone drives the geometric difference.

Model Detail And Data-To-Form Trace

01 REPRESENTATIVE MODEL CLOSE-UPS

8 ORIGINAL GLB MODELS / NORMALISED VIEW

CLIP 00019

ROOM / HUMAN_SOLO
CLOSE · RANDOM VARIABLE · 1.752 S
FOCUS / CLOSED ENVELOPE

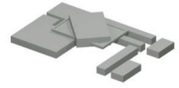
CLIP 00627

CORRIDOR / HUMAN_GROUP
MEDIUM · RANDOM VARIABLE · 0.751 S
FOCUS / GROUP EMBEDDING

CLIP 00001

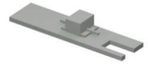
HALL / ARCHITECTURAL
MEDIUM · STATIC · 2.586 S
FOCUS / VERTICAL VOID CUT

CLIP 00002



LANDSCAPE / NATURAL
LONG · RANDOM VARIABLE · 29.863 S
FOCUS / OPEN PLAN MASS

CLIP 00026

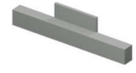


STREET / OBJECT
MEDIUM · RANDOM VARIABLE · 1.168 S
FOCUS / OBJECT INTERFACE

CLIP 01734

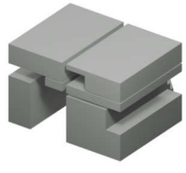
WATERFRONT / HUMAN_SOLO
LONG · LEFTWARD · 2.377 S
FOCUS / EDGE RELATION

CLIP 00274



INFRASTRUCTURE / ARCHITECTURAL
CLOSE · DOWNWARD · 1.543 S
FOCUS / LINEAR CONNECTION

CLIP 00039

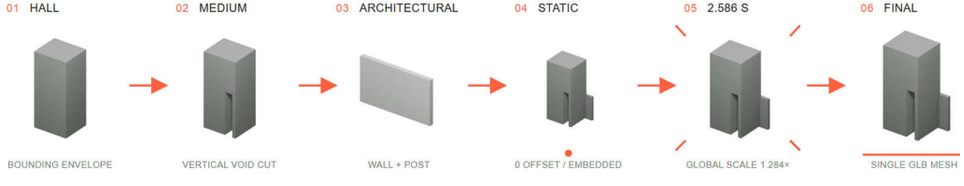


ROOM / OBJECT
LONG · RANDOM VARIABLE · 3.795 S
FOCUS / SOCKET + PLATFORM

02 ONE DATA RECORD → ONE MODEL

TOMORROWLAND / CLIP 0000

SCENE	COMPONENT	SHOT SIZE	MOVEMENT	DURATION	COLOUR
Hall	architectural_element	Medium	Static	2.586 s	gray_dark METADATA ONLY

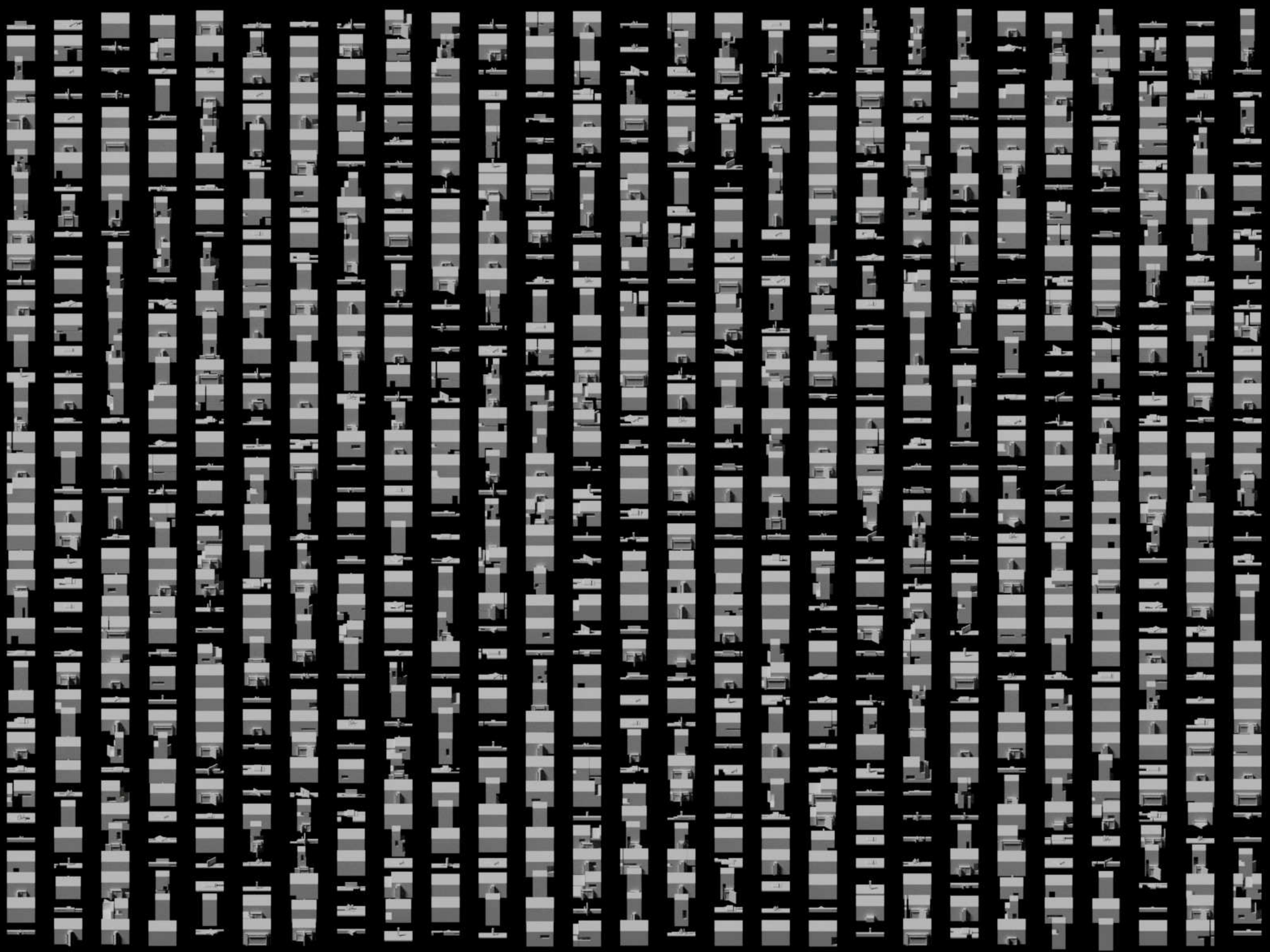


● HALL + ARCHITECTURAL_ELEMENT + MEDIUM + STATIC + 2.586 S → FINAL 3D FORM

GEOMETRIC READING

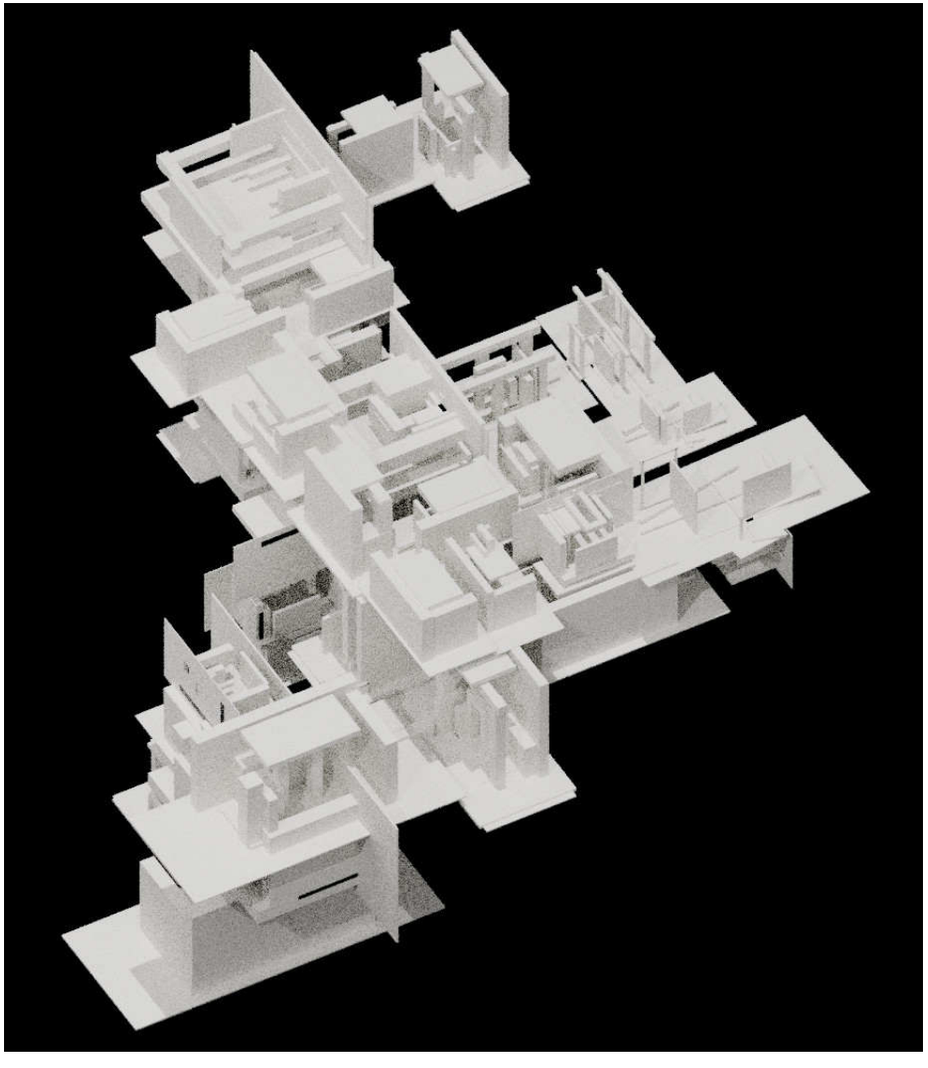
Tall hall mass → medium vertical cut → wall-and-post component → embedded interface → duration scaling

The colour label remains attached to the GLB metadata while the active model material stays neutral clay.



From Clip to Spatial Potential

In this step, each clip in the generated sequence is translated into a basic geometric form according to the rules established in the Clip Model Atlas. The labelled attributes of each clip define its initial spatial characteristics, including the primary geometry, spatial anchor, degree of openness, movement tendency, material reference and relative scale. At this stage, these translations remain simple and abstract. They are intended to establish the basic geometric potential of each clip rather than to produce a complete architectural space. The individual clip models are then arranged and connected according to their order within the generated 50-clip sequence, forming an initial sequence-based spatial structure. This stage therefore focuses only on translating the sequence into its first geometric spatial form. The Kuleshov Effect has not yet been introduced: the relationships between neighbouring clips and segments have not yet been used to reinterpret or transform the existing geometry.



Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQAx588GgV4KTJ9q72D6xAtHAQDyU5xt3eg_YJZjouZ-VrMQ?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYilsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=RirVZu

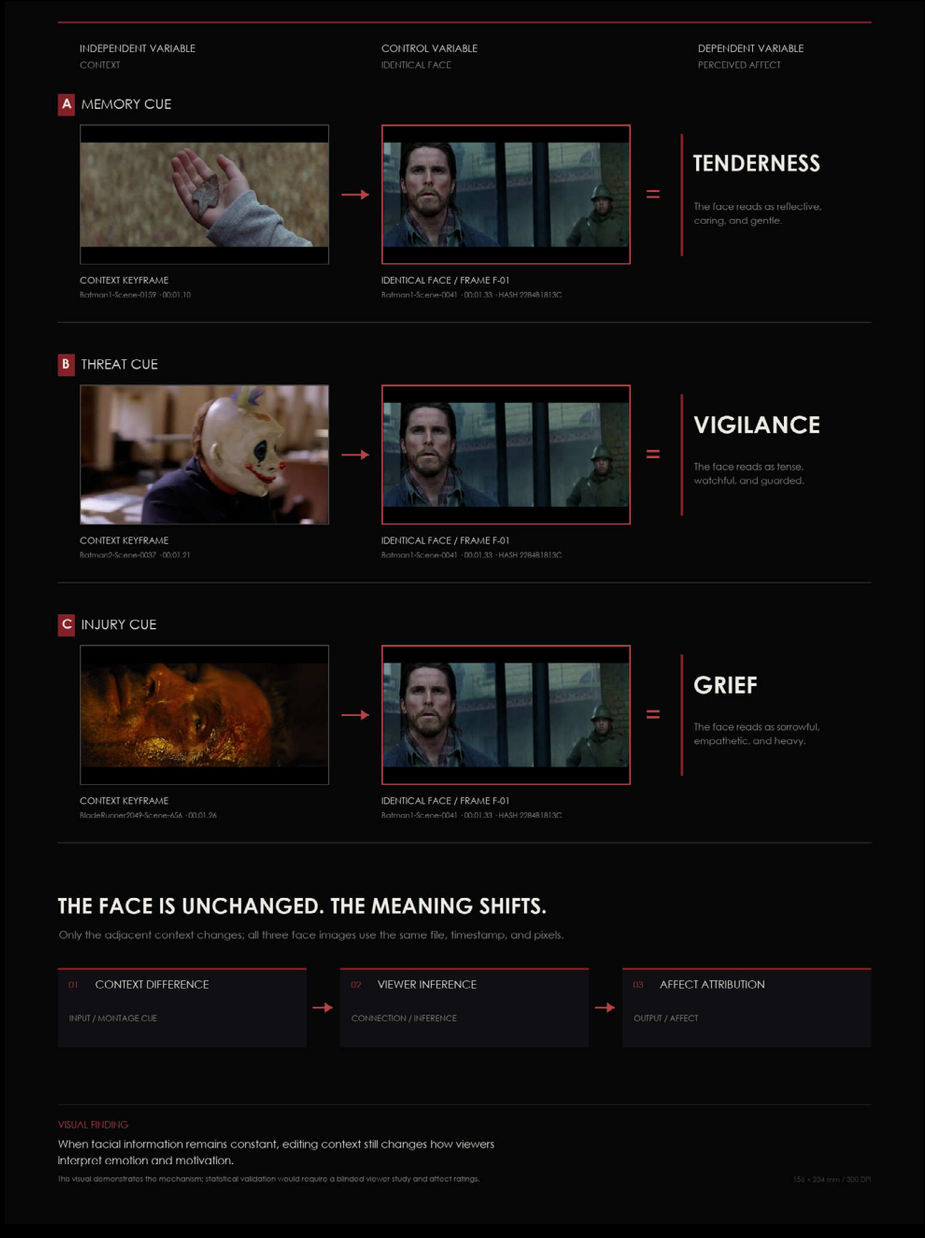
From Spatial Potential to Kuleshov Space

This step introduces the spatial logic of the Kuleshov Effect by transforming individual clip potentials into segment-based spaces. Consecutive clips with the same Scene Category are combined into one segment and treated as a shared spatial condition. However, differences in other labels, such as Basic Component, Shot Size, Movement and Duration, continue to reshape the geometry of each clip within that shared space. The transformation therefore first occurs at the clip level, where individual models are scaled, shifted, connected and reorganised to form a segment. It then operates at the segment level, where neighbouring segments further modify boundaries, thresholds, directions, connections and internal spatial organisation. Through these two levels of transformation, the initial sequence geometry gradually develops into a relational Kuleshov Space.

Spatial Kuleshov Logic

As discussed previously, the Kuleshov Effect shows that meaning is produced through the relationship between fragments rather than by a fragment alone. In this project, this logic is translated through two levels: the film layer and the spatial layer.

Film Layer



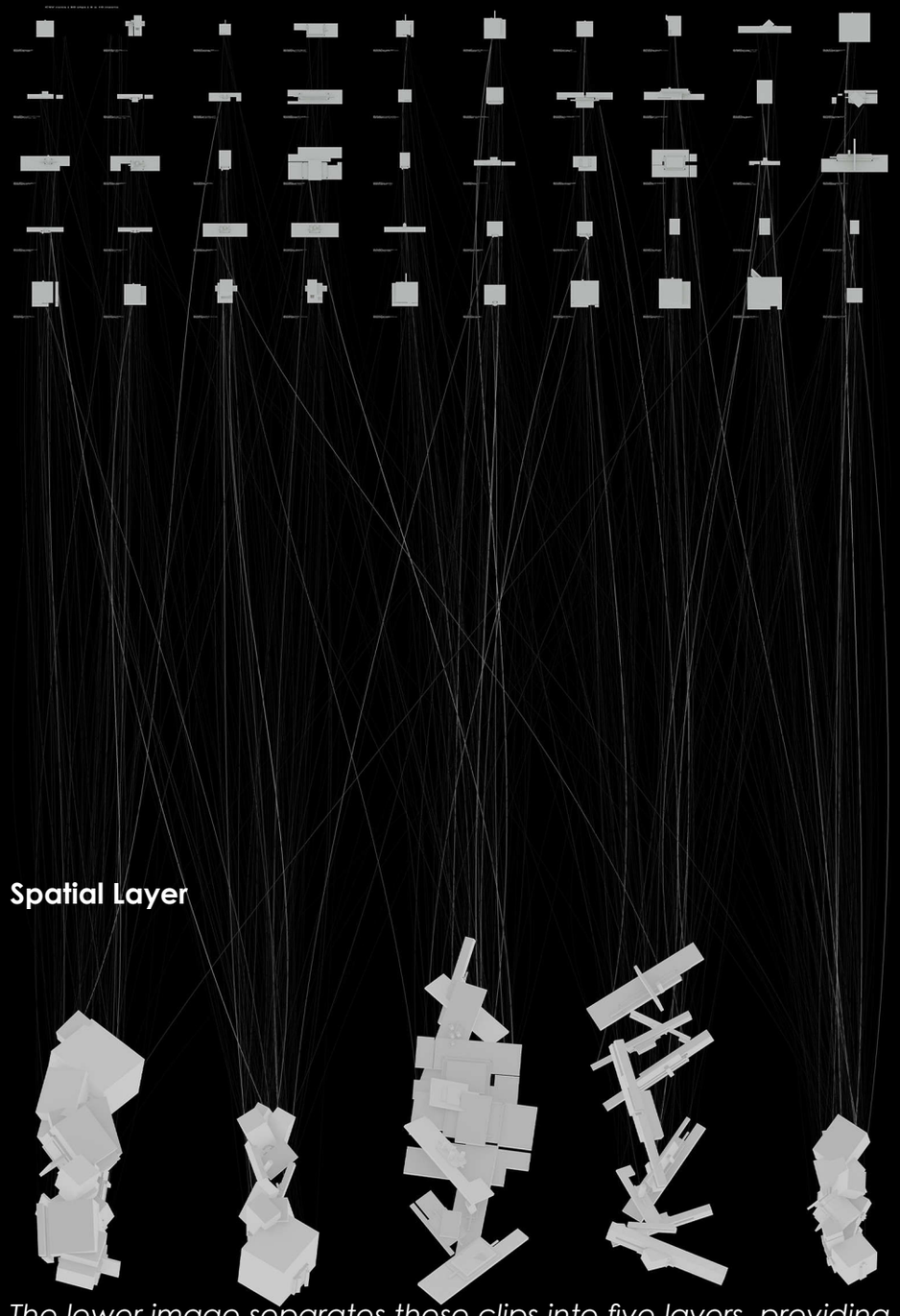
Film Layer — Kuleshov Effect between Clips

In film, the same clip can be interpreted differently when it is placed before or after different clips. Although the original image remains unchanged, its meaning, atmosphere and narrative role are altered by its neighbouring clips.

Spatial Layer — Kuleshov Effect between Spaces

The same relational logic is then translated into space. When clip models are combined into segments and placed next to different spatial conditions, their spatial meaning and organisation also change. Boundaries, openings, direction, scale, connections and internal geometry can therefore be adjusted according to the relationship between neighbouring clips and segments.

The upper image shows the spatial translation of individual clips before the Kuleshov Effect is applied.

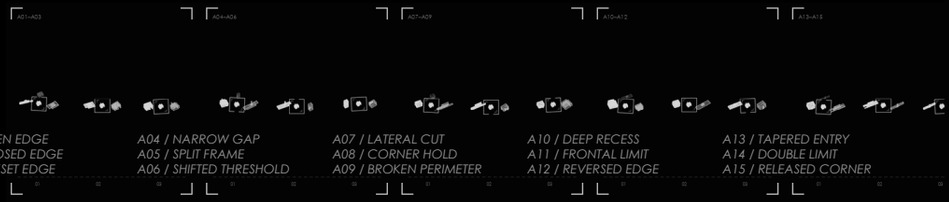


The lower image separates these clips into five layers, providing an abstract representation of how the model is transformed after the Kuleshov Effect occurs.

PRINCIPLE

MODEL X REMAINS FIXED; CHANGING ITS NEIGHBOUR, BOUNDARY OR OPENING CHANGES THE S READING.

A/BOUNDARY FRAGMENTS



A01 / OPEN EDGE
A02 / CLOSED EDGE
A03 / OFFSET EDGE

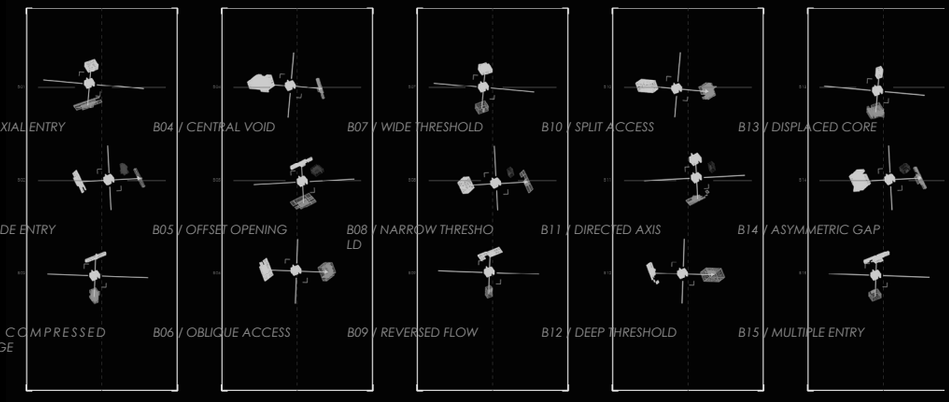
A04 / NARROW GAP
A05 / SPLIT FRAME
A06 / SHIFTED THRESHOLD

A07 / LATERAL CUT
A08 / CORNER HOLD
A09 / BROKEN PERIMETER

A10 / DEEP RECESS
A11 / FRONTAL LIMIT
A12 / REVERSED EDGE

A13 / TAPERED ENTRY
A14 / DOUBLE LIMIT
A15 / RELEASED CORNER

A/BOUNDARY FRAGMENTS THREE STAGES TEST HOW ACCESS CHANGES AROUND X; AXIS, ARROW AND RECTANGLE RECORD DIRECTION AND CONTINUITY.

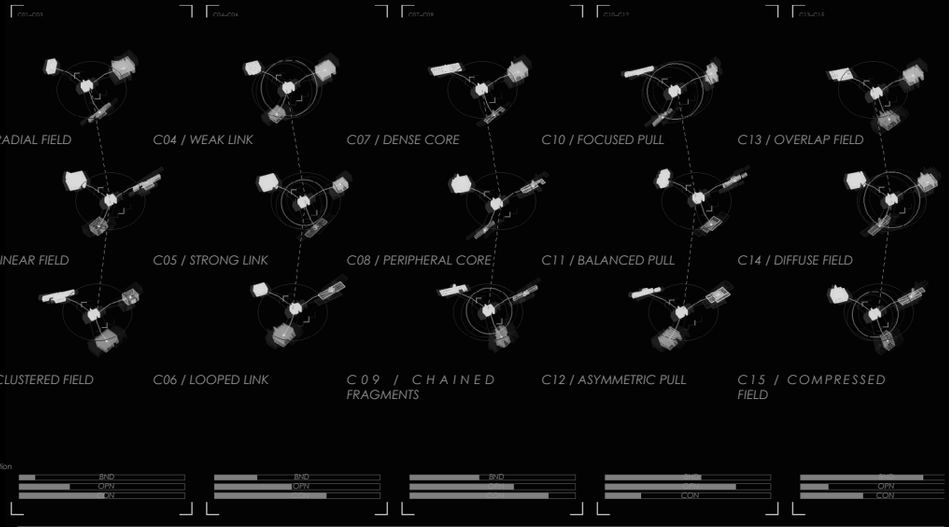


B01 / AXIAL ENTRY
B04 / CENTRAL VOID
B07 / WIDE THRESHOLD
B10 / SPLIT ACCESS
B13 / DISPLACED CORE

B02 / SIDE ENTRY
B05 / OFFSET OPENING
B08 / NARROW THRESHOLD
B11 / DIRECTED AXIS
B14 / ASYMMETRIC GAP

B03 / COMPRESSED PASSAGE
B06 / OBLIQUE ACCESS
B09 / REVERSED FLOW
B12 / DEEP THRESHOLD
B15 / MULTIPLE ENTRY

C / CONNECTION FIELDS NEIGHBOURS ACCUMULATE AROUND X; LINKS SHOW ADJACENCY, RINGS SHOW INFLUENCE, BARS COMPARE INTENSITY.

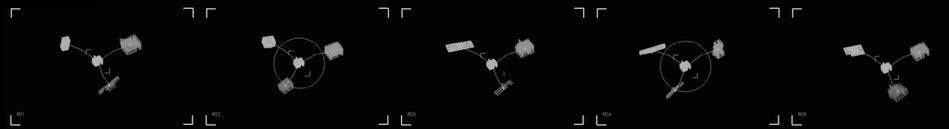


C01 / RADIAL FIELD
C04 / WEAK LINK
C07 / DENSE CORE
C10 / FOCUSED PULL
C13 / OVERLAP FIELD

C02 / LINEAR FIELD
C05 / STRONG LINK
C08 / PERIPHERAL CORE
C11 / BALANCED PULL
C14 / DIFFUSE FIELD

C03 / CLUSTERED FIELD
C06 / LOOPED LINK
C09 / CHAINED FRAGMENTS
C12 / ASYMMETRIC PULL
C15 / COMPRESSED FIELD

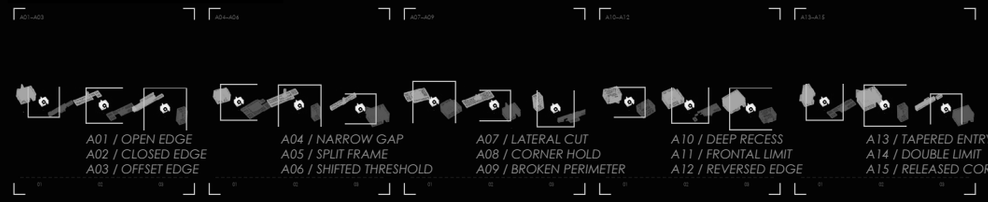
RELATIONAL OUTPUT



R01 / ENCLOSURE SHIFT
R02 / DIRECTED FLOW
R03 / THRESHOLD DEPTH
R04 / RELATIONAL CENTRE
R05 / BOUNDARY RELEASE

X / FIXED + NEIGHBOUR = SPATIAL READING

A/BOUNDARY FRAGMENTS



A01 / OPEN EDGE
A02 / CLOSED EDGE
A03 / OFFSET EDGE

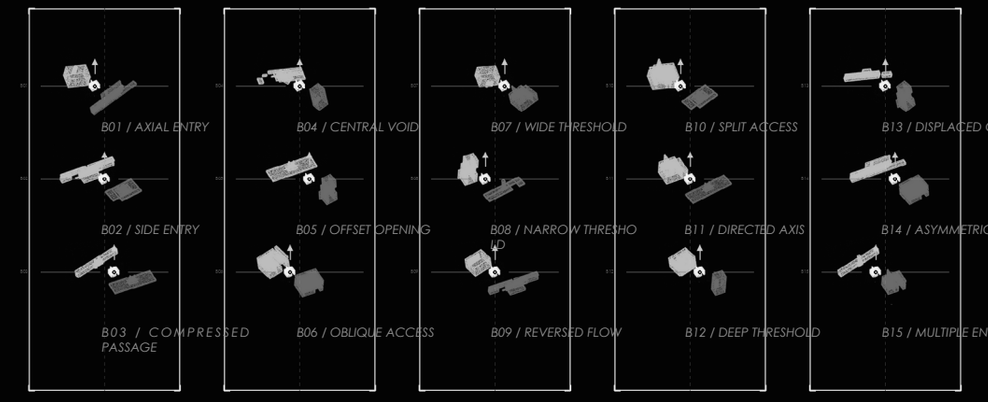
A04 / NARROW GAP
A05 / SPLIT FRAME
A06 / SHIFTED THRESHOLD

A07 / LATERAL CUT
A08 / CORNER HOLD
A09 / BROKEN PERIMETER

A10 / DEEP RECESS
A11 / FRONTAL LIMIT
A12 / REVERSED EDGE

A13 / TAPERED ENTRY
A14 / DOUBLE LIMIT
A15 / RELEASED CORNER

B / OPENING SEQUEN. *A/BOUNDARY FRAGMENTS THREE STAGES TEST HOW ACCESS CHANGES AROUND X; AXIS, ARROW AND RECTANGLE RECORD DIRECTION AND CONTINUITY.*

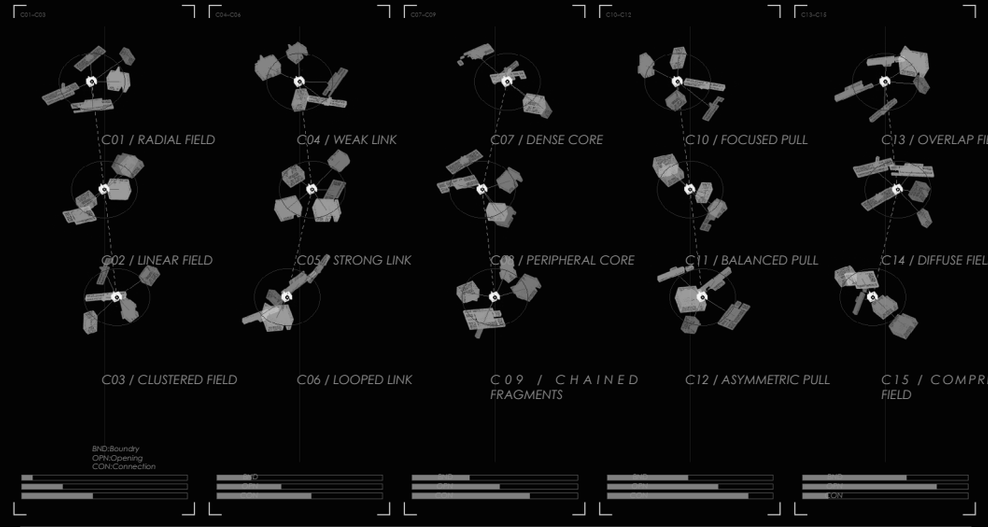


B01 / AXIAL ENTRY
B04 / CENTRAL VOID
B07 / WIDE THRESHOLD
B10 / SPLIT ACCESS
B13 / DISPLACED CORE

B02 / SIDE ENTRY
B05 / OFFSET OPENING
B08 / NARROW THRESHOLD
B11 / DIRECTED AXIS
B14 / ASYMMETRIC GAP

B03 / COMPRESSED PASSAGE
B06 / OBLIQUE ACCESS
B09 / REVERSED FLOW
B12 / DEEP THRESHOLD
B15 / MULTIPLE ENTRY

C / CONNECTION FIEI *C / CONNECTION FIELDS NEIGHBOURS ACCUMULATE AROUND X; LINKS SHOW ADJACENCY, RINGS SHOW INFLUENCE, BARS COMPARE INTENSITY.*

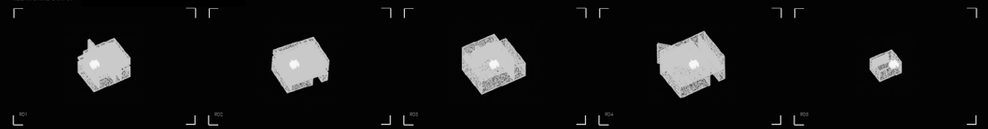


C01 / RADIAL FIELD
C04 / WEAK LINK
C07 / DENSE CORE
C10 / FOCUSED PULL
C13 / OVERLAP FIELD

C02 / LINEAR FIELD
C05 / STRONG LINK
C08 / PERIPHERAL CORE
C11 / BALANCED PULL
C14 / DIFFUSE FIELD

C03 / CLUSTERED FIELD
C06 / LOOPED LINK
C09 / CHAINED FRAGMENTS
C12 / ASYMMETRIC PULL
C15 / COMPRESSED FIELD

RELATIONAL OUTPUT *RELATIONAL OUTPUT*



R01 / ENCLOSURE SHIFT
R02 / DIRECTED FLOW
R03 / THRESHOLD DEPTH
R04 / RELATIONAL CENTRE
R05 / BOUNDARY RELEASE

X / FIXED + NEIGHBOUR = SPATIAL READING

Clip to Segment

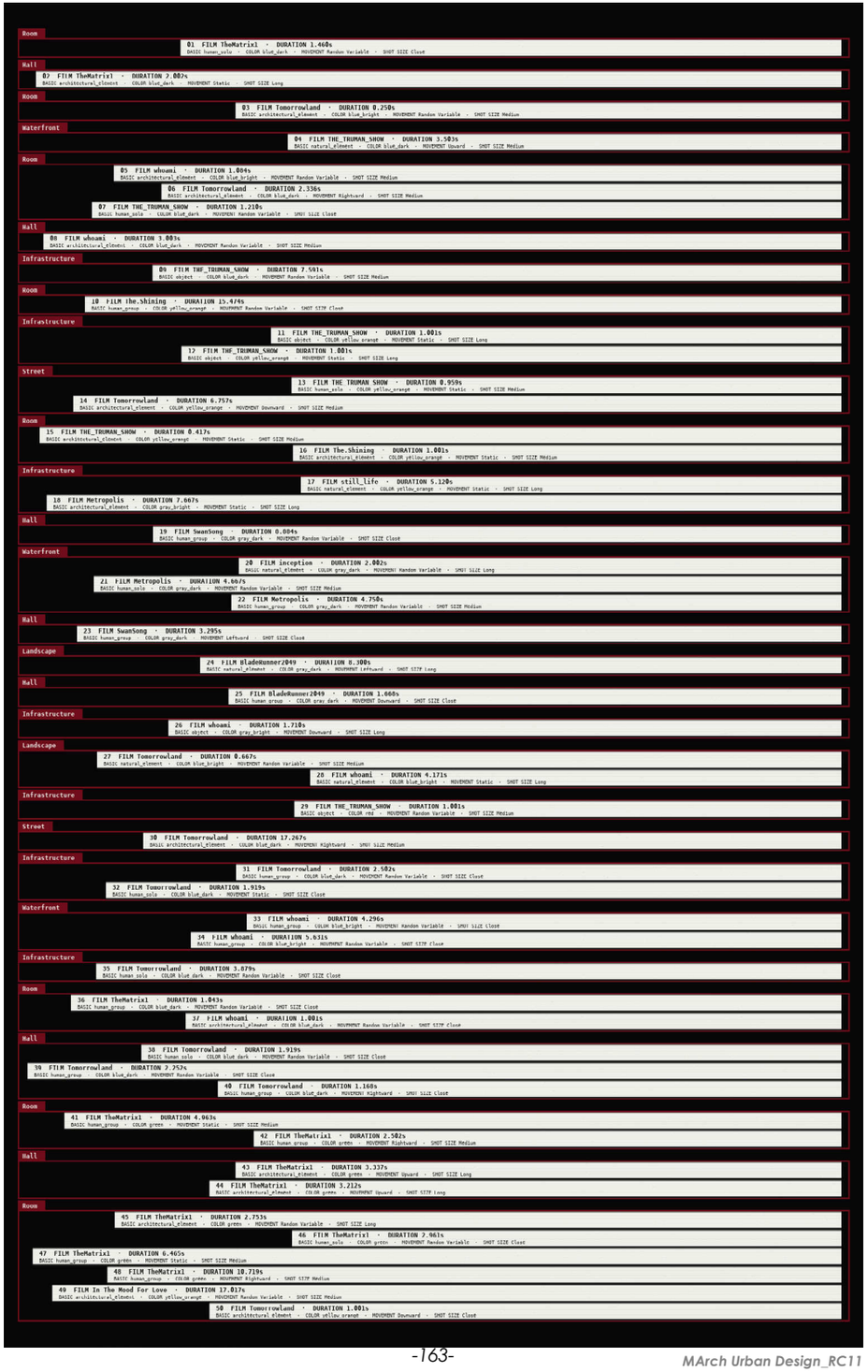
What is a segment?

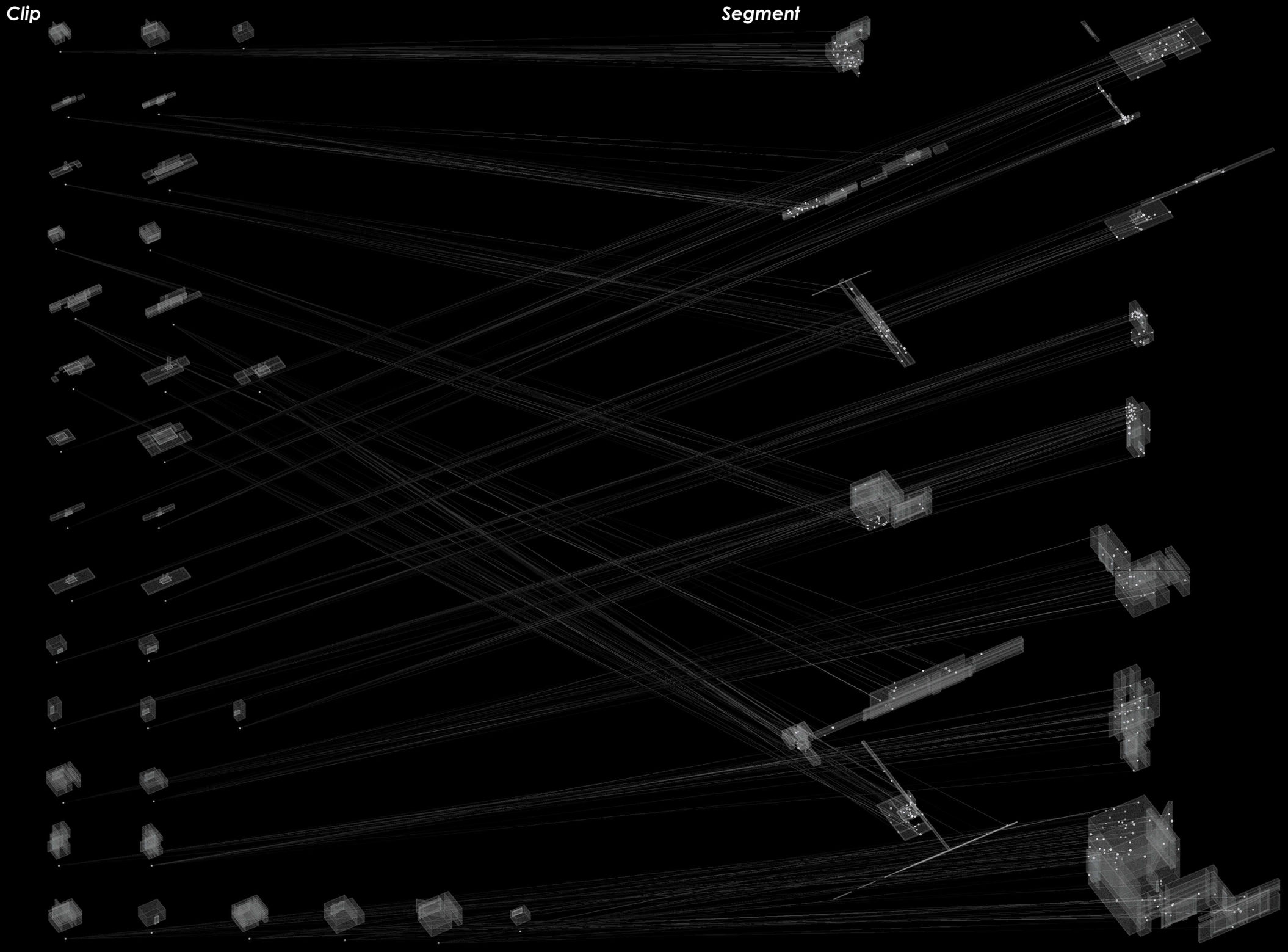
Based on the clip labels in the sequence, consecutive clips with the same Scene Category are identified and grouped as one segment.

This is because the Scene Category defines the primary spatial context and basic geometric condition of each clip, while the other labels act as modifiers within that condition. When consecutive clips remain within the same Scene Category, their fundamental spatial type does not change; instead of treating them as separate spaces, they can therefore be understood as different spatial variations within one shared space. Differences in Basic Component, Shot Size, Movement, Duration and Colour are retained and used to modify the internal geometry of the segment. This allows each segment to become a larger spatial unit that can later interact with adjacent segments through the Kuleshov Effect.

A sequence contains 50 clips

The red box indicates a group of clips forming one segment.




Clip
Segment

Key code *for dividing clips into segments based on scene category*

```
def segment_by_scene(sequence):
    segments = []
    start = 0

    for i in range(1, len(sequence) + 1):
        reached_end = i == len(sequence)

        scene_changed = (
            not reached_end
            and sequence[i]["scene"]
            != sequence[start]["scene"]
        )

        if reached_end or scene_changed:
            clips = sequence[start:i]

            segments.append({
                "segment_id": len(segments) + 1,
                "scene": clips[0]["scene"],
                "start_clip": clips[0]["idx"],
                "end_clip": clips[-1]["idx"],
                "clip_ids": [
                    clip["idx"]
                    for clip in clips
                ],
            })

            start = i

    return segments
```


Define Rules

scene_category
→ Determines which Clips make up a Segment

duration
→ Determines the intensity of each Clip's impact

shot_size
→ Determines spatial scale and framing

movement
→Determines direction, rotation, and connections

basic_component
→ Determines the content of the internal space

colour
→ Determines the visual atmosphere

sequence_index
→ Determines the temporal and spatial order

S29 Room

C45 | *2.753s/6.7%* | *Long* | *Random Variable* | *architectural_element* | *green*

C46 | *2.961s/7.2%* | *Close* | *Random Variable* | *human_solo* | *green*

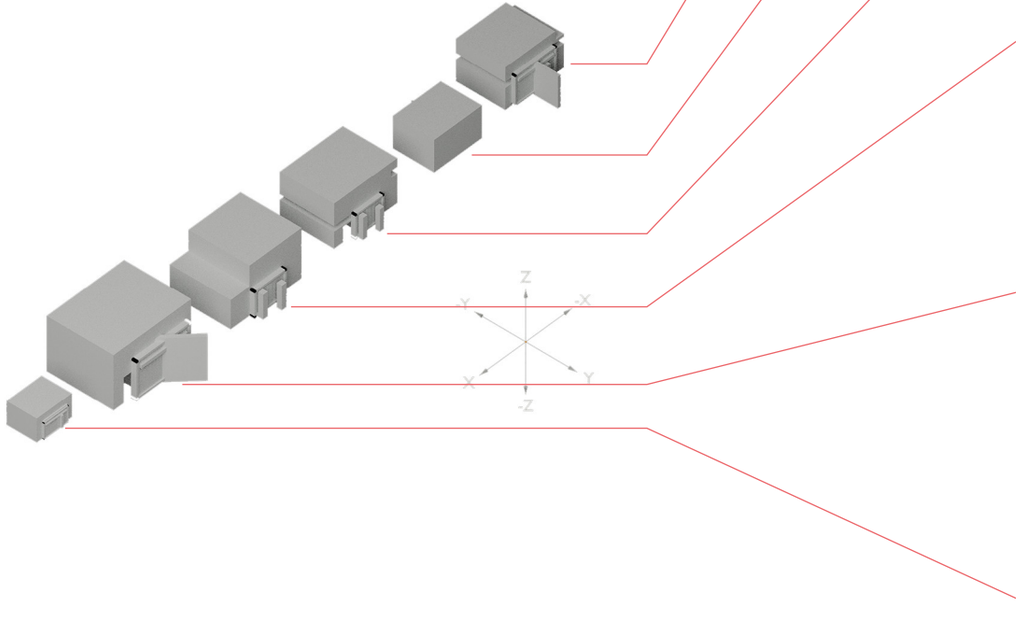
C47 | *6.465s/15.8%* | *Medium* | *Static* | *human_group* | *green*

C48 | *10.719s/26.2%* | *Medium* | *Rightward* | *human_group* | *green*

C49 | *17.017s/41.6%* | *Medium* | *Random Variable* | *architectural_element* | *yellow_orange*

C50 | *1.001s/2.4%* | *Close* | *Downward* | *architectural_element* | *yellow_orange*

***Clip** C45-C50*



Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQCJAx2tzRdeTK7MNQNfrhYKAY9Kfgmz8iFPVNcw-f5t9HAM?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbG-F0Zm9ybSI6IldlYilsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=BqQ1dg

The film clip corresponding to C45–C50

S29 **ROOM**

C45 · THE MATRIX — CLIP 0284

01

C46 · THE MATRIX — CLIP 0467

02

C47 · THE MATRIX — CLIP 0564

03

C48 · THE MATRIX — CLIP 0281

04

C49 · IN THE MOOD FOR LOVE — CLIP 0043

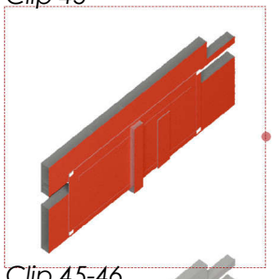
05

C50 · TOMORROWLAND — CLIP 1409

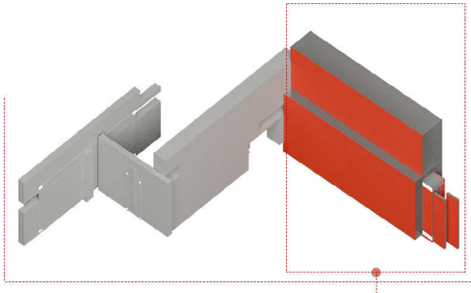
06

Rule-Based Transformation Process for Each Segment

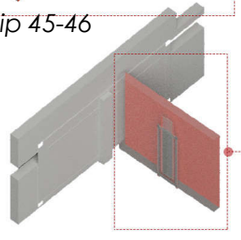
Clip 45



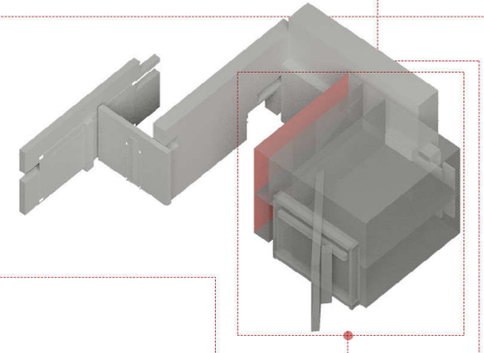
Clip 47-48



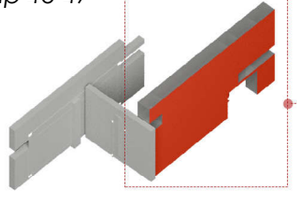
Clip 45-46



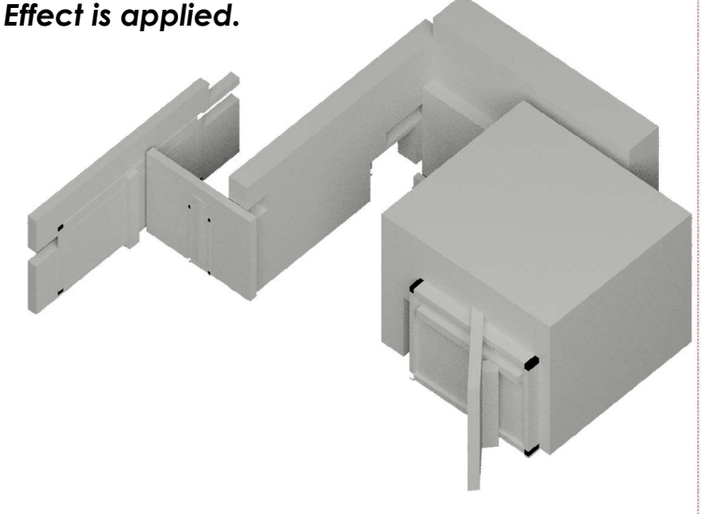
Clip 48-49



Clip 46-47



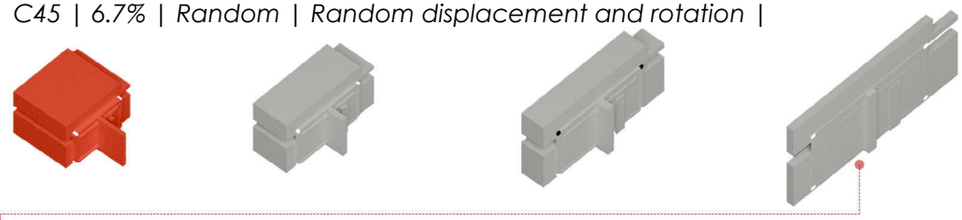
The skeleton model of Segment 29** (Clip 45-50) **after the Kuleshov Effect is applied.



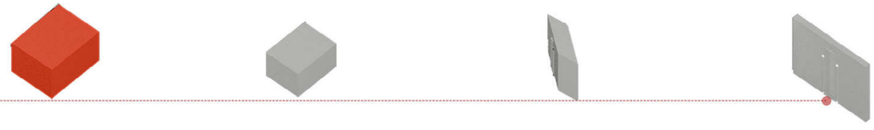
Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQB5ocSejZvqRYi3ZR43BJkdAY-hUJokLsShWdMV91axX-fs?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYilsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=rAkPjV

Single-Clip Transformation Process

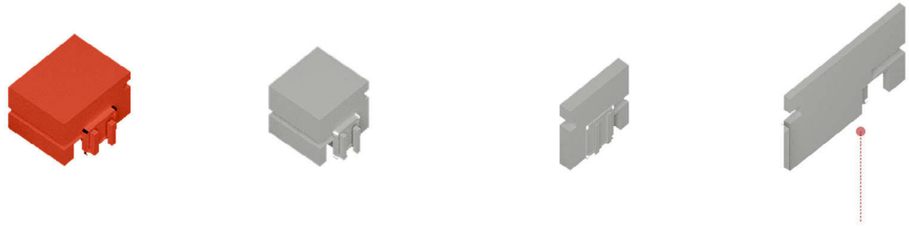
C45 | 6.7% | Random | Random displacement and rotation |



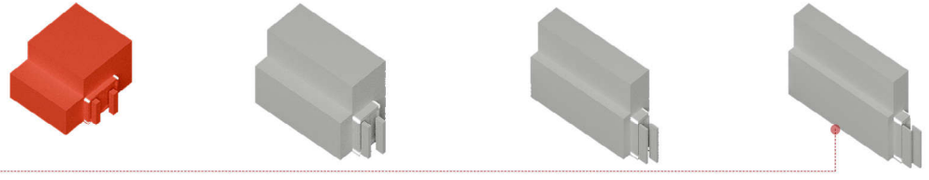
C46 | 7.2% | Random | Random displacement and rotation |



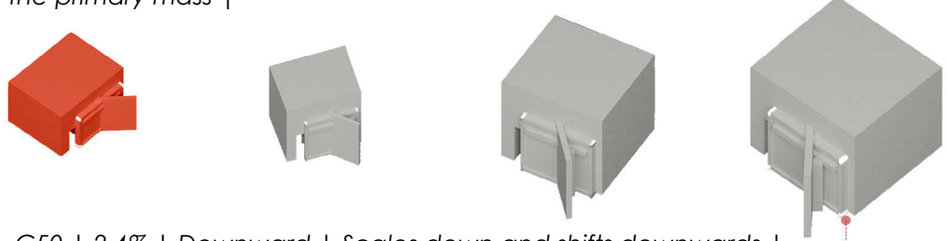
C47 | 15.8% | Static | Remains stable and acts as an internal anchor |



C48 | 26.2% | Rightward | Extends towards the right |



C49 | 41.6% | Random | Makes the largest spatial contribution and forms the primary mass |



C50 | 2.4% | Downward | Scales down and shifts downwards |

***Key code** showing how clip transforms the segment model based on the definition*

```
def clip_to_model(clip):
    # Scene
    dimensions = SCENE_SIZE[clip[“scene”]]

    # Shot Size
    shot_scale = {
        “Close”:  (0.92, 0.92, 0.58),
        “Medium”: (1.00, 1.00, 1.00),
        “Long”:   (1.18, 1.08, 1.45),
    }[clip[“shot”]]

    # Duration
    duration_scale = (
        0.78 + min(0.62, 0.16 * sqrt(clip[“duration”]))
    )

    dimensions = dimensions * shot_scale * duration_scale

    # Basic Component Movement
    geometry = {
        “human_solo”: “single_element”,
        “human_group”: “cluster”,
        “architectural_element”: “wall_frame”,
        “natural_element”: “ground_mass”,
        “object”: “solid_volume”,
    }[clip[“basic”]]

    direction = {
        “Static”: “central”,
        “Rightward”: “right”,
        “Leftward”: “left”,
        “Upward”: “up”,
        “Downward”: “down”,
        “Random Variable”: “crossed”,
    }[clip[“movement”]]

    return dimensions, geometry, direction, clip[“color”]
```


Segment to Segment

After the clips are grouped into segments, the next step is to connect these shared spatial conditions according to their order in the generated sequence. The connection between two segments is not treated as a simple linear attachment. The final clip of the previous segment provides the primary direction of transition, while the spatial characteristics of the following segment influence how the boundary, opening and threshold are formed. Different Scene Category transitions therefore produce different spatial relationships. For example, a transition from Hall to Room may create compression and a more defined threshold, while Room to Hall may produce opening and spatial release. Through these connections, the sequence begins to operate as a relational spatial system rather than a collection of independent segments.

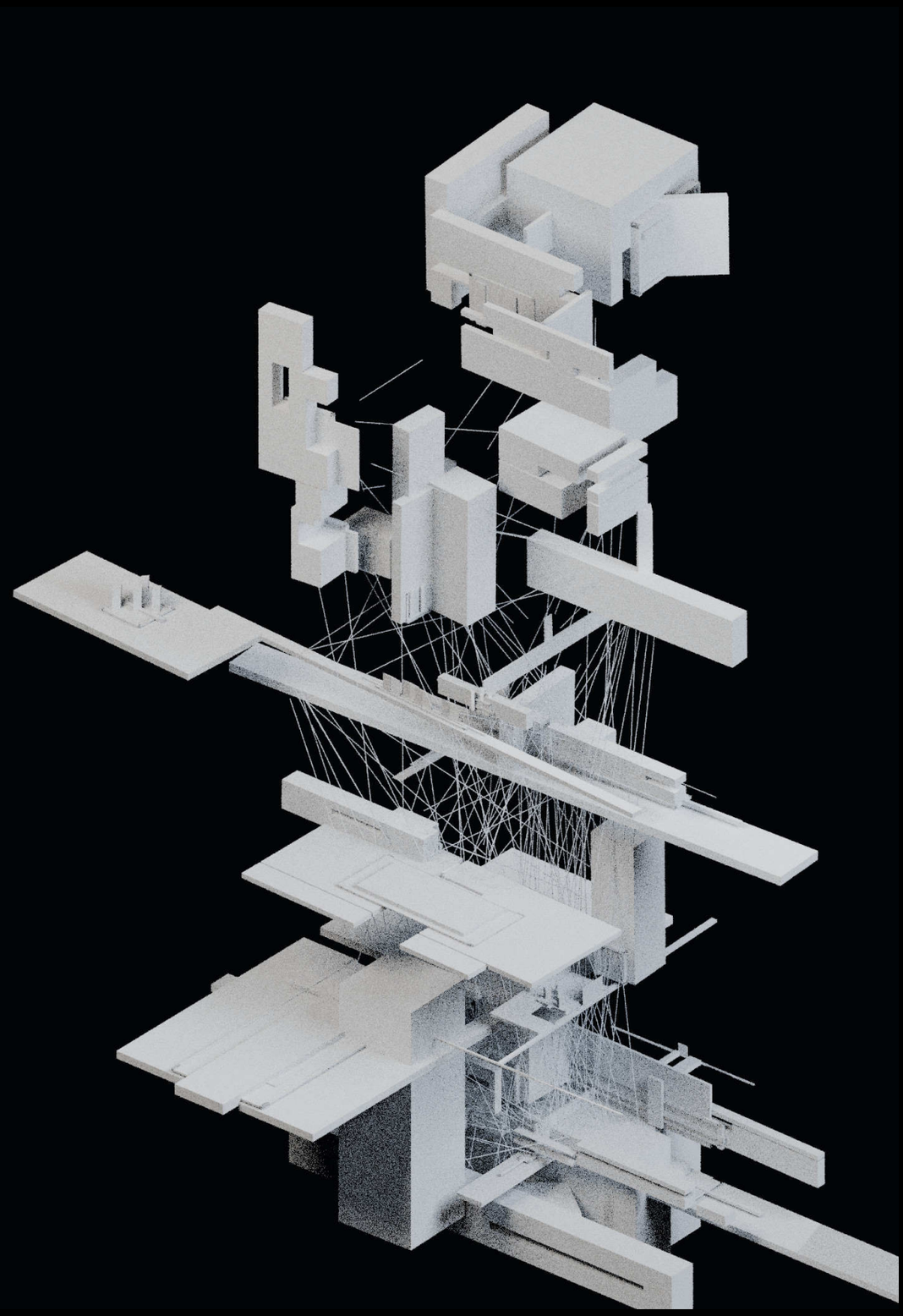
Segment to Segment Connection Rules

Join segments based on sequence
→ Segments are connected in the original sequence order.
→ Each segment has a spatial weight based on its duration and host dominance.
→ The last clip of the previous segment provides the movement direction for transition.
→ Segments connect through bounding box and actual boundary contact, not as a linear chain.
→ The system tests multiple positions and selects the one that keeps the overall model compact, folded, and connected.

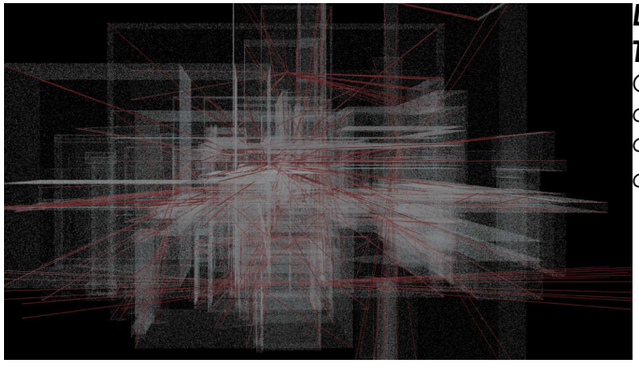
Movement Direction

Rightward → +X
Leftward → -X
Forward → +Y
Backward → -Y
Upward → +Z
Downward → -Z
Random Variable / Static → +X, +Y, -X, -Y

Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQCdlj3sfGFSSrdUIbqZD0zLAUvQwl_p8x-c3Xya41qSL9io?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcH-BQbGF0Zm9ybSI6IldlYilsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=aWQosd



The final skeletal model of sequence

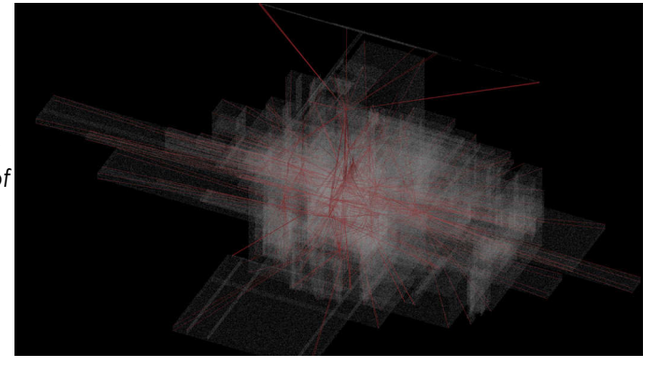


Layered Spatial Traces

Overlapping clip geometries reveal spatial density and directional accumulation.

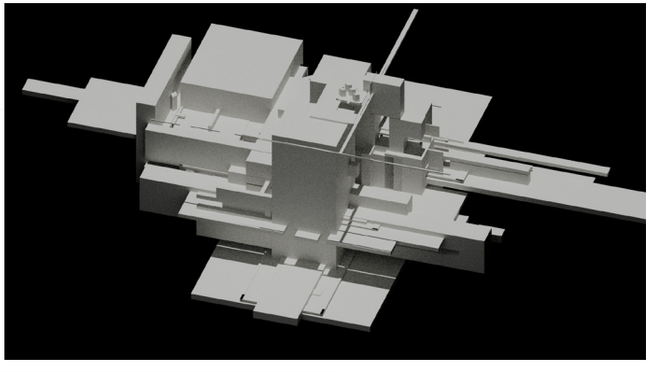
Movement and Connection Structure

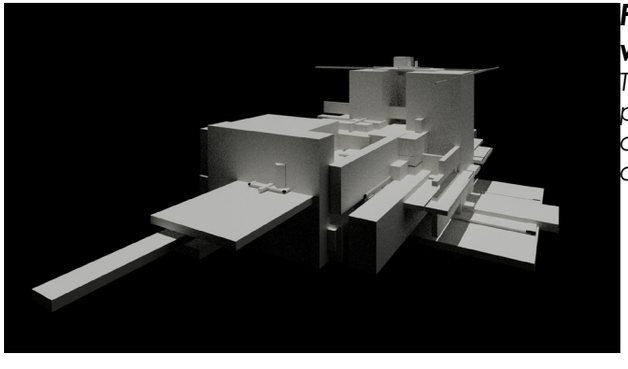
Red vectors indicate the principal directions guiding the assembly of the sequence.



Consolidated Skeletal Model

The layered geometries are merged into a single skeletal spatial structure.





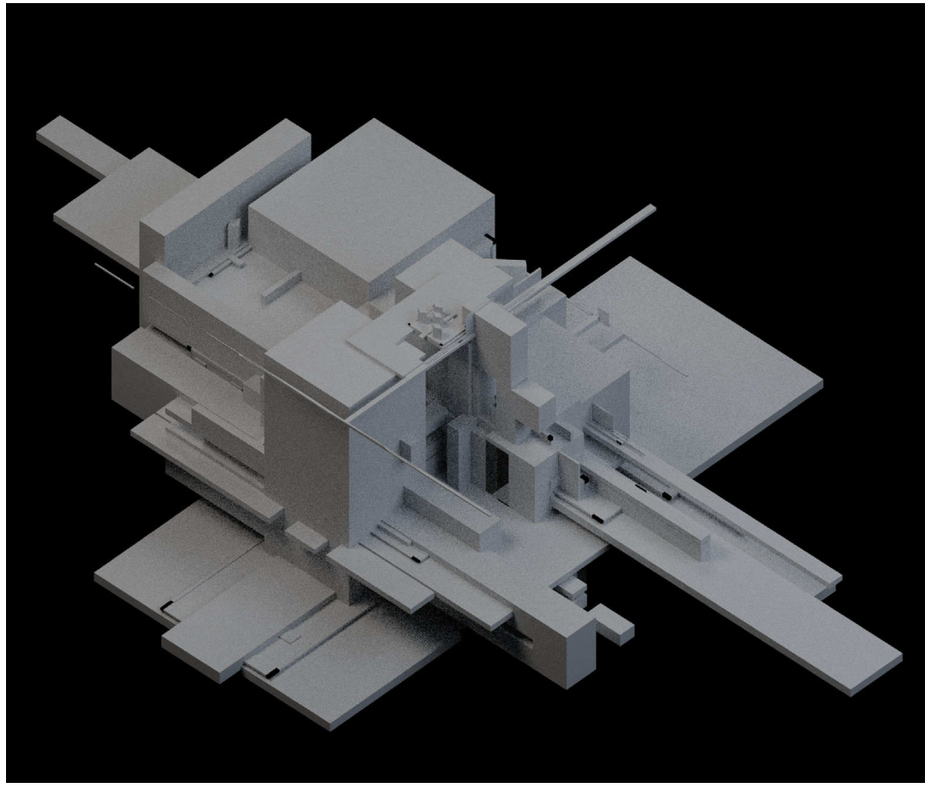
Final Spatial Framework

The resulting model presents the overall sequence as a coherent abstract framework.

Key code *for connecting segments*

```
def connect_segments(segments):
    direction_vector = {
        “right”:    Vector(( 1,  0,  0)),
        “left”:     Vector((-1,  0,  0)),
        “forward”:  Vector(( 0,  1,  0)),
        “backward”: Vector(( 0, -1,  0)),
        “up”:       Vector(( 0,  0,  1)),
        “down”:     Vector(( 0,  0, -1)),
    }

    segments[0][“position”] = Vector((0, 0, 0))

    for previous, current in zip(segments[:-1], seg-
ments[1:]):

        movement = current[“dominant_movement”]
        direction = direction_vector.get(
            movement,
            Vector((1, 0, 0)),
        )


        previous_extent = segment_extent(previous, direc-
tion)
        current_extent = segment_extent(current, direc-
tion)


        overlap = min(
            0.22,
            min(previous_extent, current_extent) * 0.5,
        )

        contact_distance = (
            previous_extent
            + current_extent
            - overlap
        )


        current[“position”] = (
            previous[“position”]
            + direction * contact_distance
        )

        current[“rotation_z”] = atan2(
            direction.y,
            direction.x,
        )

    return segments
```

The final skeletal model of sequence



Spatial Articulation

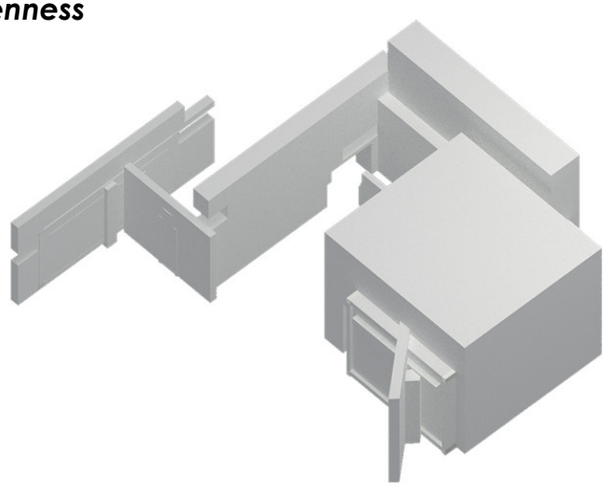
Calibrating Openness
Shot Size and Duration are translated into an openness ratio, which controls the balance between solid wall, glazing and open void, gradually transforming the spatial skeleton into an architectural envelope.

Introducing Architectural Elements
Architecture and Infrastructure labels are used to introduce architectural fragments and structural elements into the opened spatial model, further developing it into an architectural condition.

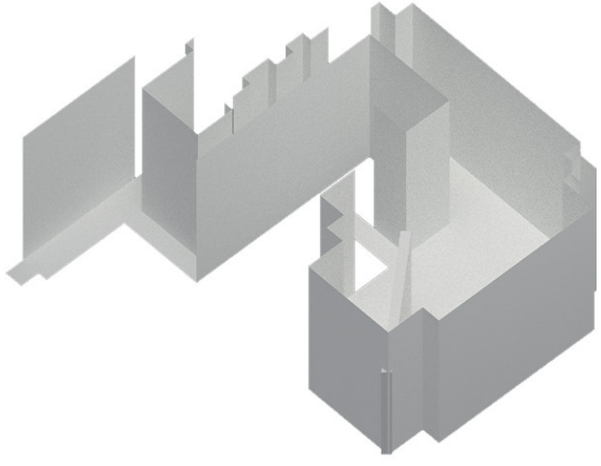
The spatial skeleton is architecturalised by calibrating openness through Shot Size and Duration, then introducing architectural and infrastructural elements based on the corresponding labels.

Taking segment29 as an example

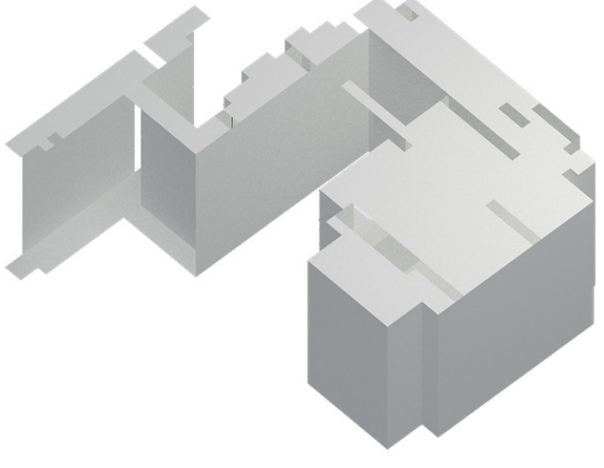
Calibrating Openness



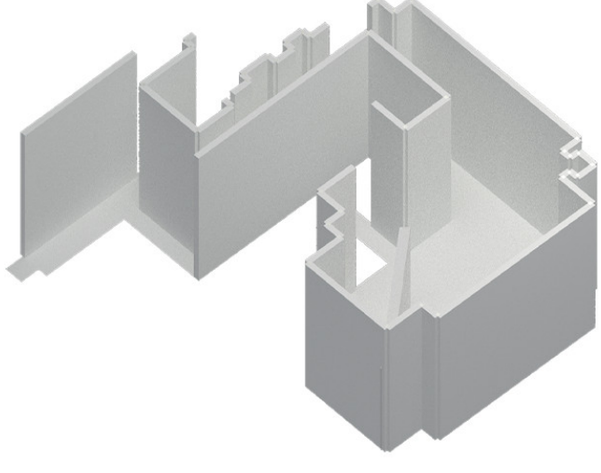
Spatial Skeleton
Start from the spatial skeleton generated in the previous step.



Hide the Roof
Temporarily remove the roof to reveal the internal spatial organisation.

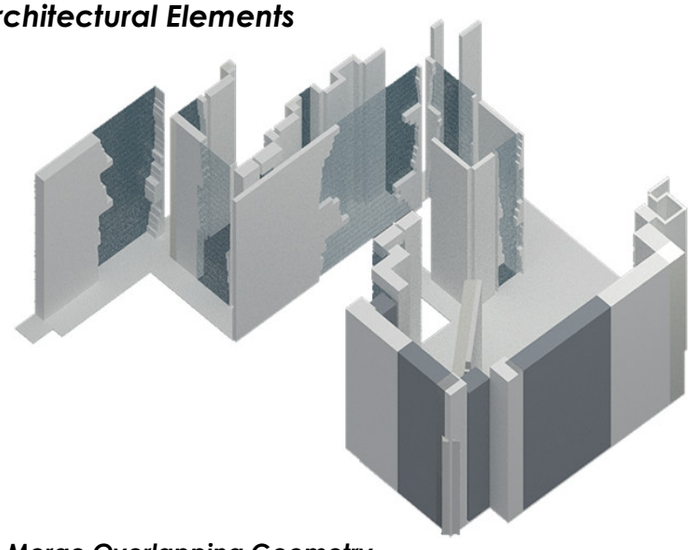


Merge Overlapping Geometry
Combine overlapping components to create a continuous spatial structure.

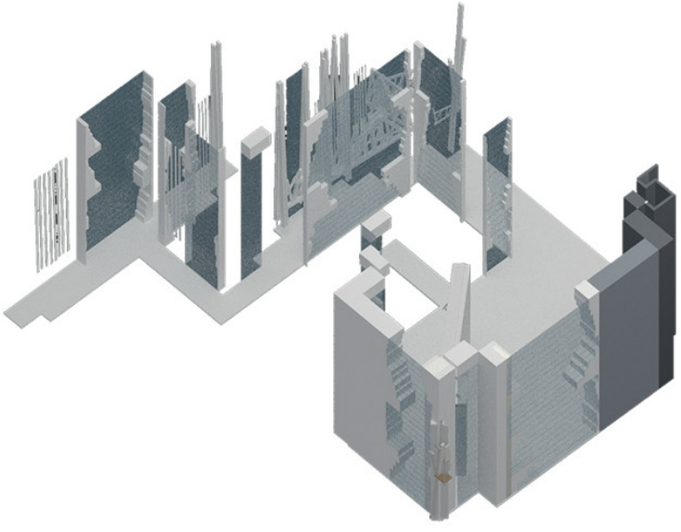


Generate Wall Thickness
Generate the walls, with their thickness determined by the proportional Duration value of each Clip within the Segment.

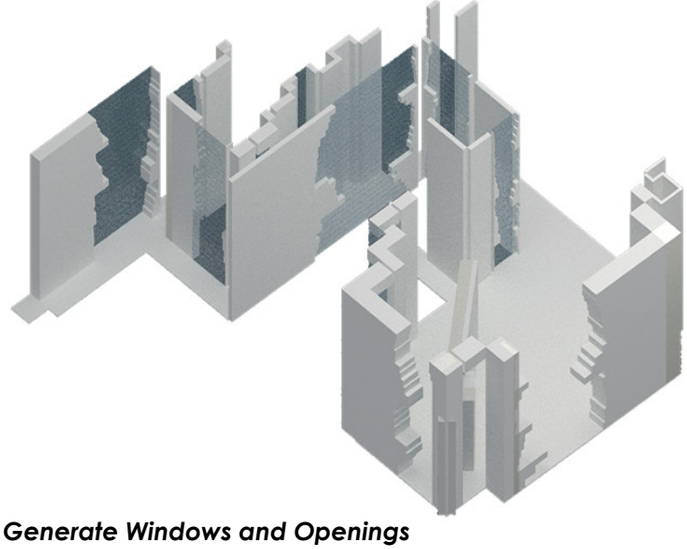
Introducing Architectural Elements



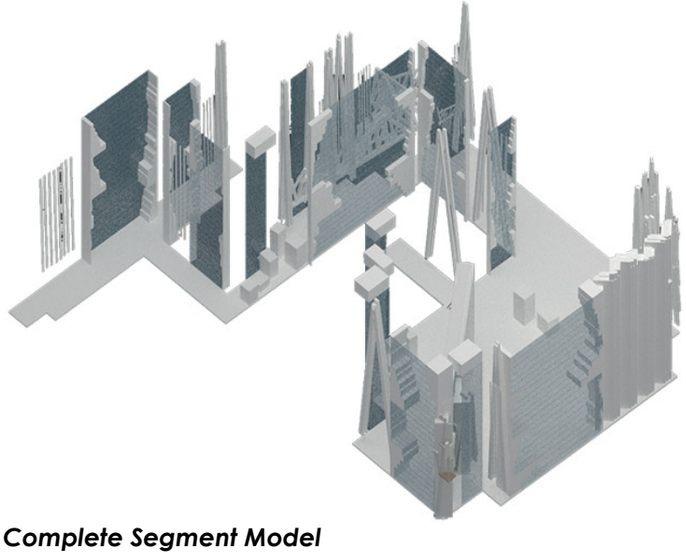
Merge Overlapping Geometry
Combine overlapping components to create a continuous spatial structure.



Generate Architectural Elements
Introduce architectural and infrastructural elements within the defined areas to further articulate the spatial structure.



Generate Windows and Openings
Create the window structures and open voids within the selected areas.

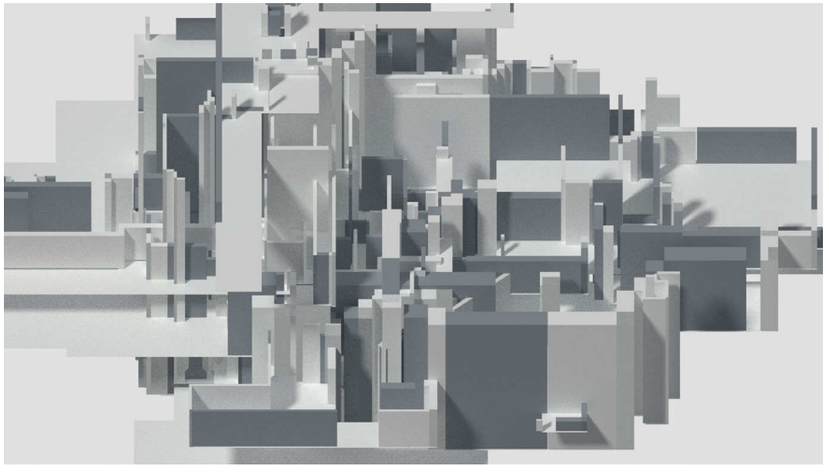


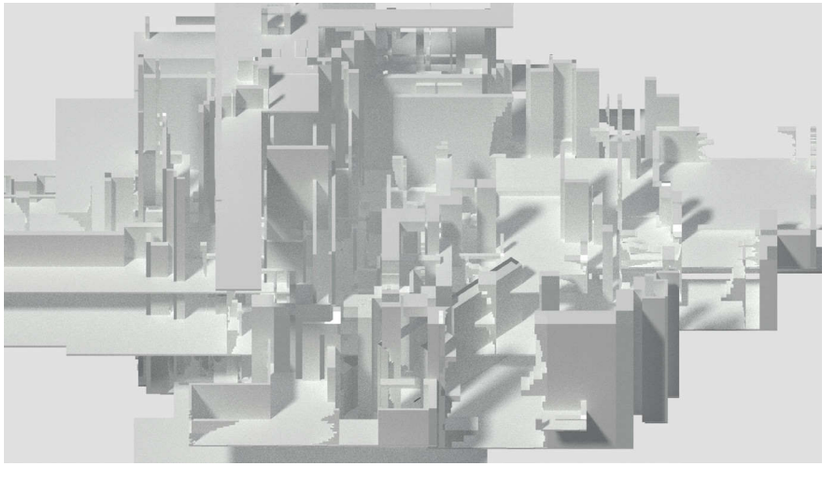
Complete Segment Model
Integrate all transformations to produce the final architectural model of the Segment.

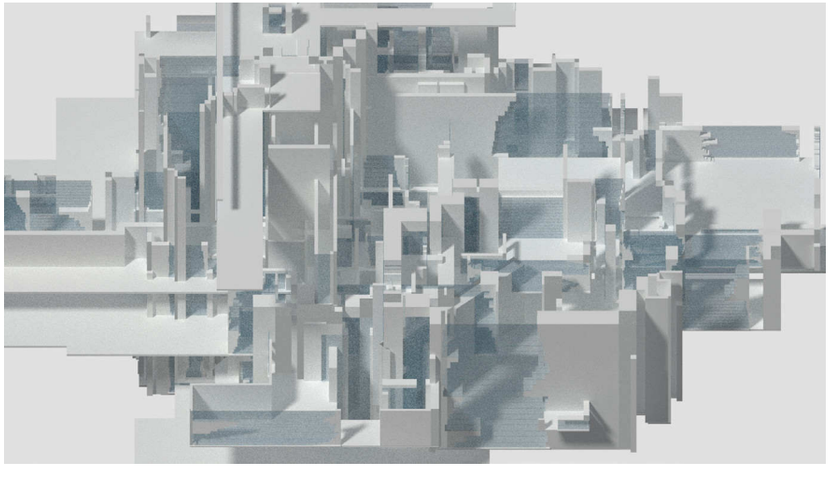
Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQDXQ0pjfQcOQrxs7A_Gi8mRAddtp95mG5yoiAmetoXWf1s?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcmI2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQb-GF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=WPLJVo

Calibrating Openness

Read the spatial skeleton
→ Map each Clip to its corresponding Segment
→ Shot Size defines the initial degree of openness
→ Duration determines each Clip's influence
→ Calculate the solid–glass–void ratio
→ Cut openings into the existing walls
→ Insert stepped and non-uniform window systems
→ Generate the final windowed base model



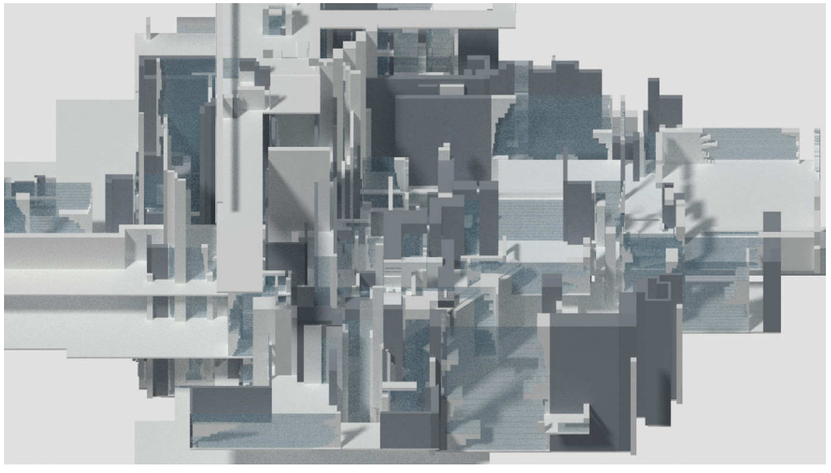


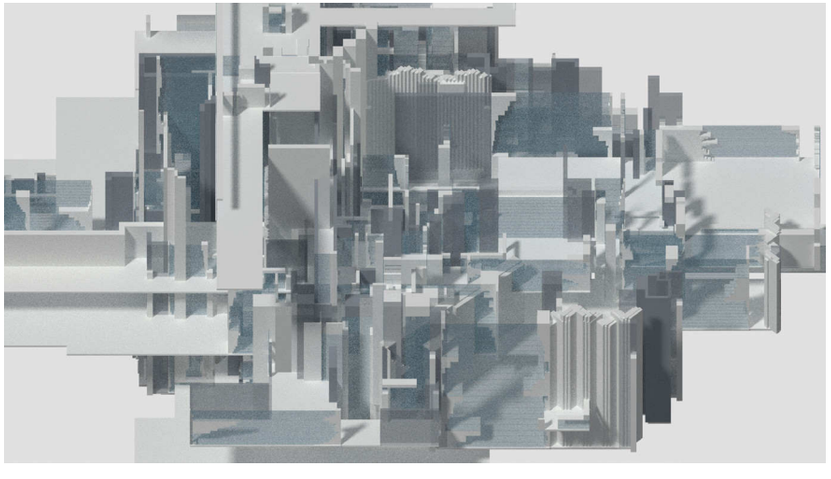


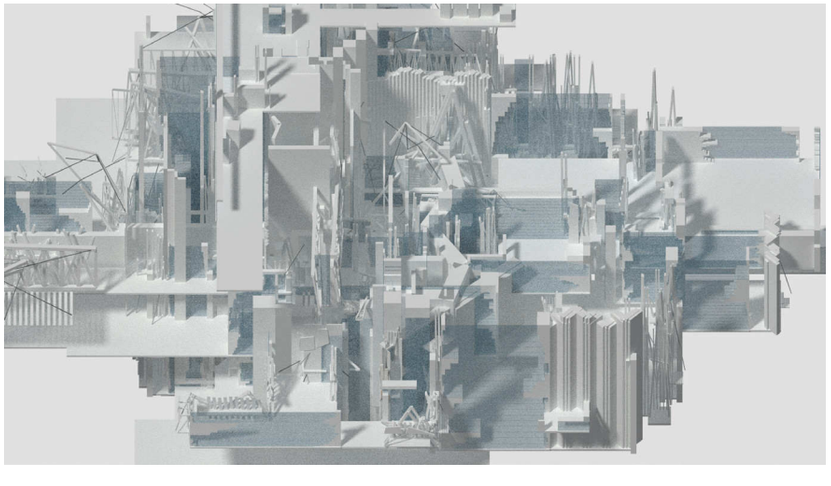
Introducing Architectural Elements

Identify walls, floors, windows and openings
→ Select primary surfaces based on Architecture/Infrastructure weights
→ Architecture: frames, trusses, folded plates and cantilevered structures
→ Infrastructure: columns, ribs, cables, platforms and node systems
→ Hybrid labels: combine both systems
→ Duration controls scale, density and complexity
→ Check structural support, collisions and opening clearance
→ Generate the final architecturalised model

Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQDNiMFXDxw4RKKWdf6iZtAiAfrum06FqtfAQYZwZ-BLulN0?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=SLIcyF

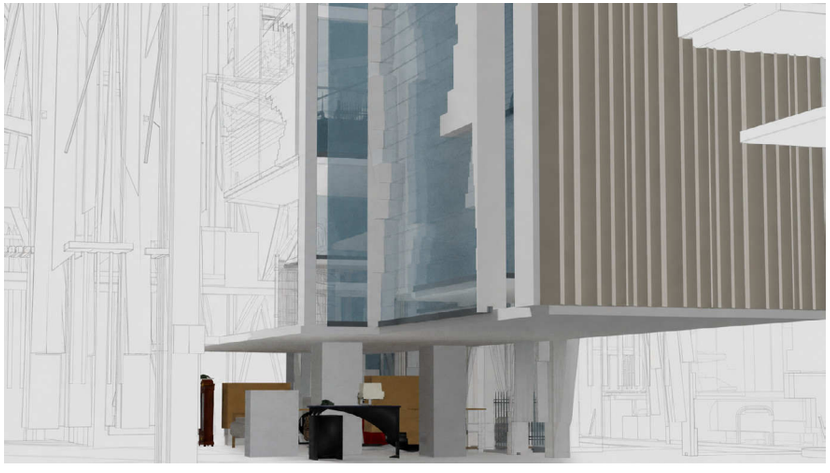


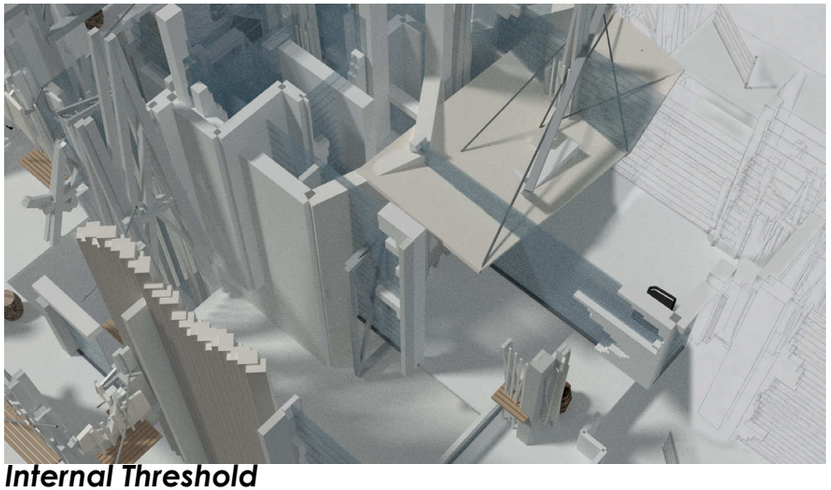




Urban Interface

The generated structure meets the existing urban edge through layered façades, openings and transitional spaces.



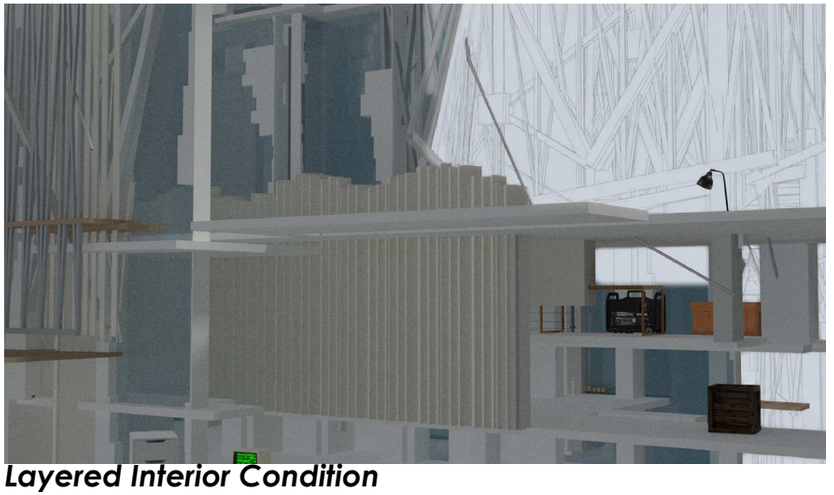


Internal Threshold

Overlapping segments create changing levels of enclosure and visual connection within the sequence.

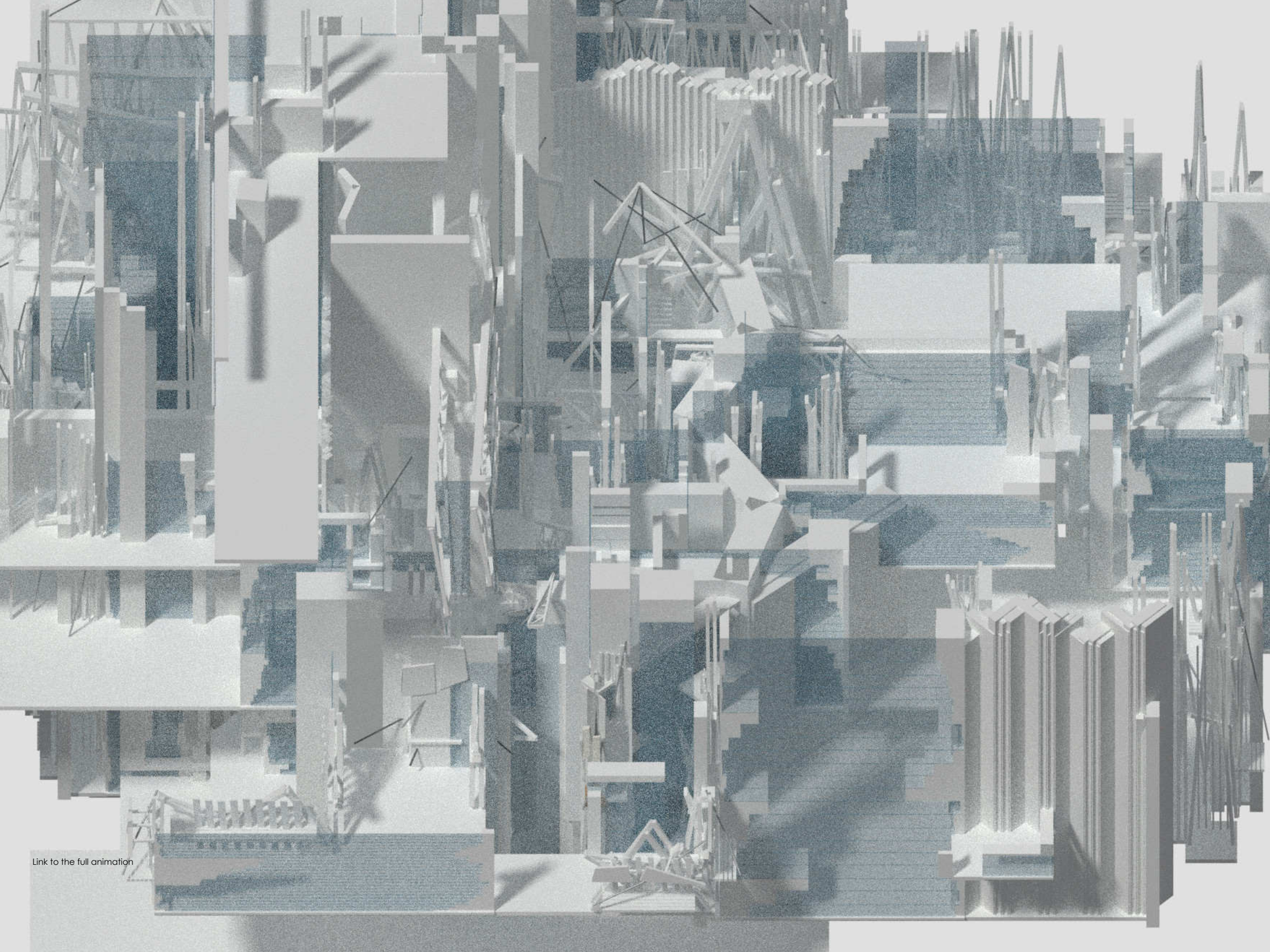
Vertical Spatial Connection

Circulation and level changes connect different segment conditions into a continuous spatial experience.

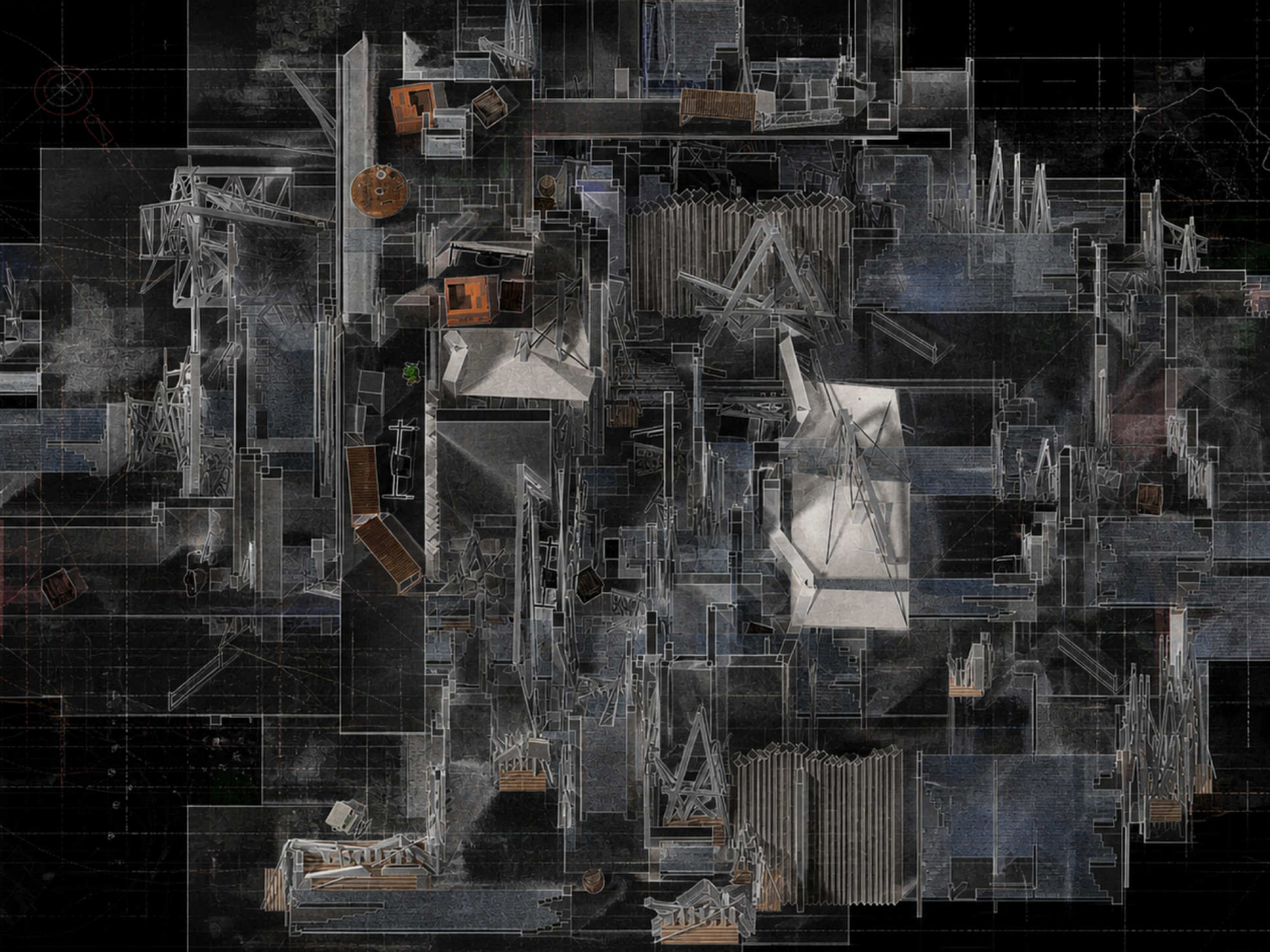


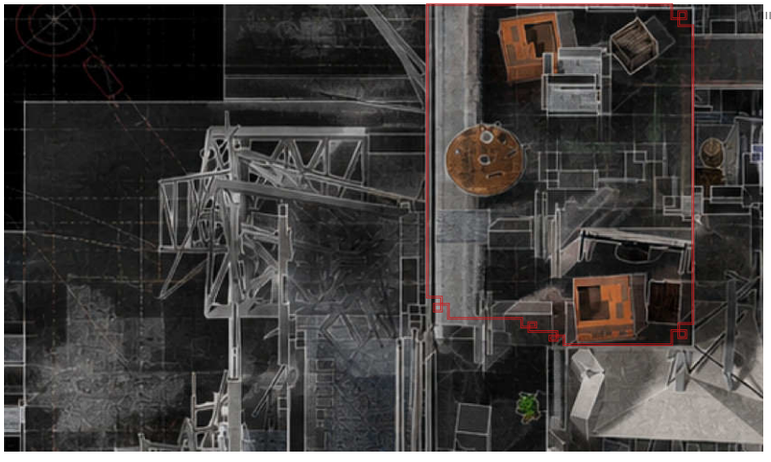
Layered Interior Condition

Walls, glazing and architectural fragments overlap to produce depth, transparency and spatial compression.



Link to the full animation



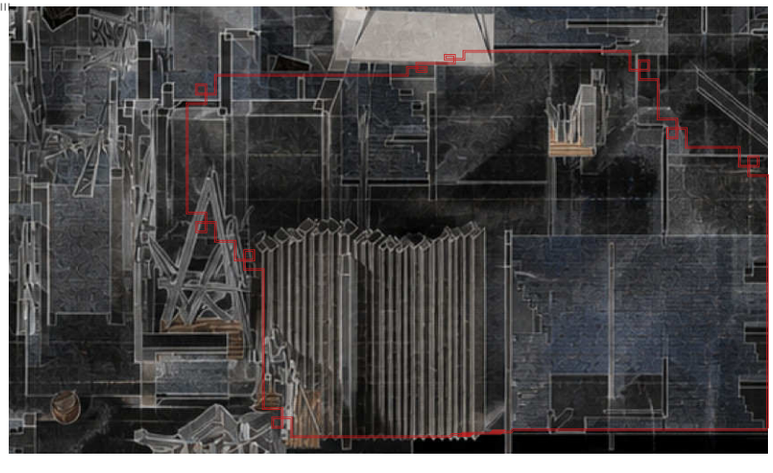


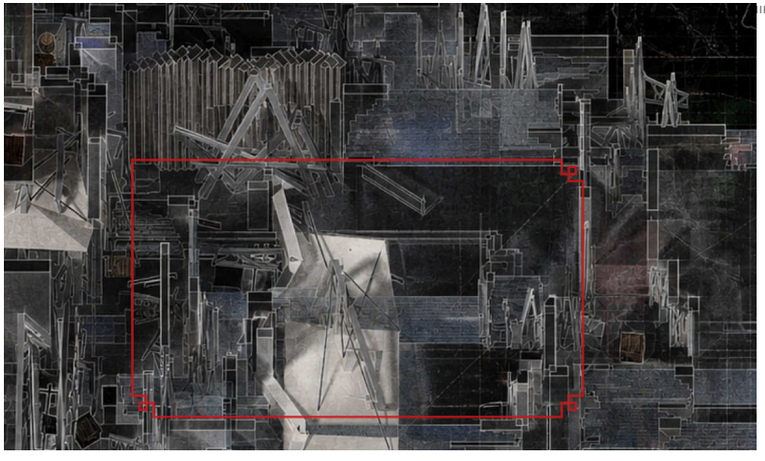
Segment 14 (Waterfront): C20-C22
clip labels:
- C20_Waterfront_Long_RandomVariable_natural_element_2.002s
- C21_Waterfront_Medium_RandomVariable_human_solo_4.667s
- C22_Waterfront_Medium_RandomVariable_human_group_4.750s

Kuleshov-effect spatial translation:
The sequence from an open natural field to individual contraction and collective occupation is translated into a continuous waterfront communal space. Personal niches, shared platforms, terraces, and open boundaries coexist within one connected spatial system rather than forming separate rooms.

Segment 29 (Room): C45- C50
Specific clip labels:
- C45_Room_Long_RandomVariable_architectural_element_2.753s
- C46_Room_Close_RandomVariable_human_solo_2.961s
- C47_Room_Medium_Static_human_group_6.465s
- C48_Room_Medium_Rightward_human_group_10.719s
- C49_Room_Medium_RandomVariable_architectural_element_17.017s
- C50_Room_Close_Downward_architectural_element_1.001s

Kuleshov-effect spatial translation:
The sequence moves from architectural scale to personal contraction, a stable group centre, rightward expansion, architectural reframing, and final downward compression. It is translated into one continuous room containing a collective platform, a personal stay point, lateral openings, structural columns and beams, and a lowered terminal floor. The previous Hall boundary is reinterpreted as the entrance threshold to the Room.





Segment S23 (Waterfront): C33-C34
Specific clip labels:
- C33_Waterfront_Close_RandomVariable_human_group_4.296s
- C34_Waterfront_Close_RandomVariable_human_group_5.631s

Kuleshov-effect spatial translation:
The repetition of two close-range group scenes is translated into an intimate waterfront space centred on collective occupation. Both clips contribute to one continuous communal platform with an open upper boundary, a waterfront edge, and a framed transition toward the following infrastructure space.

Segment 21 (Street): C30
Specific clip label:
- C30_Street_Medium_Rightward_architectural_element_17.267s

Kuleshov-effect spatial translation:
The sustained rightward movement of the architectural element is translated into a linear street space that extends laterally across the site. The result is one continuous open route defined by side walls, portal frames, structural columns, and a release strip that transitions toward the following infrastructure space.

Material Identification

This chapter establishes a traceable translation system from cinematic imagery to architectural materiality. Following a Markov sequence, the program processes 50 film clips and extracts one representative keyframe from each. Avoiding human faces, it identifies material cues from walls, floors, structures, glazing, and objects. These cues are classified as Primary, Secondary, or Detail materials, matched with corresponding assets from a material library, and mapped onto the model according to the sequence. Without altering the original geometry, the model accumulates colours, textures, and atmospheres derived from different cinematic spaces. The connecting lines document the complete lineage from keyframe to material and architectural component.

Define Rules

scene_category
→ Determines the material family

colour
→ Determines tone, brightness, and atmosphere

basic_component
→ Determines structural or detailed material use

shot_size
→ Determines texture scale and visual density

movement
→ Preserves directional and geometric information

duration
→ Preserves the influence weight of each Clip

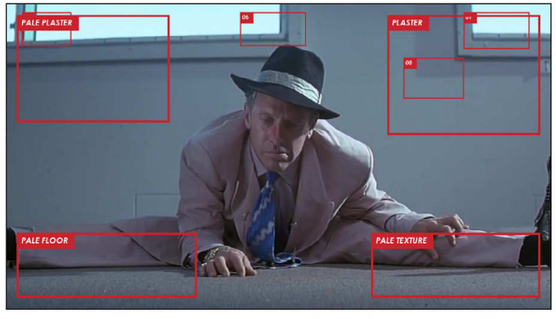
sequence_index
→ Links each material to its original Film Clip

material_panel
→ Organises primary, secondary, and detail materials

source_clip_id
→ Transfers materials into the final architectural model

Extracting Textures from Film Keyframes

Identifying Materials in Movie Clips



C05 / 50 — WHOAMI

SCENE Room | COLOUR blue_bright | TIME 1.08s

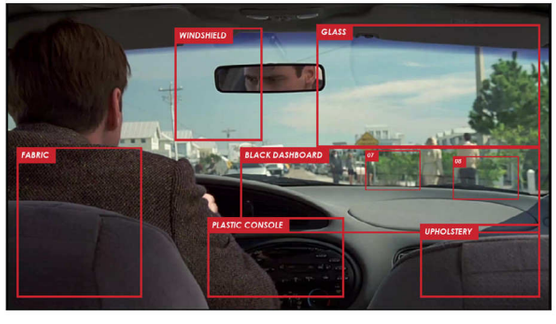
pale room with table and suited figure

bright / smooth / dry / diffuse

PRIMARY	PALE PLASTER Plaster External Coarse Semigloss	1.250
SECONDARY	WHITE TABLETOP Glowing White	0.670
DETAIL	PALE TEXTILE Castle Wall V2	0.820

EXCLUDED face / hands / small accessories

FRAME → 3 MATERIALS



C09 / 50 — THE TRUMAN SHOW

SCENE Infrastructure | COLOUR blue_dark | TIME 7.59s

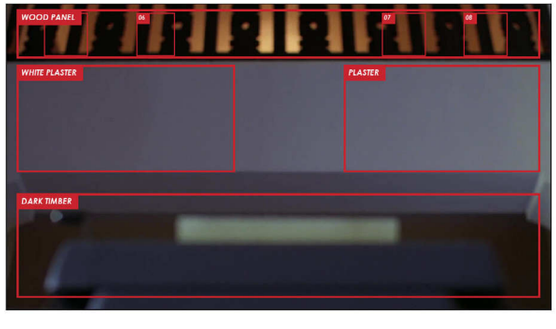
car interior looking toward street

smooth / synthetic / reflective / automotive

PRIMARY	BLACK PLASTIC DASHBOARD Geometric pattern Black Plastic PL	1.105
SECONDARY	WINDSHIELD GLASS Smuged Glass	1.125
DETAIL	GRAY FABRIC UPHOLSTERY Fabric 05	1.154

EXCLUDED sky / distant exterior / human head

FRAME → 3 MATERIALS



C10 / 50 — THE SHINING

SCENE Room | COLOUR yellow_orange | TIME 13.47s

hotel interior with polished floor and warm room

warm / polished / theatrical / interior

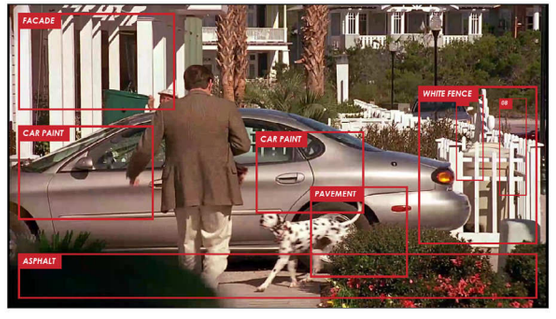
PRIMARY	POLISHED TIMBER FLOOR Wet Forest Floor	0.915
SECONDARY	WHITE PLASTER COLUMNS plaster white	1.250
DETAIL	DARK WOOD PANELING Wood panel verze2	1.200

EXCLUDED face / clothing foreground

FRAME → 3 MATERIALS

METHOD **VISUAL REGION → SURFACE QUALITY → SEARCH QUERY → SELECTED MATERIAL**

A film clip produces a material palette only after human-readable semantic interpretation.



C13 / 50 — THE TRUMAN SHOW

SCENE Street | COLOUR yellow_orange | TIME 0.96s

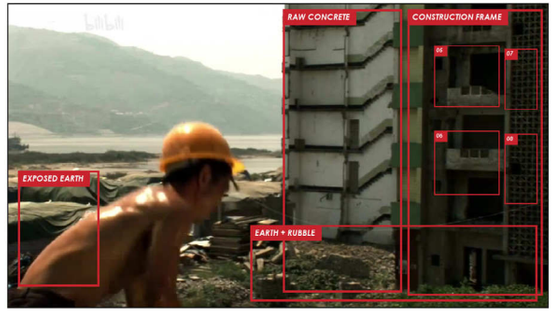
suburban street with car fence and facade

warm / suburban / smooth / exterior

PRIMARY	ASPHALT / PAVEMENT Asphalt Damaged	1.020
SECONDARY	BEIGE CAR PAINT Procedural Car Paint	1.250
DETAIL	WHITE PICKET FENCE Old White Paint scan	0.942

EXCLUDED human figure / dog as event

FRAME → 3 MATERIALS



C17 / 50 — STILL LIFE

SCENE Infrastructure | COLOUR yellow_orange | TIME 5.12s

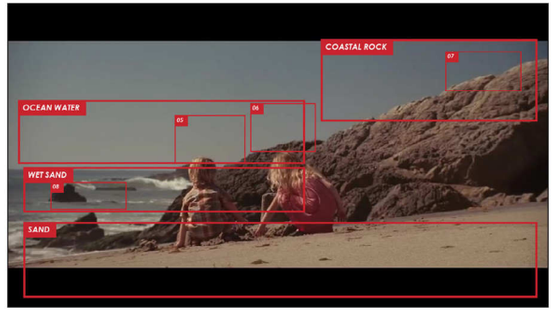
construction site beside unfinished tower

rough / unfinished / dry / infrastructural

PRIMARY	RAW CONCRETE Concrete	1.230
SECONDARY	EXPOSED EARTH / RUBBLE Planet	0.599
DETAIL	CONSTRUCTION METAL Iridescent metal	1.050

EXCLUDED human skin / helmet / distant landscape

FRAME → 3 MATERIALS



C20 / 50 — INCEPTION

SCENE Waterfront | COLOUR gray_dark | TIME 2.00s

rocky beach and ocean

rough / dry-wet contrast / natural

PRIMARY	SAND Sand 02	1.028
SECONDARY	WEATHERED COASTAL ROCK Rock core	0.884
DETAIL	OCEAN WATER Water fake caustics	1.033

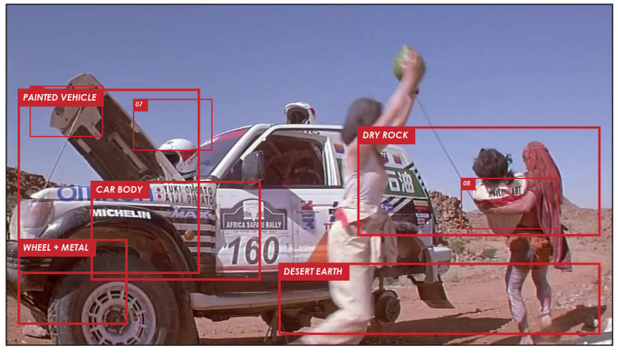
EXCLUDED people as small figures / distant sky

FRAME → 3 MATERIALS

METHOD **VISUAL REGION → SURFACE QUALITY → SEARCH QUERY → SELECTED MATERIAL**

A film clip produces a material palette only after human-readable semantic interpretation.

Identifying Materials in Movie Clips



C28 / 50 — WHOAMI

SCENE Landscape | COLOUR blue_bright | TIME 4.17s

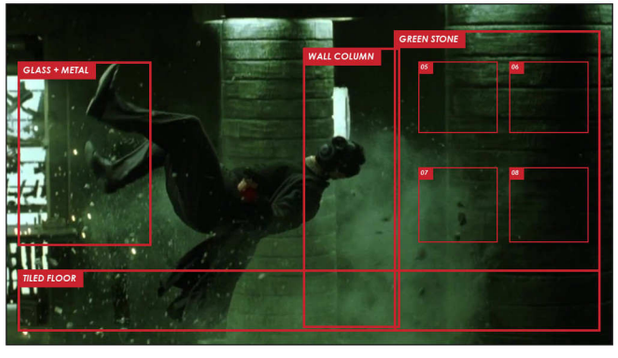
desert scene with vehicle and people

dry / rough / sunlit / exterior

PRIMARY	DESERT SAND Sand 02	0.933
SECONDARY	DRY ROCK / EARTH Alien-rocks	0.779
DETAIL	PAINTED VEHICLE METAL Peeling Paint	1.117

EXCLUDED people / clothing / sky

FRAME → 3 MATERIALS



C44 / 50 — THEMATRIX1

SCENE Hall | COLOUR green | TIME 3.21s

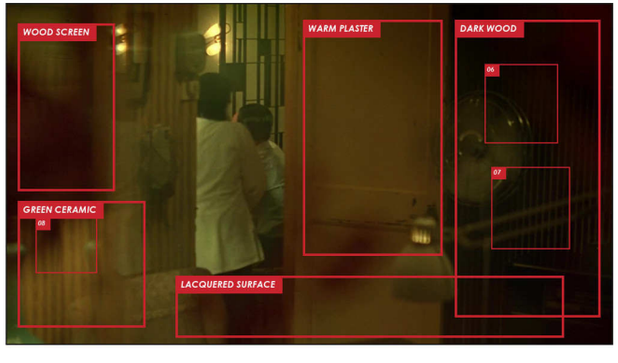
green hall action with wall columns and smoke

hard / cold / gridded / smoky

PRIMARY	GREEN STONE / CONCRETE WALL Stylized Stone Wall with Moss	1.250
SECONDARY	TILED FLOOR Cork Floor Tiles	1.036
DETAIL	GLASS / METAL ENTRANCE Prosedural Glass	1.029

EXCLUDED smoke / falling body / light effects

FRAME → 3 MATERIALS



C49 / 50 — IN THE MOOD FOR LOVE

SCENE Room | COLOUR yellow_orange | TIME 17.02s

warm interior corridor with lamp and wood

warm / humid / intimate / lacquered

PRIMARY	WARM PLASTER Plaster External Coarse Semigloss	1.250
SECONDARY	DARK WOOD SCREEN / DOOR Dark Wood	1.146
DETAIL	GREEN CERAMIC LAMP Pearl ceramic	1.049

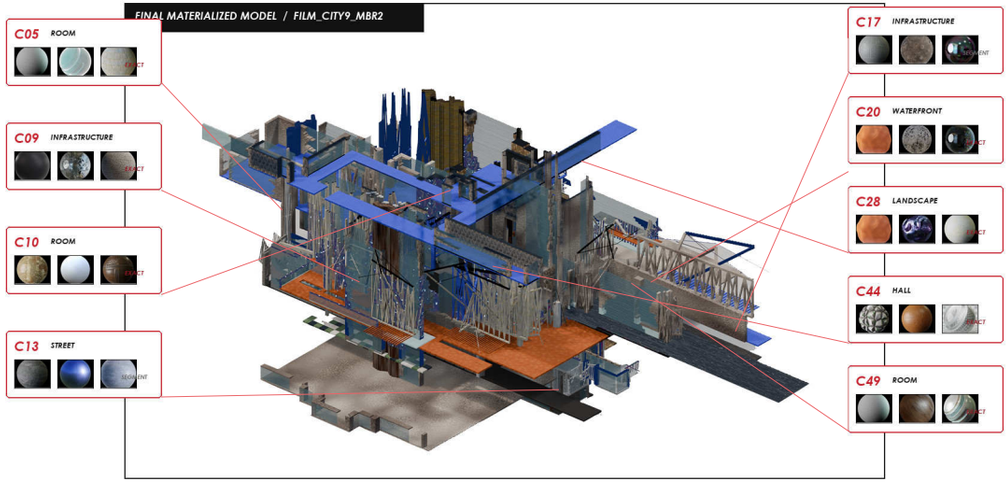
EXCLUDED blurred foreground body / curtain

FRAME → 3 MATERIALS

METHOD **VISUAL REGION → SURFACE QUALITY → SEARCH QUERY → SELECTED MATERIAL**

A film clip produces a material palette only after human-readable semantic interpretation.

PRIMARY 0—62% | SECONDARY 62—86% | DETAIL 86—100%



04.1 NINE-CLIP PROVENANCE — DASHED LINES = CLIP-TO-MODEL MATERIAL LINKS — 7 EXACT / 2 SEGMENT FALLBACKS

PALETTE INDEX

C05	C09	C10	C13	C17	C20	C28	C44	C49
EXACT	EXACT	EXACT	FALLBACK	FALLBACK	EXACT	EXACT	EXACT	EXACT

04.2 SYSTEM FLOW

9 FILM CLIPS (27 material roles) → 9 PANELS (25 unique assets) → FACE SLOTS (62 / 24 / 14%) → SEGMENTS (exact + fallback) → FINAL MODEL (9 palettes active)

SELECTED SET / AUDIT

9 film clips	**27** material-role assignments	**25** unique assets
7 exact source mappings	**2** segment fallbacks	**9 / 9** palettes active on final faces

GEOMETRY PRESERVED / MATERIAL PROVENANCE ADDED

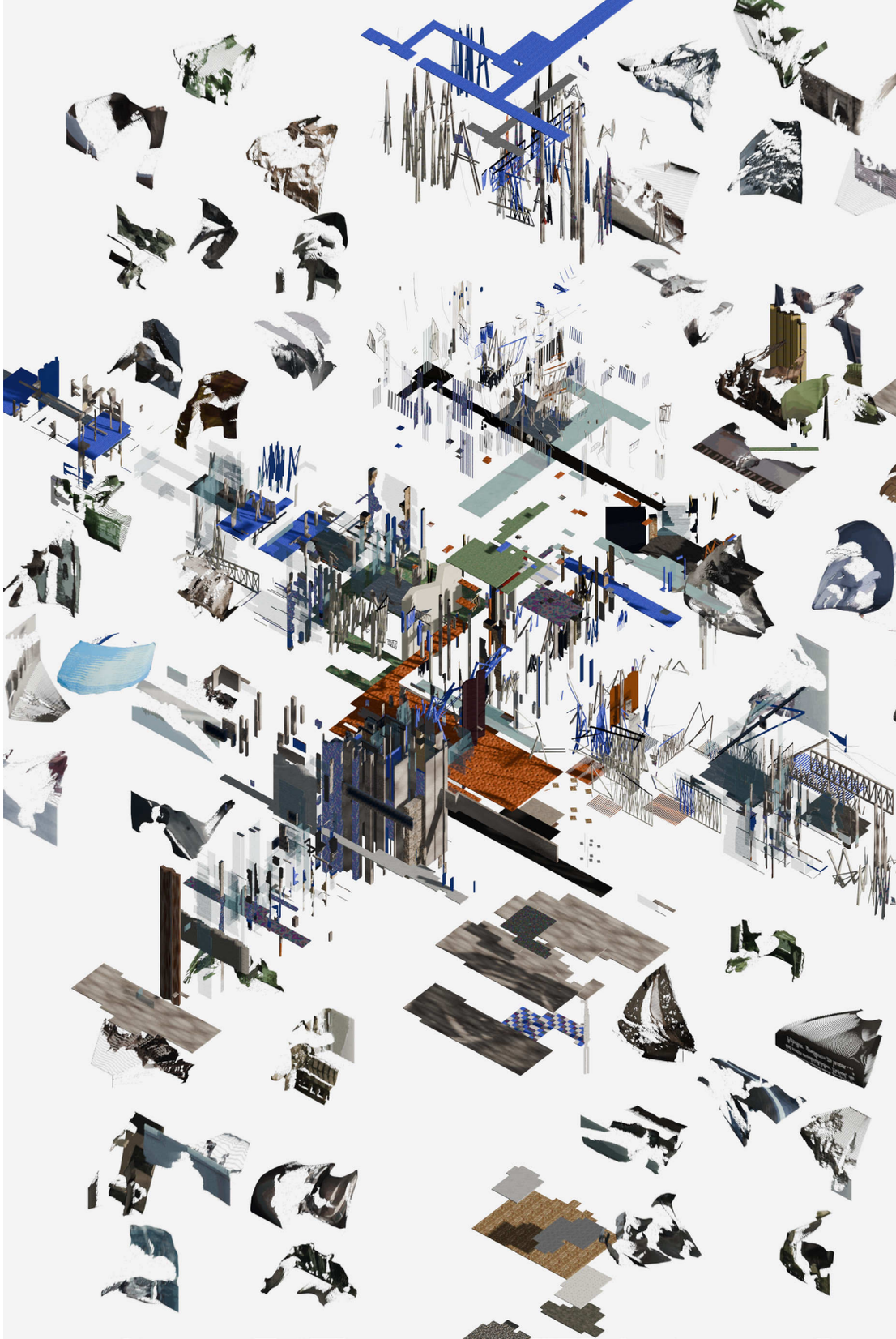
RESULT **9 FILM CLIPS → 27 MATERIAL ROLES → ONE TRACEABLE MATERIALIZED MODEL**

The model carries multiple cinematic material histories without changing its geometry.

Material Fragment Atlas

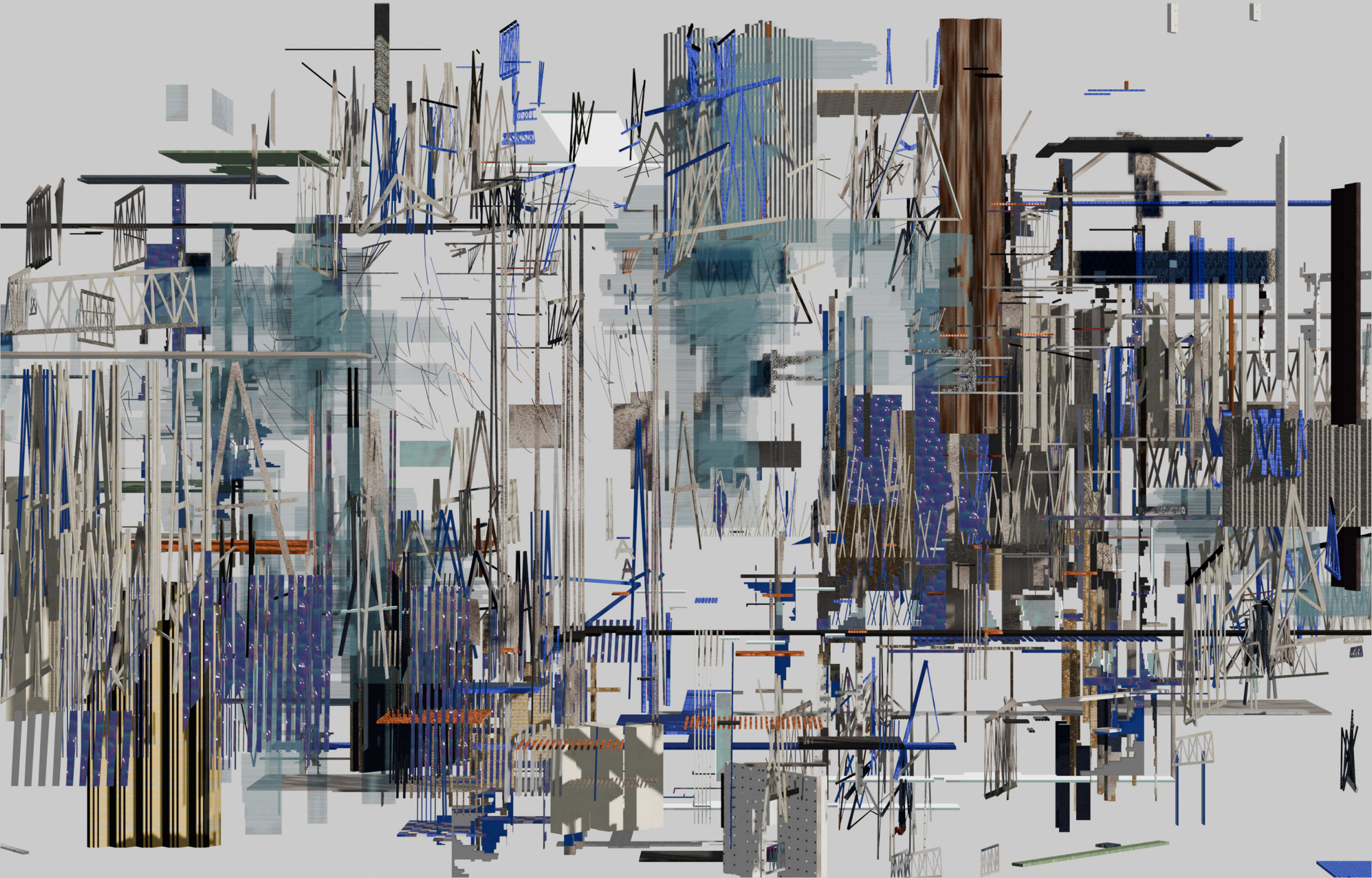
Materials extracted from the original film clips are applied to individual spatial elements. The model is then disassembled to show how each fragment retains its own cinematic colour, texture and material character.

The extracted film materials are mapped back onto the spatial fragments and reassembled according to the sequence structure. The exploded view reveals how individual materialised elements accumulate into a larger architectural composition while retaining their cinematic origins.

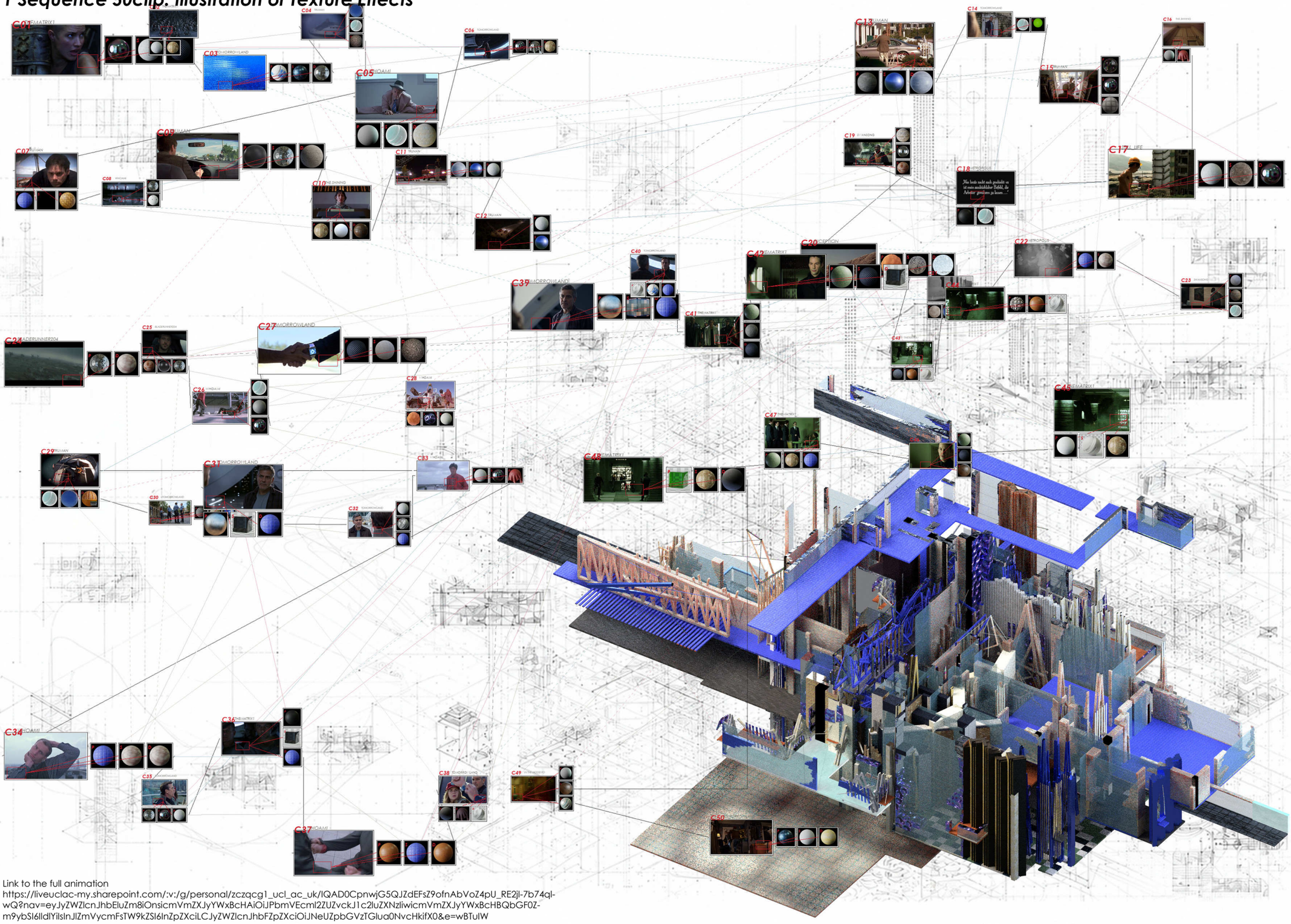


Materialised Spatial Field

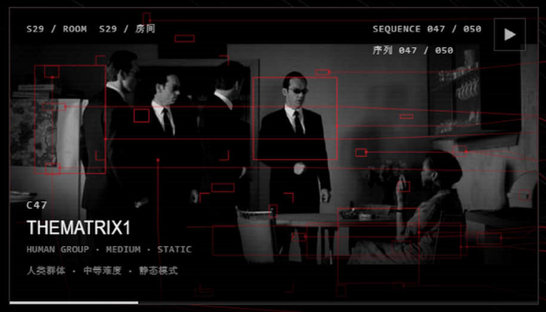
The individual materialised fragments are brought together into a dense spatial field, revealing how different cinematic textures, structural elements and surfaces overlap within the final sequence. The assembled view shows the complexity produced when multiple clip-based fragments coexist within one architectural system.



1 Sequence 50clip: Illustration of Texture Effects



Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQAD0CpnwjG5QJZdEFsZ9ofnAbVoZ4pU_RE2jl-7b74ql-wQ?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYilsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=wBTuIW



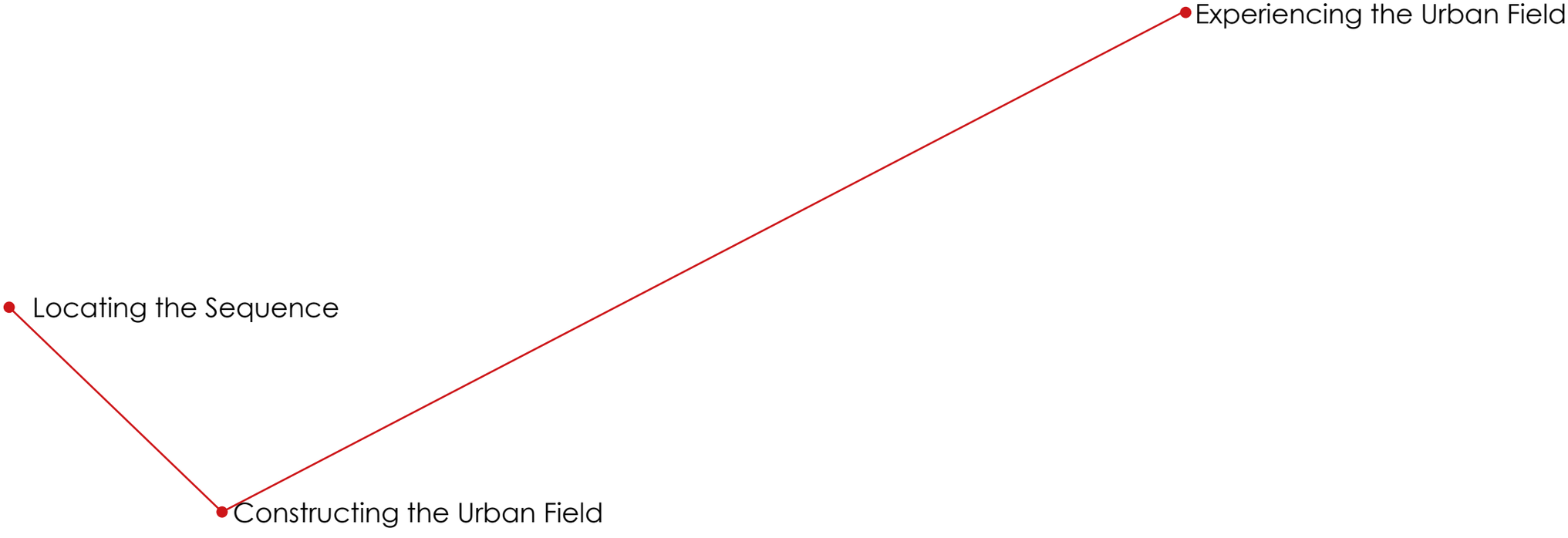

```
            records.append({
                "clip": clip,
                "source_name": obj.name,
                "verts_world": verts_world,
                "faces": faces,
                "mat_indices": mat_indices,
                "materials": [slot.material for slot in obj.material_slots],
            })
            points.extend(verts_world)
    mn, mx = bbox_from_points(points)
    center = (mn + mx) * 0.5
    volume = bbox_volume(mn, mx)
    total_existing_volume += volume
    assets.append({
        "segment_id": sid,
        "clip_range": clip_range,
        "scene_category": first["scene_category"],
        "first_seq": first["sequence_index"],
        "last_seq": last["sequence_index"],
        "duration": first["segment_total_duration"],
        "source_min": mn,
        "source_max": mx,
        "source_center": center,
        "source_volume": volume,
        "sequence_direction": segment_data["sequence_direction"],
        "records": records,
    })
return assets, total_existing_volume


def compute_scaled_placements(ns, assets, total_existing_volume, total_duration):
    placement_entries = []
    for asset in assets:
        ratio = asset["duration"] / total_duration if total_duration else 1.0 / len(assets)
        target_volume = max(total_existing_volume * ratio, MIN_VOLUME)
        scale = math.pow(target_volume / asset["source_volume"], 1.0 / 3.0)
        local_points = []
        for record in asset["records"]:
            local_points.extend([(p - asset["source_center"]) * scale for p in record["verts_world"]
        local_min, local_max = bbox_from_points(local_points)
        asset["duration_ratio"] = ratio
        asset["target_volume"] = target_volume
        asset["duration_scale"] = scale
        entry = {
            "clip": {
                "scene_category": asset["scene_category"],
                "movement": asset["records"][-1]["clip"]["movement"],
```


Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQCy3c-rPs3gT46ztD_sLNHOAa7e9Aq_XV9Gh-7Whz7mBdFk?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcH-BQbGF0Zm9ybSI6IldlYilsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=dFjMUg

CHAPTER 06

Sequence in the city

→ → → →



Locating the Sequence

The site strategy is developed through the urban network of the International Film Festival Rotterdam (IFFR). Rather than selecting a single isolated site, the project identifies 12 locations across Rotterdam that are connected to the festival and its cultural activities. These venues are distributed across different parts of the city and exist within different urban conditions, forming a wider network that reflects how the film festival operates through Rotterdam rather than within one building. Each location becomes a potential host for the generated spatial sequences, allowing the same design system to respond to different site boundaries, access conditions, surrounding buildings and urban contexts. By using the film festival as a curatorial framework for site selection, the project extends the logic of cinematic sequencing from the film scale to the urban scale, connecting the generated spaces with the existing cultural geography of Rotterdam.

Together, these 12 sites form a distributed urban field in which the generated sequences can be tested under different spatial and contextual conditions. Rather than producing one fixed architectural object, the project establishes a network of interventions connected through the shared logic of film, sequence and curation. The city therefore becomes the final scale of the system, where cinematic relationships are translated into a series of spatial encounters across Rotterdam.

IFFR / CINEMATIC GEOGRAPHIES OF ROTTERDAM

I

ROTTERDAM CENTRAAL
Stationsplein, 3013 AJ Rotterdam
Arrival point and station-hall Art Directions location.

II

FESTIVAL HEART / SCHOUWBURGPLEIN
Kruisplein 40 / Schouwburgplein, Rotterdam
de Doelen, Pathé and Theater Rotterdam festival cluster.

III

CINERAMA / WORM / V2_
Westblaak 18 / Boomgaardsstraat / Eendrachtsstraat
Cinema screenings, sound/vision and unstable-media programme.

IV

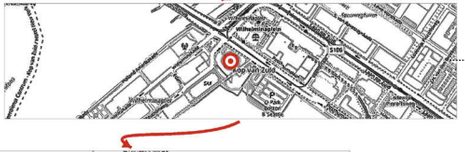
KINO
Gouvernestraat 129–133, Rotterdam
Independent cinema and official IFFR screening venue.

V

OUDE LUXOR THEATER
Kruiskade 10, Rotterdam
Historic city-centre theatre and IFFR screening venue.

VI

LANTARENVENSTER
Otto Reuchlinweg 996, Rotterdam
Film and music venue on the Wilhelminapier waterfront.

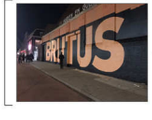


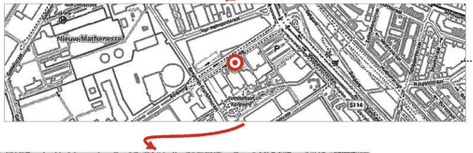
VII

NIEUWE LUXOR THEATER
Posthumalaan 1, Rotterdam
Large auditorium and festival programme location.

VIII

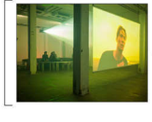
FENIX / PLEIN
Paul Nijghkade 5, Rotterdam
IFFR screenings, workshops and Filmclub events on Plein.

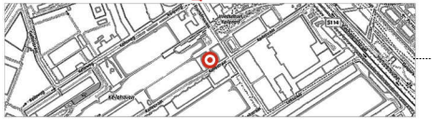




IX

BRUTUS
Keileweg 10–18, Rotterdam
Industrial art complex and Art Directions programme site.





X

KATOENHUIS
Keilestraat 9C, Rotterdam
IFFR Art Directions hub for installations and immersive media.

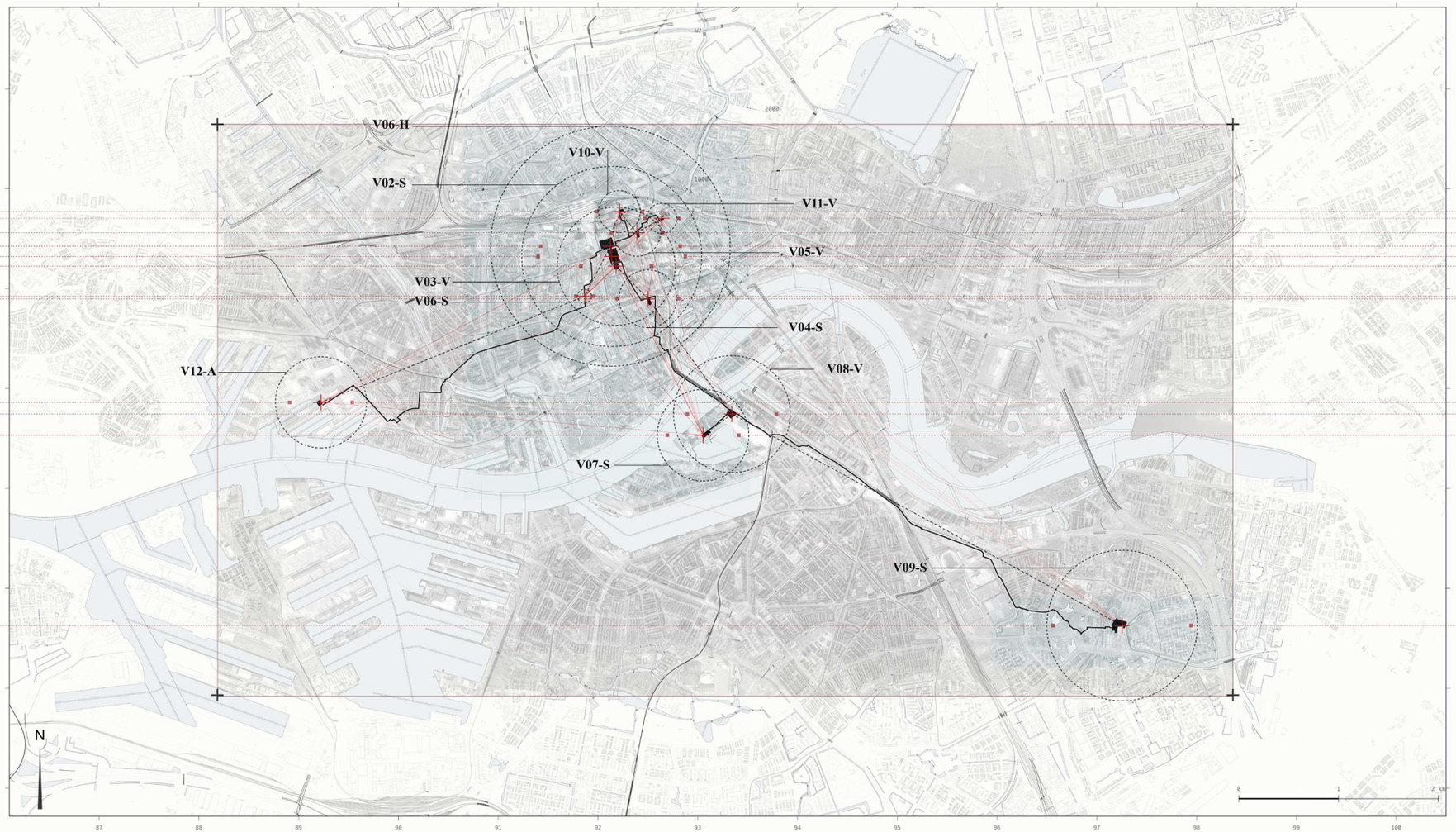
ROTTERDAM GEOGRAPHIC MONTAGE MAPPING

the relationships formed between multiple site segments placed side by side.

ANCHOR / BAG YEAR

MEAN WALK TO THE OTHER ELEVEN

/ Rotterdam: twelve fragments /



ROODKAPJE V10 2014 — 27.7 MIN

MUZIEKWERF V11 1950 — 28.6 MIN

OUDE LUXOR V05 1928 — 25.2 MIN

DE DOELEN & DE DOELEN STUD V01 1966 — 24.5 MIN

PATHÉ SCHOUWBURGPLEIN V02 2006 — 24.1 MIN

THEATER ROTTERDAM V03 1987 — 24.6 MIN

KINO V06 1930 — 27.4 MIN

CINERAMA FILMTHEATER V04 1957 — 25.3 MIN

KATOENHUIS V12 1949 — 64.8 MIN

NIEUWE LUXOR V08 2001 — 35.4 MIN

LANTARENVENSTER V07 2010 — 39.4 MIN

PODIUM ISLEMUNDA V09 2011 — 94.9 MIN

87 88 89 90 91 92 93 94 95 96 97 98 99 100

RD NEW EASTING / km

H Festival heart S Screening venue V Festival / partner venue A Art directions hub

geometric adjacency, weighted by detour — straight chord the walk cannot take — walked traversal

ring area proportional to BAG registered footprint · 12,905 m² (V02) to 120 m² (V06)

ELEVEN CUTS

the walked traversal as an edit list · bar length = minutes on foot · tone = detour ratio

TOTAL 180 MIN SHORTEST 1.8 (V01–V02) LONGEST 66.1 (V08–V09) RANGE 1 : 37

01. MONTAGE ADJACENCY

[ ]

Desigualdades de adyacencia · Rotterdam V01–V12 · EPSG:28992

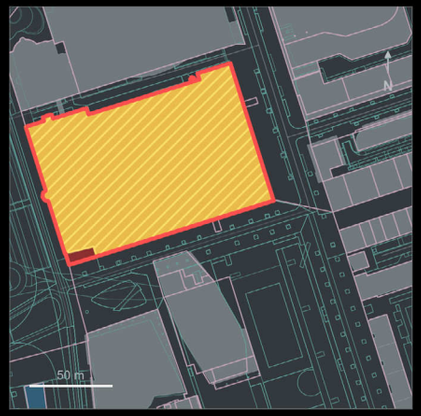
V01–V12 · Parcel- and Morphology-Following Hosting Fields

Existing venue building
Candidate Hosting Plot
BRK cadastral boundary
BGT physical surface edge

DRAFT 02 — every red boundary inherits a BRK parcel or the existing venue footprint; no rectangles, buffers, or growth geometry.

V01 De Doelen

BRK host parcel + footprint · 11,697 m²

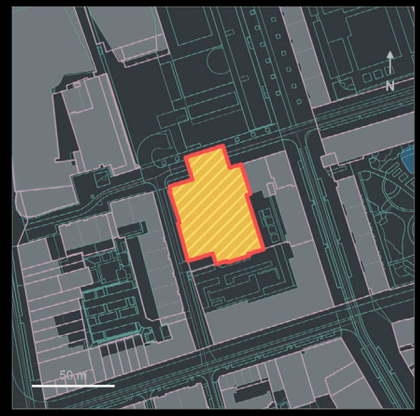


V02 Pathé Schouwburgplein

BRK host parcel + footprint · 3,191 m²

V03 Theater Rotterdam

BRK host parcel + footprint · 2,804 m²

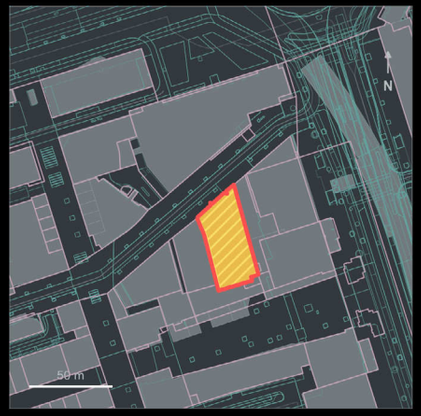


V04 Cinerama Filmtheater

BRK host parcel + footprint · 2,118 m²

V05 Oude Luxor

BRK host parcel + footprint · 1,422 m²

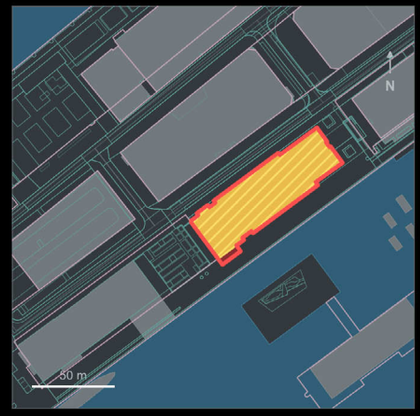


V06 KINO

BRK host parcel + footprint · 2,179 m²

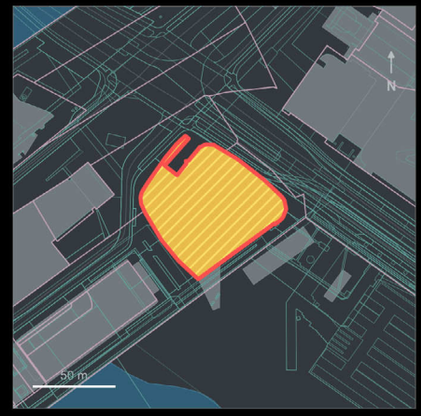
V07 LantarenVenster

BRK host parcel + footprint · 2,945 m²



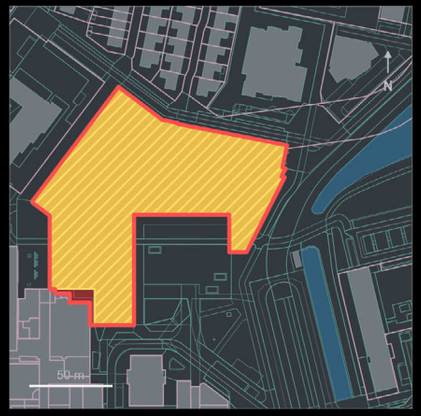
V08 Nieuwe Luxor

Venue footprint fallback · 4,535 m²



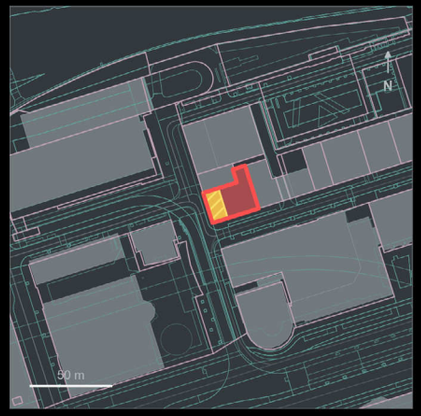
V09 Podium Islemunda

BRK host parcel + footprint · 11,484 m²



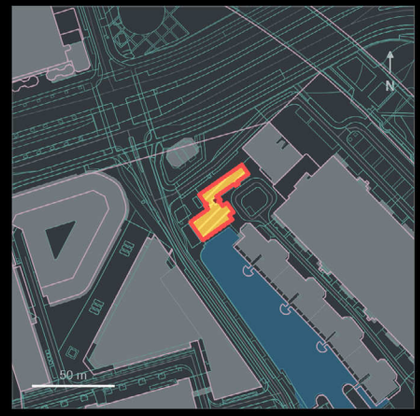
V10 Roodkapje

BRK host parcel + footprint · 653 m²



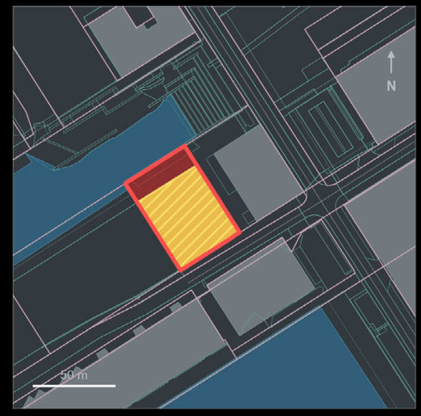
V11 Muziekwerf

BRK host parcel + footprint · 595 m²



V12 Katoenhuis

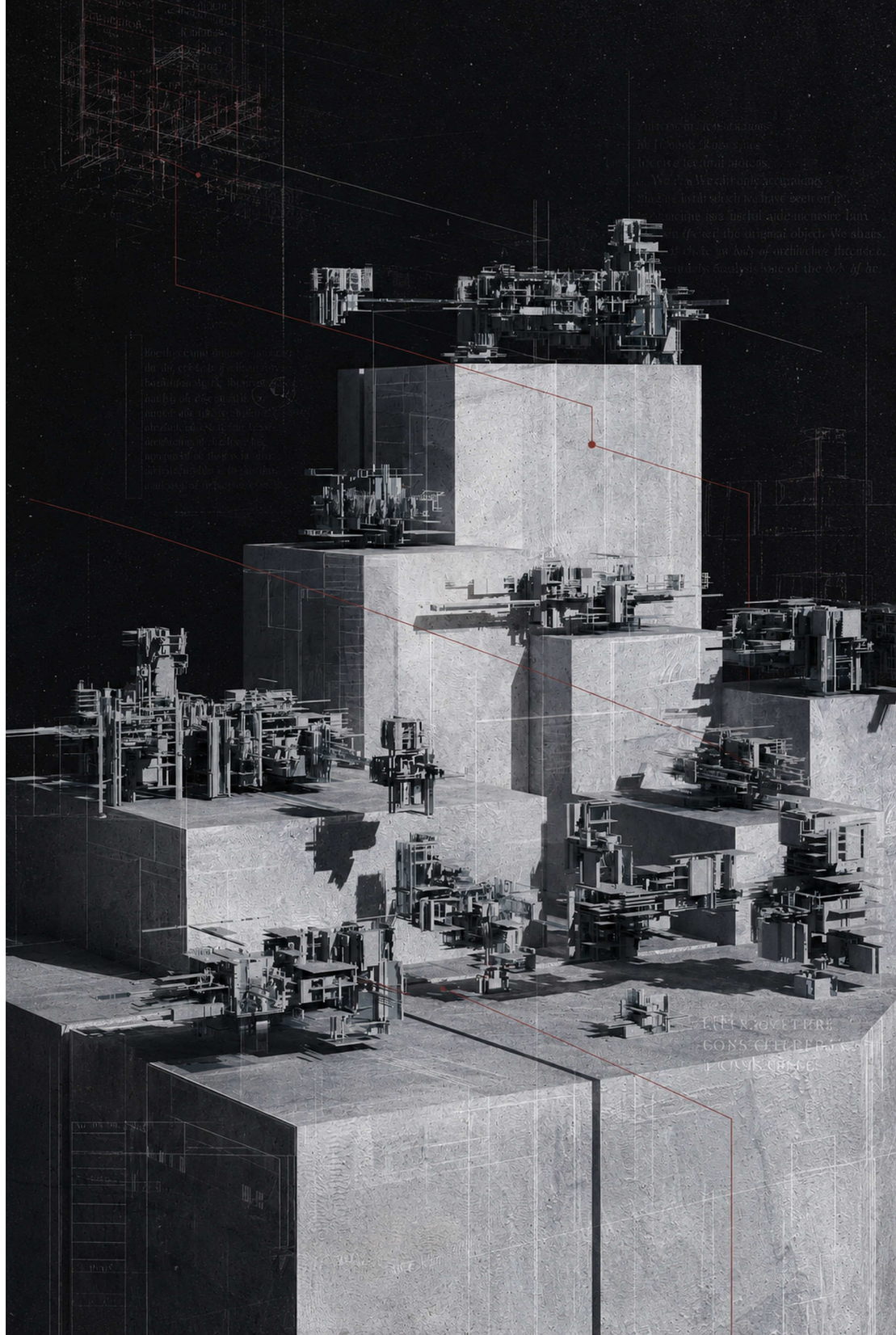
BRK host parcel + footprint · 2,793 m²



Boundary hierarchy: cadastral plot first · shared or excessively large parcel → existing venue footprint · adjacent BGT space is not added until access is verified.

Constructing the Urban Field

Following the festival-based site selection, 12 locations across Rotterdam were identified as potential intervention sites. Using the curatorial logic introduced in the previous chapter, each site is now assigned a specific combination of generated sequences. This process defines how the sequences are distributed, grouped and organised across the 12 locations, allowing each site to develop its own spatial composition while remaining part of the same overall system.

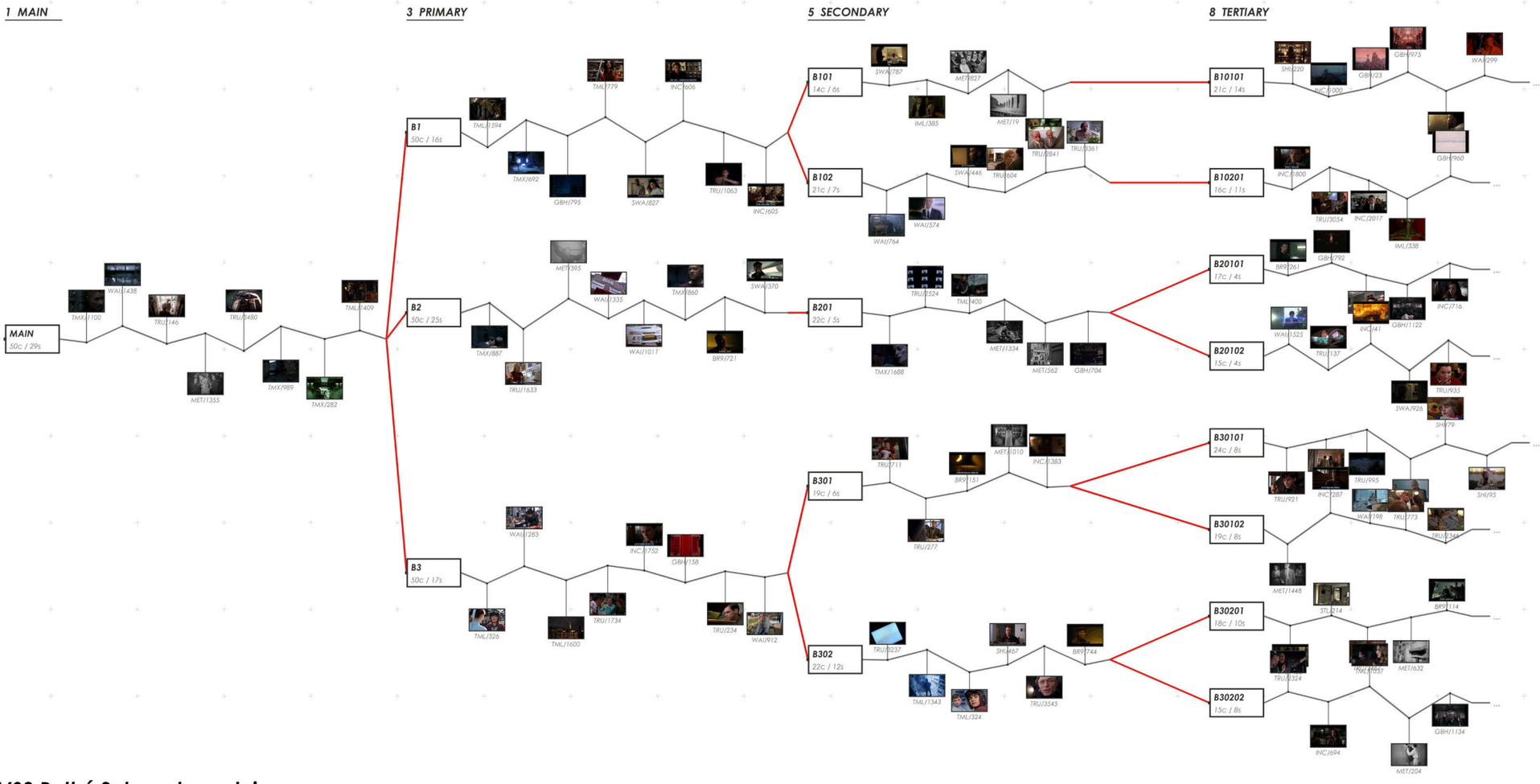


Sequences are determined according to the curatorial logic

Following the curatorial logic, each sequence is further divided into a series of branches, generating multiple possible spatial developments from the same sequence. Selected film keyframes are also included to show the cinematic sources and visual conditions associated with these branching points

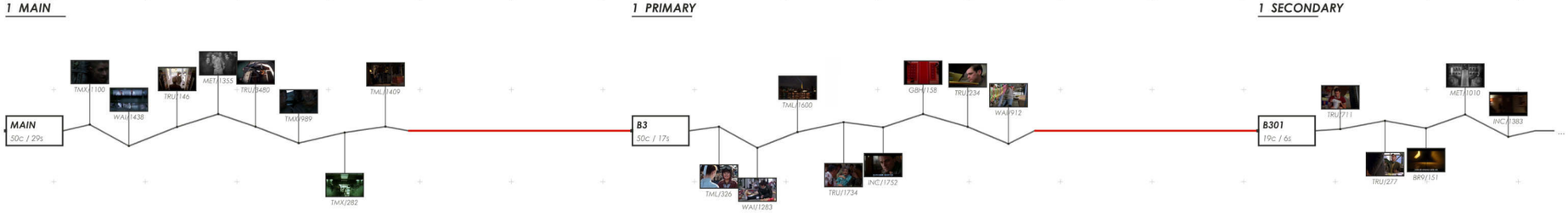
V01 De Doelen & De Doelen Studios

MAX A · 443 clips · 17 sequences · 190 segments



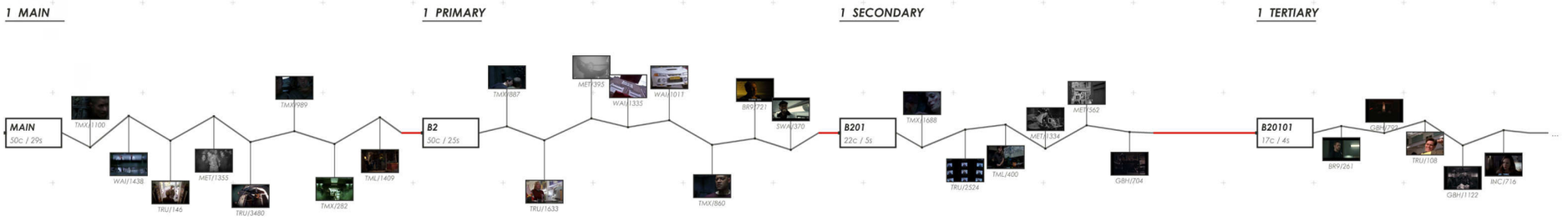
V02 Pathé Schouwburgplein

MAX A · 119 clips · 3 sequences · 52 segments



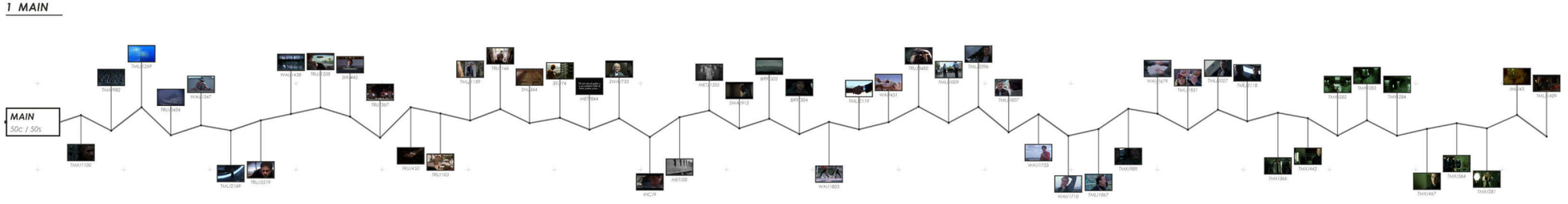
V03 Theater Rotterdam

MAX A · 139 clips · 4 sequences · 63 segments



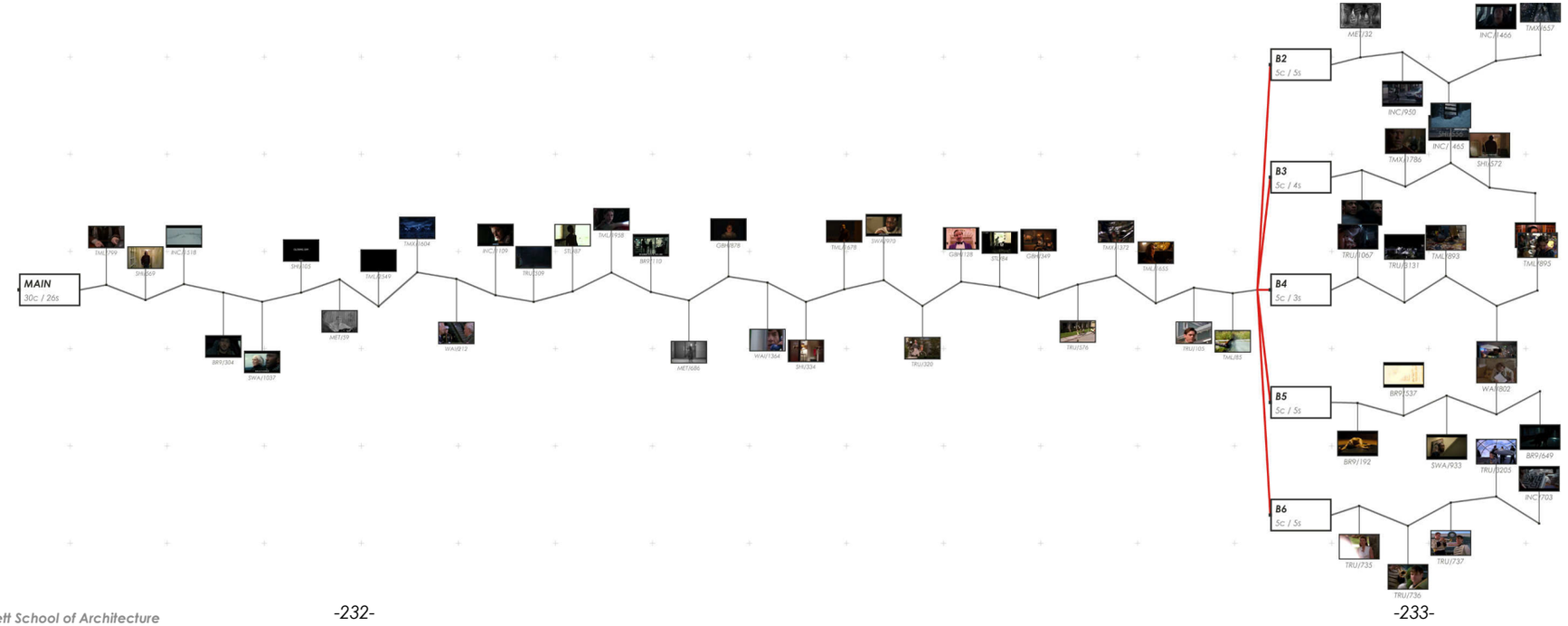
V04 Cinerama Filmtheater

MAX A · 65 clips · 8 sequences · 62 segments



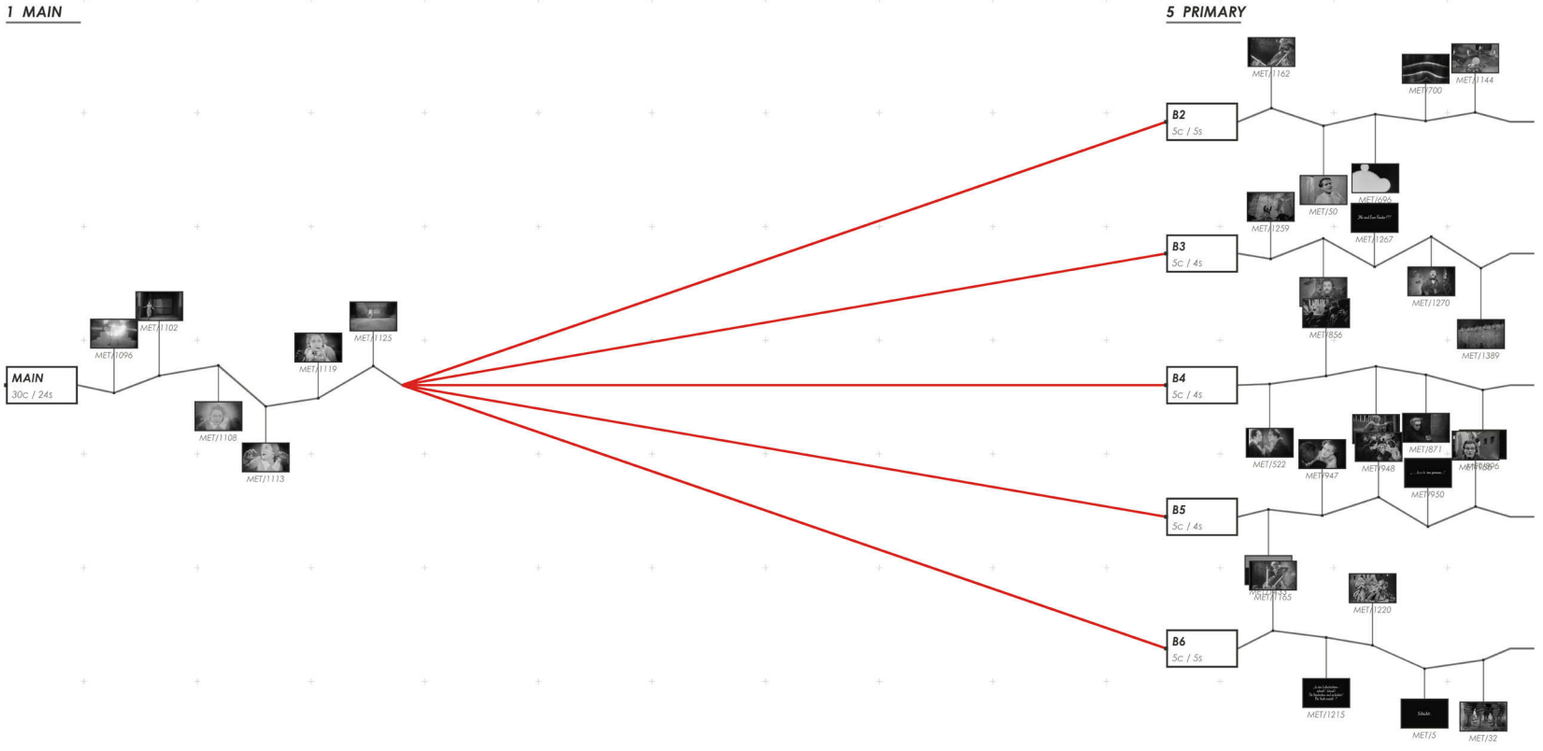
V05 Oude Luxor

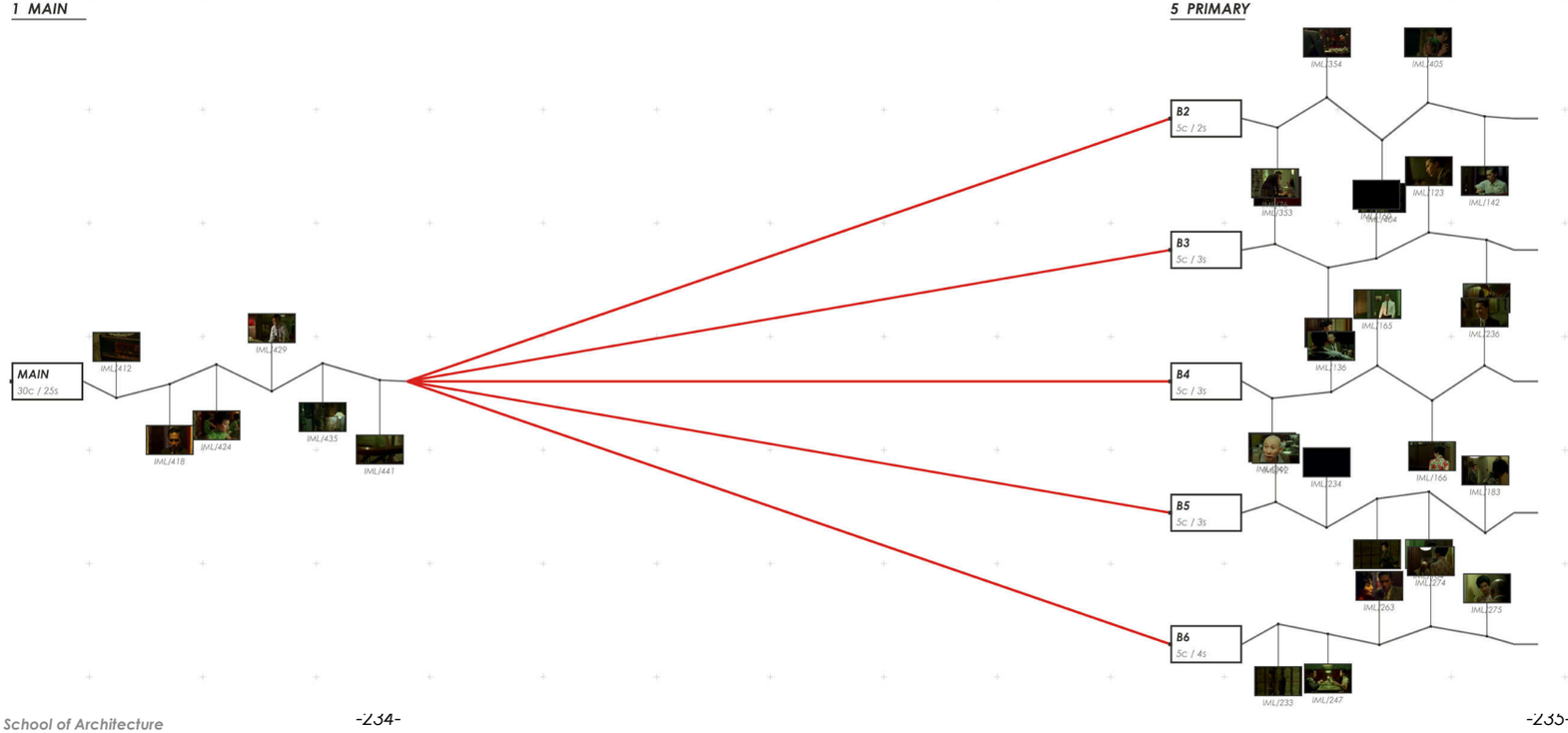
MAX A · 55 clips · 6 sequences · 52 segments



V06 KINO

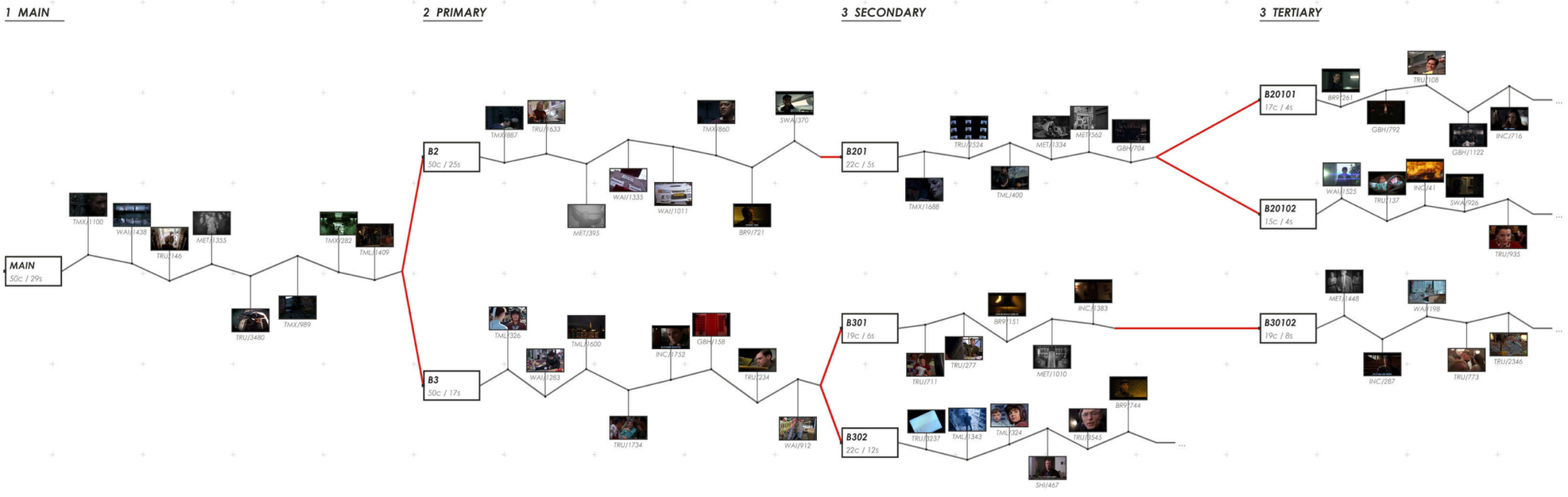
MAX B · 55 clips · 6 sequences · 51 segments





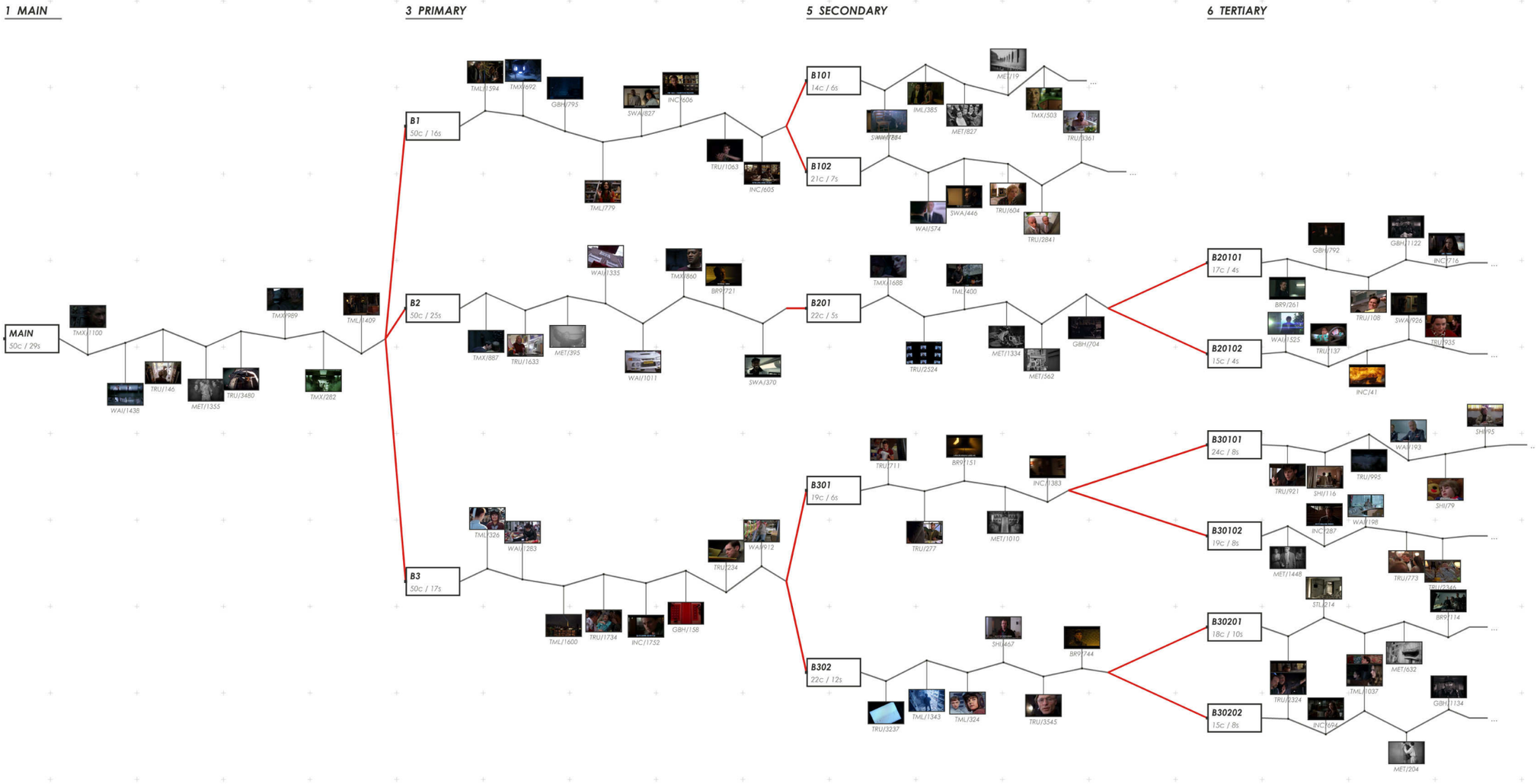
V08 Nieuwe Luxor

MAX A · 264 clips · 9 sequences · 110 segments

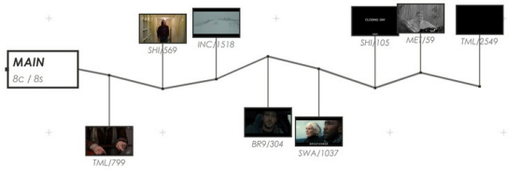


V09 Podium Islemunda

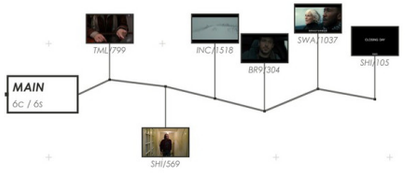
MAX A · 406 clips · 15 sequences · 165 segments



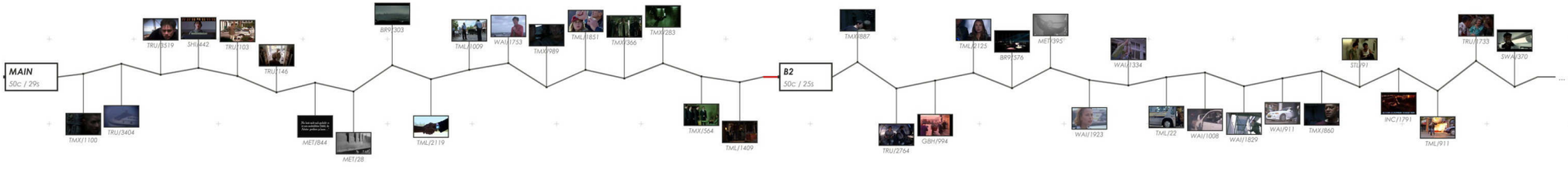
V10 Roodkapje
MAX A ·8 clips ·1 sequences ·8 segments



V11 Muziekwerf
MAX A ·6 clips ·1 sequences ·6 segments



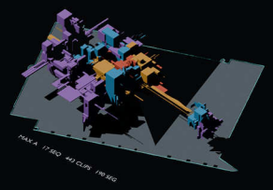
V12 Katoenhuis
MAX A ·100 clips ·2 sequences ·54 segments



Spatial Skeletons

Sequence-Generated Spatial Skeletons for the 12 Sites

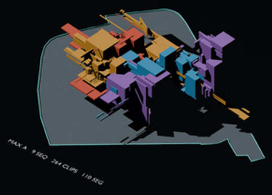
V01 De Doelen & De Doelen Studios
MAX A: 17 seq / 443 clips



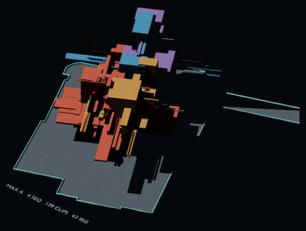
V02 Pathé Schouwburgplein
MAX A: 3 seq / 119 clips

V07 LantarenVenster
MAX A: 6 seq / 55 clips | MAX B: 6 seq / 55 clips

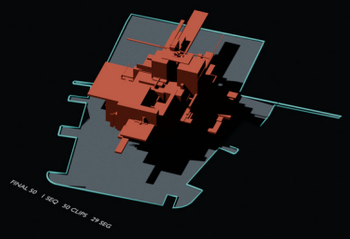
V08 Nieuwe Luxor
MAX A: 9 seq / 264 clips



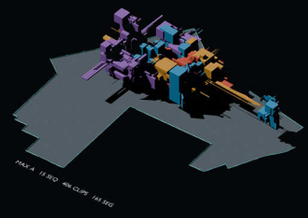
V03 Theater Rotterdam
MAX A: 4 seq / 139 clips



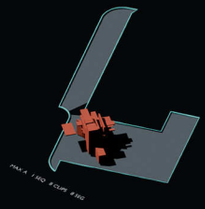
V04 Cinerama Filmtheater
MAX A: 1 seq / 50 clips



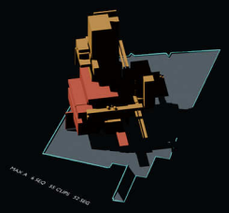
V09 Podium Islemunda
MAX A: 15 seq / 406 clips



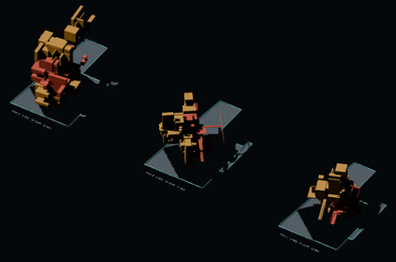
V10 Roodkapje
MAX A: 1 seq / 8 clips



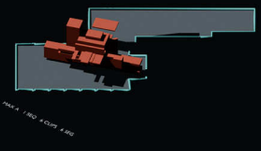
V05 Oude Luxor
MAX A: 6 seq / 55 clips



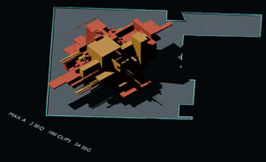
V06 KINO
MAX A: 6 seq / 55 clips | MAX B: 6 seq / 55 clips | MAX C: 6 seq / 55 clips

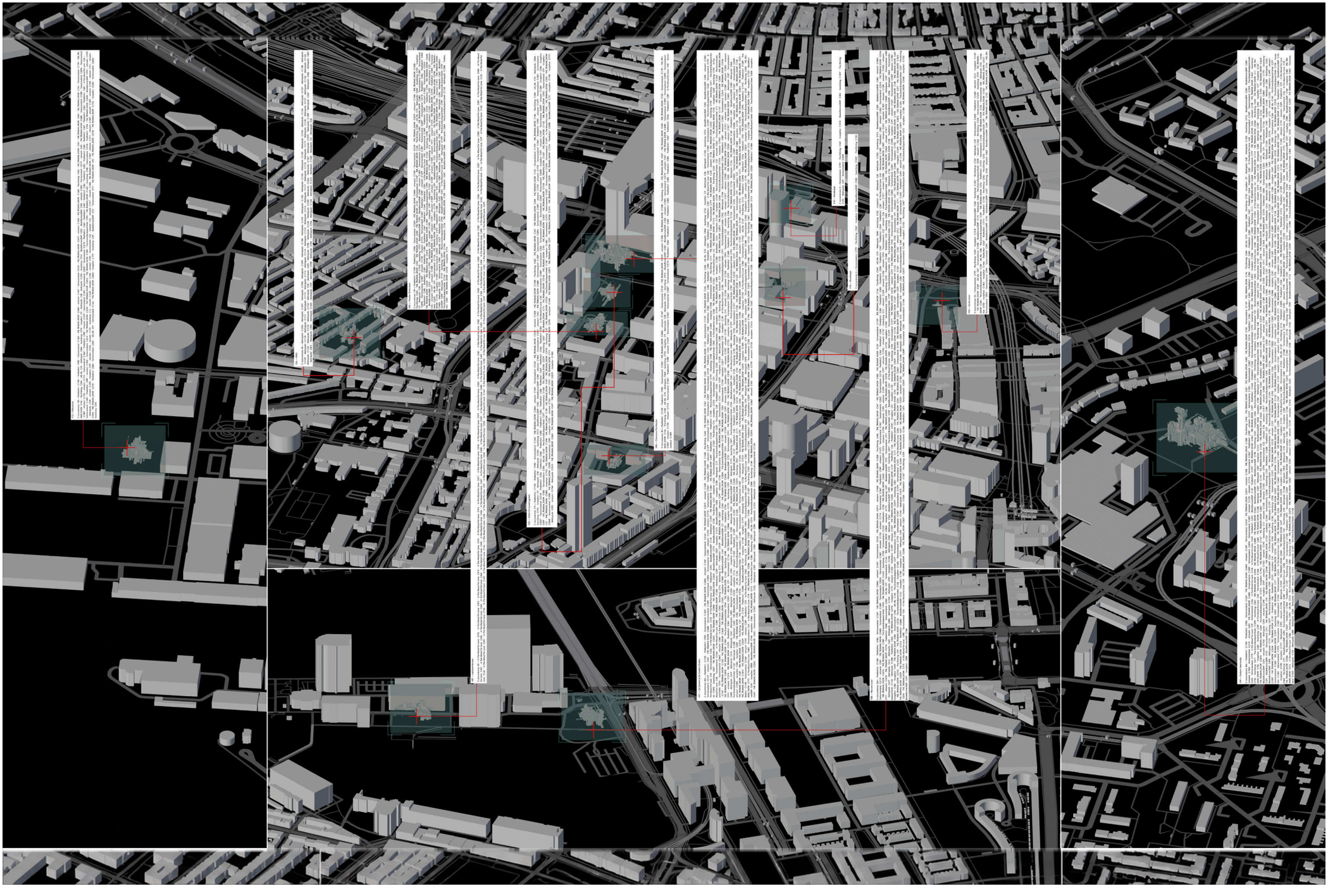


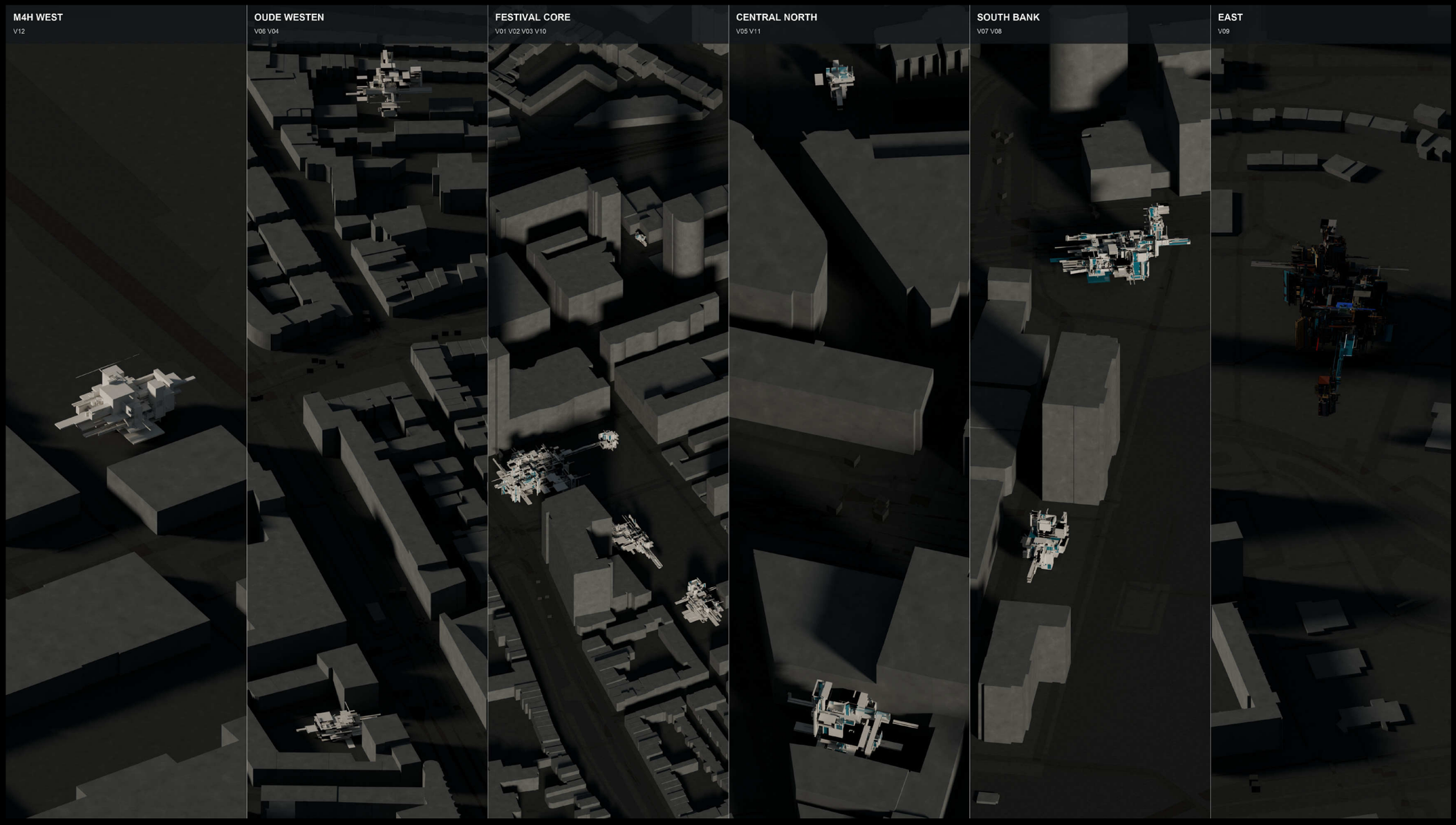
V11 Muziekwerf
MAX A: 1 seq / 6 clips



V12 Katoenhuis
MAX A: 2 seq / 100 clips



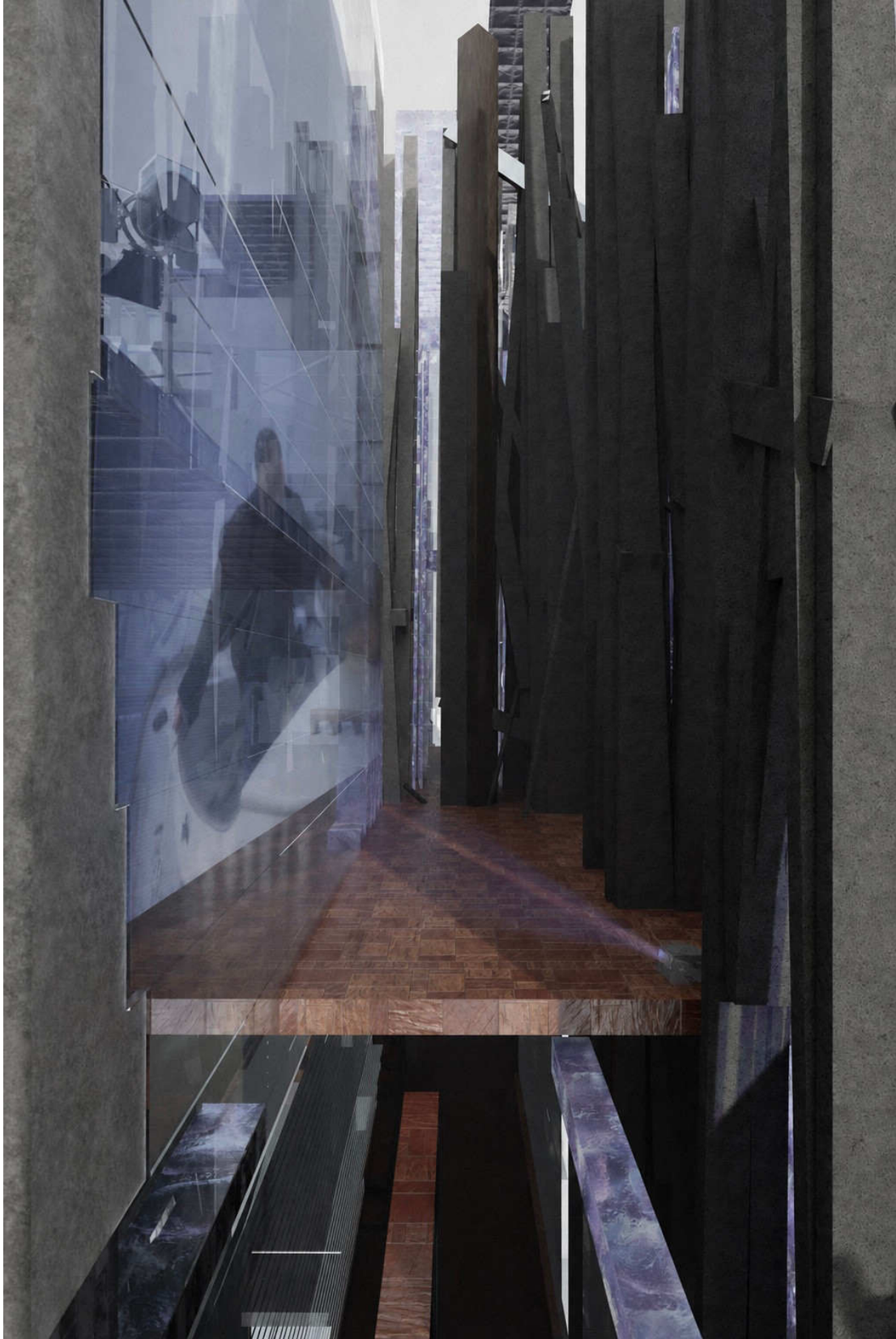



M4H WEST
V12
OUDE WESTEN
V06 V04
FESTIVAL CORE
V01 V02 V03 V10
CENTRAL NORTH
V05 V11
SOUTH BANK
V07 V08
EAST
V09

Spatial Encounters

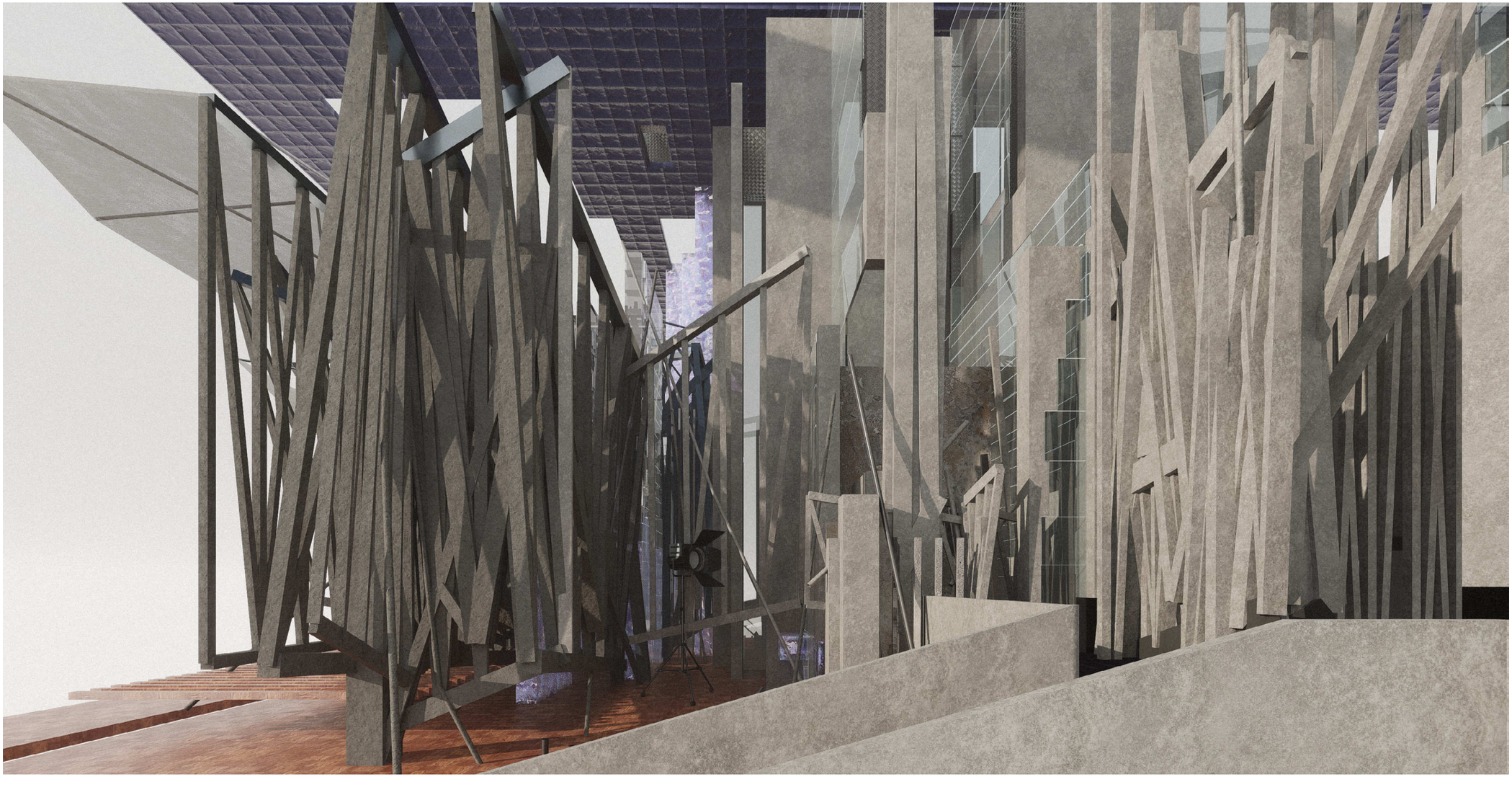
This part takes the target cluster of the v04 site as the main visual display object of the venue, hoping to display the spatial encounters of the Kurishov effect in space through this part.

Spatial Encounters presents the final spatial experience produced through the sequence-based design system. The generated structures are shown through a series of views that reveal changing thresholds, overlaps, openings, materials and spatial depth. These images focus on how the cinematic logic developed throughout the project is ultimately experienced as architecture, translating edited sequences into continuous spatial encounters.



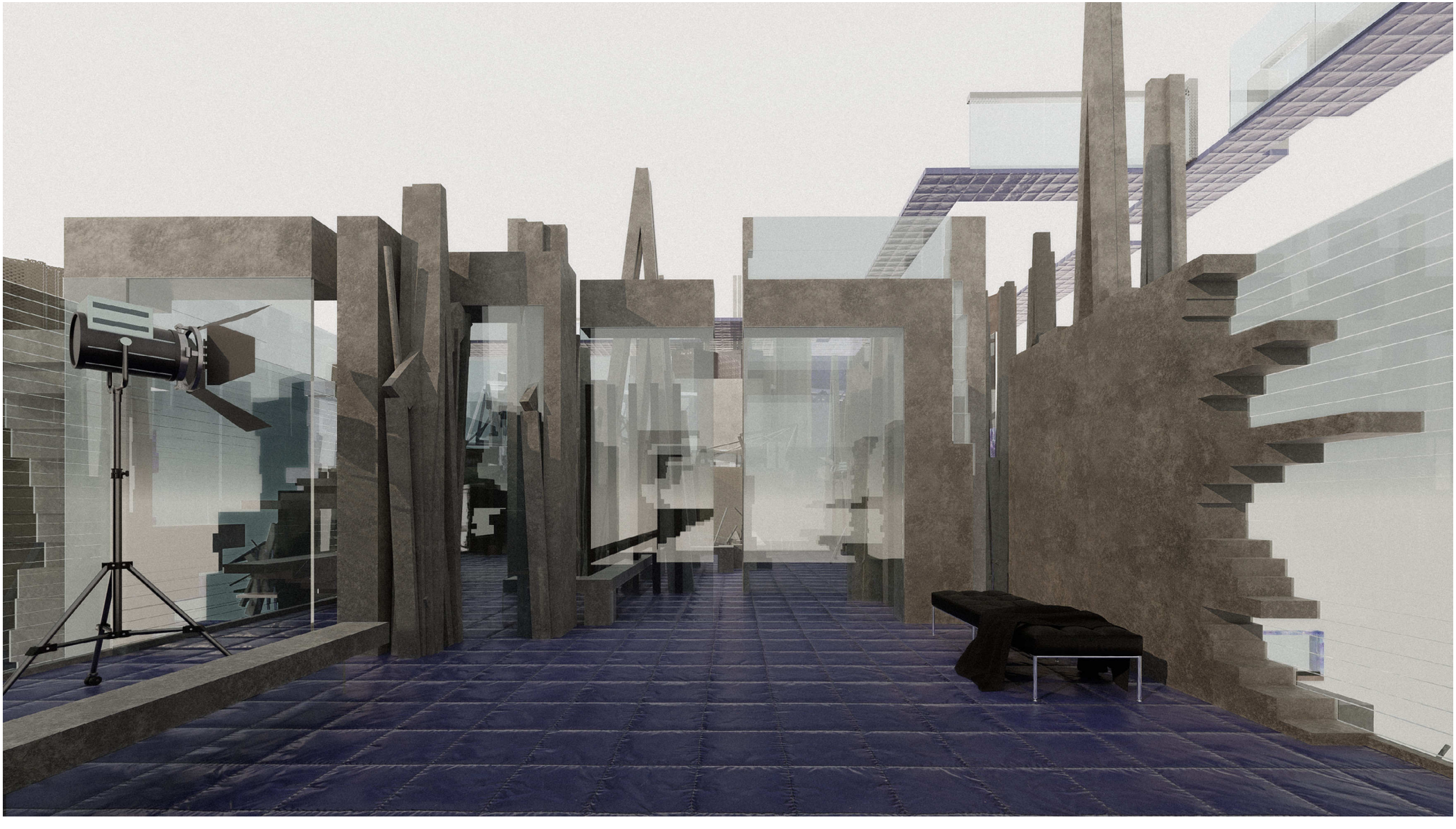
This view shows how the generated sequence, derived from individual film clips, is experienced as a layered architectural condition. Spatial characteristics extracted from the clips are translated into overlapping walls, glazing, structural fragments and changing thresholds. As the viewer moves through the space, different clip-based fragments appear, disappear and recombine, turning the original cinematic sequence into a continuous architectural experience.

This view presents the complete spatial result developed from the generated sequence, through the skeletal model and architecturalisation process, to the final application of materials and atmospheres extracted from the original film clips.

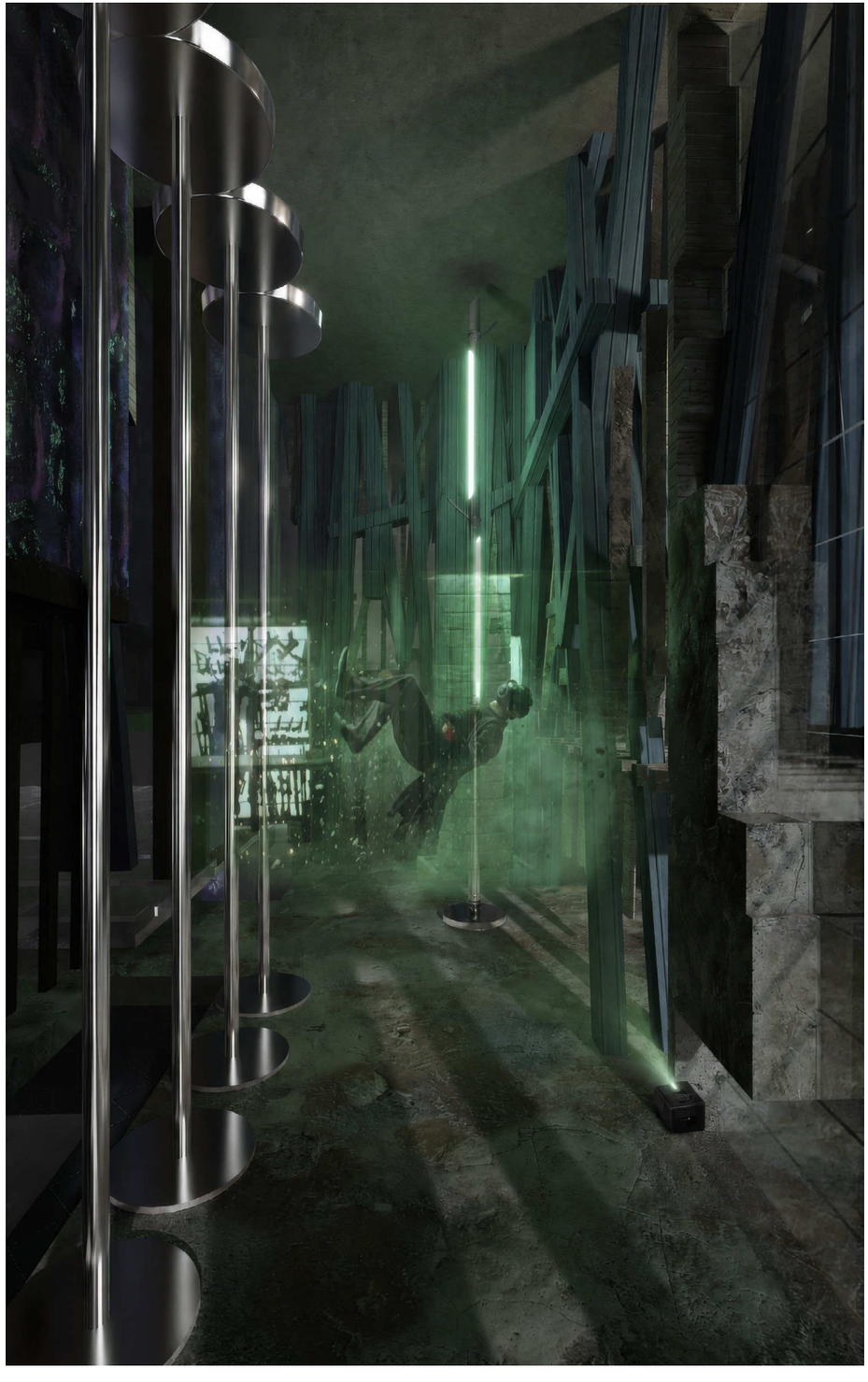


The layered structures, thresholds and material fragments reveal how cinematic information is gradually transformed into an architectural spatial condition.

This view shows the final architectural space produced through the full sequence-to-space workflow. The initial clip relationships are translated into a spatial skeleton, further articulated through walls, glazing, openings and architectural elements, and finally materialised using colours and textures extracted from the source film clips.



The result combines sequence, spatial transformation and cinematic atmosphere within one continuous architectural environment.



Final Kuleshov Spatial Experience

Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQDfZncXEOpkT6DCmamXES2yAfdw-wS3G-ohs62o08Z0g-Nk?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=55V1gZ

Final Kuleshov Spatial Experience

Link to the full animation
https://liveuclac-my.sharepoint.com/:v:/g/personal/zczqcg1_ucl_ac_uk/IQDfZncXEOpkT6DCmamXES2yAfdw-wS3G-ohs62o08Z0g-Nk?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=55V1gZ